

Library of Linux

Unified RHCSA/RHCE Linux Command Knowledge Library

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1 FINAL MASTER RHCSA / Linux Command Knowledge Library

AOIA-compatible deduplicated command knowledge base generated from local source files only.

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- File Management
- Users & Permissions
- Networking
- Storage
- Systemd
- Processes
- Bash
- SSH
- Logs
- Security
- Packages
- Automation
- RHCSA Exam Tasks

1.2 Build Metadata

- Files processed: 7
- Total command candidates found: 4205
- Unique command entries: 2559
- Duplicates removed: 1646
- Sections created: 13

1.3 Source Files

- LINUX COMMAND
- RHCSA_Command_Library.pdf
- encyklopedia_inzynierii_linux.html
- encyklopedia_inzynierii_linux.pdf
- komendy
- linux_engineering_book.md
- linux_engineering_book.pdf

1.4 File Management

1.4.1 `basename`

Nazwa pliku ze ścieżki

`basename /etc/hosts`

Sources: LINUX COMMAND

1.4.2 `cat`

Wyświetla zawartość pliku

`cat /etc/hosts`

`cat`

Sources: LINUX COMMAND, RHCSA_Command_Library.pdf

1.4.3 `cat -A file.txt`

Sekcja Komend

```
cat -A file.txt
```

Sources: RHCSA_Command_Library.pdf

1.4.4 `cat -n file`

number lines. **SAFE**

```
cat -n file
```

Sources: komendy

1.4.5 `cat -n file.txt`

Sekcja Komend

```
cat -n file.txt
```

Sources: RHCSA_Command_Library.pdf

1.4.6 `cat /etc/anacrontab`

Sekcja Komend

```
cat /etc/anacrontab
```

Sources: RHCSA_Command_Library.pdf

1.4.7 `cat /etc/bashrc`

Sekcja Komend

```
cat /etc/bashrc
```

Sources: RHCSA_Command_Library.pdf

1.4.8 `cat /etc/cron.allow`

Sekcja Komend

```
cat /etc/cron.allow
```

Sources: RHCSA_Command_Library.pdf

1.4.9 `cat /etc/cron.deny`

Sekcja Komend

```
cat /etc/cron.deny
```

Sources: RHCSA_Command_Library.pdf

1.4.10 `cat /etc/crontab`

system crontab. **SAFE**

```
cat /etc/crontab
```

Sources: RHCSA_Command_Library.pdf, komendy

1.4.11 `cat /etc/exports`

Sekcja Komend

`cat /etc/exports`

Sources: RHCSA_Command_Library.pdf

1.4.12 `cat /etc/fstab`

Trwa montowanie Samba w fstab systemctl restart Zrestartuj usługi Samba

`cat /etc/fstab`

Sources: RHCSA_Command_Library.pdf

1.4.13 `cat /etc/os-release`

canonical OS ID. **SAFE**

`cat /etc/os-release`

Sources: RHCSA_Command_Library.pdf, komendy

1.4.14 `cat /etc/profile`

Sekcja Komend

`cat /etc/profile`

Sources: RHCSA_Command_Library.pdf

1.4.15 `cat /etc/systemd/sys`

Sekcja Komend

`cat /etc/systemd/sys`

Sources: RHCSA_Command_Library.pdf

1.4.16 `cat /proc/[pid]/oom_score` # wyższy = bardziej prawdopodobne zabicie

Sprawdzenie OOM score procesu

`cat /proc/[pid]/oom_score` # *wyższy = bardziej prawdopodobne zabicie*

Sources: linux_engineering_book.md, linux_engineering_book.pdf

1.4.17 `cat /proc/[pid]/oom_score_adj` # -1000 do 1000 (ręczna korekta)

Sprawdzenie OOM score procesu

`cat /proc/[pid]/oom_score_adj` # *-1000 do 1000 (ręczna korekta)*

Sources: linux_engineering_book.md, linux_engineering_book.pdf

1.4.18 `cat /proc/cmdline`

Sekcja Komend

`cat /proc/cmdline`

Sources: RHCSA_Command_Library.pdf

1.4.19 cat /proc/cpuinfo

Sekcja Komend

```
cat /proc/cpuinfo
```

Sources: RHCSA_Command_Library.pdf

1.4.20 cat /proc/mdstat

Komentarz Komentarz w skrypcie function myfunc() { Definicja funkcji

```
cat /proc/mdstat
```

Sources: RHCSA_Command_Library.pdf

1.4.21 cat /proc/mdstat

18.1 RAID (mdadm)

```
cat /proc/mdstat #
```

Sources: linux_engineering_book.md, linux_engineering_book.pdf

1.4.22 cat /proc/meminfo

Informacje o pamięci

```
cat /proc/meminfo
```

Sources: RHCSA_Command_Library.pdf, linux_engineering_book.md, linux_engineering_book.pdf

1.4.23 cat /proc/meminfo | grep

5.2 Strony Pamięci

```
cat /proc/meminfo | grep
```

Sources: linux_engineering_book.md

1.4.24 cat /proc/mounts

Sekcja Komend

```
cat /proc/mounts
```

Sources: RHCSA_Command_Library.pdf

1.4.25 cat /proc/net/tcp

Command — klawisz : Tryb komend # Szukaj sowa pod kursorem (do tyu)

```
cat /proc/net/tcp
```

Sources: RHCSA_Command_Library.pdf

1.4.26 cat /proc/net/udp

Command — klawisz : Tryb komend # Szukaj sowa pod kursorem (do tyu)

```
cat /proc/net/udp
```

Sources: RHCSA_Command_Library.pdf

1.4.27 `cat /proc/sys/kernel`

Sekcja Komend

`cat /proc/sys/kernel`

Sources: RHCSA_Command_Library.pdf

1.4.28 `cat /proc/version`

Sekcja Komend

`cat /proc/version`

Sources: RHCSA_Command_Library.pdf

1.4.29 `cat /sys/block/sda/queue/scheduler`

14.4 Tuning Systemowy

`cat /sys/block/sda/queue/scheduler`

Sources: linux_engineering_book.md, linux_engineering_book.pdf

1.4.30 `cat /sys/devices/system/cpu/cpu0/cpufreq/scaling_governor`

14.4 Tuning Systemowy

`cat /sys/devices/system/cpu/cpu0/cpufreq/scaling_governor`

Sources: linux_engineering_book.md, linux_engineering_book.pdf

1.4.31 `cat /sys/fs/cgroup/m`

Komentarz Komentarz w skrypcie function myfunc() { Definicja funkcji

`cat /sys/fs/cgroup/m`

Sources: RHCSA_Command_Library.pdf

1.4.32 `cat /sys/fs/cgroup/mygroup/memory.pressure`

25.2 Cgroups v2 — Zaawansowane

`cat /sys/fs/cgroup/mygroup/memory.pressure`

Sources: linux_engineering_book.md, linux_engineering_book.pdf

1.4.33 `cat /sys/kernel/mm/transparent_hugepage/enabled`

Transparent Huge Pages

`cat /sys/kernel/mm/transparent_hugepage/enabled`

Sources: linux_engineering_book.md, linux_engineering_book.pdf

1.4.34 `cat /var/log/audit/a`

Sekcja Komend

`cat /var/log/audit/a`

Sources: RHCSA_Command_Library.pdf

1.4.35 cat /var/log/cron

Sekcja Komend

```
cat /var/log/cron
```

Sources: RHCSA_Command_Library.pdf

1.4.36 cat /var/log/maillog

Sekcja Komend

```
cat /var/log/maillog
```

Sources: RHCSA_Command_Library.pdf

1.4.37 cat /var/log/secure

Sekcja Komend

```
cat /var/log/secure
```

Sources: RHCSA_Command_Library.pdf

1.4.38 cat /var/spool/cron/

Sekcja Komend

```
cat /var/spool/cron/
```

Sources: RHCSA_Command_Library.pdf

1.4.39 cat <<EOF > config.txt

10.5 Zaawansowany Bash

```
cat <<EOF > config.txt
```

Sources: linux_engineering_book.md, linux_engineering_book.pdf

1.4.40 cat file.txt

Sekcja Komend

```
cat file.txt
```

Sources: RHCSA_Command_Library.pdf

1.4.41 cat file1 file2

Sekcja Komend

```
cat file1 file2
```

Sources: RHCSA_Command_Library.pdf

1.4.42 cat trace

15.2 Diagnostyka Jądra

```
cat trace
```

Sources: linux_engineering_book.md, linux_engineering_book.pdf

1.4.43 cat urls.txt

parallel -j 10 wget

```
cat urls.txt
```

Sources: linux_engineering_book.pdf

1.4.44 cat urls.txt | parallel -j 10 wget

28.1 Bash — Zaawansowane Techniki

```
cat urls.txt | parallel -j 10 wget
```

Sources: linux_engineering_book.md

1.4.45 cat ~/.bash_logout

Sekcja Komend

```
cat ~/.bash_logout
```

Sources: RHCSA_Command_Library.pdf

1.4.46 cat ~/.bash_profile

Sekcja Komend

```
cat ~/.bash_profile
```

Sources: RHCSA_Command_Library.pdf

1.4.47 cat ~/.bashrc

Sekcja Komend

```
cat ~/.bashrc
```

Sources: RHCSA_Command_Library.pdf

1.4.48 cd

safer than in automation.

```
cd /var/www
```

```
cd
```

Sources: LINUX COMMAND, komendy

1.4.49 cd \$(dirname "\$0")

without symlink check.

```
cd $(dirname "$0")
```

Sources: komendy

1.4.50 cd \$(dirname "\$0") without symlink

Dangerous Mistakes

```
cd $(dirname "$0")` without symlink
```

Sources: komendy

1.4.51 `cd -`

previous directory. **SAFE**

`cd -`

Sources: RHCSA_Command_Library.pdf, komendy

1.4.52 `cd ../../`

Sekcja Komend

`cd ../../`

Sources: RHCSA_Command_Library.pdf

1.4.53 `cd /`

Sekcja Komend

`cd /`

Sources: RHCSA_Command_Library.pdf

1.4.54 `cd /path/to/dir`

Sekcja Komend

`cd /path/to/dir`

Sources: RHCSA_Command_Library.pdf

1.4.55 `cd /sys/kernel/debug/tracing`

15.2 Diagnostyka Jądra

`cd /sys/kernel/debug/tracing`

Sources: linux_engineering_book.md, linux_engineering_book.pdf

1.4.56 `cd linux-6.8`

16.1 Pobieranie i Konfiguracja

`cd linux-6.8`

Sources: linux_engineering_book.md, linux_engineering_book.pdf

1.4.57 `cd ~`

Sekcja Komend

`cd ~`

Sources: RHCSA_Command_Library.pdf

1.4.58 `cksum`

Suma kontrolna CRC

`cksum file.txt`

Sources: LINUX COMMAND

1.4.59 cmp

Porównuje bajt po bajcie

```
cmp file1 file2
```

Sources: LINUX COMMAND

1.4.60 cmp file1 file2

byte comparison. **SAFE**

```
cmp file1 file2
```

Sources: komendy

1.4.61 cp

Kopiuje plik

```
cp file1 file2
```

Sources: LINUX COMMAND

1.4.62 cp --backup src dst

Sekcja Komend

```
cp --backup src dst
```

Sources: RHCSA_Command_Library.pdf

1.4.63 cp --backup=numbered

for critical overwrites.

```
cp --backup=numbered
```

Sources: komendy

1.4.64 cp --backup=numbered for critical overwrites.

Production-Safe Patterns

```
cp --backup=numbered` for critical overwrites.
```

Sources: komendy

1.4.65 cp -a

Archiwizuje (rekurencyjnie + atrybuty)

```
cp -a /home /backup
```

Sources: LINUX COMMAND

1.4.66 cp -a / /mnt/backup

without exclusions.

```
cp -a / /mnt/backup
```

Sources: komendy

1.4.67 `cp -a / /mnt/backup` without exclusions:

Dangerous Mistakes

```
cp -a / /mnt/backup` without exclusions.
```

Sources: komendy

1.4.68 `cp -a src/ dest/`

archive mode (preserves xattrs, links). **CAUTION** (overwrites)

```
cp -a src/ dest/
```

Sources: komendy

1.4.69 `cp -a src/ dst/`

Sekcja Komend

```
cp -a src/ dst/
```

Sources: RHCSA_Command_Library.pdf

1.4.70 `cp -i src dst`

Sekcja Komend

```
cp -i src dst
```

Sources: RHCSA_Command_Library.pdf

1.4.71 `cp -p`

Zachowuje atrybuty

```
cp -p file1 file2
```

Sources: LINUX COMMAND

1.4.72 `cp -p src dst`

Sekcja Komend

```
cp -p src dst
```

Sources: RHCSA_Command_Library.pdf

1.4.73 `cp -r`

Kopiuje katalog rekurencyjnie

```
cp -r /etc /backup
```

Sources: LINUX COMMAND

1.4.74 `cp -r src/ dst/`

Sekcja Komend

```
cp -r src/ dst/
```

Sources: RHCSA_Command_Library.pdf

1.4.75 `cp -u src dst`

Sekcja Komend

`cp -u src dst`

Sources: RHCSA_Command_Library.pdf

1.4.76 `cp -v src dst`

Sekcja Komend

`cp -v src dst`

Sources: RHCSA_Command_Library.pdf

1.4.77 `cp /boot/config-$(uname -r) .config`

Kopiowanie aktualnej konfiguracji

`cp /boot/config-$(uname -r) .config`

Sources: linux_engineering_book.md, linux_engineering_book.pdf

1.4.78 `cp source dest`

Sekcja Komend

`cp source dest`

Sources: RHCSA_Command_Library.pdf

1.4.79 `csplit`

Dzieli plik wg wzorca

`csplit file '/pattern/'`

Sources: LINUX COMMAND

1.4.80 `df`

Wolne miejsce na filesystemach

`df -h`

Sources: LINUX COMMAND

1.4.81 `df -h`

disk full check. **SAFE**

`df -h`

Sources: RHCSA_Command_Library.pdf, komendy

1.4.82 `df -hT`

disk usage by filesystem type. **SAFE**

`df -hT`

Sources: RHCSA_Command_Library.pdf, komendy

1.4.83 df -hT # zamontowane

11.4 Zarządzanie Dyskami

```
df -hT # zamontowane
```

Sources: linux_engineering_book.md, linux_engineering_book.pdf

1.4.84 df -i

Wolne inody

```
df -i
```

Sources: LINUX COMMAND

1.4.85 diff

Verification step: expected vs actual state before declaring success.

```
diff file1 file2
diff
```

Sources: LINUX COMMAND, komendy

1.4.86 diff -Naur

Generuje patch

```
diff -Naur old/ new/ > fix.patch
```

Sources: LINUX COMMAND

1.4.87 diff -qr dir1 dir2

recursive quick diff. **SAFE**

```
diff -qr dir1 dir2
```

Sources: komendy

1.4.88 diff -u

Unified diff

```
diff -u file1 file2
```

Sources: LINUX COMMAND

1.4.89 diff -u file1 file2

unified diff. **SAFE**

```
diff -u file1 file2
```

Sources: komendy

1.4.90 diff <(ls -la dir1) <(ls -la dir2)

Dodatek A: Przydatne Jednowierszowce

```
diff <(ls -la dir1) <(ls -la dir2)
```

Sources: linux_engineering_book.md, linux_engineering_book.pdf

1.4.91 diff <(ls dir1) <(ls dir2)

10.5 Zaawansowany Bash

```
diff <(ls dir1) <(ls dir2)
```

Sources: linux_engineering_book.md, linux_engineering_book.pdf

1.4.92 dirname

Katalog ze ścieżki

```
dirname /etc/hosts
```

Sources: LINUX COMMAND

1.4.93 dirs

Pokazuje stos katalogów

```
dirs -v  
dirs
```

Sources: LINUX COMMAND, RHCSA_Command_Library.pdf

1.4.94 du

Użycie dysku przez pliki/katalogi

```
du -sh /var/log
```

Sources: LINUX COMMAND

1.4.95 du -ah

Szczegółowo wszystkie pliki

```
du -ah /home \
```

Sources: LINUX COMMAND

1.4.96 du -h --max-depth=1 /var

per-subdirectory size. **SAFE**

```
du -h --max-depth=1 /var
```

Sources: komendy

1.4.97 du -sh /*

sort -h

```
tail -20
```

Sources: linux_engineering_book.pdf

1.4.98 du -sh /* | sort -h | tail -20

Dodatek B: Komendy Ratunkowe

```
du -sh /* | sort -h | tail -20
```

Sources: linux_engineering_book.md

1.4.99 `du -sh /var/*` # zużycie dysku

11.4 Zarządzanie Dyskami

`du -sh /var/*` # zużycie dysku

Sources: linux_engineering_book.md, linux_engineering_book.pdf

1.4.100 `du -sh /var/log`

directory total size. **SAFE**

`du -sh /var/log`

Sources: komendy

1.4.101 `fallocate -l 10G disk.img`

fast allocation. **SAFE**

`fallocate -l 10G disk.img`

Sources: komendy

1.4.102 `fallocate -l 4G /swapfile`

Tworzenie swap

`fallocate -l 4G /swapfile`

Sources: linux_engineering_book.md, linux_engineering_book.pdf

1.4.103 `file`

Komentarz Komentarz w skrypcie function myfunc() { Definicja funkcji

`file /bin/bash`

`file`

Sources: LINUX_COMMAND, RHCSA_Command_Library.pdf

1.4.104 `file '/start/,/end/p'`

Sekcja Komend

`file '/start/,/end/p'`

Sources: RHCSA_Command_Library.pdf

1.4.105 `file -b /bin/ls`

file type without filename. **SAFE**

`file -b /bin/ls`

Sources: komendy

1.4.106 `file -out file.enc`

Komentarz Komentarz w skrypcie function myfunc() { Definicja funkcji

`file -out file.enc`

Sources: RHCSA_Command_Library.pdf

1.4.107 file /etc/passwd

Sekcja Komend

`file /etc/passwd`

Sources: RHCSA_Command_Library.pdf

1.4.108 file 2>/dev/null

Sekcja Komend

`file 2>/dev/null`

Sources: RHCSA_Command_Library.pdf

1.4.109 file chronyd

Komentarz Komentarz w skrypcie function myfunc() { Definicja funkcji

`file chronyd`

Sources: RHCSA_Command_Library.pdf

1.4.110 file file

Sekcja Komend

`file file`

Sources: RHCSA_Command_Library.pdf

1.4.111 file grup

Sekcja Komend

`file grup`

Sources: RHCSA_Command_Library.pdf

1.4.112 file kadej linii

Sekcja Komend

`file kadej linii`

Sources: RHCSA_Command_Library.pdf

1.4.113 file regularnego

Sekcja Komend

`file regularnego`

Sources: RHCSA_Command_Library.pdf

1.4.114 file symboliczne u:user:rwx file

Sekcja Komend

`file symboliczne u:user:rwx file`

Sources: RHCSA_Command_Library.pdf

1.4.115 find

Wyszukiwanie plików

```
find / -name "*.conf"
```

Sources: LINUX COMMAND

1.4.116 find -exec

Wykonuje komendę na wynikach

```
find . -name "*.tmp" -exec rm {} \;
```

Sources: LINUX COMMAND

1.4.117 find -mtime

Modyfikacja w ciągu N dni

```
find /var/log -mtime +7
```

Sources: LINUX COMMAND

1.4.118 find -size

Filtruje rozmiar

```
find / -size +100M
```

Sources: LINUX COMMAND

1.4.119 find -type

Filtruje typ (f/d/l)

```
find /tmp -type f
```

Sources: LINUX COMMAND

1.4.120 find . -atime -1

Sekcja Komend

```
find . -atime -1
```

Sources: RHCSA_Command_Library.pdf

1.4.121 find . -ctime -1

Sekcja Komend

```
find . -ctime -1
```

Sources: RHCSA_Command_Library.pdf

1.4.122 find . -gid 1000

Sekcja Komend

```
find . -gid 1000
```

Sources: RHCSA_Command_Library.pdf

1.4.123 find . -group

Sekcja Komend

```
find . -group
```

Sources: RHCSA_Command_Library.pdf

1.4.124 find . -iname

Sekcja Komend

```
find . -iname
```

Sources: RHCSA_Command_Library.pdf

1.4.125 find . -mmin -60

Sekcja Komend

```
find . -mmin -60
```

Sources: RHCSA_Command_Library.pdf

1.4.126 find . -mtime +30

Sekcja Komend

```
find . -mtime +30
```

Sources: RHCSA_Command_Library.pdf

1.4.127 find . -mtime -7

Sekcja Komend

```
find . -mtime -7
```

Sources: RHCSA_Command_Library.pdf

1.4.128 find . -name "*.jpg"

parallel convert {} -resize 800x600 resized/{}
find . -name "*.jpg"

```
find . -name "*.jpg"
```

Sources: linux_engineering_book.pdf

1.4.129 find . -name "*.jpg" | parallel convert {} -resize 800x600 resized/{} 28.1 Bash — Zaawansowane Techniki

28.1 Bash — Zaawansowane Techniki

```
find . -name "*.jpg" | parallel convert {} -resize 800x600 resized/{}  
Sources: linux_engineering_book.md
```

Sources: linux_engineering_book.md

1.4.130 find . -name "*.log"

xargs rm

```
find . -name "*.log"
```

Sources: linux_engineering_book.pdf

1.4.131 `find . -name "*.log" | xargs rm`

10.4 Przetwarzanie Tekstu

`find . -name "*.log" | xargs rm`

Sources: linux_engineering_book.md

1.4.132 `find . -name "*.txt"`

`xargs -I {} cp {} /backup/`

`find . -name "*.txt"`

Sources: linux_engineering_book.pdf

1.4.133 `find . -name "*.txt" | xargs -I {} cp {} /backup/`

10.4 Przetwarzanie Tekstu

`find . -name "*.txt" | xargs -I {} cp {} /backup/`

Sources: linux_engineering_book.md

1.4.134 `find . -name '*.log'`

Sekcja Komend

`find . -name '*.log'`

Sources: RHCSA_Command_Library.pdf

1.4.135 `find . -newer`

Sekcja Komend

`find . -newer`

Sources: RHCSA_Command_Library.pdf

1.4.136 `find . -nouser`

Sekcja Komend

`find . -nouser`

Sources: RHCSA_Command_Library.pdf

1.4.137 `find . -perm -1000`

Sekcja Komend

`find . -perm -1000`

Sources: RHCSA_Command_Library.pdf

1.4.138 `find . -perm -2000`

Sekcja Komend

`find . -perm -2000`

Sources: RHCSA_Command_Library.pdf

1.4.139 `find . -perm -4000`

Szukaj konfiguracji w /etc

`find . -perm -4000`

Sources: RHCSA_Command_Library.pdf

1.4.140 `find . -perm -644`

Sekcja Komend

`find . -perm -644`

Sources: RHCSA_Command_Library.pdf

1.4.141 `find . -perm /644`

Sekcja Komend

`find . -perm /644`

Sources: RHCSA_Command_Library.pdf

1.4.142 `find . -perm 644`

Sekcja Komend

`find . -perm 644`

Sources: RHCSA_Command_Library.pdf

1.4.143 `find . -size +100M`

Sekcja Komend

`find . -size +100M`

Sources: RHCSA_Command_Library.pdf

1.4.144 `find . -size +1G`

Sekcja Komend

`find . -size +1G`

Sources: RHCSA_Command_Library.pdf

1.4.145 `find . -size -10k`

Sekcja Komend

`find . -size -10k`

Sources: RHCSA_Command_Library.pdf

1.4.146 `find . -size 512c`

Sekcja Komend

`find . -size 512c`

Sources: RHCSA_Command_Library.pdf

1.4.147 `find . -type f -exec chmod 644 {} +`

Prefer over .

`find . -type f -exec chmod 644 {} +`

Sources: komendy

1.4.148 `find . -type f -exec md5sum {} \;`

sort

`uniq -w32 -dD`

Sources: linux_engineering_book.pdf

1.4.149 `find . -type f -exec md5sum {} \; | sort | uniq -w32 -dD`

Dodatek A: Przydatne Jednowierszowce

`find . -type f -exec md5sum {} \; | sort | uniq -w32 -dD`

Sources: linux_engineering_book.md

1.4.150 `find . -uid 1000`

Sekcja Komend

`find . -uid 1000`

Sources: RHCSA_Command_Library.pdf

1.4.151 `find . -user`

Sekcja Komend

`find . -user`

Sources: RHCSA_Command_Library.pdf

1.4.152 `find / -name`

Sekcja Komend

`find / -name`

Sources: RHCSA_Command_Library.pdf

1.4.153 `find / -type b`

Sekcja Komend

`find / -type b`

Sources: RHCSA_Command_Library.pdf

1.4.154 `find / -type c`

Sekcja Komend

`find / -type c`

Sources: RHCSA_Command_Library.pdf

1.4.155 `find / -type d -name`

Sekcja Komend

`find / -type d -name`

Sources: RHCSA_Command_Library.pdf

1.4.156 `find / -type f -mmin -60 2>/dev/null`

Znajdź pliki modyfikowane w ostatniej godzinie

`find / -type f -mmin -60 2>/dev/null`

Sources: linux_engineering_book.md, linux_engineering_book.pdf

1.4.157 `find / -type f -name`

Sekcja Komend

`find / -type f -name`

Sources: RHCSA_Command_Library.pdf

1.4.158 `find / -type f -size +100M -exec ls -lh {} \; 2>/dev/null`

Dodatek A: Przydatne Jednowierszowce

`find / -type f -size +100M -exec ls -lh {} \; 2>/dev/null`

Sources: linux_engineering_book.md, linux_engineering_book.pdf

1.4.159 `find / -type f -size +100M 2>/dev/null`

Dodatek B: Komendy Ratunkowe

`find / -type f -size +100M 2>/dev/null`

Sources: linux_engineering_book.md, linux_engineering_book.pdf

1.4.160 `find / -type l`

Sekcja Komend

`find / -type l`

Sources: RHCSA_Command_Library.pdf

1.4.161 `findmnt`

mounted filesystems as tree. **SAFE**

`findmnt`

Sources: LINUX_COMMAND, RHCSA_Command_Library.pdf, komendy

1.4.162 `findmnt --fstab`

Sekcja Komend

`findmnt --fstab`

Sources: RHCSA_Command_Library.pdf

1.4.163 findmnt --kernel

Kernel mountinfo

```
findmnt --kernel
```

Sources: LINUX COMMAND

1.4.164 findmnt --poll

Monitor mounts

```
findmnt --poll
```

Sources: LINUX COMMAND

1.4.165 findmnt --target

Find mount for path

```
findmnt --target /var
```

Sources: LINUX COMMAND

1.4.166 findmnt -t xfs

Sekcja Komend

```
findmnt -t xfs
```

Sources: RHCSA_Command_Library.pdf

1.4.167 findmnt /mnt

verify mount options. **SAFE**

```
findmnt /mnt
```

Sources: RHCSA_Command_Library.pdf, komendy

1.4.168 head

Pierwsze N linii

```
head -n 20 file.log
```

Sources: LINUX COMMAND

1.4.169 head -c 100 file.txt

Sekcja Komend

```
head -c 100 file.txt
```

Sources: RHCSA_Command_Library.pdf

1.4.170 head -n 20 file

first lines. **SAFE**

```
head -n 20 file
```

Sources: komendy

1.4.171 `head -n 20 file.txt`

Sekcja Komend

`head -n 20 file.txt`

Sources: RHCSA_Command_Library.pdf

1.4.172 `head file.txt`

Sekcja Komend

`head file.txt`

Sources: RHCSA_Command_Library.pdf

1.4.173 `install`

Kopiuje z uprawnieniami

`install -m 755 script.sh /usr/local/bin/`

Sources: LINUX COMMAND

1.4.174 `less`

Przeglądarka stronicowa

`less /var/log/secure`

Sources: LINUX COMMAND

1.4.175 `less +/pattern file`

Sekcja Komend

`less +/pattern file`

Sources: RHCSA_Command_Library.pdf

1.4.176 `less +F file`

follow mode. **SAFE**

`less +F file`

Sources: komendy

1.4.177 `less +G file.txt`

Sekcja Komend

`less +G file.txt`

Sources: RHCSA_Command_Library.pdf

1.4.178 `less file.txt`

Sekcja Komend

`less file.txt`

Sources: RHCSA_Command_Library.pdf

1.4.179 ln

Tworzy twardy link

```
ln file1 file2
```

Sources: LINUX COMMAND

1.4.180 ln -s

Tworzy symboliczny link

```
ln -s /etc/hosts /tmp/hosts
```

Sources: LINUX COMMAND

1.4.181 ln -s target linkname

symbolic link. **SAFE**

```
ln -s target linkname
```

Sources: komendy

1.4.182 ln target linkname

hard link. **SAFE**

```
ln target linkname
```

Sources: komendy

1.4.183 locate

Szybkie wyszukiwanie (baza danych)

```
locate httpd.conf
```

Sources: LINUX COMMAND

1.4.184 losetup -fP disk.img

attach loop with partition scan. **CAUTION**

```
losetup -fP disk.img
```

Sources: komendy

1.4.185 ls

Wyświetla zawartość katalogu

```
ls -la /etc
```

```
ls
```

Sources: LINUX COMMAND, RHCSA_Command_Library.pdf

1.4.186 ls *(.) # tylko pliki

28.2 Zsh — Unikalne Funkcje

```
ls *(.) # tylko pliki
```

Sources: linux_engineering_book.md, linux_engineering_book.pdf

1.4.187 `ls *(.L+1M) # pliki > 1MB`

28.2 Zsh — Unikalne Funkcje

`ls *(.L+1M) # pliki > 1MB`

Sources: linux_engineering_book.md, linux_engineering_book.pdf

1.4.188 `ls *(.m-7) # pliki modyfikowane w ostatnich 7 dniach`

28.2 Zsh — Unikalne Funkcje

`ls *(.m-7) # pliki modyfikowane w ostatnich 7 dniach`

Sources: linux_engineering_book.md, linux_engineering_book.pdf

1.4.189 `ls */( ) # tylko katalogi`

28.2 Zsh — Unikalne Funkcje

`ls */( ) # tylko katalogi`

Sources: linux_engineering_book.md, linux_engineering_book.pdf

1.4.190 `ls **/*.txt # recursive glob`

28.2 Zsh — Unikalne Funkcje

`ls **/*.txt # recursive glob`

Sources: linux_engineering_book.md, linux_engineering_book.pdf

1.4.191 `ls --color=auto`

Sekcja Komend

`ls --color=auto`

Sources: RHCSA_Command_Library.pdf

1.4.192 `ls -R`

Rekurencyjnie listuje pliki

`ls -R /var/log`

`ls -R`

Sources: LINUX_COMMAND, RHCSA_Command_Library.pdf

1.4.193 `ls -Z`

Pokazuje konteksty

`ls -Z /var/www`

Sources: LINUX_COMMAND

1.4.194 `ls -Z /var/www/html # pokaż konteksty plików`

9.3 SELinux

`ls -Z /var/www/html # pokaż konteksty plików`

Sources: linux_engineering_book.md, linux_engineering_book.pdf

1.4.195 `ls -Z dir/`

Sekcja Komend

`ls -Z dir/`

Sources: RHCSA_Command_Library.pdf

1.4.196 `ls -Z file`

Sekcja Komend

`ls -Z file`

Sources: RHCSA_Command_Library.pdf

1.4.197 `ls -d */`

Sekcja Komend

`ls -d */`

Sources: RHCSA_Command_Library.pdf

1.4.198 `ls -dZ dir/`

Sekcja Komend

`ls -dZ dir/`

Sources: RHCSA_Command_Library.pdf

1.4.199 `ls -i`

Pokazuje numery i-węzłów (inodów)

`ls -i /etc/passwd`

`ls -i`

Sources: LINUX COMMAND, RHCSA_Command_Library.pdf

1.4.200 `ls -l`

Sekcja Komend

`ls -l`

Sources: RHCSA_Command_Library.pdf

1.4.201 `ls -lS`

Sekcja Komend

`ls -lS`

Sources: RHCSA_Command_Library.pdf

1.4.202 `ls -la`

Sekcja Komend

`ls -la`

Sources: RHCSA_Command_Library.pdf

1.4.203 `ls -la /proc/[pid]/ns/`

Listowanie namespaces

`ls -la /proc/[pid]/ns/`

Sources: linux_engineering_book.md, linux_engineering_book.pdf

1.4.204 `ls -ld`

Informacje o katalogu, nie o jego zawartości

`ls -ld /home`

Sources: LINUX COMMAND

1.4.205 `ls -lh`

Sekcja Komend

`ls -lh`

Sources: RHCSA_Command_Library.pdf

1.4.206 `ls -lt`

Sekcja Komend

`ls -lt`

Sources: RHCSA_Command_Library.pdf

1.4.207 `ls -ltr`

Sortuje po czasie, odwrócony porządek

`ls -ltr /tmp`

Sources: LINUX COMMAND

1.4.208 `ls /boot/`

Sekcja Komend

`ls /boot/`

Sources: RHCSA_Command_Library.pdf

1.4.209 `ls /boot/grub2/`

Sekcja Komend

`ls /boot/grub2/`

Sources: RHCSA_Command_Library.pdf

1.4.210 `ls /etc/cron.d/`

cron snippets. **SAFE**

`ls /etc/cron.d/`

Sources: RHCSA_Command_Library.pdf, komendy

1.4.211 `ls /etc/cron.daily/`

Sekcja Komend

`ls /etc/cron.daily/`

Sources: RHCSA_Command_Library.pdf

1.4.212 `ls /etc/cron.hourly/`

Sekcja Komend

`ls /etc/cron.hourly/`

Sources: RHCSA_Command_Library.pdf

1.4.213 `ls /etc/cron.weekly/`

Sekcja Komend

`ls /etc/cron.weekly/`

Sources: RHCSA_Command_Library.pdf

1.4.214 `ls /etc/profile.d/`

Sekcja Komend

`ls /etc/profile.d/`

Sources: RHCSA_Command_Library.pdf

1.4.215 `ls /sys/fs/cgroup/`

Komentarz Komentarz w skrypcie function myfunc() { Definicja funkcji

`ls /sys/fs/cgroup/`

Sources: RHCSA_Command_Library.pdf

1.4.216 `md5sum`

Sumy MD5

`md5sum file.iso`

Sources: LINUX COMMAND

1.4.217 `md5sum file`

Komentarz Komentarz w skrypcie function myfunc() { Definicja funkcji

`md5sum file`

Sources: RHCSA_Command_Library.pdf

1.4.218 `md5sum file >`

Komentarz Komentarz w skrypcie function myfunc() { Definicja funkcji

`md5sum file >`

Sources: RHCSA_Command_Library.pdf

1.4.219 mkdir

Tworzy katalog

```
mkdir /backup
```

Sources: LINUX COMMAND

1.4.220 mkdir -p

Tworzy katalogi nadrzędne

```
mkdir -p /a/b/c
```

Sources: LINUX COMMAND

1.4.221 mkdir -p /a/b/c

recursive creation. **SAFE**

```
mkdir -p /a/b/c
```

Sources: komendy

1.4.222 mkdir -p /var/log/journal

enable persistent journal. **SAFE**

```
mkdir -p /var/log/journal
```

Sources: komendy

1.4.223 mkdir /sys/fs/cgroup/mygroup

4.6 Cgroups (Control Groups)

```
mkdir /sys/fs/cgroup/mygroup
```

Sources: linux_engineering_book.md, linux_engineering_book.pdf

1.4.224 mktemp

Tworzy tymczasowy plik/katalog

```
mktemp -d /tmp/XXXXXX
```

Sources: LINUX COMMAND

1.4.225 mktemp -t backup.XXXXXX

secure temp file. **SAFE**

```
mktemp -t backup.XXXXXX
```

Sources: komendy

1.4.226 more

Prostsza przeglądarka

```
more file.txt
```

Sources: LINUX COMMAND

1.4.227 `more file.txt`

Sekcja Komend

`more file.txt`

Sources: RHCSA_Command_Library.pdf

1.4.228 `mount`

Montuje filesystem

`mount /dev/sda1 /mnt`

`mount`

Sources: LINUX COMMAND, RHCSA_Command_Library.pdf

1.4.229 `mount --bind`

Bind mount

`mount --bind /var/www /home/user/www`

Sources: LINUX COMMAND

1.4.230 `mount --bind /dev /mnt/dev`

19.2 Disaster Recovery

`mount --bind /dev /mnt/dev`

Sources: linux_engineering_book.md, linux_engineering_book.pdf

1.4.231 `mount --bind /dev /mnt/sysimage/dev`

bind mount devices. **SAFE**

`mount --bind /dev /mnt/sysimage/dev`

Sources: komendy

1.4.232 `mount --bind /proc /mnt/proc`

19.2 Disaster Recovery

`mount --bind /proc /mnt/proc`

Sources: linux_engineering_book.md, linux_engineering_book.pdf

1.4.233 `mount --bind /proc /mnt/sysimage/proc`

bind mount proc. **SAFE**

`mount --bind /proc /mnt/sysimage/proc`

Sources: komendy

1.4.234 `mount --bind /source /destination`

6.3 Montowanie i fstab

`mount --bind /source /destination`

Sources: linux_engineering_book.md, linux_engineering_book.pdf

1.4.235 `mount --bind /sys /mnt/sys`

19.2 Disaster Recovery

`mount --bind /sys /mnt/sys`

Sources: linux_engineering_book.md, linux_engineering_book.pdf

1.4.236 `mount --bind /sys /mnt/sysimage/sys`

bind mount sys. **SAFE**

`mount --bind /sys /mnt/sysimage/sys`

Sources: komendy

1.4.237 `mount --make-private`

Private mount

`mount --make-private /mnt`

Sources: LINUX COMMAND

1.4.238 `mount --make-shared`

Shared mount

`mount --make-shared /mnt`

Sources: LINUX COMMAND

1.4.239 `mount --make-slave`

Slave mount

`mount --make-slave /mnt`

Sources: LINUX COMMAND

1.4.240 `mount --make-unbindable`

Unbindable

`mount --make-unbindable /mnt`

Sources: LINUX COMMAND

1.4.241 `mount --move`

Move mount

`mount --move /old /new`

Sources: LINUX COMMAND

1.4.242 `mount --rbind`

Recursive bind

`mount --rbind /dev /mnt/chroot/dev`

Sources: LINUX COMMAND

1.4.243 `mount -a`

Montuje wszystko z fstab

```
mount -a
```

Sources: LINUX_COMMAND, RHCSA_Command_Library.pdf

1.4.244 `mount -o`

Sekcja Komend

```
mount -o
```

Sources: RHCSA_Command_Library.pdf

1.4.245 `mount -o loop image.iso /mnt/iso`

6.3 Montowanie i fstab

```
mount -o loop image.iso /mnt/iso
```

Sources: linux_engineering_book.md, linux_engineering_book.pdf

1.4.246 `mount -o remount,ro`

Sekcja Komend

```
mount -o remount,ro
```

Sources: RHCSA_Command_Library.pdf

1.4.247 `mount -o remount,rw`

Przymontowuje do zapisu

```
mount -o remount,rw /
```

```
mount -o remount,rw
```

Sources: LINUX_COMMAND, RHCSA_Command_Library.pdf

1.4.248 `mount -o remount,rw /`

Po uruchomieniu powłoki awaryjnej, przemontowanie głównego systemu plików w tryb zapisu/odczytu

```
mount -o remount,rw /
```

Sources: encyklopedia_inzynierii_linux.html, encyklopedia_inzynierii_linux.pdf, komendy, linux_engineering_book.md, linux_engineering_book.pdf

1.4.249 `mount -o remount,rw,nosuid,nodev,noexec /tmp`

restrict /tmp. **CAUTION**

```
mount -o remount,rw,nosuid,nodev,noexec /tmp
```

Sources: komendy

1.4.250 `mount -o ro`

Sekcja Komend

```
mount -o ro
```

Sources: RHCSA_Command_Library.pdf

1.4.251 `mount -o ro,soft`

Sekcja Komend

`mount -o ro,soft`

Sources: RHCSA_Command_Library.pdf

1.4.252 `mount -o rw,noexec`

Sekcja Komend

`mount -o rw,noexec`

Sources: RHCSA_Command_Library.pdf

1.4.253 `mount -t cifs`

Komentarz Komentarz w skrypcie function myfunc() { Definicja funkcji

`mount -t cifs`

Sources: RHCSA_Command_Library.pdf

1.4.254 `mount -t ext4`

Sekcja Komend

`mount -t ext4`

Sources: RHCSA_Command_Library.pdf

1.4.255 `mount -t nfs`

Sekcja Komend

`mount -t nfs`

Sources: RHCSA_Command_Library.pdf

1.4.256 `mount -t nfs server:/share /mnt/nfs`

6.3 Montowanie i fstab

`mount -t nfs server:/share /mnt/nfs`

Sources: linux_engineering_book.md, linux_engineering_book.pdf

1.4.257 `mount -t nfs server:/srv/nfs /mnt/nfs`

18.3 NFS

`mount -t nfs server:/srv/nfs /mnt/nfs`

Sources: linux_engineering_book.md, linux_engineering_book.pdf

1.4.258 `mount -t nfs4`

Sekcja Komend

`mount -t nfs4`

Sources: RHCSA_Command_Library.pdf

1.4.259 `mount -t xfs`

Sekcja Komend

`mount -t xfs`

Sources: RHCSA_Command_Library.pdf

1.4.260 `mount /dev/sda1 /mnt`

mount filesystem. **CAUTION**

`mount /dev/sda1 /mnt`

Sources: komendy

1.4.261 `mount /dev/sda1 /mnt/boot`

19.2 Disaster Recovery

`mount /dev/sda1 /mnt/boot`

Sources: linux_engineering_book.md, linux_engineering_book.pdf

1.4.262 `mount /dev/sda2 /mnt`

19.2 Disaster Recovery

`mount /dev/sda2 /mnt`

Sources: linux_engineering_book.md, linux_engineering_book.pdf

1.4.263 `mount /dev/sdb1 /mnt`

Sekcja Komend

`mount /dev/sdb1 /mnt`

Sources: RHCSA_Command_Library.pdf

1.4.264 `mount /dev/sdb1 /mnt/data`

6.3 Montowanie i fstab

`mount /dev/sdb1 /mnt/data`

Sources: linux_engineering_book.md, linux_engineering_book.pdf

1.4.265 `mv`

Prefer to for atomic renames on same filesystem.

`mv old.txt new.txt`

`mv`

Sources: LINUX_COMMAND, komendy

1.4.266 `mv -i old new`

interactive move. **CAUTION**

`mv -i old new`

Sources: komendy

1.4.267 `mv -i src dst`

Sekcja Komend

`mv -i src dst`

Sources: RHCSA_Command_Library.pdf

1.4.268 `mv -n src dst`

Sekcja Komend

`mv -n src dst`

Sources: RHCSA_Command_Library.pdf

1.4.269 `mv -u src dst`

Sekcja Komend

`mv -u src dst`

Sources: RHCSA_Command_Library.pdf

1.4.270 `mv -v src dst`

Sekcja Komend

`mv -v src dst`

Sources: RHCSA_Command_Library.pdf

1.4.271 `mv source dest`

Sekcja Komend

`mv source dest`

Sources: RHCSA_Command_Library.pdf

1.4.272 `nl`

Numeruje linie

`nl /etc/passwd`

Sources: LINUX COMMAND

1.4.273 `nl file`

logical line numbers. **SAFE**

`nl file`

Sources: komendy

1.4.274 `popd`

Przywraca katalog ze stosu

`popd`

Sources: LINUX COMMAND, RHCSA_Command_Library.pdf

1.4.275 pushd

Zapamiętuje katalog na stosie

```
pushd /usr/local
```

Sources: LINUX COMMAND

1.4.276 pushd /path

head -20 Pierwsze 20 plików z /etc

```
pushd /path
```

Sources: RHCSA_Command_Library.pdf

1.4.277 pushd /var/log; popd

directory stack. **SAFE**

```
pushd /var/log; popd
```

Sources: komendy

1.4.278 pwd

Aktualna ścieżka

```
pwd
```

Sources: LINUX COMMAND, RHCSA_Command_Library.pdf

1.4.279 pwd -P

physical path, resolves symlinks. **SAFE**

```
pwd -P
```

Sources: komendy

1.4.280 readlink

Pokazuje cel linku symbolicznego

```
readlink -f /tmp/hosts
```

Sources: LINUX COMMAND

1.4.281 realpath

Validate all file paths with and prefix checks.

```
realpath ../file
```

```
realpath
```

Sources: LINUX COMMAND, komendy

1.4.282 rm

Uruchomienie kontenera z pełnymi uprawnieniami roota hosta i dostępem do surowego systemu

```
rm file.txt
```

```
rm
```

Sources: LINUX COMMAND, encyklopedia_inzynierii_linux.pdf

1.4.283 `rm -f /tmp/lockfile`

10.3 Funkcje

`rm -f /tmp/lockfile`

Sources: linux_engineering_book.md, linux_engineering_book.pdf

1.4.284 `rm -f file.txt`

Sekcja Komend

`rm -f file.txt`

Sources: RHCSA_Command_Library.pdf

1.4.285 `rm -i file`

Sekcja Komend

`rm -i file`

Sources: RHCSA_Command_Library.pdf

1.4.286 `rm -r dir/`

Sekcja Komend

`rm -r dir/`

Sources: RHCSA_Command_Library.pdf

1.4.287 `rm -rf`

Usuwa rekurencyjnie i na siłę

`rm -rf /tmp/olddata`

Sources: LINUX COMMAND

1.4.288 `rm -rf "$VAR/"`

if \$VAR unset, deletes root. **DANGEROUS**

`rm -rf "$VAR/"`

Sources: komendy

1.4.289 `rm -rf $VAR/`

or when unset.

`rm -rf $VAR/`

Sources: komendy

1.4.290 `rm -rf /`

package — prevents .

`rm -rf /`

Sources: komendy

1.4.291 `rm -rf /` or `rm -rf $VAR/` when `$VAR` unset.

Dangerous Mistakes

`rm -rf /`` or ``rm -rf $VAR/`` when ``$VAR`` unset.

Sources: komendy

1.4.292 `rm -rf /var/log/journal`

bypassing vacuum logic.

`rm -rf /var/log/journal`

Sources: komendy

1.4.293 `rm -rf /var/log/journal` bypassing vacuum logic.

Dangerous Mistakes

`rm -rf /var/log/journal`` bypassing vacuum logic.

Sources: komendy

1.4.294 `rm -rf /var/log/old_metrics_data/`

- Przykład produkcyjny:

`rm -rf /var/log/old_metrics_data/`

Sources: encyklopedia_inzynierii_linux.html

1.4.295 `rm -rf dir/`

recursive force remove. **DANGEROUS**

`rm -rf dir/`

Sources: RHCSA_Command_Library.pdf, komendy

1.4.296 `rm -v file`

Sekcja Komend

`rm -v file`

Sources: RHCSA_Command_Library.pdf

1.4.297 `rm file.txt`

Sekcja Komend

`rm file.txt`

Sources: RHCSA_Command_Library.pdf

1.4.298 `sudo rm -rf /`

.

`sudo rm -rf /`

Sources: komendy

1.4.299 `sudo rm -rf /`

Dangerous Mistakes

`sudo rm -rf /`.`

Sources: komendy

1.4.300 `rmdir`

Usuwa pusty katalog

`rmdir olddir`

Sources: LINUX COMMAND

1.4.301 `sha256sum`

Sumy SHA256

`sha256sum file.iso`

Sources: LINUX COMMAND

1.4.302 `sha256sum -c`

Komentarz Komentarz w skrypcie function myfunc() { Definicja funkcji

`sha256sum -c`

Sources: RHCSA_Command_Library.pdf

1.4.303 `sha256sum -c checksums.txt`

verify file integrity. **SAFE**

`sha256sum -c checksums.txt`

Sources: komendy

1.4.304 `sha256sum file`

bc Arytmetyka z bc

`sha256sum file`

Sources: RHCSA_Command_Library.pdf

1.4.305 `split`

Dzieli plik na części

`split -b 1M largefile`

Sources: LINUX COMMAND

1.4.306 `stat`

Szczegółowe metadane pliku

`stat /etc/passwd`

Sources: LINUX COMMAND

1.4.307 stat /etc/passwd

inode, access times, blocks. **SAFE**

`stat /etc/passwd`

Sources: komendy

1.4.308 tac

Odwraca kolejność linii

`tac /var/log/messages`

Sources: LINUX COMMAND

1.4.309 tac file

reverse lines. **SAFE**

`tac file`

Sources: komendy

1.4.310 tac file.txt

Sekcja Komend

`tac file.txt`

Sources: RHCSA_Command_Library.pdf

1.4.311 tail

Ostatnie N linii

`tail -f /var/log/messages`

Sources: LINUX COMMAND

1.4.312 tail -F file.txt led plik (nawet gdy zostanie column -t file.txt

Sekcja Komend

`tail -F file.txt led plik (nawet gdy zostanie column -t file.txt`

Sources: RHCSA_Command_Library.pdf

1.4.313 tail -c 200 file.txt

Sekcja Komend

`tail -c 200 file.txt`

Sources: RHCSA_Command_Library.pdf

1.4.314 tail -f /var/log/messages

follow messages (RHEL). **SAFE**

`tail -f /var/log/messages`

Sources: komendy

1.4.315 `tail -f /var/log/syslog`

follow syslog (Debian). **SAFE**

`tail -f /var/log/syslog`

Sources: komendy

1.4.316 `tail -f file.txt` led plik na bieco (logi) `unexpand file.txt`

Sekcja Komend

`tail -f file.txt` led plik na bieco (logi) `unexpand file.txt`

Sources: RHCSA_Command_Library.pdf

1.4.317 `tail -f` ledzenie logów bezpieczeństwa `sar -d 1 5`

Sekcja Komend

`tail -f` ledzenie logów bezpieczeństwa `sar -d 1 5`

Sources: RHCSA_Command_Library.pdf

1.4.318 `tail -f` ledzenie logów systemowych `sar -r 1 5`

Sekcja Komend

`tail -f` ledzenie logów systemowych `sar -r 1 5`

Sources: RHCSA_Command_Library.pdf

1.4.319 `tail -n +5 file.txt`

Sekcja Komend

`tail -n +5 file.txt`

Sources: RHCSA_Command_Library.pdf

1.4.320 `tail -n 20 file`

last lines. **SAFE**

`tail -n 20 file`

Sources: komendy

1.4.321 `tail -n 20 file.txt`

Sekcja Komend

`tail -n 20 file.txt`

Sources: RHCSA_Command_Library.pdf

1.4.322 `tail file.txt`

Sekcja Komend

`tail file.txt`

Sources: RHCSA_Command_Library.pdf

1.4.323 touch

Tworzy pusty plik lub aktualizuje czas

```
touch newfile.txt
```

Sources: LINUX COMMAND

1.4.324 touch -d

Sekcja Komend

```
touch -d
```

Sources: RHCSA_Command_Library.pdf

1.4.325 touch -t

Sekcja Komend

```
touch -t
```

Sources: RHCSA_Command_Library.pdf

1.4.326 touch file

create or update mtime. **SAFE**

```
touch file
```

Sources: komendy

1.4.327 touch file.txt

Sekcja Komend

```
touch file.txt
```

Sources: RHCSA_Command_Library.pdf

1.4.328 touch kernel logic

20. DODATKOWE NARZĘDZIA SYSTEMOWE (Rozszerzenie)

```
touch kernel logic
```

Sources: LINUX COMMAND

1.4.329 touch planner systems

20. DODATKOWE NARZĘDZIA SYSTEMOWE (Rozszerzenie)

```
touch planner systems
```

Sources: LINUX COMMAND

1.4.330 tree -dL 2 /etc

directory tree, depth-limited. **SAFE**

```
tree -dL 2 /etc
```

Sources: komendy

1.4.331 type

Określa typ komendy

`type cd`

Sources: LINUX COMMAND

1.4.332 type -a ls

command resolution. **SAFE**

`type -a ls`

Sources: komendy

1.4.333 type httpd_t;

24.1 SELinux — Tworzenie Polityk

`type httpd_t;`

Sources: linux_engineering_book.md, linux_engineering_book.pdf

1.4.334 type user_home_t;

24.1 SELinux — Tworzenie Polityk

`type user_home_t;`

Sources: linux_engineering_book.md, linux_engineering_book.pdf

1.4.335 umount

Odmontowuje

`umount /mnt`

Sources: LINUX COMMAND

1.4.336 umount -f /mnt

Sekcja Komend

`umount -f /mnt`

Sources: RHCSA_Command_Library.pdf

1.4.337 umount -l

Lazy unmount

`umount -l /mnt`

Sources: LINUX COMMAND

1.4.338 umount -l /mnt

lazy unmount. **CAUTION**

`umount -l /mnt`

Sources: RHCSA_Command_Library.pdf, komendy

1.4.339 `umount /dev/sdb1`

Sekcja Komend

`umount /dev/sdb1`

Sources: RHCSA_Command_Library.pdf

1.4.340 `umount /mnt`

unmount. **CAUTION** (fails if busy)

`umount /mnt`

Sources: RHCSA_Command_Library.pdf, komendy

1.4.341 `updatedb`

Aktualizuje bazę locate

`updatedb`

Sources: LINUX COMMAND

1.4.342 `whereis`

Lokalizuje binaria i manuale

`whereis sshd`

Sources: LINUX COMMAND

1.4.343 `whereis -b ls`

binary location. **SAFE**

`whereis -b ls`

Sources: komendy

1.4.344 `which`

Lokalizuje plik wykonywalny

`which python3`

`which`

Sources: LINUX COMMAND, komendy

1.4.345 `which python3`

path in \$PATH. **SAFE**

`which python3`

Sources: komendy

1.5 Users & Permissions

1.5.1 `chage`

Zarządzanie datą wygaśnięcia

`chage -l john`

Sources: LINUX COMMAND

1.5.2 chage -E

Data wygaśnięcia konta

```
chage -E 2025-12-31 john
```

Sources: LINUX COMMAND

1.5.3 chage -M

Maksymalny wiek hasła

```
chage -M 90 john
```

Sources: LINUX COMMAND

1.5.4 chage -M 90 -m 7 -W 14 alice

password aging. **CAUTION**

```
chage -M 90 -m 7 -W 14 alice
```

Sources: komendy

1.5.5 chage -M 90 user1 # max 90 dni ważności hasła

/etc/modprobe.d/bonding.conf

```
chage -M 90 user1 # max 90 dni ważności hasła
```

Sources: linux_engineering_book.md, linux_engineering_book.pdf

1.5.6 chage -l user

Sekcja Komend

```
chage -l user
```

Sources: RHCSA_Command_Library.pdf

1.5.7 chage -l user1 # informacje o hasle

/etc/modprobe.d/bonding.conf

```
chage -l user1 # informacje o hasle
```

Sources: linux_engineering_book.md, linux_engineering_book.pdf

1.5.8 chattr

Zmienia atrybuty rozszerzone

```
chattr +i file.txt
```

Sources: LINUX COMMAND

1.5.9 chattr +a file

append-only. **CAUTION**

```
chattr +a file
```

Sources: komendy

1.5.10 `chattr +a file # append-only`

Atrybuty plików (ext4)

```
chattr +a file # append-only
```

Sources: linux_engineering_book.md, linux_engineering_book.pdf

1.5.11 `chattr +i`

on critical configs.

```
chattr +i
```

Sources: komendy

1.5.12 `chattr +i /etc/resolv.conf`

prevent DNS tampering. **CAUTION**

```
chattr +i /etc/resolv.conf
```

Sources: komendy

1.5.13 `chattr +i file`

immutable. **CAUTION** (root cannot modify without removal)

```
chattr +i file
```

Sources: komendy

1.5.14 `chattr +i file # immutable - nie można zmienić/usunąć`

Atrybuty plików (ext4)

```
chattr +i file # immutable - nie można zmienić/usunąć
```

Sources: linux_engineering_book.md, linux_engineering_book.pdf

1.5.15 `chfn`

Zmiana danych GECOS

```
chfn -f "John"
```

Sources: LINUX COMMAND

1.5.16 `chgrp`

Zmienia grupę

```
chgrp developers file.txt
```

Sources: LINUX COMMAND

1.5.17 `chgrp group file`

change group only. **CAUTION**

```
chgrp group file
```

Sources: komendy

1.5.18 chmod

Zmienia uprawnienia

```
chmod 755 script.sh  
chmod
```

Sources: LINUX COMMAND, RHCSA_Command_Library.pdf

1.5.19 chmod +t /tmp

sticky bit. **SAFE**

```
chmod +t /tmp
```

Sources: komendy

1.5.20 chmod +t dir

Sekcja Komend

```
chmod +t dir
```

Sources: RHCSA_Command_Library.pdf

1.5.21 chmod +t directory

Uprawnienia plików

```
chmod +t directory #
```

Sources: linux_engineering_book.md, linux_engineering_book.pdf

1.5.22 chmod +x

Nadaje prawo wykonania

```
chmod +x script.sh
```

Sources: LINUX COMMAND

1.5.23 chmod +x script.sh

Komentarz Komentarz w skrypcie function myfunc() { Definicja funkcji

```
chmod +x script.sh
```

Sources: RHCSA_Command_Library.pdf

1.5.24 chmod -R

Rekurencyjnie

```
chmod -R 755 /var/www  
chmod -R
```

Sources: LINUX COMMAND, komendy

1.5.25 chmod -R 750 directory # rekursywnie

Uprawnienia plików

```
chmod -R 750 directory # rekursywnie
```

Sources: linux_engineering_book.md, linux_engineering_book.pdf

1.5.26 `chmod -R 755 dir/`

Sekcja Komend

`chmod -R 755 dir/`

Sources: RHCSA_Command_Library.pdf

1.5.27 `chmod 1755 dir`

Sekcja Komend

`chmod 1755 dir`

Sources: RHCSA_Command_Library.pdf

1.5.28 `chmod 2755 dir`

Sekcja Komend

`chmod 2755 dir`

Sources: RHCSA_Command_Library.pdf

1.5.29 `chmod 4755 file`

Sekcja Komend

`chmod 4755 file`

Sources: RHCSA_Command_Library.pdf

1.5.30 `chmod 600 /swapfile`

Tworzenie swap

`chmod 600 /swapfile`

Sources: linux_engineering_book.md, linux_engineering_book.pdf

1.5.31 `chmod 600 file`

Sekcja Komend

`chmod 600 file`

Sources: RHCSA_Command_Library.pdf

1.5.32 `chmod 644 file`

Sekcja Komend

`chmod 644 file`

Sources: RHCSA_Command_Library.pdf

1.5.33 `chmod 700 /home/*`

enforce private home dirs. **CAUTION**

`chmod 700 /home/*`

Sources: komendy

1.5.34 `chmod 750 dir`

`rwxr-x—`. CAUTION

`chmod 750 dir`

Sources: komendy

1.5.35 `chmod 755 file`

Sekcja Komend

`chmod 755 file`

Sources: RHCSA_Command_Library.pdf

1.5.36 `chmod 755 file # rwxr-xr-x`

Uprawnienia plików

`chmod 755 file # rwxr-xr-x`

Sources: linux_engineering_book.md, linux_engineering_book.pdf

1.5.37 `chmod 777`

anywhere in production.

`chmod 777`

Sources: komendy

1.5.38 `chmod 777 /etc`

or .

`chmod 777 /etc`

Sources: komendy

1.5.39 `chmod 777 /etc` or `chown -R user:user /`

Dangerous Mistakes

`chmod 777 /etc`` or ``chown -R user:user /``.

Sources: komendy

1.5.40 `chmod 777 file`

Sekcja Komend

`chmod 777 file`

Sources: RHCSA_Command_Library.pdf

1.5.41 `chmod a+r file`

Sekcja Komend

`chmod a+r file`

Sources: RHCSA_Command_Library.pdf

1.5.42 chmod g+s

Setgid bit

`chmod g+s /shared`

Sources: LINUX COMMAND

1.5.43 chmod g+s /shared

setgid on directory. **CAUTION**

`chmod g+s /shared`

Sources: komendy

1.5.44 chmod g+s dir

Sekcja Komend

`chmod g+s dir`

Sources: RHCSA_Command_Library.pdf

1.5.45 chmod g+s directory

Uprawnienia plików

`chmod g+s directory #`

Sources: linux_engineering_book.md, linux_engineering_book.pdf

1.5.46 chmod g-w

Odbiera write dla grupy

`chmod g-w file.txt`

Sources: LINUX COMMAND

1.5.47 chmod g-w file

Sekcja Komend

`chmod g-w file`

Sources: RHCSA_Command_Library.pdf

1.5.48 chmod g-w file # usuń write dla grupy

Uprawnienia plików

`chmod g-w file # usuń write dla grupy`

Sources: linux_engineering_book.md, linux_engineering_book.pdf

1.5.49 chmod o+t

Sticky bit

`chmod o+t /tmp`

Sources: LINUX COMMAND

1.5.50 `chmod o-r file`

Sekcja Komend

`chmod o-r file`

Sources: RHCSA_Command_Library.pdf

1.5.51 `chmod o= file # usuń wszystko dla others`

Uprawnienia plików

`chmod o= file # usuń wszystko dla others`

Sources: linux_engineering_book.md, linux_engineering_book.pdf

1.5.52 `chmod o=r`

Ustawia read dla innych

`chmod o=r file.txt`

Sources: LINUX COMMAND

1.5.53 `chmod u+s`

Setuid bit

`chmod u+s /usr/bin/passwd`

Sources: LINUX COMMAND

1.5.54 `chmod u+s /bin/ping`

setuid. **DANGEROUS**

`chmod u+s /bin/ping`

Sources: komendy

1.5.55 `chmod u+s file`

Kopiuj ACL między katalogami

`chmod u+s file`

Sources: RHCSA_Command_Library.pdf

1.5.56 `chmod u+s file #`

Uprawnienia plików

`chmod u+s file #`

Sources: linux_engineering_book.md, linux_engineering_book.pdf

1.5.57 `chmod u+x`

Dodaje execute dla właściciela

`chmod u+x script.sh`

Sources: LINUX COMMAND

1.5.58 `chmod u+x file`

add owner execute. **SAFE**

`chmod u+x file`

Sources: RHCSA_Command_Library.pdf, komendy

1.5.59 `chmod u+x file # dodaj execute dla właściciela`

Uprawnienia plików

`chmod u+x file # dodaj execute dla właściciela`

Sources: linux_engineering_book.md, linux_engineering_book.pdf

1.5.60 `chmod u=rwx,g=rx,o=r`

Sekcja Komend

`chmod u=rwx,g=rx,o=r`

Sources: RHCSA_Command_Library.pdf

1.5.61 `chown`

Zmienia właściciela

`chown john file.txt`

Sources: LINUX COMMAND

1.5.62 `chown -R`

Rekurencyjnie

`chown -R apache:apache /var/www`

Sources: LINUX COMMAND

1.5.63 `chown -R user:group`

Sekcja Komend

`chown -R user:group`

Sources: RHCSA_Command_Library.pdf

1.5.64 `chown -R user:user /`

or .

`chown -R user:user /`

Sources: komendy

1.5.65 `chown :group file`

Sekcja Komend

`chown :group file`

Sources: RHCSA_Command_Library.pdf

1.5.66 chown user file

Sekcja Komend

`chown user file`

Sources: RHCSA_Command_Library.pdf

1.5.67 chown user:group

Właściciel i grupa

`chown john:developers file.txt`

`chown user:group`

Sources: LINUX COMMAND, RHCSA_Command_Library.pdf

1.5.68 chown user:group file

change owner and group. **CAUTION**

`chown user:group file`

Sources: komendy

1.5.69 chpasswd

Masowa zmiana haseł

`echo "john:pass123" \`

Sources: LINUX COMMAND

1.5.70 chsh

Zmiana powłoki

`chsh -s /bin/zsh john`

Sources: LINUX COMMAND

1.5.71 faillock

Blokada po nieudanych logowaniach

`faillock --user john`

Sources: LINUX COMMAND

1.5.72 faillock --reset

Resetuje licznik

`faillock --user john --reset`

Sources: LINUX COMMAND

1.5.73 faillock --user alice

check lockout status. **SAFE**

`faillock --user alice`

Sources: komendy

1.5.74 faillock --user alice --reset

reset lockout. **CAUTION**

```
faillock --user alice --reset
```

Sources: komendy

1.5.75 getfacl

Wyświetla ACL

```
getfacl file.txt
```

Sources: LINUX COMMAND

1.5.76 getfacl file

display ACL. **SAFE**

```
getfacl file
```

Sources: komendy

1.5.77 getfacl file # pokaż

ACL (Access Control Lists)

```
getfacl file # pokaż
```

Sources: linux_engineering_book.md, linux_engineering_book.pdf

1.5.78 getfattr

Rozszerzone atrybuty (xattr)

```
getfattr -d file.txt
```

Sources: LINUX COMMAND

1.5.79 getfattr -n

Get specific xattr

```
getfattr -n user.comment file
```

Sources: LINUX COMMAND

1.5.80 gpasswd

Hasło grupy/administratorzy

```
gpasswd -a john wheel
```

Sources: LINUX COMMAND

1.5.81 gpasswd -a user

Sekcja Komend

```
gpasswd -a user
```

Sources: RHCSA_Command_Library.pdf

1.5.82 gpasswd -d user

Sekcja Komend

`gpasswd -d user`

Sources: RHCSA_Command_Library.pdf

1.5.83 groupadd

Tworzy grupę

`groupadd developers`

Sources: LINUX COMMAND

1.5.84 groupadd -g 1500

Sekcja Komend

`groupadd -g 1500`

Sources: RHCSA_Command_Library.pdf

1.5.85 groupadd -r sysgroup

Sekcja Komend

`groupadd -r sysgroup`

Sources: RHCSA_Command_Library.pdf

1.5.86 groupadd groupname

Sekcja Komend

`groupadd groupname`

Sources: RHCSA_Command_Library.pdf

1.5.87 groupdel

Usuwa grupę

`groupdel developers`

Sources: LINUX COMMAND

1.5.88 groupdel groupname

Sekcja Komend

`groupdel groupname`

Sources: RHCSA_Command_Library.pdf

1.5.89 groupmod

Modyfikuje grupę

`groupmod -n newname oldname`

Sources: LINUX COMMAND

1.5.90 groupmod -g 1600

Sekcja Komend

groupmod -g 1600

Sources: RHCSA_Command_Library.pdf

1.5.91 groupmod -n newname

Sekcja Komend

groupmod -n newname

Sources: RHCSA_Command_Library.pdf

1.5.92 groups

Grupy użytkownika

groups john

Sources: LINUX COMMAND

1.5.93 grpck

verify group file integrity. **SAFE**

grpck

Sources: LINUX COMMAND, komendy

1.5.94 id

Informacje o tożsamości

id john

id -u john

Sources: LINUX COMMAND

1.5.95 id -Z

Kontekst użytkownika

id -Z

Sources: LINUX COMMAND, RHCSA_Command_Library.pdf

1.5.96 last

recent logins. **SAFE**

last

last -x

Sources: LINUX COMMAND, komendy

1.5.97 lastb

failed login attempts. **SAFE**

lastb

Sources: LINUX COMMAND, komendy

1.5.98 lastlog

last login all users. **SAFE**

lastlog

lastlog **-u** john

Sources: LINUX COMMAND, komendy

1.5.99 lsattr

Wyświetla atrybuty

lsattr file.txt

Sources: LINUX COMMAND

1.5.100 lsattr file

list attributes. **SAFE**

lsattr file

Sources: komendy

1.5.101 lsattr file # pokaż atrybuty

Atrybuty plików (ext4)

lsattr file **# pokaż atrybuty**

Sources: linux_engineering_book.md, linux_engineering_book.pdf

1.5.102 newgrp

Zmiana grupy efektywnej

newgrp developers

Sources: LINUX COMMAND

1.5.103 passwd

Zmienia hasło

passwd john

Sources: LINUX COMMAND

1.5.104 passwd --status user

Sekcja Komend

passwd **--status** user

Sources: RHCSA_Command_Library.pdf

1.5.105 passwd -d

Usuwa hasło (puste)

passwd **-d** john

Sources: LINUX COMMAND

1.5.106 `passwd -d user`

Sekcja Komend

`passwd -d user`

Sources: RHCSA_Command_Library.pdf

1.5.107 `passwd -e user`

Sekcja Komend

`passwd -e user`

Sources: RHCSA_Command_Library.pdf

1.5.108 `passwd -i 10 user`

Sekcja Komend

`passwd -i 10 user`

Sources: RHCSA_Command_Library.pdf

1.5.109 `passwd -l`

Blokuje hasło

`passwd -l john`

Sources: LINUX COMMAND

1.5.110 `passwd -l alice`

lock password. **CAUTION**

`passwd -l alice`

Sources: komendy

1.5.111 `passwd -l user`

Sekcja Komend

`passwd -l user`

Sources: RHCSA_Command_Library.pdf

1.5.112 `passwd -n 7 user`

Sekcja Komend

`passwd -n 7 user`

Sources: RHCSA_Command_Library.pdf

1.5.113 `passwd -u`

Odblokowuje hasło

`passwd -u john`

Sources: LINUX COMMAND

1.5.114 `passwd -u user`

Sekcja Komend

`passwd -u user`

Sources: RHCSA_Command_Library.pdf

1.5.115 `passwd -w 14 user`

Sekcja Komend

`passwd -w 14 user`

Sources: RHCSA_Command_Library.pdf

1.5.116 `passwd -x 90 user`

Sekcja Komend

`passwd -x 90 user`

Sources: RHCSA_Command_Library.pdf

1.5.117 `passwd root`

change root password. **DANGEROUS**

`passwd root`

Sources: encyklopedia_inzynierii_linux.html, encyklopedia_inzynierii_linux.pdf, komendy, linux_engineering_book.m
linux_engineering_book.pdf

1.5.118 `passwd user`

Sekcja Komend

`passwd user`

Sources: RHCSA_Command_Library.pdf

1.5.119 `passwd user1 # zmień hasło`

/etc/modprobe.d/bonding.conf

`passwd user1 # zmień hasło`

Sources: linux_engineering_book.md, linux_engineering_book.pdf

1.5.120 `pwck`

verify passwd file integrity. **SAFE**

`pwck`

Sources: LINUX_COMMAND, komendy

1.5.121 `setfacl`

Ustawia ACL

`setfacl -m u:john:rw file.txt`

Sources: LINUX_COMMAND

1.5.122 `setfacl --set-file=-`

Sekcja Komend

```
setfacl --set-file=-
```

Sources: RHCSA_Command_Library.pdf

1.5.123 `setfacl -R`

Rekurencyjne ACL

```
setfacl -R -m u:john:rx /shared
```

Sources: LINUX COMMAND

1.5.124 `setfacl -R -m d:u:alice:rwx dir`

default ACL recursive. **CAUTION**

```
setfacl -R -m d:u:alice:rwx dir
```

Sources: komendy

1.5.125 `setfacl -b`

Usuwa wszystkie ACL

```
setfacl -b file.txt
```

Sources: LINUX COMMAND

1.5.126 `setfacl -b file`

remove all ACLs. **CAUTION**

```
setfacl -b file
```

Sources: komendy

1.5.127 `setfacl -b file # usuń wszystkie`

ACL (Access Control Lists)

```
setfacl -b file # usuń wszystkie
```

Sources: linux_engineering_book.md, linux_engineering_book.pdf

1.5.128 `setfacl -d`

Domyślne ACL (katalogi)

```
setfacl -d -m u:john:rx /shared
```

Sources: LINUX COMMAND

1.5.129 `setfacl -d -m g:sec_auditors:rX /var/log/audit/`

Ustawienie domyślnej maski ACL (Inheritance), aby każdy nowy plik automatycznie dziedziczył dostęp

```
setfacl -d -m g:sec_auditors:rX /var/log/audit/
```

Sources: encyklopedia_inzynierii_linux.html, encyklopedia_inzynierii_linux.pdf

1.5.130 `setfacl -m`

Modyfikuje ACL

```
setfacl -m g:developers:rwX dir
```

Sources: LINUX COMMAND

1.5.131 `setfacl -m g:devs:rx file #`

ACL (Access Control Lists)

```
setfacl -m g:devs:rx file #
```

Sources: linux_engineering_book.md, linux_engineering_book.pdf

1.5.132 `setfacl -m g:sec_auditors:rwX /var/log/audit/`

Nadanie uprawnień odczytu i zapisu dla konkretnej grupy audytorów do katalogu logów

```
setfacl -m g:sec_auditors:rwX /var/log/audit/
```

Sources: encyklopedia_inzynierii_linux.html, encyklopedia_inzynierii_linux.pdf

1.5.133 `setfacl -m u:alice:rw file`

grant user read-write. **CAUTION**

```
setfacl -m u:alice:rw file
```

Sources: komendy

1.5.134 `setfacl -m u:user1:rwX file # dodaj`

ACL (Access Control Lists)

```
setfacl -m u:user1:rwX file # dodaj
```

Sources: linux_engineering_book.md, linux_engineering_book.pdf

1.5.135 `setfacl -x`

Usuwa wpis ACL

```
setfacl -x u:john file.txt
```

Sources: LINUX COMMAND

1.5.136 `setfattr`

Ustawia xattr

```
setfattr -n user.comment -v "test" file.txt
```

Sources: LINUX COMMAND

1.5.137 `setfattr -n`

Set specific xattr

```
setfattr -n user.comment -v "value" file
```

Sources: LINUX COMMAND

1.5.138 `setfattr -x`

Remove specific xattr

```
setfattr -x user.comment file
```

Sources: LINUX COMMAND

1.5.139 `sg`

Wykonanie komendy z grupą

```
sg developers -c "touch file"
```

Sources: LINUX COMMAND

1.5.140 `su`

username' — login shell as user. **CAUTION**

```
su - john
```

```
su
```

Sources: LINUX COMMAND, komendy

1.5.141 `su -`

Zmiana z środowiskiem (login)

```
su - root
```

Sources: LINUX COMMAND

1.5.142 `su - username`

login shell as user. **CAUTION**

```
su - username
```

Sources: komendy

1.5.143 `sudo`

Use with command restrictions, not full root.

```
sudo systemctl
```

```
sudo
```

Sources: LINUX COMMAND, komendy

1.5.144 `sudoedit`

Edycja pliku z sudo

```
sudoedit /etc/hosts
```

Sources: LINUX COMMAND

1.5.145 `umask`

Domyślne uprawnienia

```
umask 0022
```

Sources: LINUX COMMAND

1.5.146 `umask -S`

Symboliczny umask

```
umask -S u=rwx,g=rx,o=rx
```

Sources: LINUX COMMAND

1.5.147 `umask 027`

default permissions. **SAFE**

```
umask 027
```

Sources: komendy

1.5.148 `useradd`

Tworzy użytkownika

```
useradd -m -s /bin/bash john
```

Sources: LINUX COMMAND

1.5.149 `useradd -G`

Dodatkowe grupy

```
useradd -G wheel john
```

Sources: LINUX COMMAND

1.5.150 `useradd -G g1,g2`

Sekcja Komend

```
useradd -G g1,g2
```

Sources: RHCSA_Command_Library.pdf

1.5.151 `useradd -M user`

Sekcja Komend

```
useradd -M user
```

Sources: RHCSA_Command_Library.pdf

1.5.152 `useradd -N user`

Sekcja Komend

```
useradd -N user
```

Sources: RHCSA_Command_Library.pdf

1.5.153 `useradd -c 'Imi`

Sekcja Komend

```
useradd -c 'Imi
```

Sources: RHCSA_Command_Library.pdf

1.5.154 useradd -d

Sekcja Komend

```
useradd -d
```

Sources: RHCSA_Command_Library.pdf

1.5.155 useradd -e

Sekcja Komend

```
useradd -e
```

Sources: RHCSA_Command_Library.pdf

1.5.156 useradd -f 30 user

Sekcja Komend

```
useradd -f 30 user
```

Sources: RHCSA_Command_Library.pdf

1.5.157 useradd -g group

Sekcja Komend

```
useradd -g group
```

Sources: RHCSA_Command_Library.pdf

1.5.158 useradd -m -s

Sekcja Komend

```
useradd -m -s
```

Sources: RHCSA_Command_Library.pdf

1.5.159 useradd -m -s /bin/bash -G sudo,docker user1 # dodaj użytkownika

/etc/modprobe.d/bonding.conf

```
useradd -m -s /bin/bash -G sudo,docker user1 # dodaj użytkownika
```

Sources: linux_engineering_book.md, linux_engineering_book.pdf

1.5.160 useradd -m -s /bin/bash -U alice

create user with home and group. **SAFE**

```
useradd -m -s /bin/bash -U alice
```

Sources: komendy

1.5.161 useradd -m username

Sekcja Komend

```
useradd -m username
```

Sources: RHCSA_Command_Library.pdf

1.5.162 `useradd -r -s /usr/sbin/nologin svc_name`

Separate service accounts: .

`useradd -r -s /usr/sbin/nologin svc_name`

Sources: komendy

1.5.163 `useradd -r sysuser`

Sekcja Komend

`useradd -r sysuser`

Sources: RHCSA_Command_Library.pdf

1.5.164 `useradd -s`

Określa powłokę

`useradd -s /sbin/nologin service`

Sources: LINUX COMMAND

1.5.165 `useradd -u 1500 user`

Sekcja Komend

`useradd -u 1500 user`

Sources: RHCSA_Command_Library.pdf

1.5.166 `useradd username`

Sekcja Komend

`useradd username`

Sources: RHCSA_Command_Library.pdf

1.5.167 `userdel`

Usuwa użytkownika

`userdel john`

Sources: LINUX COMMAND

1.5.168 `userdel -f user`

Sekcja Komend

`userdel -f user`

Sources: RHCSA_Command_Library.pdf

1.5.169 `userdel -r`

Usuwa z katalogiem domowym

`userdel -r john`

Sources: LINUX COMMAND

1.5.170 userdel -r user

Sekcja Komend

```
userdel -r user
```

Sources: RHCSA_Command_Library.pdf

1.5.171 userdel -r user1 # usuń z katalogiem domowym

/etc/modprobe.d/bonding.conf

```
userdel -r user1 # usuń z katalogiem domowym
```

Sources: linux_engineering_book.md, linux_engineering_book.pdf

1.5.172 userdel user

Sekcja Komend

```
userdel user
```

Sources: RHCSA_Command_Library.pdf

1.5.173 usermod

Modyfikuje użytkownika

```
usermod -aG wheel john
```

Sources: LINUX COMMAND

1.5.174 usermod -G g1,g2

Sekcja Komend

```
usermod -G g1,g2
```

Sources: RHCSA_Command_Library.pdf

1.5.175 usermod -L

Blokuje konto (zablokuj hasło)

```
usermod -L john
```

Sources: LINUX COMMAND

1.5.176 usermod -L user

Sekcja Komend

```
usermod -L user
```

Sources: RHCSA_Command_Library.pdf

1.5.177 usermod -U

Odblokowuje konto

```
usermod -U john
```

Sources: LINUX COMMAND

1.5.178 `usermod -U user`

Sekcja Komend

`usermod -U user`

Sources: RHCSA_Command_Library.pdf

1.5.179 `usermod -aG`

Dodaje do grupy (append)

`usermod -aG docker john`

Sources: LINUX COMMAND

1.5.180 `usermod -aG group`

Sekcja Komend

`usermod -aG group`

Sources: RHCSA_Command_Library.pdf

1.5.181 `usermod -aG sudo alice`

add to group. **CAUTION**

`usermod -aG sudo alice`

Sources: komendy

1.5.182 `usermod -aG wheel user1 # dodaj do grupy`

/etc/modprobe.d/bonding.conf

`usermod -aG wheel user1 # dodaj do grupy`

Sources: linux_engineering_book.md, linux_engineering_book.pdf

1.5.183 `usermod -c 'Nowy`

Sekcja Komend

`usermod -c 'Nowy`

Sources: RHCSA_Command_Library.pdf

1.5.184 `usermod -d /new/home`

Sekcja Komend

`usermod -d /new/home`

Sources: RHCSA_Command_Library.pdf

1.5.185 `usermod -e`

Sekcja Komend

`usermod -e`

Sources: RHCSA_Command_Library.pdf

1.5.186 `usermod -e '' user`

Sekcja Komend

`usermod -e '' user`

Sources: RHCSA_Command_Library.pdf

1.5.187 `usermod -g group`

Sekcja Komend

`usermod -g group`

Sources: RHCSA_Command_Library.pdf

1.5.188 `usermod -l newname`

Sekcja Komend

`usermod -l newname`

Sources: RHCSA_Command_Library.pdf

1.5.189 `usermod -s /bin/zsh`

Sekcja Komend

`usermod -s /bin/zsh`

Sources: RHCSA_Command_Library.pdf

1.5.190 `usermod -u 1600 user`

Sekcja Komend

`usermod -u 1600 user`

Sources: RHCSA_Command_Library.pdf

1.5.191 `vigr`

Bezpieczna edycja /etc/group

`vigr`

Sources: LINUX COMMAND

1.5.192 `vipw`

Bezpieczna edycja /etc/passwd

`vipw`

Sources: LINUX COMMAND

1.5.193 `visudo`

safe sudoers edit. **CAUTION**

`visudo`

Sources: LINUX COMMAND, komendy

1.5.194 w

Command — klawisz : Tryb komend # Szukaj sowa pod kursorem (do tyu)

w

Sources: LINUX COMMAND, RHCSA_Command_Library.pdf

1.5.195 w kronice. *writeback:

w kronice. * `writeback`:

Sources: encyklopedia_inzynierii_linux.pdf

1.5.196 w pamięci fizycznej (np. jest współdzielona z inną aplikacją), ale nie została jeszcze zmapowana w tablicy stron

Sekwencja REISUB (Synchronizacja dysków, Odmontowanie, Natychmiastowy restart)

w pamięci fizycznej (np. jest współdzielona z inną aplikacją), ale nie została jeszcze zmapowana w tablicy s

Sources: encyklopedia_inzynierii_linux.pdf

1.5.197 who

Zalogowani użytkownicy

who

who -H

Sources: LINUX COMMAND

1.5.198 whoami

Aktualny użytkownik

whoami

Sources: LINUX COMMAND

1.6 Networking

1.6.1 /etc/hostname

Stała nazwa hosta

/etc/hostname

Sources: LINUX COMMAND

1.6.2 /etc/hosts

Lokalne mapowania

/etc/hosts

Sources: LINUX COMMAND

1.6.3 /etc/resolv.conf

Serwery DNS

/etc/resolv.conf

Sources: LINUX COMMAND

1.6.4 NetworkManager

Daemon

systemctl

NetworkManager

Sources: LINUX COMMAND, linux_engineering_book.pdf

1.6.5 ab

Apache benchmark

ab -n 1000 -c 10 http://localhost/

Sources: LINUX COMMAND

1.6.6 arptables

ARP firewall

arptables -L

Sources: LINUX COMMAND

1.6.7 brctl

Legacy bridge

brctl

Sources: LINUX COMMAND

1.6.8 bridge

Bridge management

bridge link

Sources: LINUX COMMAND

1.6.9 bridge fdb

Forwarding database

bridge fdb

Sources: LINUX COMMAND

1.6.10 conntrack

Connection tracking

conntrack -L

Sources: LINUX COMMAND

1.6.11 conntrack -D

Usuwa conntrack

conntrack -D --src 192.168.1.100

Sources: LINUX COMMAND

1.6.12 curl

Transfer danych URL

```
curl -O http://example.com/file
```

Sources: LINUX COMMAND

1.6.13 curl --cookie

Cookie

```
curl --cookie "session=abc" http://localhost
```

Sources: LINUX COMMAND

1.6.14 curl --cookie-jar

Zapisuje cookies

```
curl --cookie-jar cookies.txt http://localhost
```

Sources: LINUX COMMAND

1.6.15 curl -F

Multipart form

```
curl -F "file=@upload.txt" http://localhost/upload
```

Sources: LINUX COMMAND

1.6.16 curl -H

Custom header

```
curl -H "X-Custom: value" http://localhost
```

Sources: LINUX COMMAND

1.6.17 curl -I

Sekcja Komend

```
curl -I http://localhost
```

```
curl -I
```

Sources: LINUX COMMAND, RHCSA_Command_Library.pdf

1.6.18 curl -I https://example.com

headers only. **SAFE**

```
curl -I https://example.com
```

Sources: komendy

1.6.19 curl -L

Follow redirects

```
curl -L http://bit.ly/xxx
```

Sources: LINUX COMMAND

1.6.20 curl -X

HTTP method

```
curl -X
```

Sources: LINUX COMMAND

1.6.21 curl -d

POST data

```
curl -d "name=john" http://localhost/form
```

Sources: LINUX COMMAND

1.6.22 curl -k

Ignore SSL

```
curl -k https://localhost
```

Sources: LINUX COMMAND

1.6.23 curl -o file.iso https://example.com/file.iso

download. **SAFE**

```
curl -o file.iso https://example.com/file.iso
```

Sources: komendy

1.6.24 curl -o file.zip https://example.com/file # pobieranie

8.4 Diagnostyka Sieci

```
curl -o file.zip https://example.com/file # pobieranie
```

Sources: linux_engineering_book.md, linux_engineering_book.pdf

1.6.25 curl -u

Basic auth

```
curl -u user:pass http://localhost
```

Sources: LINUX COMMAND

1.6.26 curl -v http://localhost:80

HTTP test. **SAFE**

```
curl -v http://localhost:80
```

Sources: komendy

1.6.27 curl -v https://example.com # verbose

8.4 Diagnostyka Sieci

```
curl -v https://example.com # verbose
```

Sources: linux_engineering_book.md, linux_engineering_book.pdf

1.6.28 dhclient

DHCP client

```
dhclient -r eth0
```

Sources: LINUX COMMAND

1.6.29 dig

Szczegółowe zapytania DNS

```
dig @8.8.8.8 google.com
```

Sources: LINUX COMMAND

1.6.30 dig +short

Krótki wynik

```
dig +short google.com
```

Sources: LINUX COMMAND

1.6.31 dig +short google.com

DNS A record. **SAFE**

```
dig +short google.com
```

Sources: komendy

1.6.32 dig +short hostname

Sekcja Komend

```
dig +short hostname
```

Sources: RHCSA_Command_Library.pdf

1.6.33 dig +trace

Trace DNS resolution

```
dig +trace google.com
```

Sources: LINUX COMMAND

1.6.34 dig +trace hostname led cae zapytanie

Sekcja Komend

```
dig +trace hostname led cae zapytanie
```

Sources: RHCSA_Command_Library.pdf

1.6.35 dig -x 8.8.8.8

Sekcja Komend

```
dig -x 8.8.8.8
```

Sources: RHCSA_Command_Library.pdf

1.6.36 dig @8.8.8.8

Sekcja Komend

dig @8.8.8.8

Sources: RHCSA_Command_Library.pdf

1.6.37 dig @8.8.8.8 example.com

Diagnostyka

dig @8.8.8.8 example.com

Sources: linux_engineering_book.md, linux_engineering_book.pdf

1.6.38 dig @8.8.8.8 google.com

specific resolver. **SAFE**

dig @8.8.8.8 google.com

Sources: komendy

1.6.39 dig example.com # zapytanie

Diagnostyka

dig example.com # *zapytanie*

Sources: linux_engineering_book.md, linux_engineering_book.pdf

1.6.40 dig example.com →

Dodatek B: Komendy Ratunkowe

dig example.com →

Sources: linux_engineering_book.md, linux_engineering_book.pdf

1.6.41 dig hostname

Sekcja Komend

dig hostname

Sources: RHCSA_Command_Library.pdf

1.6.42 dnsmasq

DNS/DHCP server

dnsmasq --*interface*=eth0

Sources: LINUX COMMAND

1.6.43 ebtables

Ethernet bridge firewall

ebtables -*L*

Sources: LINUX COMMAND

1.6.44 ethtool

Ustawienia karty sieciowej

```
ethtool eth0
```

Sources: LINUX COMMAND

1.6.45 ethtool -S eth0

NIC statistics. **SAFE**

```
ethtool -S eth0
```

Sources: komendy

1.6.46 ethtool -S eth0 # statystyki sterownika

8.4 Diagnostyka Sieci

```
ethtool -S eth0 # statystyki sterownika
```

Sources: linux_engineering_book.md, linux_engineering_book.pdf

1.6.47 ethtool -s

Zmiana parametrów

```
ethtool -s eth0 speed 1000 duplex full
```

Sources: LINUX COMMAND

1.6.48 ethtool eth0

driver/hardware settings. **SAFE**

```
ethtool eth0
```

Sources: komendy

1.6.49 ethtool eth0 # parametry karty sieciowej

7.4 Narzędzia Diagnostyczne Sprzętu

```
ethtool eth0 # parametry karty sieciowej
```

Sources: linux_engineering_book.md, linux_engineering_book.pdf

1.6.50 exportfs

Manage NFS exports

```
exportfs -a
```

```
exportfs
```

Sources: LINUX COMMAND, RHCSA_Command_Library.pdf

1.6.51 exportfs -a

Sekcja Komend

```
exportfs -a
```

Sources: RHCSA_Command_Library.pdf

1.6.52 exportfs -r

Sekcja Komend

`exportfs -r`

Sources: LINUX COMMAND, RHCSA_Command_Library.pdf

1.6.53 exportfs -ra

/etc/exports

`exportfs -ra`

Sources: linux_engineering_book.md, linux_engineering_book.pdf

1.6.54 exportfs -s

Sekcja Komend

`exportfs -s`

Sources: RHCSA_Command_Library.pdf

1.6.55 exportfs -u

Un-export

`exportfs -u /export`

Sources: LINUX COMMAND

1.6.56 exportfs -ua

Sekcja Komend

`exportfs -ua`

Sources: RHCSA_Command_Library.pdf

1.6.57 exportfs -v

Sekcja Komend

`exportfs -v`

Sources: RHCSA_Command_Library.pdf

1.6.58 firewall-cmd

Komentarz Komentarz w skrypcie function myfunc() { Definicja funkcji

`firewall-cmd --state`

`firewall-cmd`

Sources: LINUX COMMAND, RHCSA_Command_Library.pdf

1.6.59 firewall-cmd --add-interface

Dodaje interfejs do strefy

`firewall-cmd --add-interface=eth1 --zone=internal`

Sources: LINUX COMMAND

1.6.60 firewall-cmd --add-p

Sekcja Komend

```
firewall-cmd --add-p
```

Sources: RHCSA_Command_Library.pdf

1.6.61 firewall-cmd --add-port

Dodaje port

```
firewall-cmd --add-port=8080/tcp
```

Sources: LINUX COMMAND

1.6.62 firewall-cmd --add-service

Dodaje usługę (runtime)

```
firewall-cmd --add-service=http
```

```
firewall-cmd --add-service=https
```

Sources: LINUX COMMAND

1.6.63 firewall-cmd --add-service=ssh

temporary add. **CAUTION**

```
firewall-cmd --add-service=ssh
```

Sources: komendy

1.6.64 firewall-cmd --add-service=ssh --permanent

permanent add. **CAUTION**

```
firewall-cmd --add-service=ssh --permanent
```

Sources: komendy

1.6.65 firewall-cmd --add-source

Dodaje źródło do strefy

```
firewall-cmd --add-source=192.168.1.0/24 --zone=trusted
```

Sources: LINUX COMMAND

1.6.66 firewall-cmd --change-interface

Zmienia strefę interfejsu

```
firewall-cmd --change-interface=eth1 --zone=dmz
```

Sources: LINUX COMMAND

1.6.67 firewall-cmd --get-active-zones

firewalld (RHEL/Fedora)

```
firewall-cmd --get-active-zones
```

Sources: linux_engineering_book.md, linux_engineering_book.pdf

1.6.68 firewall-cmd --get-default-zone

default zone. **SAFE**

```
firewall-cmd --get-default-zone
```

Sources: LINUX COMMAND, komendy

1.6.69 firewall-cmd --get-services

Lista znanych usług

```
firewall-cmd --get-services
```

Sources: LINUX COMMAND

1.6.70 firewall-cmd --get-zones

Dostępne strefy

```
firewall-cmd --get-zones
```

Sources: LINUX COMMAND

1.6.71 firewall-cmd --list-all

active zone rules. **SAFE**

```
firewall-cmd --list-all
```

Sources: LINUX COMMAND, komendy, linux_engineering_book.md, linux_engineering_book.pdf

1.6.72 firewall-cmd --list-services

Lista aktywnych usług

```
firewall-cmd --list-services
```

Sources: LINUX COMMAND

1.6.73 firewall-cmd --panic-off

Wyłącza panic

```
firewall-cmd --panic-off
```

Sources: LINUX COMMAND

1.6.74 firewall-cmd --panic-on

Panic mode (blokuje wszystko)

```
firewall-cmd --panic-on
```

Sources: LINUX COMMAND

1.6.75 firewall-cmd --permanent

Zmiana trwała

```
firewall-cmd --add-service=http --permanent
```

Sources: LINUX COMMAND

1.6.76 firewall-cmd --query-panic

Stan panic

```
firewall-cmd --query-panic
```

Sources: LINUX COMMAND

1.6.77 firewall-cmd --query-service

Sprawdza usługę

```
firewall-cmd --query-service=ssh
```

Sources: LINUX COMMAND

1.6.78 firewall-cmd --reload

reload. **CAUTION**

```
firewall-cmd --reload
```

Sources: LINUX COMMAND, komendy, linux_engineering_book.md, linux_engineering_book.pdf

1.6.79 firewall-cmd --reload losing temporary rescue rules.

Dangerous Mistakes

```
firewall-cmd --reload` losing temporary rescue rules.
```

Sources: komendy

1.6.80 firewall-cmd --remov

Sekcja Komend

```
firewall-cmd --remov
```

Sources: RHCSA_Command_Library.pdf

1.6.81 firewall-cmd --remove-service

Usuwa usługę

```
firewall-cmd --remove-service=http
```

Sources: LINUX COMMAND

1.6.82 firewall-cmd --remove-source

Usuwa źródło

```
firewall-cmd --remove-source=192.168.1.0/24
```

Sources: LINUX COMMAND

1.6.83 firewall-cmd --runtime-to-permanent

Zapisuje runtime na stałe

```
firewall-cmd --runtime-to-permanent
```

Sources: LINUX COMMAND

1.6.84 firewall-cmd --set-d

Sekcja Komend

```
firewall-cmd --set-d
```

Sources: RHCSA_Command_Library.pdf

1.6.85 firewall-cmd --set-default-zone

Ustawia domyślną

```
firewall-cmd --set-default-zone=drop
```

Sources: LINUX COMMAND

1.6.86 firewall-cmd --state

firewalld (RHEL/Fedora)

```
firewall-cmd --state
```

Sources: RHCSA_Command_Library.pdf, linux_engineering_book.md, linux_engineering_book.pdf

1.6.87 firewall-cmd --zone=public

Wybór strefy

```
firewall-cmd --zone=public --list-all
```

Sources: LINUX COMMAND

1.6.88 firewall-cmd --zone=public --add-port=8080/tcp --permanent

firewalld (RHEL/Fedora)

```
firewall-cmd --zone=public --add-port=8080/tcp --permanent
```

Sources: linux_engineering_book.md, linux_engineering_book.pdf

1.6.89 firewall-cmd --zone=public --add-service=http --permanent

firewalld (RHEL/Fedora)

```
firewall-cmd --zone=public --add-service=http --permanent
```

Sources: linux_engineering_book.md, linux_engineering_book.pdf

1.6.90 ftp

FTP client

```
ftp ftp.example.com
```

Sources: LINUX COMMAND

1.6.91 fusermount

Unmount FUSE

```
fusermount -u /mnt
```

Sources: LINUX COMMAND

1.6.92 haproxy

Load balancer

```
haproxy -f /etc/haproxy/haproxy.cfg
```

Sources: LINUX COMMAND

1.6.93 host

Proste zapytanie DNS

```
host google.com
```

Sources: LINUX COMMAND

1.6.94 host -t

Sekcja Komend

```
host -t
```

Sources: LINUX COMMAND, RHCSA_Command_Library.pdf

1.6.95 host example.com # proste zapytanie

Diagnostyka

```
host example.com # proste zapytanie
```

Sources: linux_engineering_book.md, linux_engineering_book.pdf

1.6.96 host hostname

Sekcja Komend

```
host hostname
```

Sources: RHCSA_Command_Library.pdf

1.6.97 host image

Sekcja Komend

```
host image
```

Sources: RHCSA_Command_Library.pdf

1.6.98 hostapd

Access point daemon

```
hostapd /etc/hostapd.conf
```

Sources: LINUX COMMAND

1.6.99 httpd -S

Apache vhosts

```
httpd -S
```

Sources: LINUX COMMAND

1.6.100 httpd / apachectl

Apache

apachectl configtest

Sources: LINUX COMMAND

1.6.101 ip

Uruchomienie kontenera z pełnymi uprawnieniami roota hosta i dostępem do surowego systemu

ip addr

ip

Sources: LINUX COMMAND, encyklopedia_inzynierii_linux.pdf, komendy

1.6.102 ip -6 addr

Sekcja Komend

ip -6 addr

Sources: RHCSA_Command_Library.pdf

1.6.103 ip -6 route

Sekcja Komend

ip -6 route

Sources: RHCSA_Command_Library.pdf

1.6.104 ip -s link

Statystyki interfejsu

ip -s link

Sources: LINUX COMMAND

1.6.105 ip addr

interface addresses. **SAFE**

ip addr

Sources: LINUX COMMAND, RHCSA_Command_Library.pdf, komendy, linux_engineering_book.md, linux_engineering_book.pdf

1.6.106 ip addr add

Sekcja Komend

ip addr add

Sources: RHCSA_Command_Library.pdf

1.6.107 ip addr add 10.0.0.1/24 dev veth0

/etc/modprobe.d/bonding.conf

ip addr add 10.0.0.1/24 dev veth0

Sources: linux_engineering_book.md, linux_engineering_book.pdf

1.6.108 `ip addr add 10.0.100.1/24 dev eth0.100`

`/etc/modprobe.d/bonding.conf`

`ip addr add 10.0.100.1/24 dev eth0.100`

Sources: linux_engineering_book.md, linux_engineering_book.pdf

1.6.109 `ip addr add 192.168.1.1/24 dev br0`

`/etc/modprobe.d/bonding.conf`

`ip addr add 192.168.1.1/24 dev br0`

Sources: linux_engineering_book.md, linux_engineering_book.pdf

1.6.110 `ip addr add 192.168.1.10/24 dev bond0`

`/etc/modprobe.d/bonding.conf`

`ip addr add 192.168.1.10/24 dev bond0`

Sources: linux_engineering_book.md, linux_engineering_book.pdf

1.6.111 `ip addr add 192.168.1.10/24 dev eth0 # dodaj adres`

Narzędzia ip (iproute2)

`ip addr add 192.168.1.10/24 dev eth0 # dodaj adres`

Sources: linux_engineering_book.md, linux_engineering_book.pdf

1.6.112 `ip addr del`

Sekcja Komend

`ip addr del`

Sources: RHCSA_Command_Library.pdf

1.6.113 `ip addr del 192.168.1.10/24 dev eth0 # usuń adres`

Narzędzia ip (iproute2)

`ip addr del 192.168.1.10/24 dev eth0 # usuń adres`

Sources: linux_engineering_book.md, linux_engineering_book.pdf

1.6.114 `ip addr flush dev eth0`

Natychmiastowe usunięcie wszystkich adresów IP przypisanych do interfejsu produkcyjnego

`ip addr flush dev eth0`

Sources: encyklopedia_inzynierii_linux.html, encyklopedia_inzynierii_linux.pdf

1.6.115 `ip link`

Dodatek B: Komendy Ratunkowe

`ip link set eth0 up`

`ip link`

Sources: LINUX_COMMAND, RHCSA_Command_Library.pdf, linux_engineering_book.md, linux_engineering_book.pdf

1.6.116 `ip link add bond0 type bond mode 802.3ad`

`/etc/modprobe.d/bonding.conf`

`ip link add bond0 type bond mode 802.3ad`

Sources: linux_engineering_book.md, linux_engineering_book.pdf

1.6.117 `ip link add br0 type bridge`

`/etc/modprobe.d/bonding.conf`

`ip link add br0 type bridge`

Sources: linux_engineering_book.md, linux_engineering_book.pdf

1.6.118 `ip link add link eth0 name eth0.100 type vlan id 100`

`/etc/modprobe.d/bonding.conf`

`ip link add link eth0 name eth0.100 type vlan id 100`

Sources: linux_engineering_book.md, linux_engineering_book.pdf

1.6.119 `ip link add type bridge`

Tworzy bridge

`ip link add br0 type bridge`

Sources: LINUX COMMAND

1.6.120 `ip link add veth0 type veth peer name veth1`

`/etc/modprobe.d/bonding.conf`

`ip link add veth0 type veth peer name veth1`

Sources: linux_engineering_book.md, linux_engineering_book.pdf

1.6.121 `ip link add vxlan0 type vxlan id 42 remote 1.2.3.4 dstport 4789`

Narzędzia ip (iproute2)

`ip link add vxlan0 type vxlan id 42 remote 1.2.3.4 dstport 4789`

Sources: linux_engineering_book.md, linux_engineering_book.pdf

1.6.122 `ip link set bond0 up`

`/etc/modprobe.d/bonding.conf`

`ip link set bond0 up`

Sources: linux_engineering_book.md, linux_engineering_book.pdf

1.6.123 `ip link set br0 up`

`/etc/modprobe.d/bonding.conf`

`ip link set br0 up`

Sources: linux_engineering_book.md, linux_engineering_book.pdf

1.6.124 `ip link set dev eth0 xdp obj xdp_prog.o`

Programy XDP działają na poziomie sterownika NIC

```
ip link set dev eth0 xdp obj xdp_prog.o
```

Sources: linux_engineering_book.md, linux_engineering_book.pdf

1.6.125 `ip link set dev eth0 xdp off`

Programy XDP działają na poziomie sterownika NIC

```
ip link set dev eth0 xdp off
```

Sources: linux_engineering_book.md, linux_engineering_book.pdf

1.6.126 `ip link set eth0 down`

on primary remote interface.

```
ip link set eth0 down
```

Sources: RHCSA_Command_Library.pdf, komendy

1.6.127 `ip link set eth0 master bond0`

/etc/modprobe.d/bonding.conf

```
ip link set eth0 master bond0
```

Sources: linux_engineering_book.md, linux_engineering_book.pdf

1.6.128 `ip link set eth0 master br0`

/etc/modprobe.d/bonding.conf

```
ip link set eth0 master br0
```

Sources: linux_engineering_book.md, linux_engineering_book.pdf

1.6.129 `ip link set eth0 mtu`

Sekcja Komend

```
ip link set eth0 mtu
```

Sources: RHCSA_Command_Library.pdf

1.6.130 `ip link set eth0 mtu 9000 # ustaw`

Narzędzia ip (iproute2)

```
ip link set eth0 mtu 9000 # ustaw
```

Sources: linux_engineering_book.md, linux_engineering_book.pdf

1.6.131 `ip link set eth0 up`

bring interface up. **CAUTION**

```
ip link set eth0 up
```

Sources: RHCSA_Command_Library.pdf, komendy

1.6.132 `ip link set eth0 up/down # włącz/wyłącz`

Narzędzia ip (iproute2)

`ip link set eth0 up/down # włącz/wyłącz`

Sources: linux_engineering_book.md, linux_engineering_book.pdf

1.6.133 `ip link set eth0.100 up`

/etc/modprobe.d/bonding.conf

`ip link set eth0.100 up`

Sources: linux_engineering_book.md, linux_engineering_book.pdf

1.6.134 `ip link set eth1 master bond0`

/etc/modprobe.d/bonding.conf

`ip link set eth1 master bond0`

Sources: linux_engineering_book.md, linux_engineering_book.pdf

1.6.135 `ip link set eth1 master br0`

/etc/modprobe.d/bonding.conf

`ip link set eth1 master br0`

Sources: linux_engineering_book.md, linux_engineering_book.pdf

1.6.136 `ip link set veth0 up`

/etc/modprobe.d/bonding.conf

`ip link set veth0 up`

Sources: linux_engineering_book.md, linux_engineering_book.pdf

1.6.137 `ip link set veth1 netns ns1`

/etc/modprobe.d/bonding.conf

`ip link set veth1 netns ns1`

Sources: linux_engineering_book.md, linux_engineering_book.pdf

1.6.138 `ip maddr`

Multicast addresses

`ip maddr`

Sources: LINUX COMMAND

1.6.139 `ip monitor`

Monitoruje zmiany

`ip monitor all`

Sources: LINUX COMMAND

1.6.140 ip neigh

ARP table check. **SAFE**

ip neigh

Sources: LINUX COMMAND, RHCSA_Command_Library.pdf, komendy, linux_engineering_book.md, linux_engineering_book.pdf

1.6.141 ip neigh add 192.168.1.1 lladdr aa:bb:cc:dd:ee:ff dev eth0

Narzędzia ip (iproute2)

ip neigh add 192.168.1.1 lladdr aa:bb:cc:dd:ee:ff dev eth0

Sources: linux_engineering_book.md, linux_engineering_book.pdf

1.6.142 ip neigh flush all

Sekcja Komend

ip neigh flush all

Sources: RHCSA_Command_Library.pdf

1.6.143 ip netns

Komentarz Komentarz w skrypcie function myfunc() { Definicja funkcji

ip netns add myns

ip netns

Sources: LINUX COMMAND, RHCSA_Command_Library.pdf, komendy

1.6.144 ip netns add myns

Komentarz Komentarz w skrypcie function myfunc() { Definicja funkcji

ip netns add myns

Sources: RHCSA_Command_Library.pdf

1.6.145 ip netns add ns1

/etc/modprobe.d/bonding.conf

ip netns add ns1

Sources: linux_engineering_book.md, linux_engineering_book.pdf

1.6.146 ip netns delete myns

Komentarz Komentarz w skrypcie function myfunc() { Definicja funkcji

ip netns delete myns

Sources: RHCSA_Command_Library.pdf

1.6.147 ip netns exec

Wykonuje w namespace

ip netns exec myns bash

Sources: LINUX COMMAND

1.6.148 ip netns exec myns

Komentarz Komentarz w skrypcie function myfunc() { Definicja funkcji

ip netns exec myns

Sources: RHCSA_Command_Library.pdf

1.6.149 ip netns exec ns1 ip addr

run in namespace. **CAUTION**

ip netns exec ns1 ip addr

Sources: komendy

1.6.150 ip netns exec ns1 ip addr add 10.0.0.2/24 dev veth1

/etc/modprobe.d/bonding.conf

ip netns exec ns1 ip addr add 10.0.0.2/24 dev veth1

Sources: linux_engineering_book.md, linux_engineering_book.pdf

1.6.151 ip netns exec ns1 ip link set lo up

/etc/modprobe.d/bonding.conf

ip netns exec ns1 ip link set lo up

Sources: linux_engineering_book.md, linux_engineering_book.pdf

1.6.152 ip netns exec ns1 ip link set veth1 up

/etc/modprobe.d/bonding.conf

ip netns exec ns1 ip link set veth1 up

Sources: linux_engineering_book.md, linux_engineering_book.pdf

1.6.153 ip ntable

Neighbor table

ip ntable

Sources: LINUX COMMAND

1.6.154 ip route

routing table. **SAFE**

ip route add 10.0.0.0/8 via 192.168.1.1

ip route

Sources: LINUX COMMAND, RHCSA_Command_Library.pdf, komendy, linux_engineering_book.md, linux_engineering_book.pdf

1.6.155 ip route add

Sekcja Komend

ip route add default via 192.168.1.1

ip route add

Sources: LINUX COMMAND, RHCSA_Command_Library.pdf

1.6.156 `ip route add 10.0.0.0/8 via 192.168.1.1`

add route. **CAUTION**

`ip route add 10.0.0.0/8 via 192.168.1.1`

Sources: komendy

1.6.157 `ip route add 10.0.0.0/8 via 192.168.1.1 # dodaj trasę`

Narzędzia ip (iproute2)

`ip route add 10.0.0.0/8 via 192.168.1.1 # dodaj trasę`

Sources: linux_engineering_book.md, linux_engineering_book.pdf

1.6.158 `ip route add 10.0.0.0/8 via 192.168.1.1 dev eth0`

routingu i adresacją IP. * **Przykład produkcyjny:** #

`ip route add 10.0.0.0/8 via 192.168.1.1 dev eth0`

Sources: encyklopedia_inzynierii_linux.html, encyklopedia_inzynierii_linux.pdf

1.6.159 `ip route add default`

Sekcja Komend

`ip route add default`

Sources: RHCSA_Command_Library.pdf

1.6.160 `ip route add default via 192.168.1.1 # brama domyślna`

Narzędzia ip (iproute2)

`ip route add default via 192.168.1.1 # brama domyślna`

Sources: linux_engineering_book.md, linux_engineering_book.pdf

1.6.161 `ip route del`

Sekcja Komend

`ip route del`

Sources: RHCSA_Command_Library.pdf

1.6.162 `ip route del 10.0.0.0/8 # usuń trasę`

Narzędzia ip (iproute2)

`ip route del 10.0.0.0/8 # usuń trasę`

Sources: linux_engineering_book.md, linux_engineering_book.pdf

1.6.163 `ip route del default`

Sekcja Komend

`ip route del default`

Sources: RHCSA_Command_Library.pdf

1.6.164 ip route get 8.8.8.8

Sekcja Komend

```
ip route get 8.8.8.8
```

Sources: RHCSA_Command_Library.pdf

1.6.165 ip tunnel

IP tunnels

```
ip tunnel
```

Sources: LINUX COMMAND

1.6.166 ip tunnel add gre1 mode gre remote 1.2.3.4 local 5.6.7.8

Narzędzia ip (iproute2)

```
ip tunnel add gre1 mode gre remote 1.2.3.4 local 5.6.7.8
```

Sources: linux_engineering_book.md, linux_engineering_book.pdf

1.6.167 ip tuntap

TUN/TAP

```
ip tuntap add mode tun tun0
```

Sources: LINUX COMMAND

1.6.168 ipset

Zbiory IP dla firewalla

```
ipset
```

Sources: LINUX COMMAND

1.6.169 ipset add

Dodaje do zbioru

```
ipset add myset 192.168.1.100
```

Sources: LINUX COMMAND

1.6.170 iptables

Legacy firewall

```
iptables -L -n -v
```

Sources: LINUX COMMAND

1.6.171 iptables (legacy)

Diagnostyka

```
iptables (legacy)
```

Sources: linux_engineering_book.pdf

1.6.172 iptables -A

append rule. **CAUTION**

`iptables -A`

Sources: LINUX COMMAND, komendy, linux_engineering_book.md, linux_engineering_book.pdf

1.6.173 iptables -D

delete rule by number. **CAUTION**

`iptables -D`

Sources: LINUX COMMAND, komendy

1.6.174 iptables -F

flush all rules. **DANGEROUS**

`iptables -F`

Sources: LINUX COMMAND, komendy

1.6.175 iptables -F flushing all rules remotely:

Dangerous Mistakes

`iptables -F`` flushing all rules remotely.

Sources: komendy

1.6.176 iptables -F on remote server without console access:

Dangerous Mistakes

`iptables -F`` on remote server without console access.

Sources: komendy

1.6.177 iptables -I

Wstawia regułę

`iptables -I`

Sources: LINUX COMMAND

1.6.178 iptables -L -n -v # filter

Wyświetlanie reguł

`iptables -L -n -v # filter`

Sources: linux_engineering_book.md, linux_engineering_book.pdf

1.6.179 iptables -L -v -n

list rules with counters. **SAFE**

`iptables -L -v -n`

Sources: komendy

1.6.180 iptables -L → firewall blokuje?

Dodatek B: Komendy Ratunkowe

iptables -L → firewall blokuje?

Sources: linux_engineering_book.md, linux_engineering_book.pdf

1.6.181 iptables -N

Nowy chain

iptables -N

Sources: LINUX COMMAND

1.6.182 iptables -P

Ustawia policy

iptables -P

Sources: LINUX COMMAND

1.6.183 iptables -R

Zastępuje regułę

iptables -R

Sources: LINUX COMMAND

1.6.184 iptables -X

Usuwa łańcuchy user-defined

iptables -X

Sources: LINUX COMMAND

1.6.185 iptables -j

Ciche odrzucenie

iptables -A

Sources: LINUX COMMAND

1.6.186 iptables -m conntrack

Connection tracking

iptables -A

Sources: LINUX COMMAND

1.6.187 iptables -m iprange

Zakres IP

iptables -A

Sources: LINUX COMMAND

1.6.188 iptables -m limit

Rate limiting

```
iptables -A
```

Sources: LINUX COMMAND

1.6.189 iptables -m mac

MAC address

```
iptables -A
```

Sources: LINUX COMMAND

1.6.190 iptables -m multiport

Wiele portów

```
iptables -A
```

Sources: LINUX COMMAND

1.6.191 iptables -m recent

Recent connections

```
iptables -A
```

Sources: LINUX COMMAND

1.6.192 iptables -m string

String matching

```
iptables -A
```

Sources: LINUX COMMAND

1.6.193 iptables -m time

Time-based rules

```
iptables -A
```

Sources: LINUX COMMAND

1.6.194 iptables -t mangle

Komentarz Komentarz w skrypcie function myfunc() { Definicja funkcji

```
iptables -t mangle -L
```

```
iptables -t mangle
```

Sources: LINUX COMMAND, RHCSA_Command_Library.pdf

1.6.195 iptables -t nat

NAT table

```
iptables -t nat -L
```

Sources: LINUX COMMAND

1.6.196 iptables -t nat -A

iptables (legacy)

```
iptables -t nat -A
```

Sources: linux_engineering_book.md, linux_engineering_book.pdf

1.6.197 iptables -t nat -L

Komentarz Komentarz w skrypcie function myfunc() { Definicja funkcji

```
iptables -t nat -L
```

Sources: RHCSA_Command_Library.pdf

1.6.198 iptables -t nat -L -n -v

Wyświetlanie reguł

```
iptables -t nat -L -n -v #
```

Sources: linux_engineering_book.md, linux_engineering_book.pdf

1.6.199 iptables -t raw

Raw table

```
iptables -t raw -L
```

Sources: LINUX COMMAND

1.6.200 iptables-restore

Przywracanie iptables

```
iptables-restore < rules.txt
```

```
iptables-restore < /etc/sysconfig/iptables
```

Sources: LINUX COMMAND

1.6.201 iptables-restore < /etc/iptables/rules.v4

iptables (legacy)

```
iptables-restore < /etc/iptables/rules.v4
```

Sources: linux_engineering_book.md, linux_engineering_book.pdf

1.6.202 iptables-restore < /root/rules.backup

restore. **CAUTION**

```
iptables-restore < /root/rules.backup
```

Sources: komendy

1.6.203 iptables-save

Zapisuje reguły

```
iptables-save > rules.txt
```

```
iptables-save > /etc/sysconfig/iptables
```

Sources: LINUX COMMAND

1.6.204 iptables-save > /etc/iptables/rules.v4

iptables (legacy)

iptables-save > /etc/iptables/rules.v4

Sources: linux_engineering_book.md, linux_engineering_book.pdf

1.6.205 iptables-save > /root/rules.backup

backup. **SAFE**

iptables-save > /root/rules.backup

Sources: komendy

1.6.206 ipvsadm

IP Virtual Server (LVS)

ipvsadm -Ln

Sources: LINUX COMMAND

1.6.207 iw

Wireless configuration

iw dev wlan0 scan

Sources: LINUX COMMAND

1.6.208 iw dev wlan0 scan

scan networks. **SAFE**

iw dev wlan0 scan

Sources: komendy

1.6.209 iwconfig

legacy wireless. **SAFE**

iwconfig wlan0

iwconfig

Sources: LINUX COMMAND, komendy

1.6.210 iwconfig wlan0 # parametry

7.4 Narzędzia Diagnostyczne Sprzętu

iwconfig wlan0 # *parametry*

Sources: linux_engineering_book.md, linux_engineering_book.pdf

1.6.211 iwevent

Wireless events

iwevent

Sources: LINUX COMMAND

1.6.212 iwlist

Wireless list

```
iwlist wlan0 scan
```

Sources: LINUX COMMAND

1.6.213 keepalived

VRRP/HA

```
systemctl
```

Sources: LINUX COMMAND

1.6.214 lftp

Advanced FTP client

```
lftp ftp.example.com
```

Sources: LINUX COMMAND

1.6.215 mount.cifs

Mount CIFS/SMB

```
mount -t cifs //server/share /mnt -o username=user
```

Sources: LINUX COMMAND

1.6.216 mount.nfs

Mount NFS

```
mount -t nfs server:/export /mnt
```

Sources: LINUX COMMAND

1.6.217 mount.nfs4

Mount NFSv4

```
mount -t nfs4 server:/export /mnt
```

Sources: LINUX COMMAND

1.6.218 mtr

Dynamiczny traceroute

```
mtr google.com
```

Sources: LINUX COMMAND

1.6.219 mtr 8.8.8.8 # interaktywny traceroute

8.4 Diagnostyka Sieci

```
mtr 8.8.8.8 # interaktywny traceroute
```

Sources: linux_engineering_book.md, linux_engineering_book.pdf

1.6.220 `mtr google.com`

dynamic path analysis. **SAFE**

`mtr google.com`

Sources: komendy

1.6.221 `mtr hostname`

Sekcja Komend

`mtr hostname`

Sources: RHCSA_Command_Library.pdf

1.6.222 `named-checkconf`

Sprawdza `named.conf`

`named-checkconf`

Sources: LINUX COMMAND

1.6.223 `named-checkzone`

Sprawdza strefę DNS

`named-checkzone example.com db.example.com`

Sources: LINUX COMMAND

1.6.224 `nc -l -p 1234`

listen on port. **CAUTION**

`nc -l -p 1234`

Sources: komendy

1.6.225 `nc -zv host 80`

port test. **SAFE**

`nc -zv host 80`

Sources: komendy

1.6.226 `nc / netcat`

Narzędzie sieciowe

`nc -zv localhost 22`

Sources: LINUX COMMAND

1.6.227 `ncat`

Nmap netcat (nowoczesne)

`ncat -l 8080`

Sources: LINUX COMMAND

1.6.228 networkctl

Zarządzanie siecią (systemd)

networkctl

Sources: LINUX COMMAND, linux_engineering_book.md, linux_engineering_book.pdf

1.6.229 nfct

Netfilter conntrack tool

nfct

Sources: LINUX COMMAND

1.6.230 nfsiostat

NFS I/O stats

nfsiostat

Sources: LINUX COMMAND

1.6.231 nfsstat

NFS statistics

nfsstat

Sources: LINUX COMMAND

1.6.232 nft

Uruchomienie kontenera z pełnymi uprawnieniami roota hosta i dostępem do surowego systemu

nft

Sources: LINUX COMMAND, encyklopedia_inzynierii_linux.html, encyklopedia_inzynierii_linux.pdf, komendy, linux_engineering_book.md, linux_engineering_book.pdf

1.6.233 nft add chain

Dodaje chain

```
nft add chain inet filter input { type filter hook input priority 0 \; }
```

Sources: LINUX COMMAND

1.6.234 nft add chain inet edge_firewall inbound { type filter hook input priority 0 \; policy drop

Dodanie bazowego łańcucha wejściowego z domyślną polityką odrzucania ruchu (Drop)

```
nft add chain inet edge_firewall inbound { type filter hook input priority 0 \; policy drop
```

Sources: encyklopedia_inzynierii_linux.pdf

1.6.235 nft add chain inet edge_firewall inbound { type filter hook input priority 0 \; policy drop \; }

Dodanie bazowego łańcucha wejściowego z domyślną polityką odrzucania ruchu (Drop)

```
nft add chain inet edge_firewall inbound { type filter hook input priority 0 \; policy drop \; }
```

Sources: encyklopedia_inzynierii_linux.html

1.6.236 `nft add chain inet filter input { type filter hook input priority 0 \;`

Tworzenie tabeli i łańcucha

`nft add chain inet filter input { type filter hook input priority 0 \;`

Sources: linux_engineering_book.pdf

1.6.237 `nft add chain inet filter input { type filter hook input priority 0 \; policy drop \; }`

nftables (nowoczesny)

`nft add chain inet filter input { type filter hook input priority 0 \; policy drop \; }`

Sources: linux_engineering_book.md

1.6.238 `nft add chain inet filter input { type filter hook input priority 0; policy drop; }`

create chain. **CAUTION**

`nft add chain inet filter input { type filter hook input priority 0; policy drop; }`

Sources: komendy

1.6.239 `nft add rule`

Dodaje regułę

`nft add rule inet filter input tcp dport 22 accept`

Sources: LINUX COMMAND

1.6.240 `nft add rule inet edge_firewall inbound ct state established,related accept`

Dodanie reguł zezwalających na ruch powiązany (Established) oraz ruch na interfejsie lokalnym (Loopback)

`nft add rule inet edge_firewall inbound ct state established,related accept`

Sources: encyklopedia_inzynierii_linux.html, encyklopedia_inzynierii_linux.pdf

1.6.241 `nft add rule inet edge_firewall inbound iifname "lo" accept`

Dodanie reguł zezwalających na ruch powiązany (Established) oraz ruch na interfejsie lokalnym (Loopback)

`nft add rule inet edge_firewall inbound iifname "lo" accept`

Sources: encyklopedia_inzynierii_linux.html, encyklopedia_inzynierii_linux.pdf

1.6.242 `nft add rule inet filter input ct state established,related accept`

nftables (nowoczesny)

`nft add rule inet filter input ct state established,related accept`

Sources: linux_engineering_book.md, linux_engineering_book.pdf

1.6.243 `nft add rule inet filter input iif lo accept`

nftables (nowoczesny)

`nft add rule inet filter input iif lo accept`

Sources: linux_engineering_book.md, linux_engineering_book.pdf

1.6.244 `nft add rule inet filter input tcp dport 22 accept`

add rule. **CAUTION**

`nft add rule inet filter input tcp dport 22 accept`

Sources: komendy, linux_engineering_book.md, linux_engineering_book.pdf

1.6.245 `nft add rule inet filter input tcp dport {80, 443} accept`

nftables (nowoczesny)

`nft add rule inet filter input tcp dport {80, 443} accept`

Sources: linux_engineering_book.md, linux_engineering_book.pdf

1.6.246 `nft add table`

Dodaje tabelę

`nft add table inet filter`

Sources: LINUX COMMAND

1.6.247 `nft add table inet edge_firewall`

Utworzenie nowej zunifikowanej tabeli dla protokołów IPv4 i IPv6

`nft add table inet edge_firewall`

Sources: encyklopedia_inzynierii_linux.html, encyklopedia_inzynierii_linux.pdf

1.6.248 `nft add table inet filter`

create table. **CAUTION**

`nft add table inet filter`

Sources: komendy, linux_engineering_book.md, linux_engineering_book.pdf

1.6.249 `nft delete rule`

Usuwa regułę

`nft delete rule inet filter input handle 10`

Sources: LINUX COMMAND

1.6.250 `nft delete table inet filter`

flush table. **DANGEROUS**

`nft delete table inet filter`

Sources: komendy

1.6.251 `nft flush`

Czyści

`nft flush ruleset`

Sources: LINUX COMMAND

1.6.252 nginx

Web server/reverse proxy

```
nginx -t  
nginx
```

Sources: LINUX COMMAND, RHCSA_Command_Library.pdf

1.6.253 nginx -s

Reload nginx

```
nginx -s
```

Sources: LINUX COMMAND

1.6.254 nmap

Skaner portów

```
nmap -sT localhost
```

Sources: LINUX COMMAND

1.6.255 nmap -sV 192.168.1.0/24 # skanowanie portów

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```
nmap -sV 192.168.1.0/24 # skanowanie portów
```

Sources: linux_engineering_book.md, linux_engineering_book.pdf

1.6.256 nmblookup

NetBIOS lookup

```
nmblookup server
```

Sources: LINUX COMMAND

1.6.257 nmblookup -A server

Komentarz Komentarz w skrypcie function myfunc() { Definicja funkcji

```
nmblookup -A server
```

Sources: RHCSA_Command_Library.pdf

1.6.258 nmcli

NetworkManager CLI

```
nmcli dev
```

Sources: LINUX COMMAND

1.6.259 nmcli con

Lista połączeń

```
nmcli con
```

Sources: LINUX COMMAND

1.6.260 nmcli con add type bond

Tworzy bond

```
nmcli con add type bond con-name bond0 ifname bond0
```

Sources: LINUX COMMAND

1.6.261 nmcli con add type bridge

Tworzy bridge NM

```
nmcli con add type bridge con-name br0 ifname br0
```

Sources: LINUX COMMAND

1.6.262 nmcli con add type team

Tworzy team w NM

```
nmcli con add type team con-name team0 ifname team0
```

Sources: LINUX COMMAND

1.6.263 nmcli con add type vlan

Tworzy VLAN NM

```
nmcli con add type vlan con-name vlan10 ifname eth0.10 dev eth0 id 10
```

Sources: LINUX COMMAND

1.6.264 nmcli con down

Deaktywuje

```
nmcli con down "Wired connection 1"
```

Sources: LINUX COMMAND

1.6.265 nmcli con mod

Modyfikuje połączenie

```
nmcli con mod "eth0" ipv4.addresses 192.168.1.10/24
```

Sources: LINUX COMMAND

1.6.266 nmcli con up

Aktywuje połączenie

```
nmcli con up "Wired connection 1"
```

Sources: LINUX COMMAND

1.6.267 nmcli con up id "Wired connection 1"

activate connection. **CAUTION**

```
nmcli con up id "Wired connection 1"
```

Sources: komendy

1.6.268 nmcli connection

Komentarz Komentarz w skrypcie function myfunc() { Definicja funkcji

nmcli connection

Sources: RHCSA_Command_Library.pdf, linux_engineering_book.md, linux_engineering_book.pdf

1.6.269 nmcli connection add type ethernet con-name "LAN" ifname eth0

NetworkManager

nmcli connection add type ethernet con-name "LAN" ifname eth0

Sources: linux_engineering_book.md, linux_engineering_book.pdf

1.6.270 nmcli connection modify "LAN" ipv4.addresses 192.168.1.10/24

NetworkManager

nmcli connection modify "LAN" ipv4.addresses 192.168.1.10/24

Sources: linux_engineering_book.md, linux_engineering_book.pdf

1.6.271 nmcli connection modify "LAN" ipv4.dns "8.8.8.8 8.8.4.4"

NetworkManager

nmcli connection modify "LAN" ipv4.dns "8.8.8.8 8.8.4.4"

Sources: linux_engineering_book.md, linux_engineering_book.pdf

1.6.272 nmcli connection modify "LAN" ipv4.gateway 192.168.1.1

NetworkManager

nmcli connection modify "LAN" ipv4.gateway 192.168.1.1

Sources: linux_engineering_book.md, linux_engineering_book.pdf

1.6.273 nmcli connection modify "LAN" ipv4.method manual

NetworkManager

nmcli connection modify "LAN" ipv4.method manual

Sources: linux_engineering_book.md, linux_engineering_book.pdf

1.6.274 nmcli connection up "LAN"

NetworkManager

nmcli connection up "LAN"

Sources: linux_engineering_book.md, linux_engineering_book.pdf

1.6.275 nmcli dev

Szczegóły urządzenia

nmcli dev

Sources: LINUX COMMAND

1.6.276 nmcli device

NetworkManager status. **SAFE**

`nmcli device`

Sources: komendy, linux_engineering_book.md, linux_engineering_book.pdf

1.6.277 nmcli device connect

Łączy

`nmcli device connect eth0`

Sources: LINUX COMMAND

1.6.278 nmcli device disconnect

Rozłącza

`nmcli device disconnect eth0`

Sources: LINUX COMMAND

1.6.279 nmcli device reapply

Reaplikuje konfigurację

`nmcli device reapply eth0`

Sources: LINUX COMMAND

1.6.280 nmcli device wifi

WiFi scan

`nmcli device wifi`

Sources: LINUX COMMAND

1.6.281 nmcli device wifi connect

WiFi connect

`nmcli device wifi connect "SSID" password "pass"`

Sources: LINUX COMMAND

1.6.282 nmcli networking

Włącza/wyłącza networking

`nmcli networking off`

Sources: LINUX COMMAND

1.6.283 nmcli radio

Zarządzanie radiem

`nmcli radio wifi off`

Sources: LINUX COMMAND

1.6.284 nmstatectl

Deklaratywna konfiguracja (RHEL 9+)

nmstatectl

Sources: LINUX COMMAND

1.6.285 nmtui

TUI NetworkManager

nmtui

Sources: LINUX COMMAND

1.6.286 nslookup

Zapytania DNS

nslookup google.com

Sources: LINUX COMMAND

1.6.287 nslookup -type=NS

Rekord NS

nslookup -type=NS example.com

Sources: LINUX COMMAND

1.6.288 nslookup example.com # proste zapytanie

Diagnostyka

nslookup example.com # *proste zapytanie*

Sources: linux_engineering_book.md, linux_engineering_book.pdf

1.6.289 nslookup host

DNS resolution. **SAFE**

nslookup host

Sources: komendy

1.6.290 nslookup hostname

Sekcja Komend

nslookup hostname

Sources: RHCSA_Command_Library.pdf

1.6.291 pdbedit

Samba user DB

pdbedit -L

Sources: LINUX COMMAND

1.6.292 ping

Use with , , to avoid DNS delays in scripts.

```
ping -c 4 8.8.8.8  
ping
```

Sources: LINUX COMMAND, komendy

1.6.293 ping -c 4 8.8.8.8

ICMP echo. **SAFE**

```
ping -c 4 8.8.8.8
```

Sources: komendy

1.6.294 ping -c 4 8.8.8.8

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```
ping -c 4 8.8.8.8 #
```

Sources: linux_engineering_book.md, linux_engineering_book.pdf

1.6.295 ping -c 4 gateway

L3 connectivity. **SAFE**

```
ping -c 4 gateway
```

Sources: komendy

1.6.296 ping -c 4 hostname

Sekcja Komend

```
ping -c 4 hostname
```

Sources: RHCSA_Command_Library.pdf

1.6.297 ping -i 0.2 hostname

Sekcja Komend

```
ping -i 0.2 hostname
```

Sources: RHCSA_Command_Library.pdf

1.6.298 ping -s 1400

Sekcja Komend

```
ping -s 1400
```

Sources: RHCSA_Command_Library.pdf

1.6.299 ping gateway → łączność

Dodatek B: Komendy Ratunkowe

```
ping gateway → łączność
```

Sources: linux_engineering_book.md, linux_engineering_book.pdf

1.6.300 ping hostname

Sekcja Komend

ping hostname

Sources: RHCSA_Command_Library.pdf

1.6.301 ping6

Test ICMPv6

ping6 -c 4 google.com

Sources: LINUX COMMAND

1.6.302 ping6 hostname

Sekcja Komend

ping6 hostname

Sources: RHCSA_Command_Library.pdf

1.6.303 resolvectl

17.4 Networkd i Resolved

resolvectl query google.com

resolvectl

Sources: LINUX COMMAND, linux_engineering_book.md, linux_engineering_book.pdf

1.6.304 resolvectl dns eth0 8.8.8.8 8.8.4.4

systemd-resolved

resolvectl dns eth0 8.8.8.8 8.8.4.4

Sources: linux_engineering_book.md, linux_engineering_book.pdf

1.6.305 resolvectl flush-caches

Czyści cache DNS

resolvectl flush-caches

Sources: LINUX COMMAND

1.6.306 resolvectl query

Query DNS

resolvectl query google.com

Sources: LINUX COMMAND

1.6.307 resolvectl query example.com

17.4 Networkd i Resolved

resolvectl query example.com

Sources: linux_engineering_book.md, linux_engineering_book.pdf

1.6.308 systemd-resolve

Legacy resolvectl

`systemd-resolve --status`

Sources: LINUX COMMAND

1.6.309 rndc

BIND control

`rndc`

Sources: LINUX COMMAND

1.6.310 route

Legacy route (deprecated)

`route -n`

`route`

Sources: LINUX COMMAND, komendy

1.6.311 route -n

legacy routing. **SAFE** (read-only)

`route -n`

Sources: komendy

1.6.312 rpc.mountd

NFS mount daemon

`systemctl`

Sources: LINUX COMMAND

1.6.313 rpc.nfsd

NFS daemon

`systemctl`

Sources: LINUX COMMAND

1.6.314 rpcbind

RPC portmapper

`systemctl`

Sources: LINUX COMMAND

1.6.315 rpcclient

RPC client

`rpcclient -U user server`

Sources: LINUX COMMAND

1.6.316 rpcinfo

RPC info

```
rpcinfo -p
```

Sources: LINUX COMMAND

1.6.317 rpcinfo -p server

Sekcja Komend

```
rpcinfo -p server
```

Sources: RHCSA_Command_Library.pdf

1.6.318 showmount

NFS exports

```
showmount -e server
```

Sources: LINUX COMMAND

1.6.319 showmount -a server

Sekcja Komend

```
showmount -a server
```

Sources: RHCSA_Command_Library.pdf

1.6.320 showmount -e server

Sekcja Komend

```
showmount -e server
```

Sources: RHCSA_Command_Library.pdf

1.6.321 siege

HTTP load tester

```
siege -c 10 -t 1M http://localhost
```

Sources: LINUX COMMAND

1.6.322 smbclient

Komentarz Komentarz w skrypcie function myfunc() { Definicja funkcji

```
smbclient //server/share -U user
```

```
smbclient
```

Sources: LINUX COMMAND, RHCSA_Command_Library.pdf

1.6.323 smbclient -L

Komentarz Komentarz w skrypcie function myfunc() { Definicja funkcji

```
smbclient -L
```

Sources: RHCSA_Command_Library.pdf

1.6.324 smbpasswd

SMB password

```
smbpasswd -a john
```

Sources: LINUX COMMAND

1.6.325 smbstatus

SMB status

```
smbstatus
```

Sources: LINUX COMMAND

1.6.326 socat

TCP relay. **CAUTION**

```
socat
```

Sources: LINUX COMMAND, komendy

1.6.327 ss

Use with , , to avoid DNS delays in scripts.

```
ss -tlnp
```

```
ss
```

Sources: LINUX COMMAND, komendy

1.6.328 ss -4

Tylko IPv4

```
ss -4
```

Sources: LINUX COMMAND

1.6.329 ss -4 -t

Sekcja Komend

```
ss -4 -t
```

Sources: RHCSA_Command_Library.pdf

1.6.330 ss -6

Tylko IPv6

```
ss -6
```

Sources: LINUX COMMAND

1.6.331 ss -6 -t

Sekcja Komend

```
ss -6 -t
```

Sources: RHCSA_Command_Library.pdf

1.6.332 ss -K

kills connections without warning.

ss -K

Sources: komendy

1.6.333 ss -K dst 192.168.1.10 dport = 80

close matching sockets. **DANGEROUS**

ss -K dst 192.168.1.10 dport = 80

Sources: komendy

1.6.334 ss -K kills connections without warning.

Dangerous Mistakes

ss -K kills connections without warning.

Sources: komendy

1.6.335 ss -Z

Sekcja Komend

ss -Z

Sources: RHCSA_Command_Library.pdf

1.6.336 ss -anp

Sekcja Komend

ss -anp

Sources: RHCSA_Command_Library.pdf

1.6.337 ss -i

TCP internal info. **SAFE**

ss -i

Sources: komendy

1.6.338 ss -lntp

Sekcja Komend

ss -lntp

Sources: RHCSA_Command_Library.pdf

1.6.339 ss -lnup

Sekcja Komend

ss -lnup

Sources: RHCSA_Command_Library.pdf

1.6.340 ss -s

socket statistics summary. **SAFE**

```
ss -s
```

Sources: LINUX_COMMAND, RHCSA_Command_Library.pdf, komendy

1.6.341 ss -s # statystyki socketów

8.4 Diagnostyka Sieci

```
ss -s # statystyki socketów
```

Sources: linux_engineering_book.md, linux_engineering_book.pdf

1.6.342 ss -t state

Sekcja Komend

```
ss -t state
```

Sources: RHCSA_Command_Library.pdf

1.6.343 ss -tlnp

Command — klawisz : Tryb komend # Szukaj sowa pod kursorem (do tyu)

```
ss -tlnp
```

Sources: LINUX_COMMAND, RHCSA_Command_Library.pdf

1.6.344 ss -tlnp → usługa nasłuchuje?

Dodatek B: Komendy Ratunkowe

```
ss -tlnp → usługa nasłuchuje?
```

Sources: linux_engineering_book.md, linux_engineering_book.pdf

1.6.345 ss -tn

```
awk '{print $5}'
```

```
cut -d: -f1
```

Sources: linux_engineering_book.pdf

1.6.346 ss -tn | awk '{print \$5}' | cut -d: -f1 | sort | uniq -c | sort -rn | head

Dodatek A: Przydatne Jednowierszowce

```
ss -tn | awk '{print $5}' | cut -d: -f1 | sort | uniq -c | sort -rn | head
```

Sources: linux_engineering_book.md

1.6.347 ss -tulnp

listening TCP/UDP with processes. **SAFE**

```
ss -tulnp
```

Sources: RHCSA_Command_Library.pdf, komendy

1.6.348 `ss -tulnp` # otwarte porty (zastępuje

Łączność

`ss -tulnp` # *otwarte porty (zastępuje*

Sources: linux_engineering_book.pdf

1.6.349 `ss -tulnp` # otwarte porty (zastępuje netstat)

8.4 Diagnostyka Sieci

`ss -tulnp` # *otwarte porty (zastępuje netstat)*

Sources: linux_engineering_book.md

1.6.350 `ss -tulpna`

Analiza aktywnych nasłuchujących gniazd TCP z uwzględnieniem identyfikatorów PID procesów i liczników buforów

`ss -tulpna`

Sources: encyklopedia_inzynierii_linux.html, encyklopedia_inzynierii_linux.pdf

1.6.351 `ss -ulnp`

UDP listening + procesy

`ss -ulnp`

Sources: LINUX COMMAND

1.6.352 `ss dst 192.168.1.100`

Sekcja Komend

`ss dst 192.168.1.100`

Sources: RHCSA_Command_Library.pdf

1.6.353 `ss sport = :22`

Sekcja Komend

`ss sport = :22`

Sources: RHCSA_Command_Library.pdf

1.6.354 `sshfs`

SSH filesystem

`sshfs user@host:/ /mnt`

Sources: LINUX COMMAND

1.6.355 `tc`

Traffic control (QoS)

`tc qdisc add dev eth0 root fq_codel`
`tc`

Sources: LINUX COMMAND, linux_engineering_book.md

1.6.356 tc class

Traffic classes

tc class

Sources: LINUX COMMAND

1.6.357 tc class add dev eth0 parent 1: classid 1:1 htb rate 100mbit

HTB (Hierarchical Token Bucket)

tc class add dev eth0 parent 1: classid 1:1 htb rate 100mbit

Sources: linux_engineering_book.md, linux_engineering_book.pdf

1.6.358 tc class add dev eth0 parent 1:1 classid 1:10 htb rate 80mbit ceil 100mbit

HTB (Hierarchical Token Bucket)

tc class add dev eth0 parent 1:1 classid 1:10 htb rate 80mbit ceil 100mbit

Sources: linux_engineering_book.md, linux_engineering_book.pdf

1.6.359 tc class add dev eth0 parent 1:1 classid 1:30 htb rate 20mbit ceil 50mbit

HTB (Hierarchical Token Bucket)

tc class add dev eth0 parent 1:1 classid 1:30 htb rate 20mbit ceil 50mbit

Sources: linux_engineering_book.md, linux_engineering_book.pdf

1.6.360 tc filter

Filters

tc filter

Sources: LINUX COMMAND

1.6.361 tc qdisc

traffic control. **SAFE**

tc qdisc

Sources: LINUX COMMAND, komendy

1.6.362 tc qdisc add dev eth0 root handle 1: htb default 30

HTB (Hierarchical Token Bucket)

tc qdisc add dev eth0 root handle 1: htb default 30

Sources: linux_engineering_book.md, linux_engineering_book.pdf

1.6.363 tc qdisc add dev eth0 root netem delay 100ms 20ms # 100ms ±20ms

Symulacja opóźnień (netem)

tc qdisc add dev eth0 root netem delay 100ms 20ms # 100ms ±20ms

Sources: linux_engineering_book.md, linux_engineering_book.pdf

1.6.364 `tc qdisc add dev eth0 root netem duplicate 1% # duplikacja`

Symulacja opóźnień (netem)

`tc qdisc add dev eth0 root netem duplicate 1% # duplikacja`

Sources: linux_engineering_book.md, linux_engineering_book.pdf

1.6.365 `tc qdisc add dev eth0 root netem loss 1% # 1% packet loss`

Symulacja opóźnień (netem)

`tc qdisc add dev eth0 root netem loss 1% # 1% packet loss`

Sources: linux_engineering_book.md, linux_engineering_book.pdf

1.6.366 `tc qdisc add dev eth0 root tbf rate 100mbit burst 32kbit latency 400ms`

23.1 Traffic Control (tc)

`tc qdisc add dev eth0 root tbf rate 100mbit burst 32kbit latency 400ms`

Sources: linux_engineering_book.md, linux_engineering_book.pdf

1.6.367 `tc - traffic control`

some avg10=0.00 avg60=0.00 avg300=0.00 total=0

`tc - traffic control`

Sources: linux_engineering_book.pdf

1.6.368 `tcpdump`

Sniffer pakietów

`tcpdump -i eth0 port 80`

Sources: LINUX COMMAND

1.6.369 `tcpdump -i`

Sniff na interfejsie

`tcpdump -i eth0`

Sources: LINUX COMMAND

1.6.370 `tcpdump -i any -nn port 80`

capture HTTP. **SAFE**

`tcpdump -i any -nn port 80`

Sources: komendy

1.6.371 `tcpdump -i any -w`

Komentarz Komentarz w skrypcie function myfunc() { Definicja funkcji

`tcpdump -i any -w`

Sources: RHCSA_Command_Library.pdf

1.6.372 `tcpdump -i any -w capture.pcap # zapis do pliku`

8.4 Diagnostyka Sieci

`tcpdump -i any -w capture.pcap # zapis do pliku`

Sources: linux_engineering_book.md, linux_engineering_book.pdf

1.6.373 `tcpdump -i any host problematic_ip`

targeted capture. **SAFE**

`tcpdump -i any host problematic_ip`

Sources: komendy

1.6.374 `tcpdump -i eth0 -w capture.pcap`

write to file. **SAFE**

`tcpdump -i eth0 -w capture.pcap`

Sources: komendy

1.6.375 `tcpdump -i eth0 port 80 # przechwytywanie pakietów`

8.4 Diagnostyka Sieci

`tcpdump -i eth0 port 80 # przechwytywanie pakietów`

Sources: linux_engineering_book.md, linux_engineering_book.pdf

1.6.376 `tcpdump -nn`

Bez resolving DNS/portów

`tcpdump -nn port 53`

Sources: LINUX COMMAND

1.6.377 `tcpdump -r`

Komentarz Komentarz w skrypcie function myfunc() { Definicja funkcji

`tcpdump -r capture.pcap`

`tcpdump -r`

Sources: LINUX COMMAND, RHCSA_Command_Library.pdf

1.6.378 `tcpdump -w`

Zapisuje do pliku

`tcpdump -w capture.pcap`

Sources: LINUX COMMAND

1.6.379 `teamd`

Network teaming

`teamd -g -t team0 -c '{"runner": {"name": "activebackup"}}'`

Sources: LINUX COMMAND

1.6.380 teamctl

Kontrola teamu

```
teamctl team0 state
```

Sources: LINUX COMMAND

1.6.381 teamnl

Team netlink

```
teamnl team0 ports
```

Sources: LINUX COMMAND

1.6.382 telnet

Remote login (insecure)

```
telnet localhost 80
```

Sources: LINUX COMMAND

1.6.383 testparm

Check smb.conf

```
testparm
```

Sources: LINUX COMMAND

1.6.384 tracepath

Śledzenie bez roota

```
tracepath google.com
```

Sources: LINUX COMMAND

1.6.385 tracepath google.com

no-root traceroute. **SAFE**

```
tracepath google.com
```

Sources: komendy

1.6.386 tracepath hostname

Sekcja Komend

```
tracepath hostname
```

Sources: RHCSA_Command_Library.pdf

1.6.387 traceroute

Use with , , to avoid DNS delays in scripts.

```
traceroute google.com
```

```
traceroute
```

Sources: LINUX COMMAND, komendy

1.6.388 `traceroute -I google.com`

trace with ICMP. **SAFE**

`traceroute -I google.com`

Sources: komendy

1.6.389 `traceroute -n`

Sekcja Komend

`traceroute -n`

Sources: RHCSA_Command_Library.pdf

1.6.390 `traceroute 8.8.8.8 # śledzenie trasy`

8.4 Diagnostyka Sieci

`traceroute 8.8.8.8 # śledzenie trasy`

Sources: linux_engineering_book.md, linux_engineering_book.pdf

1.6.391 `traceroute host`

path trace. **SAFE**

`traceroute host`

Sources: komendy

1.6.392 `traceroute hostname`

Sekcja Komend

`traceroute hostname`

Sources: RHCSA_Command_Library.pdf

1.6.393 `tshark`

Wireshark CLI

`tshark -i eth0`

Sources: LINUX COMMAND

1.6.394 `tshark -i eth0`

Komentarz Komentarz w skrypcie function myfunc() { Definicja funkcji

`tshark -i eth0`

Sources: RHCSA_Command_Library.pdf

1.6.395 `tshark -i eth0 -f "port 443" #`

8.4 Diagnostyka Sieci

`tshark -i eth0 -f "port 443" #`

Sources: linux_engineering_book.md, linux_engineering_book.pdf

1.6.396 `tshark -i eth0 -f "tcp port 443"`

Wireshark CLI. **SAFE**

`tshark -i eth0 -f "tcp port 443"`

Sources: komendy

1.6.397 `tshark -r cap.pcap`

Komentarz Komentarz w skrypcie function myfunc() { Definicja funkcji

`tshark -r cap.pcap`

Sources: RHCSA_Command_Library.pdf

1.6.398 `vconfig`

Legacy VLAN

`vconfig add eth0 10`

Sources: LINUX COMMAND

1.6.399 `wbinfo`

Winbind info

`wbinfo -u`

Sources: LINUX COMMAND

1.6.400 `wget`

Pobieranie plików

`wget http://example.com/file`

Sources: LINUX COMMAND

1.6.401 `wget --limit-rate`

Limit prędkości

`wget --limit-rate=200k http://example.com/file`

Sources: LINUX COMMAND

1.6.402 `wget --mirror`

Mirror site

`wget --mirror http://example.com`

Sources: LINUX COMMAND

1.6.403 `wget --user`

Auth wget

`wget --user=user --password=pass http://localhost`

Sources: LINUX COMMAND

1.6.404 `wget -c https://example.com/file # pobieranie z wznowieniem`

8.4 Diagnostyka Sieci

`wget -c https://example.com/file # pobieranie z wznowieniem`

Sources: linux_engineering_book.md, linux_engineering_book.pdf

1.6.405 `wget -c https://example.com/file.iso`

resume download. **SAFE**

`wget -c https://example.com/file.iso`

Sources: komendy

1.6.406 `wget https://cdn.kernel.org/pub/linux/kernel/v6.x/linux-6.8.tar.xz`

16.1 Pobieranie i Konfiguracja

`wget https://cdn.kernel.org/pub/linux/kernel/v6.x/linux-6.8.tar.xz`

Sources: linux_engineering_book.md, linux_engineering_book.pdf

1.6.407 `whois`

Informacje WHOIS

`whois google.com`

Sources: LINUX COMMAND

1.6.408 `whois google.com`

registry lookup. **SAFE**

`whois google.com`

Sources: komendy

1.6.409 `wpa_supplicant`

WiFi supplicant

`wpa_supplicant -i wlan0 -c /etc/wpa_supplicant.conf`

Sources: LINUX COMMAND

1.7 Storage

1.7.1 `/proc`

to wirtualny system plików dostarczający informacji o procesach i jądrze

`/proc`

Sources: linux_engineering_book.md

1.7.2 `blkid`

UUID i typ systemu plików

`blkid /dev/sda1`

`blkid -s`

`blkid`

Sources: LINUX COMMAND, RHCSA_Command_Library.pdf

1.7.3 blkid

11.4 Zarządzanie Dyskami

blkid #

Sources: linux_engineering_book.md, linux_engineering_book.pdf

1.7.4 blkid -t

Sekcja Komend

blkid -t

Sources: RHCSA_Command_Library.pdf

1.7.5 blkid /dev/sda1

UUID and type. **SAFE**

blkid /dev/sda1

Sources: RHCSA_Command_Library.pdf, komendy

1.7.6 blockdev

Operacje na urządzeniach blokowych

blockdev --getsize64 /dev/sda

Sources: LINUX COMMAND

1.7.7 cfdisk

Curses fdisk

cfdisk /dev/sda

Sources: LINUX COMMAND

1.7.8 cryptsetup

Zarządzanie LUKS

cryptsetup luksFormat /dev/sdb1

Sources: LINUX COMMAND

1.7.9 cryptsetup close

Zamyka (zamyka mapping)

cryptsetup close secret_disk

Sources: LINUX COMMAND

1.7.10 cryptsetup luksFormat

Formatuje jako LUKS

cryptsetup luksFormat /dev/sdb1

Sources: LINUX COMMAND

1.7.11 cryptsetup open

Otwiera (odmappingowuje)

```
cryptsetup open /dev/sdb1 secret_disk
```

Sources: LINUX COMMAND

1.7.12 dd

Uruchomienie kontenera z pełnymi uprawnieniami roota hosta i dostępem do surowego systemu

```
dd if=/dev/sda of=/backup.img bs=1M
dd
```

Sources: LINUX COMMAND, RHCSA_Command_Library.pdf, encyklopedia_inzynierii_linux.pdf, komendy

1.7.13 dd if=/dev/sda

```
gzip > /backup/disk.img.gz
```

```
dd if=/dev/sda
```

Sources: linux_engineering_book.pdf

1.7.14 dd if=/dev/sda of=/backup/disk.img bs=4M status=progress

19.1 Narzędzia Backup

```
dd if=/dev/sda of=/backup/disk.img bs=4M status=progress
```

Sources: linux_engineering_book.md, linux_engineering_book.pdf

1.7.15 dd if=/dev/sda of=/dev/sdb bs=4M status=progress

clone disk. **DANGEROUS**

```
dd if=/dev/sda of=/dev/sdb bs=4M status=progress
```

Sources: encyklopedia_inzynierii_linux.html, encyklopedia_inzynierii_linux.pdf, komendy

1.7.16 dd if=/dev/sda | gzip > /backup/disk.img.gz

19.1 Narzędzia Backup

```
dd if=/dev/sda | gzip > /backup/disk.img.gz
```

Sources: linux_engineering_book.md

1.7.17 dd if=/dev/zero

Command — klawisz : Tryb komend # Szukaj sowa pod kursorem (do tyu)

```
dd if=/dev/zero
```

Sources: RHCSA_Command_Library.pdf

1.7.18 dd if=/dev/zero of=/dev/sda bs=1M

wipe disk. **DANGEROUS**

```
dd if=/dev/zero of=/dev/sda bs=1M
```

Sources: komendy

1.7.19 `dd if=/dev/zero of=test bs=1M count=1024 oflag=direct`

11.4 Zarządzanie Dyskami

`dd if=/dev/zero of=test bs=1M count=1024 oflag=direct`

Sources: linux_engineering_book.md, linux_engineering_book.pdf

1.7.20 `dd if=/tmp/test`

Command — klawisz : Tryb komend # Szukaj sowa pod kursorem (do tyu)

`dd if=/tmp/test`

Sources: RHCSA_Command_Library.pdf

1.7.21 `debugfs`

Debugowanie ext

`debugfs /dev/sda1`

Sources: LINUX COMMAND

1.7.22 `debugfs /dev/sda1`

ext2/3/4 debugger. **DANGEROUS**

`debugfs /dev/sda1`

Sources: komendy

1.7.23 `dmsetup`

Zarządzanie device-mapper

`dmsetup ls`

Sources: LINUX COMMAND

1.7.24 `dumpe2fs`

Informacje o superbloku ext

`dumpe2fs /dev/sda1`

Sources: LINUX COMMAND

1.7.25 `e2fsck`

Sprawdza ext2/3/4

`e2fsck -f /dev/sda1`

Sources: LINUX COMMAND

1.7.26 `e2fsck -b 32768 -y /dev/sdb1`

Wymuszenie naprawy struktury za pomocą wskaźnika do czystego superbloku zapasowego (np. blok 32768)

`e2fsck -b 32768 -y /dev/sdb1`

Sources: encyklopedia_inzynierii_linux.html, encyklopedia_inzynierii_linux.pdf

1.7.27 e2fsck -f /dev/sda1

force check ext. **DANGEROUS**

```
e2fsck -f /dev/sda1
```

Sources: komendy

1.7.28 e2fsck -f /dev/sdb1 # sprawdzenie

ext4 (Fourth Extended Filesystem)

```
e2fsck -f /dev/sdb1 # sprawdzenie
```

Sources: linux_engineering_book.md, linux_engineering_book.pdf

1.7.29 e2label

Etykieta ext

```
e2label /dev/sda1 "data"
```

Sources: LINUX COMMAND

1.7.30 e2label /dev/sda1

ext label. **SAFE**

```
e2label /dev/sda1
```

Sources: komendy

1.7.31 edquota

Edytuje quota

```
edquota -u john
```

Sources: LINUX COMMAND

1.7.32 fdisk

Partycjonowanie MBR/GPT

```
fdisk /dev/sda
```

Sources: LINUX COMMAND

1.7.33 fdisk -l

Lista partycji

```
fdisk -l
```

Sources: LINUX COMMAND, RHCSA_Command_Library.pdf

1.7.34 fdisk -l /dev/sda

list partitions. **SAFE**

```
fdisk -l /dev/sda
```

Sources: RHCSA_Command_Library.pdf, komendy

1.7.35 `fdisk -u /dev/sda` `fdisk` z jednostkami sektorów `hdparm -I /dev/sda`

Sekcja Komend

`fdisk -u /dev/sda` `fdisk` z jednostkami sektorów `hdparm -I /dev/sda`

Sources: `RHCSA_Command_Library.pdf`

1.7.36 `fdisk /dev/sda`

interactive MBR/GPT. **DANGEROUS**

`fdisk /dev/sda`

Sources: `RHCSA_Command_Library.pdf`, komendy

1.7.37 `fdisk /dev/sdb #`

11.4 Zarządzanie Dyskami

`fdisk /dev/sdb #`

Sources: `linux_engineering_book.md`, `linux_engineering_book.pdf`

1.7.38 `findfs`

Znajduje FS po UUID/etykiecie

`findfs`

Sources: `LINUX COMMAND`

1.7.39 `fsck`

on mounted read-write filesystem.

`fsck /dev/sda1`

`fsck`

Sources: `LINUX COMMAND`, komendy

1.7.40 `fsck -n`

Use first to assess damage.

`fsck -n`

Sources: komendy

1.7.41 `fsck -n /dev/sda1`

dry-run filesystem check. **SAFE**

`fsck -n /dev/sda1`

Sources: komendy

1.7.42 `fsck -y`

Always have backups before running or .

`fsck -y`

Sources: komendy

1.7.43 `fsck -y /dev/sda1`

auto-repair. **DANGEROUS** (may cause data loss)

```
fsck -y /dev/sda1
```

Sources: komendy

1.7.44 `fsck /dev/sda1`

check and repair. **DANGEROUS** (unmount first)

```
fsck /dev/sda1
```

Sources: komendy

1.7.45 `gdisk`

GPT partycjonowanie

```
gdisk /dev/sda
```

Sources: LINUX COMMAND

1.7.46 `gdisk -l /dev/sda`

Sekcja Komend

```
gdisk -l /dev/sda
```

Sources: RHCSA_Command_Library.pdf

1.7.47 `gdisk /dev/sda`

GPT partitioning. **DANGEROUS**

```
gdisk /dev/sda
```

Sources: RHCSA_Command_Library.pdf, komendy

1.7.48 `gdisk /dev/sdb #`

11.4 Zarządzanie Dyskami

```
gdisk /dev/sdb #
```

Sources: linux_engineering_book.md, linux_engineering_book.pdf

1.7.49 `kpartx`

Mapowanie partycji (multipath)

```
kpartx -a /dev/sda
```

Sources: LINUX COMMAND

1.7.50 `lsblk`

Drzewo urządzeń blokowych

```
lsblk -f
```

```
lsblk
```

Sources: LINUX COMMAND, RHCSA_Command_Library.pdf

1.7.51 lsblk -d

Sekcja Komend

lsblk -d

Sources: RHCSA_Command_Library.pdf

1.7.52 lsblk -f

block devices + filesystems. **SAFE**

lsblk -f

Sources: RHCSA_Command_Library.pdf, komendy

1.7.53 lsblk -f # drzewo urządzeń z

11.4 Zarządzanie Dyskami

lsblk -f # *drzewo urządzeń z*

Sources: linux_engineering_book.md, linux_engineering_book.pdf

1.7.54 lsblk -f # urządzenia blokowe

7.4 Narzędzia Diagnostyczne Sprzętu

lsblk -f # *urządzenia blokowe*

Sources: linux_engineering_book.md, linux_engineering_book.pdf

1.7.55 lsblk -o

Sekcja Komend

lsblk -o

Sources: RHCSA_Command_Library.pdf

1.7.56 lvchange

Zmienia atrybuty LV

lvchange -a n /dev/vg_data/lv_www

Sources: LINUX COMMAND

1.7.57 lvchange -ay /dev/vg0/lv_data

activate LV. **SAFE**

lvchange -ay /dev/vg0/lv_data

Sources: komendy

1.7.58 lvconvert

Konwertuje (mirror/thin/snapshot)

lvconvert --type snapshot ...

Sources: LINUX COMMAND

1.7.59 `lvconvert --merge /dev/vg0/lv_snap`

merge snapshot. **CAUTION**

`lvconvert --merge /dev/vg0/lv_snap`

Sources: komendy

1.7.60 `lvcreate`

Tworzy Logical Volume

`lvcreate -L 10G -n lv_www vg_data`

Sources: LINUX COMMAND

1.7.61 `lvcreate -L 100G -n lv_data vg0`

create LV. **SAFE**

`lvcreate -L 100G -n lv_data vg0`

Sources: komendy

1.7.62 `lvcreate -L 100G -n lv_data vg_data`

6.4 LVM (Logical Volume Manager)

`lvcreate -L 100G -n lv_data vg_data`

Sources: linux_engineering_book.md, linux_engineering_book.pdf

1.7.63 `lvcreate -L 10G -s -n lv_snap /dev/vg0/lv_data`

snapshot. **SAFE**

`lvcreate -L 10G -s -n lv_snap /dev/vg0/lv_data`

Sources: komendy

1.7.64 `lvcreate -L 10G -s -n snap_data /dev/vg_data/lv_data`

6.4 LVM (Logical Volume Manager)

`lvcreate -L 10G -s -n snap_data /dev/vg_data/lv_data`

Sources: linux_engineering_book.md, linux_engineering_book.pdf

1.7.65 `lvcreate -l`

Używa %VG lub %FREE

`lvcreate -l 100%FREE -n lv_tmp vg_data`

Sources: LINUX COMMAND

1.7.66 `lvdisplay`

Szczegóły LV

`lvdisplay /dev/vg_data/lv_www`

Sources: LINUX COMMAND

1.7.67 `lvdisplay /dev/vg0/lv_data`

details. **SAFE**

```
lvdisplay /dev/vg0/lv_data
```

Sources: komendy

1.7.68 `lvextend`

Rozszerza LV

```
lvextend -L +5G /dev/vg_data/lv_www
```

Sources: LINUX COMMAND

1.7.69 `lvextend --resizefs -L +500G /dev/mapper/vg_prod_storage-lv_db`

Dynamiczne rozszerzenie wolumenu logicznego wraz z jednoczesnym przeskalowaniem systemu plików XFS online

```
lvextend --resizefs -L +500G /dev/mapper/vg_prod_storage-lv_db
```

Sources: [encyklopedia_inzynierii_linux.html](#), [encyklopedia_inzynierii_linux.pdf](#)

1.7.70 `lvextend -L +50G /dev/vg0/lv_data`

extend LV. **CAUTION**

```
lvextend -L +50G /dev/vg0/lv_data
```

Sources: komendy

1.7.71 `lvextend -L +50G /dev/vg_data/lv_data`

6.4 LVM (Logical Volume Manager)

```
lvextend -L +50G /dev/vg_data/lv_data
```

Sources: [linux_engineering_book.md](#), [linux_engineering_book.pdf](#)

1.7.72 `lvextend -r`

Use to extend filesystem in one step.

```
lvextend -r
```

Sources: komendy

1.7.73 `lvmanconf`

Konfiguracja LVM

```
lvmanconf --enable-halvm
```

Sources: LINUX COMMAND

1.7.74 `lvreduce`

Snapshot before or risky operations.

```
lvreduce -L 5G /dev/vg_data/lv_www
```

```
lvreduce
```

Sources: LINUX COMMAND, komendy

1.7.75 `lvreduce -L -10G /dev/vg0/lv_data`

reduce LV. **DANGEROUS**

`lvreduce -L -10G /dev/vg0/lv_data`

Sources: komendy

1.7.76 `lvremove`

Usuwa LV

`lvremove /dev/vg_data/lv_www`

Sources: LINUX COMMAND

1.7.77 `lvremove /dev/vg0/lv_data`

remove LV. **DANGEROUS**

`lvremove /dev/vg0/lv_data`

Sources: komendy

1.7.78 `lvrename`

Zmienia nazwę LV

`lvrename vg_data lv_old lv_new`

Sources: LINUX COMMAND

1.7.79 `lvresize`

Zmienia rozmiar LV

`lvresize -L 15G /dev/vg_data/lv_www`

Sources: LINUX COMMAND

1.7.80 `lvs`

summary. **SAFE**

`lvs`

Sources: LINUX COMMAND, komendy

1.7.81 `lvscan`

Skazuje LV

`lvscan`

Sources: LINUX COMMAND

1.7.82 `mke2fs`

Tworzy ext2/3/4

`mke2fs -t ext4 /dev/sda1`

Sources: LINUX COMMAND

1.7.83 mke2fs -n /dev/sdb1

Odnalezienie lokalizacji zapasowych superbloków rozmieszczonych w strukturze urządzenia

```
mke2fs -n /dev/sdb1
```

Sources: [encyklopedia_inzynierii_linux.html](#), [encyklopedia_inzynierii_linux.pdf](#)

1.7.84 mkfs

Tworzy system plików

```
mkfs -t ext4 /dev/sda1
```

```
mkfs
```

Sources: LINUX COMMAND, komendy

1.7.85 mkfs.ext4

Formatuje ext4

```
mkfs.ext4 /dev/sda1
```

Sources: LINUX COMMAND

1.7.86 mkfs.ext4 -L

column -t Lista zamontowanych FS

```
mkfs.ext4 -L
```

Sources: [RHCSA_Command_Library.pdf](#)

1.7.87 mkfs.ext4 -L "data" /dev/sdb1

ext4 (Fourth Extended Filesystem)

```
mkfs.ext4 -L "data" /dev/sdb1
```

Sources: [linux_engineering_book.md](#), [linux_engineering_book.pdf](#)

1.7.88 mkfs.ext4 -L data /dev/sda1

create ext4. **DANGEROUS**

```
mkfs.ext4 -L data /dev/sda1
```

Sources: komendy

1.7.89 mkfs.ext4 -m 1

Sekcja Komend

```
mkfs.ext4 -m 1
```

Sources: [RHCSA_Command_Library.pdf](#)

1.7.90 mkfs.ext4 /dev/sdb1

Sekcja Komend

```
mkfs.ext4 /dev/sdb1
```

Sources: [RHCSA_Command_Library.pdf](#)

1.7.91 mkfs.ext4 /dev/vg_data/lv_data

6.4 LVM (Logical Volume Manager)

mkfs.ext4 /dev/vg_data/lv_data

Sources: linux_engineering_book.md, linux_engineering_book.pdf

1.7.92 mkfs.vfat

Formatuje FAT32

mkfs.vfat /dev/sdb1

Sources: LINUX COMMAND

1.7.93 mkfs.vfat /dev/sdb1

Sekcja Komend

mkfs.vfat /dev/sdb1

Sources: RHCSA_Command_Library.pdf

1.7.94 mkfs.xfs

Formatuje XFS

mkfs.xfs /dev/sda1

Sources: LINUX COMMAND

1.7.95 mkfs.xfs -L

Sekcja Komend

mkfs.xfs -L

Sources: RHCSA_Command_Library.pdf

1.7.96 mkfs.xfs -L "data" /dev/sdb1

Zaawansowane

mkfs.xfs -L "data" /dev/sdb1

Sources: linux_engineering_book.md, linux_engineering_book.pdf

1.7.97 mkfs.xfs -L data /dev/sda1

create XFS. **DANGEROUS**

mkfs.xfs -L data /dev/sda1

Sources: komendy

1.7.98 mkfs.xfs -f

Sekcja Komend

mkfs.xfs -f

Sources: RHCSA_Command_Library.pdf

1.7.99 mkfs.xfs /dev/sdb1

Sekcja Komend

mkfs.xfs /dev/sdb1

Sources: RHCSA_Command_Library.pdf

1.7.100 mkswap

Formatuje swap

mkswap /dev/sda2

Sources: LINUX COMMAND

1.7.101 mkswap /dev/sda2

create swap. **DANGEROUS**

mkswap /dev/sda2

Sources: komendy

1.7.102 mkswap /dev/sdb2

Sekcja Komend

mkswap /dev/sdb2

Sources: RHCSA_Command_Library.pdf

1.7.103 mkswap /swapfile

Tworzenie swap

mkswap /swapfile

Sources: linux_engineering_book.md, linux_engineering_book.pdf

1.7.104 parted

Zaawansowane partycjonowanie

parted /dev/sda print

Sources: LINUX COMMAND

1.7.105 parted -l

Sekcja Komend

parted -l

Sources: RHCSA_Command_Library.pdf

1.7.106 parted /dev/sda

interactive partitioning. **DANGEROUS**

parted /dev/sda

Sources: RHCSA_Command_Library.pdf, komendy

1.7.107 parted /dev/sda print

print table. **SAFE**

```
parted /dev/sda print
```

Sources: komendy

1.7.108 parted /dev/sda rm 1

Sekcja Komend

```
parted /dev/sda rm 1
```

Sources: RHCSA_Command_Library.pdf

1.7.109 parted /dev/sdb

11.4 Zarządzanie Dyskami

```
parted /dev/sdb #
```

Sources: linux_engineering_book.md, linux_engineering_book.pdf

1.7.110 partprobe

Use after partition changes instead of reboot.

```
partprobe /dev/sda
```

```
partprobe
```

Sources: LINUX COMMAND, RHCSA_Command_Library.pdf, komendy

1.7.111 partprobe /dev/sda

re-read table. **CAUTION**

```
partprobe /dev/sda
```

Sources: RHCSA_Command_Library.pdf, komendy

1.7.112 partx

Informuje kernel o partycjach

```
partx -a /dev/sda
```

Sources: LINUX COMMAND

1.7.113 partx -u

Aktualizuje tablicę partycji

```
partx -u /dev/sda
```

Sources: LINUX COMMAND

1.7.114 pvcreate

on active filesystem partition.

```
pvcreate /dev/sdb1
```

```
pvcreate
```

Sources: LINUX COMMAND, komendy

1.7.115 pvcreate /dev/sda1

initialize PV. **DANGEROUS**

`pvcreate /dev/sda1`

Sources: komendy

1.7.116 pvcreate /dev/sdb

Sekcja Komend

`pvcreate /dev/sdb`

Sources: RHCSA_Command_Library.pdf

1.7.117 pvcreate /dev/sdb /dev/sdc

6.4 LVM (Logical Volume Manager)

`pvcreate /dev/sdb /dev/sdc`

Sources: linux_engineering_book.md, linux_engineering_book.pdf

1.7.118 pvcreate /dev/sdc

Inicjalizacja nowego fizycznego urządzenia dyskowego dla menedżera LVM

`pvcreate /dev/sdc`

Sources: encyklopedia_inzynierii_linux.html, encyklopedia_inzynierii_linux.pdf

1.7.119 pvdisplay

Sekcja Komend

`pvdisplay /dev/sdb1`

`pvdisplay`

Sources: LINUX_COMMAND, RHCSA_Command_Library.pdf

1.7.120 pvdisplay / vgdisplay / lvdisplay

6.4 LVM (Logical Volume Manager)

`pvdisplay / vgdisplay / lvdisplay`

Sources: linux_engineering_book.md, linux_engineering_book.pdf

1.7.121 pvdisplay /dev/sda1

details. **SAFE**

`pvdisplay /dev/sda1`

Sources: komendy

1.7.122 pvdisplay /dev/sdb

Sekcja Komend

`pvdisplay /dev/sdb`

Sources: RHCSA_Command_Library.pdf

1.7.123 pvmove

Przenosi dane PV

`pvmove /dev/sdb1`

Sources: LINUX COMMAND

1.7.124 pvmove /dev/sdb

Sekcja Komend

`pvmove /dev/sdb`

Sources: RHCSA_Command_Library.pdf

1.7.125 pvremove

Usuwa PV

`pvremove /dev/sdb1`

Sources: LINUX COMMAND

1.7.126 pvremove /dev/sda1

remove PV. **DANGEROUS**

`pvremove /dev/sda1`

Sources: komendy

1.7.127 pvremove /dev/sdb

Sekcja Komend

`pvremove /dev/sdb`

Sources: RHCSA_Command_Library.pdf

1.7.128 pvresize

Zmienia rozmiar PV

`pvresize /dev/sdb1`

Sources: LINUX COMMAND

1.7.129 pvresize /dev/sda1

resize PV after partition change. **CAUTION**

`pvresize /dev/sda1`

Sources: komendy

1.7.130 pvresize /dev/sdb

Sekcja Komend

`pvresize /dev/sdb`

Sources: RHCSA_Command_Library.pdf

1.7.131 pvs

summary. **SAFE**

pvs

Sources: LINUX COMMAND, RHCSA_Command_Library.pdf, komendy

1.7.132 pvs / vgs / lvs

6.4 LVM (Logical Volume Manager)

pvs / vgs / lvs

Sources: linux_engineering_book.md, linux_engineering_book.pdf

1.7.133 pvscan

Sekcja Komend

pvscan

Sources: LINUX COMMAND, RHCSA_Command_Library.pdf

1.7.134 qemu-img

Zarządzanie obrazami QEMU

qemu-img

Sources: LINUX COMMAND

1.7.135 qemu-img convert

Konwersja formatów

qemu-img convert -f raw -O qcow2 disk.raw disk.qcow2

Sources: LINUX COMMAND

1.7.136 qemu-img info

Info o obrazie

qemu-img info disk.qcow2

Sources: LINUX COMMAND

1.7.137 qemu-img resize

Zmiana rozmiaru

qemu-img resize disk.qcow2 +10G

Sources: LINUX COMMAND

1.7.138 qemu-img snapshot

Snapshot obrazu

qemu-img snapshot -c snap1 disk.qcow2

Sources: LINUX COMMAND

1.7.139 quota

Limity dyskowe użytkownika

```
quota -u john
```

Sources: LINUX COMMAND

1.7.140 quotacheck

Sprawdza quota

```
quotacheck -cum /home
```

Sources: LINUX COMMAND

1.7.141 quotaoff

Wyłącza quota

```
quotaoff -v /home
```

Sources: LINUX COMMAND

1.7.142 quotaon

Włącza quota

```
quotaon -v /home
```

Sources: LINUX COMMAND

1.7.143 repquota

Raport quota

```
repquota -a
```

Sources: LINUX COMMAND

1.7.144 rsync

Synchronizacja (różnicowa)

```
rsync -avz /src/ /dest/
```

```
rsync -avz --delete /src/ user@server:/dest/
```

Sources: LINUX COMMAND

1.7.145 rsync --backup

Backup changed files

```
rsync --backup /src/ /dest/
```

Sources: LINUX COMMAND

1.7.146 rsync --bwlimit

Bandwidth limit

```
rsync --bwlimit=1000 /src/ /dest/
```

Sources: LINUX COMMAND

1.7.147 `rsync --delete`

Usuwa pliki nieistniejące w źródle

```
rsync -avz --delete /src/ /dest/  
rsync --delete /src/ /dest/
```

Sources: LINUX COMMAND

1.7.148 `rsync --dry-run`

flags: , , .

```
rsync --dry-run -avz /src/ /dest/  
rsync --dry-run
```

Sources: LINUX COMMAND, komendy

1.7.149 `rsync --exclude`

Exclude pattern

```
rsync --exclude='*.tmp' /src/ /dest/
```

Sources: LINUX COMMAND

1.7.150 `rsync --inplace`

Atomic updates with or .

```
rsync --inplace /src/ /dest/  
rsync --inplace
```

Sources: LINUX COMMAND, komendy

1.7.151 `rsync --link-dest`

Hardlink backups

```
rsync --link-dest=/prev /src/ /dest/
```

Sources: LINUX COMMAND

1.7.152 `rsync --partial`

Resume partial

```
rsync --partial /src/ /dest/
```

Sources: LINUX COMMAND

1.7.153 `rsync --progress`

Show progress

```
rsync --progress /src/ /dest/
```

Sources: LINUX COMMAND

1.7.154 `rsync -a /backup/ /`

restoring without verification.

```
rsync -a /backup/ /
```

Sources: komendy

1.7.155 `rsync -a /backup/ /` restoring without verification.

Dangerous Mistakes

```
rsync -a /backup/ /` restoring without verification.
```

Sources: komendy

1.7.156 `rsync -av --dry-run`

before real sync.

```
rsync -av --dry-run
```

Sources: komendy

1.7.157 `rsync -av --dry-run` before real sync.

Production-Safe Patterns

```
rsync -av --dry-run` before real sync.
```

Sources: komendy

1.7.158 `rsync -av /backup/ /`

restore from backup. **DANGEROUS** (overwrites live system)

```
rsync -av /backup/ /
```

Sources: komendy

1.7.159 `rsync -avz`

Archive, verbose, compress

```
rsync -avz /src/ /dest/
```

Sources: LINUX COMMAND

1.7.160 `rsync -avz --delete /source/ /dest/ # mirror` (usuń nadmiarowe)

rsync — synchronizacja plików

```
rsync -avz --delete /source/ /dest/ # mirror (usuń nadmiarowe)
```

Sources: linux_engineering_book.md, linux_engineering_book.pdf

1.7.161 `rsync -avz --delete src/ dest/`

sync with deletion. **DANGEROUS**

```
rsync -avz --delete src/ dest/
```

Sources: komendy

1.7.162 `rsync -avz --exclude='*.log' /source/ /dest/`

rsync — synchronizacja plików

```
rsync -avz --exclude='*.log' /source/ /dest/
```

Sources: linux_engineering_book.md, linux_engineering_book.pdf

1.7.163 `rsync -avz -e "ssh -p 2222" /local/ user@host:/remote/`

sync over SSH. **CAUTION**

`rsync -avz -e "ssh -p 2222" /local/ user@host:/remote/`

Sources: komendy

1.7.164 `rsync -avz -e ssh /source/ user@host:/dest/ # zdalna kopia`

rsync — synchronizacja plików

`rsync -avz -e ssh /source/ user@host:/dest/ # zdalna kopia`

Sources: linux_engineering_book.md, linux_engineering_book.pdf

1.7.165 `rsync -avz /source/ /destination/ # lokalna kopia`

rsync — synchronizacja plików

`rsync -avz /source/ /destination/ # lokalna kopia`

Sources: linux_engineering_book.md, linux_engineering_book.pdf

1.7.166 `rsync -e ssh`

Rsync przez SSH

`rsync -avz -e "ssh -p 2222" /src/ host:/dest/`

Sources: LINUX COMMAND

1.7.167 `setquota`

Ustawia quota z CLI

`setquota -u john 100000 120000 0 0 /home`

Sources: LINUX COMMAND

1.7.168 `sfdisk`

Scriptable fdisk

`sfdisk -d /dev/sda > parts.txt`

`sfdisk /dev/sda < backup.sfdisk`

Sources: LINUX COMMAND

1.7.169 `sfdisk -d`

Dump partycji

`sfdisk -d /dev/sda > backup.sfdisk`

Sources: LINUX COMMAND

1.7.170 `sgdisk`

GPT manipulacja

`sgdisk -R /dev/sda /dev/sdb`

Sources: LINUX COMMAND

1.7.171 `sgdisk -n 1:0:+100G /dev/sdb #`

11.4 Zarządzanie Dyskami

`sgdisk -n 1:0:+100G /dev/sdb #`

Sources: linux_engineering_book.md, linux_engineering_book.pdf

1.7.172 `sgdisk -p`

Print GPT

`sgdisk -p /dev/sda`

Sources: LINUX COMMAND

1.7.173 `swapoff`

Wyłącza swap

`swapoff /dev/sda2`

Sources: LINUX COMMAND

1.7.174 `swapoff -a`

Sekcja Komend

`swapoff -a`

Sources: RHCSA_Command_Library.pdf

1.7.175 `swapoff /dev/sda2`

deactivate. **CAUTION**

`swapoff /dev/sda2`

Sources: komendy

1.7.176 `swapoff /dev/sdb2`

Sekcja Komend

`swapoff /dev/sdb2`

Sources: RHCSA_Command_Library.pdf

1.7.177 `swapon`

Włącza swap

`swapon /dev/sda2`

Sources: LINUX COMMAND

1.7.178 `swapon --show`

Monitoring

`swapon --show`

Sources: linux_engineering_book.md, linux_engineering_book.pdf

1.7.179 `swapon -a`

activate all from fstab. **CAUTION**

`swapon -a`

Sources: RHCSA_Command_Library.pdf, komendy

1.7.180 `swapon -s`

swap summary. **SAFE**

`swapon -s`

Sources: RHCSA_Command_Library.pdf, komendy

1.7.181 `swapon /dev/sda2`

activate swap. **CAUTION**

`swapon /dev/sda2`

Sources: komendy

1.7.182 `swapon /dev/sdb2`

Sekcja Komend

`swapon /dev/sdb2`

Sources: RHCSA_Command_Library.pdf

1.7.183 `swapon /swapfile`

Tworzenie swap

`swapon /swapfile`

Sources: linux_engineering_book.md, linux_engineering_book.pdf

1.7.184 `tune2fs`

Tuning ext2/3/4

`tune2fs -m 0 /dev/sda1`

Sources: LINUX COMMAND

1.7.185 `tune2fs -L`

Etykieta ext

`tune2fs -L "data" /dev/sda1`

Sources: LINUX COMMAND

1.7.186 `tune2fs -l /dev/sda1`

ext superblock. **SAFE**

`tune2fs -l /dev/sda1`

Sources: komendy

1.7.187 `tune2fs -l /dev/sdb1 # informacje o`

ext4 (Fourth Extended Filesystem)

`tune2fs -l /dev/sdb1 # informacje o`

Sources: linux_engineering_book.md, linux_engineering_book.pdf

1.7.188 `vgchange`

Zmienia atrybuty VG

`vgchange -a n vg_data`

Sources: LINUX COMMAND

1.7.189 `vgchange -an vgname`

Sekcja Komend

`vgchange -an vgname`

Sources: RHCSA_Command_Library.pdf

1.7.190 `vgchange -ay vgname`

Sekcja Komend

`vgchange -ay vgname`

Sources: RHCSA_Command_Library.pdf

1.7.191 `vgcreate`

Tworzy Volume Group

`vgcreate vg_data /dev/sdb1 /dev/sdc1`

Sources: LINUX COMMAND

1.7.192 `vgcreate -s 16M`

Sekcja Komend

`vgcreate -s 16M`

Sources: RHCSA_Command_Library.pdf

1.7.193 `vgcreate vg0 /dev/sda1 /dev/sdb1`

create VG. **CAUTION**

`vgcreate vg0 /dev/sda1 /dev/sdb1`

Sources: komendy

1.7.194 `vgcreate vg_data /dev/sdb /dev/sdc`

6.4 LVM (Logical Volume Manager)

`vgcreate vg_data /dev/sdb /dev/sdc`

Sources: linux_engineering_book.md, linux_engineering_book.pdf

1.7.195 vgcreate vname

Sekcja Komend

`vgcreate vname`

Sources: RHCSA_Command_Library.pdf

1.7.196 vgdisplay

Sekcja Komend

`vgdisplay vg_data`

`vgdisplay`

Sources: LINUX COMMAND, RHCSA_Command_Library.pdf

1.7.197 vgdisplay vg0

details. **SAFE**

`vgdisplay vg0`

Sources: komendy

1.7.198 vgdisplay vname

Sekcja Komend

`vgdisplay vname`

Sources: RHCSA_Command_Library.pdf

1.7.199 vgextend

Rozszerza VG o PV

`vgextend vg_data /dev/sdd1`

Sources: LINUX COMMAND

1.7.200 vgextend vg0 /dev/sdc1

extend VG. **CAUTION**

`vgextend vg0 /dev/sdc1`

Sources: komendy

1.7.201 vgextend vg_prod_storage /dev/sdc

Rozszerzenie istniejącej grupy wolumenów o nowy dysk fizyczny

`vgextend vg_prod_storage /dev/sdc`

Sources: encyklopedia_inzynierii_linux.html, encyklopedia_inzynierii_linux.pdf

1.7.202 vgextend vname

Sekcja Komend

`vgextend vname`

Sources: RHCSA_Command_Library.pdf

1.7.203 **vgreduce**

Zmniejsza VG

```
vgreduce vg_data /dev/sdd1
```

Sources: LINUX COMMAND

1.7.204 **vgreduce vg0 /dev/sdc1**

reduce VG. **CAUTION** (ensure PV empty)

```
vgreduce vg0 /dev/sdc1
```

Sources: komendy

1.7.205 **vgreduce vname**

Sekcja Komend

```
vgreduce vname
```

Sources: RHCSA_Command_Library.pdf

1.7.206 **vgremove**

Usuwa VG

```
vgremove vg_data
```

Sources: LINUX COMMAND

1.7.207 **vgremove vg0**

remove VG. **DANGEROUS**

```
vgremove vg0
```

Sources: komendy

1.7.208 **vgremove vname**

Sekcja Komend

```
vgremove vname
```

Sources: RHCSA_Command_Library.pdf

1.7.209 **vgrename**

Zmienia nazwę VG

```
vgrename vg_old vg_new
```

Sources: LINUX COMMAND

1.7.210 **vgrename oldname**

Sekcja Komend

```
vgrename oldname
```

Sources: RHCSA_Command_Library.pdf

1.7.211 vgs

summary. **SAFE**

vgs

Sources: LINUX COMMAND, RHCSA_Command_Library.pdf, komendy

1.7.212 vgscan

Sekcja Komend

vgscan

Sources: LINUX COMMAND, RHCSA_Command_Library.pdf

1.7.213 wipefs

Czyści sygnatury FS

wipefs -a /dev/sda1

Sources: LINUX COMMAND

1.7.214 xfs_admin

Etykieta XFS

xfs_admin -L "data" /dev/sda1

Sources: LINUX COMMAND

1.7.215 xfs_admin -L

Etykieta XFS

xfs_admin -L "data" /dev/sda1

Sources: LINUX COMMAND

1.7.216 xfs_admin -l /dev/sda1

XFS label. **SAFE**

xfs_admin -l /dev/sda1

Sources: komendy

1.7.217 xfs_check

Sprawdza XFS (deprecated)

xfs_check /dev/sda1

Sources: LINUX COMMAND

1.7.218 xfs_growfs

Rozszerza XFS online

xfs_growfs /mnt

Sources: LINUX COMMAND

1.7.219 xfs_growfs /mnt

grow XFS online. **CAUTION**

xfs_growfs /mnt

Sources: komendy

1.7.220 xfs_growfs /mountpoint

Zaawansowane

xfs_growfs /mountpoint

Sources: linux_engineering_book.md, linux_engineering_book.pdf

1.7.221 xfs_growfs /mountpoint

6.4 LVM (Logical Volume Manager)

xfs_growfs /mountpoint #

Sources: linux_engineering_book.md, linux_engineering_book.pdf

1.7.222 xfs_info

Informacje XFS

xfs_info /dev/sda1

Sources: LINUX COMMAND

1.7.223 xfs_info /dev/sdb1

Zaawansowane

xfs_info /dev/sdb1

Sources: linux_engineering_book.md, linux_engineering_book.pdf

1.7.224 xfs_info /mnt

XFS geometry. **SAFE**

xfs_info /mnt

Sources: komendy

1.7.225 xfs_quota

Quota dla XFS

xfs_quota -x -c "report -h" /home

Sources: LINUX COMMAND

1.7.226 xfs_repair

Always have backups before running or .

xfs_repair /dev/sda1

xfs_repair

Sources: LINUX COMMAND, komendy

1.7.227 `xfs_repair -n /dev/sda1`

no-modify check. **SAFE**

`xfs_repair -n /dev/sda1`

Sources: komendy

1.7.228 `xfs_repair /dev/sda1`

repair XFS. **DANGEROUS**

`xfs_repair /dev/sda1`

Sources: komendy

1.7.229 `xfs_repair /dev/sdb1`

Zaawansowane

`xfs_repair /dev/sdb1`

Sources: linux_engineering_book.md, linux_engineering_book.pdf

1.8 Systemd

1.8.1 bootctl

Zarządzanie bootloaderem

`bootctl`

Sources: LINUX COMMAND

1.8.2 hostnamectl

static/transient/pretty hostname, chassis, OS. **SAFE**

`hostnamectl set-hostname srv1`

`hostnamectl`

Sources: LINUX COMMAND, RHCSA_Command_Library.pdf, komendy

1.8.3 hostnamectl set-hostname srv01

hostname. **CAUTION**

`hostnamectl set-hostname srv01`

Sources: komendy

1.8.4 journalctl

all logs since boot. **SAFE**

`journalctl -xe`

`journalctl`

Sources: LINUX COMMAND, RHCSA_Command_Library.pdf, komendy

1.8.5 journalctl --disk-usage

journal size. **SAFE**

`journalctl --disk-usage`

Sources: komendy

1.8.6 journalctl --disk-usage # zużycie dysku

Journald (logi)

```
journalctl --disk-usage # zużycie dysku
```

Sources: linux_engineering_book.md, linux_engineering_book.pdf

1.8.7 journalctl --list-boots

available boots. **SAFE**

```
journalctl --list-boots
```

Sources: komendy

1.8.8 journalctl --since

Od konkretnego czasu

```
journalctl --since "1 hour ago"
```

```
journalctl --since
```

Sources: LINUX COMMAND, RHCSA_Command_Library.pdf

1.8.9 journalctl --since "1 hour ago"

time window. **SAFE**

```
journalctl --since "1 hour ago"
```

Sources: komendy

1.8.10 journalctl --since "1 hour ago" # od godziny

Journald (logi)

```
journalctl --since "1 hour ago" # od godziny
```

Sources: linux_engineering_book.md, linux_engineering_book.pdf

1.8.11 journalctl --since today --until "2 hours ago"

range. **SAFE**

```
journalctl --since today --until "2 hours ago"
```

Sources: komendy

1.8.12 journalctl --vacuum-size=100M

Dodatek B: Komendy Ratunkowe

```
journalctl --vacuum-size=100M
```

Sources: linux_engineering_book.md, linux_engineering_book.pdf

1.8.13 journalctl --vacuum-size=1G

limit size. **CAUTION**

```
journalctl --vacuum-size=1G
```

Sources: komendy

1.8.14 `journalctl --vacuum-size=500M # ogranicz rozmiar`

Journald (logi)

`journalctl --vacuum-size=500M # ogranicz rozmiar`

Sources: linux_engineering_book.md, linux_engineering_book.pdf

1.8.15 `journalctl --vacuum-time=1s`

deleting all logs.

`journalctl --vacuum-time=1s`

Sources: komendy

1.8.16 `journalctl --vacuum-time=1s deleting all logs.`

Dangerous Mistakes

`journalctl --vacuum-time=1s` deleting all logs.`

Sources: komendy

1.8.17 `journalctl --vacuum-time=30d`

retain 30 days. **CAUTION**

`journalctl --vacuum-time=30d`

Sources: komendy

1.8.18 `journalctl -b`

current boot only. **SAFE**

`journalctl -b`

Sources: LINUX COMMAND, RHCSA_Command_Library.pdf, komendy

1.8.19 `journalctl -b -1`

previous boot. **SAFE**

`journalctl -b -1`

Sources: komendy

1.8.20 `journalctl -b -1 # poprzedni boot`

Journald (logi)

`journalctl -b -1 # poprzedni boot`

Sources: linux_engineering_book.md, linux_engineering_book.pdf

1.8.21 `journalctl -f`

Na żywo (follow)

`journalctl -f`

Sources: LINUX COMMAND

1.8.22 journalctl -f # follow (jak tail -f)

Journald (logi)

```
journalctl -f # follow (jak tail -f)
```

Sources: linux_engineering_book.md, linux_engineering_book.pdf

1.8.23 journalctl -f ledzenie logów na ywo aureport -l

Sekcja Komend

```
journalctl -f ledzenie logów na ywo aureport -l
```

Sources: RHCSA_Command_Library.pdf

1.8.24 journalctl -k

kernel logs only. **SAFE**

```
journalctl -k
```

Sources: komendy, linux_engineering_book.pdf

1.8.25 journalctl -k | grep -i oom

5.4 OOM Killer

```
journalctl -k | grep -i oom
```

Sources: linux_engineering_book.md

1.8.26 journalctl -o

Sekcja Komend

```
journalctl -o
```

Sources: RHCSA_Command_Library.pdf

1.8.27 journalctl -p

Po prioritycie (err, warning)

```
journalctl -p err
```

Sources: LINUX COMMAND

1.8.28 journalctl -p err

priority filter. **SAFE**

```
journalctl -p err
```

Sources: RHCSA_Command_Library.pdf, komendy

1.8.29 journalctl -p err # tylko błędy

Journald (logi)

```
journalctl -p err # tylko błędy
```

Sources: linux_engineering_book.md, linux_engineering_book.pdf

1.8.30 journalctl -u

Logi konkretnej usługi

```
journalctl -u sshd
```

Sources: LINUX COMMAND

1.8.31 journalctl -u nginx

unit-specific logs. **SAFE**

```
journalctl -u nginx
```

Sources: komendy

1.8.32 journalctl -u nginx # logi usługi

Journald (logi)

```
journalctl -u nginx # logi usługi
```

Sources: linux_engineering_book.md, linux_engineering_book.pdf

1.8.33 journalctl -u nginx -f

follow unit logs. **SAFE**

```
journalctl -u nginx -f
```

Sources: komendy

1.8.34 journalctl -u sshd

Sekcja Komend

```
journalctl -u sshd
```

Sources: RHCSA_Command_Library.pdf

1.8.35 journalctl -xb

current boot logs. **SAFE**

```
journalctl -xb
```

Sources: komendy

1.8.36 journalctl -xe

recent errors with context. **SAFE**

```
journalctl -xe
```

Sources: RHCSA_Command_Library.pdf, komendy

1.8.37 kernel-install

Instalacja jądra

```
kernel-install add 5.14...
```

Sources: LINUX COMMAND

1.8.38 localectl

Ustawienia lokalizacji

```
localectl set-locale  
localectl
```

Sources: LINUX COMMAND

1.8.39 loginctl

Zarządzanie sesjami

```
loginctl show-user john
```

Sources: LINUX COMMAND

1.8.40 loginctl show-user alice

user session info. **SAFE**

```
loginctl show-user alice
```

Sources: komendy

1.8.41 machinectl

Zarządzanie maszynami (VM/LXC)

```
machinectl
```

Sources: LINUX COMMAND

1.8.42 systemctl

Uruchomienie kontenera z pełnymi uprawnieniami roota hosta i dostępem do surowego systemu

```
systemctl
```

Sources: LINUX COMMAND, RHCSA_Command_Library.pdf, encyklopedia_inzynierii_linux.html, encyklopedia_inzynierii_li
komendy, linux_engineering_book.md, linux_engineering_book.pdf

1.8.43 systemctl --failed

failed units. **SAFE**

```
systemctl --failed
```

Sources: komendy

1.8.44 systemctl --user enable --now web.service

Podman — drop-in replacement dla Docker (bez daemon)

```
systemctl --user enable --now web.service
```

Sources: linux_engineering_book.md, linux_engineering_book.pdf

1.8.45 systemctl cat

Wyświetla definicję usługi

```
systemctl cat httpd  
systemctl cat
```

Sources: LINUX COMMAND, RHCSA_Command_Library.pdf

1.8.46 `systemctl cat nginx`

show unit file. **SAFE**

```
systemctl cat nginx
```

Sources: komendy

1.8.47 `systemctl daemon-reload`

Przeładowanie konfiguracji menedżera systemd i uruchomienie nowej usługi bezpiecznej

```
systemctl daemon-reload
```

Sources: LINUX COMMAND, [encyklopedia_inzynierii_linux.html](#), [encyklopedia_inzynierii_linux.pdf](#), komendy

1.8.48 `systemctl daemon-reload # przeładuj konfigurację`

Zarządzanie usługami

```
systemctl daemon-reload # przeładuj konfigurację
```

Sources: [linux_engineering_book.md](#), [linux_engineering_book.pdf](#)

1.8.49 `systemctl disable`

Włącza autostart

```
systemctl disable httpd
```

```
systemctl disable
```

Sources: LINUX COMMAND, [RHCSA_Command_Library.pdf](#)

1.8.50 `systemctl disable nginx`

disable autostart. **CAUTION**

```
systemctl disable nginx
```

Sources: komendy

1.8.51 `systemctl edit`

Edytuje override usługi

```
systemctl edit --full httpd
```

```
systemctl edit
```

Sources: LINUX COMMAND, [RHCSA_Command_Library.pdf](#), komendy

1.8.52 `systemctl edit --full nginx`

override unit. **CAUTION**

```
systemctl edit --full nginx
```

Sources: komendy

1.8.53 ‘`systemctl edit`’

Operational Notes

```
systemctl edit`
```

Sources: komendy

1.8.54 `systemctl emergency`

emergency shell. **DANGEROUS**

```
systemctl emergency
```

Sources: komendy

1.8.55 `systemctl enable`

Włącza autostart

```
systemctl enable httpd
```

```
systemctl enable
```

Sources: LINUX_COMMAND, RHCSA_Command_Library.pdf

1.8.56 `systemctl enable --now secure_processor.service`

Przeładowanie konfiguracji menedżera systemd i uruchomienie nowej usługi bezpiecznej

```
systemctl enable --now secure_processor.service
```

Sources: encyklopedia_inzynierii_linux.html, encyklopedia_inzynierii_linux.pdf

1.8.57 `systemctl enable --now systemd-networkd`

17.4 Networkd i Resolved

```
systemctl enable --now systemd-networkd
```

Sources: linux_engineering_book.md, linux_engineering_book.pdf

1.8.58 `systemctl enable backup.timer`

enable timer. **CAUTION**

```
systemctl enable backup.timer
```

Sources: komendy

1.8.59 `systemctl enable kdump`

15.2 Diagnostyka Jądra

```
systemctl enable kdump
```

Sources: linux_engineering_book.md, linux_engineering_book.pdf

1.8.60 `systemctl enable nginx`

autostart on boot. **CAUTION**

```
systemctl enable nginx
```

Sources: komendy

1.8.61 `systemctl enable wg-quick@wg0`

23.3 WireGuard VPN

```
systemctl enable wg-quick@wg0
```

Sources: linux_engineering_book.md, linux_engineering_book.pdf

1.8.62 systemctl enable/disable nginx # autostart

Zarządzanie usługami

```
systemctl enable/disable nginx # autostart
```

Sources: linux_engineering_book.md, linux_engineering_book.pdf

1.8.63 systemctl get-default

current target. **SAFE**

```
systemctl get-default
```

Sources: komendy

1.8.64 systemctl halt

Sekcja Komend

```
systemctl halt
```

Sources: RHCSA_Command_Library.pdf

1.8.65 systemctl hibernate

Sekcja Komend

```
systemctl hibernate
```

Sources: RHCSA_Command_Library.pdf

1.8.66 systemctl is-active

Czy usługa jest aktywna?

```
systemctl is-active sshd
```

```
systemctl is-active
```

Sources: LINUX COMMAND, RHCSA_Command_Library.pdf, komendy

1.8.67 systemctl is-active nginx

boolean active check. **SAFE**

```
systemctl is-active nginx
```

Sources: komendy

1.8.68 systemctl is-active nginx # czy działa

Zarządzanie usługami

```
systemctl is-active nginx # czy działa
```

Sources: linux_engineering_book.md, linux_engineering_book.pdf

1.8.69 systemctl is-active before restart.

Idempotency Checks

```
systemctl is-active` before `restart`.
```

Sources: komendy

1.8.70 `systemctl is-enabled`

Czy włączona do autostartu?

```
systemctl is-enabled httpd
systemctl is-enabled
```

Sources: LINUX COMMAND, RHCSA_Command_Library.pdf

1.8.71 `systemctl is-enabled nginx`

boolean enabled check. **SAFE**

```
systemctl is-enabled nginx
```

Sources: komendy

1.8.72 `systemctl is-enabled nginx # czy autostart`

Zarządzanie usługami

```
systemctl is-enabled nginx # czy autostart
```

Sources: linux_engineering_book.md, linux_engineering_book.pdf

1.8.73 `systemctl is-failed`

Czy usługa jest w stanie failed?

```
systemctl is-failed httpd
systemctl is-failed
```

Sources: LINUX COMMAND, RHCSA_Command_Library.pdf

1.8.74 `systemctl isolate`

Przechodzi do targetu

```
systemctl isolate multi-user.target
systemctl isolate
```

Sources: LINUX COMMAND, RHCSA_Command_Library.pdf

1.8.75 `systemctl list-timers`

active timers. **SAFE**

```
systemctl list-timers --all
systemctl list-timers
```

Sources: LINUX COMMAND, komendy

1.8.76 `systemctl list-timers --all`

scheduled timers. **SAFE**

```
systemctl list-timers --all
```

Sources: komendy

1.8.77 `systemctl list-unit-files`

Lista plików jednostek

```
systemctl list-unit-files
```

Sources: LINUX COMMAND

1.8.78 `systemctl list-unit-files #`

Zarządzanie usługami

```
systemctl list-unit-files #
```

Sources: linux_engineering_book.md, linux_engineering_book.pdf

1.8.79 `systemctl list-unit-files --state=enabled`

enabled units. **SAFE**

```
systemctl list-unit-files --state=enabled
```

Sources: komendy

1.8.80 `systemctl list-units`

Lista aktywnych jednostek

```
systemctl list-units --type=service
```

```
systemctl list-units
```

Sources: LINUX COMMAND, RHCSA_Command_Library.pdf

1.8.81 `systemctl list-units --failed`

failed boot units. **SAFE**

```
systemctl list-units --failed
```

Sources: komendy

1.8.82 `systemctl list-units --type=service #`

Zarządzanie usługami

```
systemctl list-units --type=service #
```

Sources: linux_engineering_book.md, linux_engineering_book.pdf

1.8.83 `systemctl list-units --type=service --state=running`

active services. **SAFE**

```
systemctl list-units --type=service --state=running
```

Sources: komendy

1.8.84 `systemctl mask`

Maskuje usługę (uniemożliwia start)

```
systemctl mask cups
```

```
systemctl mask
```

Sources: LINUX COMMAND, RHCSA_Command_Library.pdf

1.8.85 `systemctl mask nginx`

prevent start entirely. **CAUTION**

```
systemctl mask nginx
```

Sources: komendy

1.8.86 `systemctl poweroff`

Sekcja Komend

`systemctl poweroff`

Sources: RHCSA_Command_Library.pdf

1.8.87 `systemctl reboot`

Sekcja Komend

`systemctl reboot`

Sources: RHCSA_Command_Library.pdf

1.8.88 `systemctl rescue`

rescue target. **DANGEROUS**

`systemctl rescue`

Sources: komendy

1.8.89 `systemctl set-default multi-user.target`

CLI boot. **CAUTION**

`systemctl set-default multi-user.target`

Sources: komendy

1.8.90 `systemctl set-property myservice.service`

Systemd slice (automatyczne cgroups)

`systemctl set-property myservice.service`

Sources: linux_engineering_book.md, linux_engineering_book.pdf

1.8.91 `systemctl start/stop/restart/reload nginx`

Zarządzanie usługami

`systemctl start/stop/restart/reload nginx`

Sources: linux_engineering_book.md, linux_engineering_book.pdf

1.8.92 `systemctl suspend`

Sekcja Komend

`systemctl suspend`

Sources: RHCSA_Command_Library.pdf

1.8.93 `systemctl unmask`

Sekcja Komend

`systemctl unmask cups`

`systemctl unmask`

Sources: LINUX_COMMAND, RHCSA_Command_Library.pdf

1.8.94 `systemctl unmask nginx`

restore. **CAUTION**

`systemctl unmask nginx`

Sources: komendy

1.8.95 `systemd-analyze`

Czas bootowania

`systemd-analyze blame`

`systemd-analyze critical-chain`

Sources: LINUX COMMAND

1.8.96 `systemd-analyze plot`

Wykres bootu (SVG)

`systemd-analyze plot > boot.svg`

Sources: LINUX COMMAND

1.8.97 `systemd-analyze verify /etc/systemd/system/my.service`

Validate with .

`systemd-analyze verify /etc/systemd/system/my.service`

Sources: komendy

1.8.98 `systemd-ask-password`

Bezpieczne pytanie o hasło

`systemd-ask-password`

Sources: LINUX COMMAND

1.8.99 `systemd-cat`

Loguje do journal

`echo "msg" \`

Sources: LINUX COMMAND

1.8.100 `systemd-cgls`

Komentarz Komentarz w skrypcie function myfunc() { Definicja funkcji

`systemd-cgls`

Sources: LINUX COMMAND, RHCSA_Command_Library.pdf

1.8.101 `systemd-cgtop`

Komentarz Komentarz w skrypcie function myfunc() { Definicja funkcji

`systemd-cgtop`

Sources: LINUX COMMAND, RHCSA_Command_Library.pdf

1.8.102 systemd-cgtop # top dla cgroups

17.3 Resource Control

`systemd-cgtop # top dla cgroups`

Sources: linux_engineering_book.md, linux_engineering_book.pdf

1.8.103 systemd-delta

Różnice w konfiguracji

`systemd-delta`

Sources: LINUX COMMAND

1.8.104 systemd-detect-virt

Detekcja wirtualizacji

`systemd-detect-virt`

Sources: LINUX COMMAND

1.8.105 systemd-escape

Escapuje strings dla systemd

`systemd-escape "foo bar"`

Sources: LINUX COMMAND

1.8.106 systemd-firstboot

Inicjalizacja systemu

`systemd-firstboot --prompt`

Sources: LINUX COMMAND

1.8.107 systemd-inhibit

Blokuje wyłączenie

`systemd-inhibit --what=shutdown ./script`

Sources: LINUX COMMAND

1.8.108 systemd-machine-id-setup

Ustawia machine-id

`systemd-machine-id-setup`

Sources: LINUX COMMAND

1.8.109 systemd-mount

Montowanie (user)

`systemd-mount /dev/sdb1 /mnt`

Sources: LINUX COMMAND

1.8.110 systemd-notify

Notify systemd z app

```
systemd-notify --ready
```

Sources: LINUX COMMAND

1.8.111 systemd-path

Ścieżki specjalne

```
systemd-path systemd-system-unit
```

```
systemd-path user-binaries
```

Sources: LINUX COMMAND

1.8.112 systemd-run

Uruchamia w przejściowej jednostce

```
systemd-run --unit=myservice /bin/sleep 100
```

```
systemd-run --on-calendar="daily" /bin/backup.sh
```

Sources: LINUX COMMAND

1.8.113 systemd-run --on-calendar="Mon *--1..7 02:00" /bin/backup.sh

monthly timer. **SAFE**

```
systemd-run --on-calendar="Mon *--1..7 02:00" /bin/backup.sh
```

Sources: komendy

1.8.114 systemd-run --on-calendar="daily" /bin/backup.sh

ad-hoc timer. **SAFE**

```
systemd-run --on-calendar="daily" /bin/backup.sh
```

Sources: komendy

1.8.115 systemd-run --scope --property=CPUWeight=10

Use for resource limits.

```
systemd-run --scope --property=CPUWeight=10
```

Sources: komendy

1.8.116 systemd-run --slice=myslice --property=MemoryMax=512M command

Systemd slice (automatyczne cgroups)

```
systemd-run --slice=myslice --property=MemoryMax=512M command
```

Sources: linux_engineering_book.md, linux_engineering_book.pdf

1.8.117 systemd-tmpfiles

Zarządzanie plikami tymczasowymi

```
systemd-tmpfiles --create
```

Sources: LINUX COMMAND

1.8.118 systemd-tmpfiles --clean

Czyści tymczasowe

```
systemd-tmpfiles --clean
```

Sources: LINUX COMMAND

1.8.119 systemd-tmpfiles --create --prefix /var/log/journal

apply permissions. **SAFE**

```
systemd-tmpfiles --create --prefix /var/log/journal
```

Sources: komendy

1.8.120 systemd-umount

Odmontowanie (user)

```
systemd-umount /mnt
```

Sources: LINUX COMMAND

1.8.121 timedatectl

Zarządzanie czasem

```
timedatectl
```

Sources: LINUX COMMAND

1.8.122 timedatectl set-ntp

Włącza/wyłącza NTP

```
timedatectl set-ntp true
```

Sources: LINUX COMMAND

1.8.123 timedatectl set-timezone

timezone. **CAUTION**

```
timedatectl set-timezone
```

Sources: LINUX COMMAND, komendy

1.9 Processes

1.9.1 atop

advanced recorder. **SAFE**

```
atop
```

Sources: LINUX COMMAND, komendy

1.9.2 atop # zaawansowany monitor z historią

4.4 Narzędzia do Zarządzania Procesami

```
atop # zaawansowany monitor z historią
```

Sources: linux_engineering_book.md, linux_engineering_book.pdf

1.9.3 bg

Wznawia w tle

`bg %1`

Sources: LINUX COMMAND

1.9.4 bg %1

resume in background. **SAFE**

`bg %1`

Sources: RHCSA_Command_Library.pdf, komendy

1.9.5 cgclassify

Przenosi PID do cgroup

`cgclassify -g cpu:mygroup 1234`

Sources: LINUX COMMAND

1.9.6 cgcreate

Tworzy cgroup (v1)

`cgcreate -g cpu,memory:/mygroup`

Sources: LINUX COMMAND

1.9.7 cgcreate -g

Komentarz Komentarz w skrypcie function myfunc() { Definicja funkcji

`cgcreate -g`

Sources: RHCSA_Command_Library.pdf

1.9.8 cgdelete

Usuwa cgroup

`cgdelete cpu,memory:/mygroup`

Sources: LINUX COMMAND

1.9.9 cgexec

Uruchamia w cgroup

`cgexec -g cpu:mygroup ./app`

Sources: LINUX COMMAND

1.9.10 cgexec -g

Komentarz Komentarz w skrypcie function myfunc() { Definicja funkcji

`cgexec -g`

Sources: RHCSA_Command_Library.pdf

1.9.11 disown

Odłącza od terminala

```
disown %1
```

Sources: LINUX COMMAND

1.9.12 disown %1 # odłącz job od shella

4.4 Narzędzia do Zarządzania Procesami

```
disown %1 # odłącz job od shella
```

Sources: linux_engineering_book.md, linux_engineering_book.pdf

1.9.13 disown -h %1

remove from job table. **SAFE**

```
disown -h %1
```

Sources: komendy

1.9.14 fg

Przywraca na pierwszy plan

```
fg %1
```

Sources: LINUX COMMAND

1.9.15 fg %1

bring to foreground. **SAFE**

```
fg %1
```

Sources: RHCSA_Command_Library.pdf, komendy

1.9.16 free

Użycie pamięci

```
free -h
```

Sources: LINUX COMMAND

1.9.17 free -h

memory exhaustion check. **SAFE**

```
free -h
```

Sources: RHCSA_Command_Library.pdf, komendy, linux_engineering_book.md, linux_engineering_book.pdf

1.9.18 free -h # podsumowanie

5.6 Narzędzia Diagnostyczne Pamięci

```
free -h # podsumowanie
```

Sources: linux_engineering_book.md, linux_engineering_book.pdf

1.9.19 `free -h` → sprawdź swap usage

Dodatek B: Komendy Ratunkowe

`free -h` → sprawdź swap usage

Sources: `linux_engineering_book.md`, `linux_engineering_book.pdf`

1.9.20 `free -m`

Sekcja Komend

`free -m`

Sources: `RHCSA_Command_Library.pdf`

1.9.21 `fuser`

Procesy używające pliku/portu

`fuser -k 8080/tcp`

Sources: `LINUX COMMAND`

1.9.22 `fuser -k`

Zabija procesy używające pliku

`fuser -k /mnt`

Sources: `LINUX COMMAND`

1.9.23 `fuser -v /var/log/syslog`

processes using file. **SAFE**

`fuser -v /var/log/syslog`

Sources: `komendy`

1.9.24 `htop`

interactive viewer. **SAFE**

`htop`

Sources: `LINUX COMMAND`, `RHCSA_Command_Library.pdf`, `komendy`

1.9.25 `htop # ulepszona wersja top`

4.4 Narzędzia do Zarządzania Procesami

`htop # ulepszona wersja top`

Sources: `linux_engineering_book.md`, `linux_engineering_book.pdf`

1.9.26 `iostat`

Statystyki I/O

`iostat -x 1`

Sources: `LINUX COMMAND`

1.9.27 iostat -x -d 1 5

Command — klawisz : Tryb komend # Szukaj sowa pod kursorem (do tyu)

```
iotstat -x -d 1 5
```

Sources: RHCSA_Command_Library.pdf

1.9.28 iostat -xz 1

extended I/O stats. **SAFE**

```
iotstat -xz 1
```

Sources: komendy

1.9.29 iostat -xz 1 # per-urządzenie

14.1 Narzędzia Monitoringu

```
iotstat -xz 1 # per-urządzenie
```

Sources: linux_engineering_book.md, linux_engineering_book.pdf

1.9.30 iostat -xz 1 → sprawdź %util dysków

Dodatek B: Komendy Ratunkowe

```
iotstat -xz 1 → sprawdź %util dysków
```

Sources: linux_engineering_book.md, linux_engineering_book.pdf

1.9.31 jobs

Lista zadań powłoki

```
jobs -l
```

```
jobs
```

Sources: LINUX COMMAND, RHCSA_Command_Library.pdf

1.9.32 jobs -l

list jobs with PIDs. **SAFE**

```
jobs -l
```

Sources: komendy

1.9.33 kill

Wysła sygnał do procesu

```
kill -9 1234
```

```
kill
```

Sources: LINUX COMMAND, RHCSA_Command_Library.pdf

1.9.34 kill -15

Sekcja Komend

```
kill -15
```

Sources: RHCSA_Command_Library.pdf

1.9.35 kill -9

(init) or on database writers.

```
kill -9
```

Sources: RHCSA_Command_Library.pdf, komendy

1.9.36 kill -9 1

(init) or on database writers.

```
kill -9 1
```

Sources: komendy

1.9.37 kill -9 1234 # wyślij

Wysyłanie sygnałów

```
kill -9 1234 # wyślij
```

Sources: linux_engineering_book.md, linux_engineering_book.pdf

1.9.38 kill -9 1 (init) or kill -9 on database writers.

Dangerous Mistakes

```
kill -9 1` (init) or `kill -9` on database writers.
```

Sources: komendy

1.9.39 kill -CONT

Sekcja Komend

```
kill -CONT
```

Sources: RHCSA_Command_Library.pdf

1.9.40 kill -HUP

Przeładowanie (SIGHUP)

```
kill -HUP $(pidof nginx)
```

```
kill -HUP
```

Sources: LINUX_COMMAND, RHCSA_Command_Library.pdf

1.9.41 kill -HUP \$(pgrep nginx)

Wysłanie sygnału bezpiecznego przeładowania konfiguracji do serwera Nginx

```
kill -HUP $(pgrep nginx)
```

Sources: encyklopedia_inzynierii_linux.html, encyklopedia_inzynierii_linux.pdf

1.9.42 kill -KILL 1234

force kill. **CAUTION** (no cleanup)

```
kill -KILL 1234
```

Sources: komendy

1.9.43 kill -SIGCHLD <PPID>

Dodatek B: Komendy Ratunkowe

`kill -SIGCHLD <PPID>`

Sources: linux_engineering_book.md, linux_engineering_book.pdf

1.9.44 kill -SIGTERM 1234 # wyślij

Wysyłanie sygnałów

`kill -SIGTERM 1234 # wyślij`

Sources: linux_engineering_book.md, linux_engineering_book.pdf

1.9.45 kill -STOP

Sekcja Komend

`kill -STOP`

Sources: RHCSA_Command_Library.pdf

1.9.46 kill -TERM 1234

graceful termination. **CAUTION**

`kill -TERM 1234`

Sources: komendy

1.9.47 kill -USR1

Sygnał użytkownika

`kill -USR1 1234`

Sources: LINUX COMMAND

1.9.48 killall

Zabija procesy po nazwie

`killall httpd`

Sources: LINUX COMMAND

1.9.49 killall -9 -u username

Zaloguj się jako root (może wymagać konsoli)

`killall -9 -u username`

Sources: linux_engineering_book.md, linux_engineering_book.pdf

1.9.50 killall -9 procname

Sekcja Komend

`killall -9 procname`

Sources: RHCSA_Command_Library.pdf

1.9.51 killall -u user process

kill by name for user. **CAUTION**

```
killall -u user process
```

Sources: komendy

1.9.52 killall nginx # wyślij

Wysyłanie sygnałów

```
killall nginx # wyślij
```

Sources: linux_engineering_book.md, linux_engineering_book.pdf

1.9.53 killall procname

Sekcja Komend

```
killall procname
```

Sources: RHCSA_Command_Library.pdf

1.9.54 lsof

Uruchomienie kontenera z pełnymi uprawnieniami roota hosta i dostępem do surowego systemu

```
lsof -i :80
```

```
lsof
```

Sources: LINUX COMMAND, encyklopedia_inzynierii_linux.pdf, komendy

1.9.55 lsof +L1

deleted files still held open. **SAFE**

```
lsof +L1
```

Sources: komendy

1.9.56 lsof -i :80

processes on port 80. **SAFE**

```
lsof -i :80
```

Sources: komendy

1.9.57 lsof -i :8080

sieciowych powiązanych z procesami. * **Przykład produkcyjny:** #

```
lsof -i :8080
```

Sources: encyklopedia_inzynierii_linux.html, encyklopedia_inzynierii_linux.pdf

1.9.58 lsof -n -P

grep Porty nasuchujce ulimit -a Wywietl wszystkie limity

```
lsof -n -P
```

Sources: RHCSA_Command_Library.pdf

1.9.59 lsof -p 1234

open files by process. **SAFE**

`lsof -p 1234`

Sources: komendy

1.9.60 ltrace

Śledzi wywołania bibliotek

`ltrace ls`

Sources: LINUX COMMAND

1.9.61 ltrace -c ./program

library call summary. **CAUTION**

`ltrace -c ./program`

Sources: komendy

1.9.62 ltrace -e malloc+free ./program

`ltrace` — śledzenie wywołań bibliotek

`ltrace -e malloc+free ./program`

Sources: linux_engineering_book.md, linux_engineering_book.pdf

1.9.63 ltrace -p 1234

trace library calls. **CAUTION**

`ltrace -p 1234`

Sources: komendy

1.9.64 ltrace ./program

`ltrace` — śledzenie wywołań bibliotek

`ltrace ./program`

Sources: linux_engineering_book.md, linux_engineering_book.pdf

1.9.65 ltrace cmd led wywoania bibliotek sar -n

Command — klawisz : Tryb komend # Szukaj sowa pod kursorem (do tyu)

`ltrace cmd led wywoania bibliotek sar -n`

Sources: RHCSA_Command_Library.pdf

1.9.66 mpstat

Statystyki per-CPU

`mpstat -P`

Sources: LINUX COMMAND

1.9.67 mpstat -P

Monitorowanie statystyk użycia wszystkich rdzeni procesora z interwałem 1 sekundy

`mpstat -P`

Sources: [encyklopedia_inzynierii_linux.html](#), [encyklopedia_inzynierii_linux.pdf](#), [komendy](#), [linux_engineering_book.md](#), [linux_engineering_book.pdf](#)

1.9.68 nice

Uruchamia z priorytetem

`nice -n 10 ./script.sh`

Sources: LINUX COMMAND

1.9.69 nice -n 10 ./script.sh

lower priority. **SAFE**

`nice -n 10 ./script.sh`

Sources: [komendy](#)

1.9.70 nice -n 10 command # uruchom z niższym priorytetem

4.4 Narzędzia do Zarządzania Procesami

`nice -n 10 command # uruchom z niższym priorytetem`

Sources: [linux_engineering_book.md](#), [linux_engineering_book.pdf](#)

1.9.71 nohup

Ignoruje SIGHUP

`nohup ./script.sh &`

Sources: LINUX COMMAND

1.9.72 nohup ./script.sh &

immune to hangup. **SAFE**

`nohup ./script.sh &`

Sources: [komendy](#)

1.9.73 nohup command &

Sekcja Komend

`nohup command &`

Sources: [RHCSA_Command_Library.pdf](#)

1.9.74 nohup command & # odporny na

4.4 Narzędzia do Zarządzania Procesami

`nohup command & # odporny na`

Sources: [linux_engineering_book.md](#), [linux_engineering_book.pdf](#)

1.9.75 pgrep

Szuka PID po wzorcu

`pgrep sshd`

Sources: LINUX COMMAND

1.9.76 pgrep -l procname

Sekcja Komend

`pgrep -l procname`

Sources: RHCSA_Command_Library.pdf

1.9.77 pgrep -u root sshd

find PIDs by name. **SAFE**

`pgrep -u root sshd`

Sources: komendy

1.9.78 pgrep -u user

Sekcja Komend

`pgrep -u user`

Sources: RHCSA_Command_Library.pdf

1.9.79 pgrep procname

Sekcja Komend

`pgrep procname`

Sources: RHCSA_Command_Library.pdf

1.9.80 pidof

Znajduje PID po nazwie

`pidof sshd`

Sources: LINUX COMMAND

1.9.81 pidof nginx

PIDs of named process. **SAFE**

`pidof nginx`

Sources: komendy

1.9.82 pidof procname

Sekcja Komend

`pidof procname`

Sources: RHCSA_Command_Library.pdf

1.9.83 pidstat

Statystyki per-proces

```
pidstat -u 1
```

Sources: LINUX COMMAND

1.9.84 pidstat 1

per-process stats. **SAFE**

```
pidstat 1
```

Sources: komendy

1.9.85 pidstat 1 # per-proces

14.1 Narzędzia Monitoringu

```
pidstat 1 # per-proces
```

Sources: linux_engineering_book.md, linux_engineering_book.pdf

1.9.86 pkill

Zabija po wzorcu

```
pkill -u username
```

Sources: LINUX COMMAND

1.9.87 pkill -9 -u user

Sekcja Komend

```
pkill -9 -u user
```

Sources: RHCSA_Command_Library.pdf

1.9.88 pkill -9 python

killing unrelated services.

```
pkill -9 python
```

Sources: komendy

1.9.89 pkill -9 python killing unrelated services.

Dangerous Mistakes

```
pkill -9 python` killing unrelated services.
```

Sources: komendy

1.9.90 pkill -f "python app.py"

pattern kill. **CAUTION**

```
pkill -f "python app.py"
```

Sources: komendy

1.9.91 `pkill -f "python script" # wyślij`

Wysyłanie sygnałów

```
pkill -f "python script" # wyślij
```

Sources: linux_engineering_book.md, linux_engineering_book.pdf

1.9.92 `pkill -u user`

Sekcja Komend

```
pkill -u user
```

Sources: RHCSA_Command_Library.pdf

1.9.93 `pkill procname`

Sekcja Komend

```
pkill procname
```

Sources: RHCSA_Command_Library.pdf

1.9.94 `prlimit`

Limity procesów (nowoczesne)

```
prlimit --pid 1234 --nofile=4096
```

Sources: LINUX COMMAND

1.9.95 `prlimit --pid 1234`

process limits. **SAFE**

```
prlimit --pid 1234
```

Sources: komendy

1.9.96 `ps`

Aktualne procesy

```
ps aux
```

```
ps
```

Sources: LINUX COMMAND, RHCSA_Command_Library.pdf

1.9.97 `ps --forest`

Sekcja Komend

```
ps --forest
```

Sources: RHCSA_Command_Library.pdf

1.9.98 `ps -Z`

Sekcja Komend

```
ps -Z
```

Sources: RHCSA_Command_Library.pdf

1.9.99 ps -Zp

Sekcja Komend

```
ps -Zp
```

Sources: RHCSA_Command_Library.pdf

1.9.100 ps -eZ

grep httpd # konteksty procesów

```
ps -eZ \
```

```
ps -eZ
```

Sources: LINUX_COMMAND, linux_engineering_book.pdf

1.9.101 ps -eZ | grep httpd # konteksty procesów

9.3 SELinux

```
ps -eZ | grep httpd # konteksty procesów
```

Sources: linux_engineering_book.md

1.9.102 ps -ef

Sekcja Komend

```
ps -ef \
```

```
ps -ef
```

Sources: LINUX_COMMAND, RHCSA_Command_Library.pdf

1.9.103 ps -ef # wszystkie procesy (System

4.4 Narzędzia do Zarządzania Procesami

```
ps -ef # wszystkie procesy (System
```

Sources: linux_engineering_book.md, linux_engineering_book.pdf

1.9.104 ps -eo pid,ppid,cmd --sort=-%mem

custom columns. **SAFE**

```
ps -eo pid,ppid,cmd --sort=-%mem
```

Sources: komendy

1.9.105 ps -eo pid,ppid,user,%cpu,%mem,cmd --sort=-%cpu # custom format

4.4 Narzędzia do Zarządzania Procesami

```
ps -eo pid,ppid,user,%cpu,%mem,cmd --sort=-%cpu # custom format
```

Sources: linux_engineering_book.md, linux_engineering_book.pdf

1.9.106 ps -eo pid,ppid,user,stat,psr,comm --forest

Wyświetlenie pełnego drzewa procesów z uwzględnieniem mapowania użytkowników i kolokacji rdzeni CPU

```
ps -eo pid,ppid,user,stat,psr,comm --forest
```

Sources: encyklopedia_inzynierii_linux.html, encyklopedia_inzynierii_linux.pdf

1.9.107 `ps -eo pid,user,cmd`

Sekcja Komend

`ps -eo pid,user,cmd`

Sources: RHCSA_Command_Library.pdf

1.9.108 `ps -o`

Niestandardowe kolumny

`ps -o pid,ppid,cmd`

Sources: LINUX COMMAND

1.9.109 `ps -o pid,ppid,cmd,%`

Sekcja Komend

`ps -o pid,ppid,cmd,%`

Sources: RHCSA_Command_Library.pdf

1.9.110 `ps -p`

Sekcja Komend

`ps -p`

Sources: RHCSA_Command_Library.pdf

1.9.111 `ps -u user`

Sekcja Komend

`ps -u user`

Sources: RHCSA_Command_Library.pdf

1.9.112 `ps aux`

Sekcja Komend

`ps aux`

Sources: RHCSA_Command_Library.pdf, linux_engineering_book.pdf

1.9.113 `ps aux # wszystkie procesy (BSD format)`

4.4 Narzędzia do Zarządzania Procesami

`ps aux # wszystkie procesy (BSD format)`

Sources: linux_engineering_book.md, linux_engineering_book.pdf

1.9.114 `ps aux --sort=-%cpu`

Sekcja Komend

`ps aux --sort=-%cpu`

Sources: RHCSA_Command_Library.pdf, linux_engineering_book.pdf

1.9.115 `ps aux --sort=-%cpu | head -11`

Dodatek A: Przydatne Jednowierszowce

`ps aux --sort=-%cpu | head -11`

Sources: linux_engineering_book.md

1.9.116 `ps aux --sort=-%mem`

Sekcja Komend

`ps aux --sort=-%mem`

Sources: RHCSA_Command_Library.pdf, linux_engineering_book.pdf

1.9.117 `ps aux --sort=-%mem | head -11`

Dodatek A: Przydatne Jednowierszowce

`ps aux --sort=-%mem | head -11`

Sources: linux_engineering_book.md

1.9.118 `ps aux | grep`

Dodatek B: Komendy Ratunkowe

`ps aux | grep`

Sources: linux_engineering_book.md

1.9.119 `ps auxf`

full format forest view. **SAFE**

`ps auxf`

Sources: komendy

1.9.120 `pstree`

Drzewo procesów

`pstree -p`

`pstree`

Sources: LINUX_COMMAND, RHCSA_Command_Library.pdf

1.9.121 `pstree -p`

process tree with PIDs. **SAFE**

`pstree -p`

Sources: RHCSA_Command_Library.pdf, komendy

1.9.122 `pstree -p # drzewo procesów z`

4.4 Narzędzia do Zarządzania Procesami

`pstree -p # drzewo procesów z`

Sources: linux_engineering_book.md, linux_engineering_book.pdf

1.9.123 pstree -u

Sekcja Komend

`pstree -u`

Sources: RHCSA_Command_Library.pdf

1.9.124 renice

Zmienia priorytet

`renice -n 5 -p 1234`

Sources: LINUX COMMAND

1.9.125 renice +5 -p 1234

change priority. **CAUTION**

`renice +5 -p 1234`

Sources: komendy

1.9.126 renice -n 5 -p 1234 # zmień priorytet działającego procesu

4.4 Narzędzia do Zarządzania Procesami

`renice -n 5 -p 1234 # zmień priorytet działającego procesu`

Sources: linux_engineering_book.md, linux_engineering_book.pdf

1.9.127 sar

Use for baseline establishment.

`sar -u 1 10`

`sar -u -f /var/log/sa/sa01`

`sar`

Sources: LINUX COMMAND, komendy

1.9.128 sar -A -f

Command — klawisz : Tryb komend # Szukaj sowa pod kursorem (do tyu)

`sar -A -f`

Sources: RHCSA_Command_Library.pdf

1.9.129 sar -d

block device activity. **SAFE**

`sar -d`

Sources: komendy

1.9.130 sar -n

network interface stats. **SAFE**

`sar -n`

Sources: komendy

1.9.131 sar -r

memory utilization. **SAFE**

```
sar -r
```

Sources: komendy

1.9.132 sar -r 1 # historyczne dane pamięci

5.6 Narzędzia Diagnostyczne Pamięci

```
sar -r 1 # historyczne dane pamięci
```

Sources: linux_engineering_book.md, linux_engineering_book.pdf

1.9.133 sar -u -r -d 1

14.1 Narzędzia Monitoringu

```
sar -u -r -d 1 #
```

Sources: linux_engineering_book.md, linux_engineering_book.pdf

1.9.134 sar -u 1 10

CPU samples. **SAFE**

```
sar -u 1 10
```

Sources: komendy

1.9.135 strace

Uruchomienie kontenera z pełnymi uprawnieniami roota hosta i dostępem do surowego systemu

```
strace -p 1234
```

```
strace
```

Sources: LINUX COMMAND, encyklopedia_inzynierii_linux.pdf, komendy

1.9.136 strace -T ./program # czas każdego syscalla

strace — śledzenie syscalli

```
strace -T ./program # czas każdego syscalla
```

Sources: linux_engineering_book.md, linux_engineering_book.pdf

1.9.137 strace -c -p 1234

- Przykład produkcyjny:

```
strace -c -p 1234
```

Sources: encyklopedia_inzynierii_linux.html, encyklopedia_inzynierii_linux.pdf

1.9.138 strace -c ./program

syscall summary. **CAUTION** (overhead)

```
strace -c ./program
```

Sources: komendy

1.9.139 `strace -c ./program # statystyki (podsumowanie)`

`strace` — śledzenie syscalls

`strace -c ./program # statystyki (podsumowanie)`

Sources: linux_engineering_book.md, linux_engineering_book.pdf

1.9.140 `strace -c cmd`

Command — klawisz : Tryb komend # Szukaj sowa pod kursorem (do tyu)

`strace -c cmd`

Sources: RHCSA_Command_Library.pdf

1.9.141 `strace -e open,read,write ./prog # filtruj syscalls`

`strace` — śledzenie syscalls

`strace -e open,read,write ./prog # filtruj syscalls`

Sources: linux_engineering_book.md, linux_engineering_book.pdf

1.9.142 `strace -e openat ls led tylko openat syscall authselect current`

Komentarz Komentarz w skrypcie function myfunc() { Definicja funkcji

`strace -e openat ls led tylko openat syscall authselect current`

Sources: RHCSA_Command_Library.pdf

1.9.143 `strace -e trace=file led tylko wywołania plikowe cat`

Command — klawisz : Tryb komend # Szukaj sowa pod kursorem (do tyu)

`strace -e trace=file led tylko wywołania plikowe cat`

Sources: RHCSA_Command_Library.pdf

1.9.144 `strace -f ./program # śledź fork/clone`

`strace` — śledzenie syscalls

`strace -f ./program # śledź fork/clone`

Sources: linux_engineering_book.md, linux_engineering_book.pdf

1.9.145 `strace -fr -e trace=file,network -p $(pgrep backend_engine) -o /tmp/engine_syscalls.log`

Rejestracja wywołań systemowych powiązanych z plikami i siecią dla konkretnego PID z pomiarem czasu mikrosekundowego

`strace -fr -e trace=file,network -p $(pgrep backend_engine) -o /tmp/engine_syscalls.log`

Sources: encyklopedia_inzynierii_linux.html, encyklopedia_inzynierii_linux.pdf

1.9.146 `strace -o trace.log`

Command — klawisz : Tryb komend # Szukaj sowa pod kursorem (do tyu)

`strace -o trace.log`

Sources: RHCSA_Command_Library.pdf

1.9.147 `strace -p`

Command — klawisz : Tryb komend # Szukaj sowa pod kursorem (do tyu)

`strace -p`

Sources: RHCSA_Command_Library.pdf, komendy, linux_engineering_book.md, linux_engineering_book.pdf

1.9.148 `strace -p 1234 -e trace=network`

trace syscalls. **CAUTION** (performance)

`strace -p 1234 -e trace=network`

Sources: komendy

1.9.149 `strace -p on`

Dangerous Mistakes

`strace -p` on`

Sources: komendy

1.9.150 `strace ./program # wszystkie syscalle`

strace — śledzenie syscalli

`strace ./program # wszystkie syscalle`

Sources: linux_engineering_book.md, linux_engineering_book.pdf

1.9.151 `timeout`

Use for unattended long-running jobs.

`timeout`

Sources: LINUX COMMAND, komendy

1.9.152 `top`

Dynamiczny podgląd procesów

`top`

Sources: LINUX COMMAND, RHCSA_Command_Library.pdf

1.9.153 `top # interaktywny monitor`

4.4 Narzędzia do Zarządzania Procesami

`top # interaktywny monitor`

Sources: linux_engineering_book.md, linux_engineering_book.pdf

1.9.154 `top -bn1`

non-interactive snapshot. **SAFE**

`top -bn1`

Sources: RHCSA_Command_Library.pdf, komendy

1.9.155 top -p

Sekcja Komend

`top -p`

Sources: RHCSA_Command_Library.pdf

1.9.156 top -u user

Sekcja Komend

`top -u user`

Sources: RHCSA_Command_Library.pdf

1.9.157 top / htop / btop # interaktywne

14.1 Narzędzia Monitoringu

`top / htop / btop # interaktywne`

Sources: linux_engineering_book.md, linux_engineering_book.pdf

1.9.158 top → sprawdź load, %wa (I/O wait)

Dodatek B: Komendy Ratunkowe

`top → sprawdź load, %wa (I/O wait)`

Sources: linux_engineering_book.md, linux_engineering_book.pdf

1.9.159 ulimit

Limity zasobów powłoki

`ulimit -n 4096`

Sources: LINUX COMMAND

1.9.160 ulimit -a

shell resource limits. **SAFE**

`ulimit -a`

Sources: LINUX COMMAND, komendy

1.9.161 ulimit -c unlimited # włącz core dumps

15.1 GDB (GNU Debugger)

`ulimit -c unlimited # włącz core dumps`

Sources: linux_engineering_book.md, linux_engineering_book.pdf

1.9.162 vmstat

Statystyki wirtualnej pamięci

`vmstat 1 5`

`vmstat 1 10`

Sources: LINUX COMMAND

1.9.163 `vmstat -SM 1 10`

Inspekcja rozkładu alokacji pamięci wirtualnej i pamięci podręcznej systemu plików

`vmstat -SM 1 10`

Sources: [encyklopedia_inzynierii_linux.html](#), [encyklopedia_inzynierii_linux.pdf](#)

1.9.164 `vmstat 1 # si/so = swap in/out`

Monitoring

`vmstat 1 # si/so = swap in/out`

Sources: [linux_engineering_book.md](#), [linux_engineering_book.pdf](#)

1.9.165 `vmstat 1 # statystyki`

5.6 Narzędzia Diagnostyczne Pamięci

`vmstat 1 # statystyki`

Sources: [linux_engineering_book.md](#), [linux_engineering_book.pdf](#)

1.9.166 `vmstat 1 5`

system-wide stats. **SAFE**

`vmstat 1 5`

Sources: komendy

1.9.167 `watch`

Wykonuje komendę cyklicznie

`watch`

Sources: LINUX COMMAND

1.10 Bash

1.10.1 `/backup/xfs.dump`

Sekcja Komend

`/backup/xfs.dump`

Sources: [RHCSA_Command_Library.pdf](#)

1.10.2 `/bin/backup.sh`

Sekcja Komend

`/bin/backup.sh`

Sources: [RHCSA_Command_Library.pdf](#)

1.10.3 `/bin/task.sh`

Sekcja Komend

`/bin/task.sh`

Sources: [RHCSA_Command_Library.pdf](#)

1.10.4 /boot

- : Montuje całą strukturę katalogów systemowych (, ,) w tryb tylko do odczytu dla danej usługi.

/boot

Sources: [encyklopedia_inzynierii_linux.html](#), [encyklopedia_inzynierii_linux.pdf](#)

1.10.5 /check.sh

Sekcja Komend

/check.sh

Sources: [RHCSA_Command_Library.pdf](#)

1.10.6 /dev/md0

Komentarz Komentarz w skrypcie function myfunc() { Definicja funkcji

/dev/md0

Sources: [RHCSA_Command_Library.pdf](#)

1.10.7 /dev/null

Znakowe (c) — dostęp po znakach (terminale, porty): ,

/dev/null

Sources: [linux_engineering_book.md](#)

1.10.8 /dev/nvme0n1

Blokowe (b) — dostęp po blokach (dyski, partycje): ,

/dev/nvme0n1

Sources: [linux_engineering_book.md](#)

1.10.9 /dev/sda

Blokowe (b) — dostęp po blokach (dyski, partycje): ,

/dev/sda

Sources: [RHCSA_Command_Library.pdf](#), [linux_engineering_book.md](#)

1.10.10 /dev/sdb

Komentarz Komentarz w skrypcie function myfunc() { Definicja funkcji

/dev/sdb

Sources: [RHCSA_Command_Library.pdf](#)

1.10.11 /dev/sdb1

Sekcja Komend

/dev/sdb1

Sources: [RHCSA_Command_Library.pdf](#)

1.10.12 /dev/sdc

Komentarz Komentarz w skrypcie function myfunc() { Definicja funkcji

/dev/sdc

Sources: RHCSA_Command_Library.pdf

1.10.13 /dev/sdd

Komentarz Komentarz w skrypcie function myfunc() { Definicja funkcji

/dev/sdd

Sources: RHCSA_Command_Library.pdf

1.10.14 /dev/tty

Znakowe (c) — dostęp po znakach (terminale, porty): ,

/dev/tty

Sources: linux_engineering_book.md

1.10.15 /dev/urandom

tr -d file.xml

/dev/urandom

Sources: RHCSA_Command_Library.pdf

1.10.16 /dev/vgname/lvname

Sekcja Komend

/dev/vgname/lvname

Sources: RHCSA_Command_Library.pdf

1.10.17 /dev/vgname/snapname

Sekcja Komend

/dev/vgname/snapname

Sources: RHCSA_Command_Library.pdf

1.10.18 /etc

- : Montuje całą strukturę katalogów systemowych (, ,) w tryb tylko do odczytu dla danej usługi.

/etc

Sources: encyklopedia_inzynierii_linux.html, encyklopedia_inzynierii_linux.pdf, komendy

1.10.19 /etc/anacrontab

Anacron (dla maszyn nie zawsze włączonych)

/etc/anacrontab

Sources: linux_engineering_book.md, linux_engineering_book.pdf

1.10.20 /etc/audit/auditd.conf

Audit daemon config

`vi /etc/audit/auditd.conf`

Sources: LINUX COMMAND

1.10.21 /etc/auto.nfs

Sekcja Komend

`/etc/auto.nfs`

Sources: RHCSA_Command_Library.pdf

1.10.22 /etc/centos-release

Sekcja Komend

`/etc/centos-release`

Sources: RHCSA_Command_Library.pdf

1.10.23 /etc/chrony.conf

Konfiguracja chrony

`/etc/chrony.conf`

Sources: LINUX COMMAND

1.10.24 /etc/cron.d

Katalog zadań

`ls /etc/cron.d`

Sources: LINUX COMMAND

1.10.25 /etc/cron.daily

Codzienne zadania

`ls /etc/cron.daily`

Sources: LINUX COMMAND

1.10.26 /etc/cron.hourly

Godzinne zadania

`ls /etc/cron.hourly`

Sources: LINUX COMMAND

1.10.27 /etc/cron.monthly

Miesięczne zadania

`ls /etc/cron.monthly`

Sources: LINUX COMMAND

1.10.28 /etc/cron.weekly

Tygodniowe zadania

`ls /etc/cron.weekly`

Sources: LINUX COMMAND

1.10.29 /etc/crontab

Systemowy crontab

`cat /etc/crontab`

Sources: LINUX COMMAND

1.10.30 /etc/crypttab

Konfiguracja boot-time

`/etc/crypttab`

Sources: LINUX COMMAND

1.10.31 /etc/default/grub

Sekcja Komend

`/etc/default/grub`

Sources: RHCSA_Command_Library.pdf

1.10.32 /etc/fstab

1. Montuje systemy plików z

`/etc/fstab`

Sources: komendy, linux_engineering_book.md

1.10.33 /etc/logrotate.conf

Sekcja Komend

`/etc/logrotate.conf`

Sources: RHCSA_Command_Library.pdf

1.10.34 /etc/nftables.conf

Komentarz Komentarz w skrypcie function myfunc() { Definicja funkcji

`/etc/nftables.conf`

Sources: RHCSA_Command_Library.pdf

1.10.35 /etc/pam.d/system-auth

Modyfikacja pliku w celu blokowania słabych haseł i wymuszenia zapamiętywania historii haseł.

`/etc/pam.d/system-auth`

Sources: encyklopedia_inzynierii_linux.html, encyklopedia_inzynierii_linux.pdf

1.10.36 /etc/passwd

Sekcja Komend

/etc/passwd

Sources: RHCSA_Command_Library.pdf

1.10.37 /etc/redhat-release

Sekcja Komend

/etc/redhat-release

Sources: RHCSA_Command_Library.pdf

1.10.38 /etc/rsyslog.conf

Sekcja Komend

/etc/rsyslog.conf

Sources: RHCSA_Command_Library.pdf

1.10.39 /etc/sss/sss.conf

Komentarz Komentarz w skrypcie function myfunc() { Definicja funkcji

/etc/sss/sss.conf

Sources: RHCSA_Command_Library.pdf

1.10.40 /etc/sudoers

Always use ; never edit directly.

/etc/sudoers

Sources: komendy

1.10.41 /etc/sysctl.conf

Stałe parametry jądra

/etc/sysctl.conf

Sources: LINUX COMMAND

1.10.42 /etc/sysctl.d/99-hardening.conf

Apply sysctl via and reload.

/etc/sysctl.d/99-hardening.conf

Sources: komendy

1.10.43 /etc/systemd/system/secure_processor.service

Utworzenie pliku konfiguracyjnego usługi w

/etc/systemd/system/secure_processor.service

Sources: encyklopedia_inzynierii_linux.html, encyklopedia_inzynierii_linux.pdf

1.10.44 /etc/updatedb.conf

Sekcja Komend

/etc/updatedb.conf

Sources: RHCSA_Command_Library.pdf

1.10.45 /hostname

Sekcja Komend

/hostname

Sources: RHCSA_Command_Library.pdf

1.10.46 /mnt

Sekcja Komend

/mnt

Sources: RHCSA_Command_Library.pdf

1.10.47 /path

Sekcja Komend

/path

Sources: RHCSA_Command_Library.pdf

1.10.48 /path/to/file

Sekcja Komend

/path/to/file

Sources: RHCSA_Command_Library.pdf

1.10.49 /path/to/key.gpg

Sekcja Komend

/path/to/key.gpg

Sources: RHCSA_Command_Library.pdf

1.10.50 /path/to/module.ko

Sekcja Komend

/path/to/module.ko

Sources: RHCSA_Command_Library.pdf

1.10.51 /path/to/script.sh

Sekcja Komend

/path/to/script.sh

Sources: RHCSA_Command_Library.pdf

1.10.52 /proc/net/sockstat

Command — klawisz : Tryb komend # Szukaj sowa pod kursorem (do tyu)

/proc/net/sockstat

Sources: RHCSA_Command_Library.pdf

1.10.53 /sbin/init

Po zamontowaniu root filesystem, jądro uruchamia (symlink do systemd). Systemd

/sbin/init

Sources: linux_engineering_book.md

1.10.54 /tmp

- : Tworzy odizolowaną, prywatną przestrzeń nazw dla katalogów i , niewidoczną dla innych procesów w systemie.

/tmp

Sources: RHCSA_Command_Library.pdf, encyklopedia_inzynierii_linux.html, encyklopedia_inzynierii_linux.pdf, komendy

1.10.55 /tmp/cap.pcap

Komentarz Komentarz w skrypcie function myfunc() { Definicja funkcji

/tmp/cap.pcap

Sources: RHCSA_Command_Library.pdf

1.10.56 /usr

- : Montuje całą strukturę katalogów systemowych (, ,) w tryb tylko do odczytu dla danej usługi.

/usr

Sources: encyklopedia_inzynierii_linux.html, encyklopedia_inzynierii_linux.pdf, komendy

1.10.57 /usr/bin/python3

Komentarz Komentarz w skrypcie function myfunc() { Definicja funkcji

/usr/bin/python3

Sources: RHCSA_Command_Library.pdf

1.10.58 /usr/sbin/sshd

Sekcja Komend

/usr/sbin/sshd

Sources: RHCSA_Command_Library.pdf

1.10.59 /var

Separate state from code: , vs .

/var

Sources: RHCSA_Command_Library.pdf, komendy

1.10.60 /var/log/auth.log

Sekcja Komend

/var/log/auth.log

Sources: RHCSA_Command_Library.pdf

1.10.61 /var/log/boot.log

Sekcja Komend

/var/log/boot.log

Sources: RHCSA_Command_Library.pdf

1.10.62 /var/log/journal

Persistent journal requires or existence.

/var/log/journal

Sources: komendy

1.10.63 /var/log/messages

Sekcja Komend

/var/log/messages

Sources: RHCSA_Command_Library.pdf

1.10.64 /var/log/secure

Sekcja Komend

/var/log/secure

Sources: RHCSA_Command_Library.pdf

1.10.65 /var/tmp

- : Tworzy odizolowaną, prywatną przestrzeń nazw dla katalogów i , niewidoczną dla innych procesów w systemie.

/var/tmp

Sources: encyklopedia_inzynierii_linux.html, encyklopedia_inzynierii_linux.pdf

1.10.66 /var/www/html/file

Sekcja Komend

/var/www/html/file

Sources: RHCSA_Command_Library.pdf

1.10.67 alias

Definiuje alias

```
alias ll='ls -la'
```

Sources: LINUX COMMAND

1.10.68 `alias ll='ls -aF'`

create alias. **SAFE**

`alias ll='ls -aF'`

Sources: komendy

1.10.69 `alias rm='rm -i'`

interactive remove. **SAFE** (user-level)

`alias rm='rm -i'`

Sources: komendy

1.10.70 `apropos`

Wyszukiwanie w man pages

`apropos firewall`

Sources: LINUX COMMAND

1.10.71 `awk`

Uruchomienie kontenera z pełnymi uprawnieniami roota hosta i dostępem do surowego systemu

`awk '{print $1}' /etc/passwd`

`awk`

Sources: LINUX COMMAND, encyklopedia_inzynierii_linux.pdf, komendy

1.10.72 `awk '$3 > 100' file # filtruj po wartości`

`awk` — język przetwarzania tekstu

`awk '$3 > 100' file # filtruj po wartości`

Sources: linux_engineering_book.md, linux_engineering_book.pdf

1.10.73 `awk '/pattern/' file`

Sekcja Komend

`awk '/pattern/' file`

Sources: RHCSA_Command_Library.pdf

1.10.74 `awk 'END{print`

Sekcja Komend

`awk 'END{print`

Sources: RHCSA_Command_Library.pdf

1.10.75 `awk 'NR==5' file`

Sekcja Komend

`awk 'NR==5' file`

Sources: RHCSA_Command_Library.pdf

1.10.76 `awk 'NR==5,NR==10' file # linie 5-10`

awk — język przetwarzania tekstu

`awk 'NR==5,NR==10' file # linie 5-10`

Sources: linux_engineering_book.md, linux_engineering_book.pdf

1.10.77 `awk 'NR>=5 &&`

Sekcja Komend

`awk 'NR>=5 &&`

Sources: RHCSA_Command_Library.pdf

1.10.78 `awk '{print`

Sekcja Komend

`awk '{print`

Sources: RHCSA_Command_Library.pdf

1.10.79 `awk '{print $1, $3}' file # kolumny 1 i 3`

awk — język przetwarzania tekstu

`awk '{print $1, $3}' file # kolumny 1 i 3`

Sources: linux_engineering_book.md, linux_engineering_book.pdf

1.10.80 `awk '{print $1}'`

Sekcja Komend

`awk '{print $1}'`

Sources: RHCSA_Command_Library.pdf

1.10.81 `awk '{print $1}' file`

first field. **SAFE**

`awk '{print $1}' file`

Sources: komendy

1.10.82 `awk '{print $NF}'`

Sekcja Komend

`awk '{print $NF}'`

Sources: RHCSA_Command_Library.pdf

1.10.83 `awk '{sum+=$1}'`

sum column. **SAFE**

`awk '{sum+=$1}'`

Sources: RHCSA_Command_Library.pdf, komendy, linux_engineering_book.md, linux_engineering_book.pdf

1.10.84 `awk -F:' ' '{print $1 " -> " $6}' /etc/passwd`

podstawie zdefiniowanych kolumn i filtrów. * **Przykład produkcyjny:**

`awk -F:' ' '{print $1 " -> " $6}' /etc/passwd`

Sources: [encyklopedia_inzynierii_linux.html](#), [encyklopedia_inzynierii_linux.pdf](#)

1.10.85 `awk -F: '$3 >= 1000 {print $1}' /etc/passwd`

filter by field. **SAFE**

`awk -F: '$3 >= 1000 {print $1}' /etc/passwd`

Sources: [komendy](#)

1.10.86 `awk -F: '{print $1}'`

Sekcja Komend

`awk -F: '{print $1}'`

Sources: [RHCSA_Command_Library.pdf](#)

1.10.87 `awk -F: '{print $1}' /etc/passwd # separator :`

awk — język przetwarzania tekstu

`awk -F: '{print $1}' /etc/passwd # separator :`

Sources: [linux_engineering_book.md](#), [linux_engineering_book.pdf](#)

1.10.88 `awk -v`

Sekcja Komend

`awk -v`

Sources: [RHCSA_Command_Library.pdf](#)

1.10.89 `base64`

Koduje/dekoduje base64

`base64 file.txt`

Sources: [LINUX COMMAND](#)

1.10.90 `base64 -d file.b64`

Komentarz Komentarz w skrypcie function myfunc() { Definicja funkcji

`base64 -d file.b64`

Sources: [RHCSA_Command_Library.pdf](#)

1.10.91 `base64 file.bin`

Komentarz Komentarz w skrypcie function myfunc() { Definicja funkcji

`base64 file.bin`

Sources: [RHCSA_Command_Library.pdf](#)

1.10.92 bash

3. **Warstwa Użytkownika (User Space):** Biblioteka standardowa C (), menedżer systemu i usług (), powłoka wykonawcza () oraz aplikacje użytkownika.

```
bash script.sh
bash
```

Sources: LINUX_COMMAND, encyklopedia_inzynierii_linux.html, encyklopedia_inzynierii_linux.pdf

1.10.93 bash -c 'command'

Sekcja Komend

```
bash -c 'command'
```

Sources: RHCSA_Command_Library.pdf

1.10.94 bash -n script.sh

Sekcja Komend

```
bash -n script.sh
```

Sources: RHCSA_Command_Library.pdf

1.10.95 bash -v script.sh

Sekcja Komend

```
bash -v script.sh
```

Sources: RHCSA_Command_Library.pdf

1.10.96 bash -x script.sh

Sekcja Komend

```
bash -x script.sh
```

Sources: RHCSA_Command_Library.pdf

1.10.97 bash -x script.sh # trace

28.1 Bash — Zaawansowane Techniki

```
bash -x script.sh # trace
```

Sources: linux_engineering_book.md, linux_engineering_book.pdf

1.10.98 bash es --set

Komentarz Komentarz w skrypcie function myfunc() { Definicja funkcji

```
bash es --set
```

Sources: RHCSA_Command_Library.pdf

1.10.99 bash script.sh

Komentarz Komentarz w skrypcie function myfunc() { Definicja funkcji

```
bash script.sh
```

Sources: RHCSA_Command_Library.pdf

1.10.100 break

Komentarz Komentarz w skrypcie function myfunc() { Definicja funkcji

break

Sources: LINUX COMMAND, RHCSA_Command_Library.pdf

1.10.101 bunzip2

Dekompresuje bzip2

bunzip2 file.txt.bz2

Sources: LINUX COMMAND

1.10.102 bzip2

Wyświetla bzip2

bzip2 file.txt.bz2

Sources: LINUX COMMAND

1.10.103 bzip2

Kompresuje bzip2

bzip2 file.txt

Sources: LINUX COMMAND

1.10.104 bzip2 -d file.bz2

Sekcja Komend

bzip2 -d file.bz2

Sources: RHCSA_Command_Library.pdf

1.10.105 bzip2 file

Sekcja Komend

bzip2 file

Sources: RHCSA_Command_Library.pdf

1.10.106 cal

Kalendarz

cal -3

Sources: LINUX COMMAND

1.10.107 clear

Czyści ekran

clear

Sources: LINUX COMMAND

1.10.108 column

Formatuje w kolumny

```
column -t file.txt
```

Sources: LINUX COMMAND

1.10.109 column -t -s',' file.csv

table format. **SAFE**

```
column -t -s',' file.csv
```

Sources: komendy

1.10.110 comm

Porównuje posortowane pliki

```
comm -12 file1 file2
```

Sources: LINUX COMMAND

1.10.111 comm -12 <(sort a) <(sort b)

common lines. **SAFE**

```
comm -12 <(sort a) <(sort b)
```

Sources: komendy

1.10.112 continue

Komentarz Komentarz w skrypcie function myfunc() { Definicja funkcji

```
continue
```

Sources: LINUX COMMAND, RHCSA_Command_Library.pdf

1.10.113 cpio

Archiwizacja (np. initrd)

```
find . \
```

Sources: LINUX COMMAND

1.10.114 cpio -idmv

Sekcja Komend

```
cpio -idmv
```

Sources: RHCSA_Command_Library.pdf

1.10.115 cpio -o >

Sekcja Komend

```
cpio -o >
```

Sources: RHCSA_Command_Library.pdf

1.10.116 cut

Wycina fragmenty linii

```
cut -d: -f1 /etc/passwd
```

Sources: LINUX COMMAND

1.10.117 cut -d: -f1 /etc/passwd

delimiter field cut. **SAFE**

```
cut -d: -f1 /etc/passwd
```

Sources: komendy

1.10.118 cut -d: -f1 /etc/passwd # wytnij pole 1

10.4 Przetwarzanie Tekstu

```
cut -d: -f1 /etc/passwd # wytnij pole 1
```

Sources: linux_engineering_book.md, linux_engineering_book.pdf

1.10.119 date

Data i czas

```
date +"%Y-%m-%d %H:%M"
```

Sources: LINUX COMMAND

1.10.120 date -s

Ustawia datę

```
date -s "2026-05-24 10:00:00"
```

Sources: LINUX COMMAND

1.10.121 declare

Deklaruje zmienne

```
declare -i num=5
```

Sources: LINUX COMMAND

1.10.122 declare -A

Sekcja Komend

```
declare -A
```

Sources: RHCSA_Command_Library.pdf

1.10.123 declare -A assoc

associative array. **SAFE**

```
declare -A assoc
```

Sources: komendy

1.10.124 declare -A map

10.1 Bash — Podstawy

```
declare -A map
```

Sources: linux_engineering_book.md, linux_engineering_book.pdf

1.10.125 declare -a

Sekcja Komend

```
declare -a
```

Sources: RHCSA_Command_Library.pdf

1.10.126 declare -i

Sekcja Komend

```
declare -i
```

Sources: RHCSA_Command_Library.pdf

1.10.127 declare -i num

integer. **SAFE**

```
declare -i num
```

Sources: komendy

1.10.128 declare -p

Sekcja Komend

```
declare -p
```

Sources: RHCSA_Command_Library.pdf

1.10.129 declare -r

Sekcja Komend

```
declare -r
```

Sources: RHCSA_Command_Library.pdf

1.10.130 declare -x

Sekcja Komend

```
declare -x
```

Sources: RHCSA_Command_Library.pdf

1.10.131 echo

Require or preview for commands.

```
echo $PATH
```

```
echo
```

Sources: LINUX COMMAND, komendy

1.10.132 `echo "$line"`

10.2 Struktury Kontrolne

`echo "$line"`

Sources: linux_engineering_book.md, linux_engineering_book.pdf

1.10.133 `echo "$var"`

Use instead of for arbitrary data.

`echo "$var"`

Sources: komendy

1.10.134 `echo "- - -" > /sys/class/scsi_host/host0/scan`

Skanywanie magistrali SCSI w celu wykrycia nowo przypisanych jednostek LUN lub dysków NVMe

`echo "- - -" > /sys/class/scsi_host/host0/scan`

Sources: encyklopedia_inzynierii_linux.html, encyklopedia_inzynierii_linux.pdf

1.10.135 `echo "1 2 3"`

`xargs -n1` # po jednym argumencie

`echo "1 2 3"`

Sources: linux_engineering_book.pdf

1.10.136 `echo "1 2 3" | xargs -n1 # po jednym argumencie`

10.4 Przetwarzanie Tekstu

`echo "1 2 3" | xargs -n1 # po jednym argumencie`

Sources: linux_engineering_book.md

1.10.137 `echo "100000 100000" > /sys/fs/cgroup/mygroup/cpu.max # 100% 1`

4.6 Cgroups (Control Groups)

`echo "100000 100000" > /sys/fs/cgroup/mygroup/cpu.max # 100% 1`

Sources: linux_engineering_book.md, linux_engineering_book.pdf

1.10.138 `echo "50" > /sys/fs/cgroup/mygroup/pids.max # max 50 procesów`

4.6 Cgroups (Control Groups)

`echo "50" > /sys/fs/cgroup/mygroup/pids.max # max 50 procesów`

Sources: linux_engineering_book.md, linux_engineering_book.pdf

1.10.139 `echo "50000 100000" > /sys/fs/cgroup/mygroup/cpu.max`

25.2 Cgroups v2 — Zaawansowane

`echo "50000 100000" > /sys/fs/cgroup/mygroup/cpu.max`

Sources: linux_engineering_book.md, linux_engineering_book.pdf

1.10.140 `echo "536870912" > /sys/fs/cgroup/mygroup/memory.max # 512MB`

4.6 Cgroups (Control Groups)

`echo "536870912" > /sys/fs/cgroup/mygroup/memory.max # 512MB`

Sources: linux_engineering_book.md, linux_engineering_book.pdf

1.10.141 `echo "8:0 rbps=1048576 wbps=1048576" > /sys/fs/cgroup/mygroup/io.max`

25.2 Cgroups v2 — Zaawansowane

`echo "8:0 rbps=1048576 wbps=1048576" > /sys/fs/cgroup/mygroup/io.max`

Sources: linux_engineering_book.md, linux_engineering_book.pdf

1.10.142 `echo "Cleaned up"`

10.3 Funkcje

`echo "Cleaned up"`

Sources: linux_engineering_book.md, linux_engineering_book.pdf

1.10.143 `echo "Hello $NAME"`

10.1 Bash — Podstawy

`echo "Hello $NAME"`

Sources: linux_engineering_book.md, linux_engineering_book.pdf

1.10.144 `echo "Hello, $name"`

10.3 Funkcje

`echo "Hello, $name"`

Sources: linux_engineering_book.md, linux_engineering_book.pdf

1.10.145 `echo "Processing: $file"`

10.2 Struktury Kontrolne

`echo "Processing: $file"`

Sources: linux_engineering_book.md, linux_engineering_book.pdf

1.10.146 `echo "Value: $VAR"`

Komentarz Komentarz w skrypcie function myfunc() { Definicja funkcji

`echo "Value: $VAR"`

Sources: RHCSA_Command_Library.pdf

1.10.147 `echo "Wersja: ${BASH_VERSION}"`

10.1 Bash — Podstawy

`echo "Wersja: ${BASH_VERSION}"`

Sources: linux_engineering_book.md, linux_engineering_book.pdf

1.10.148 `echo "hello" >&${myproc[1]}`

10.5 Zaawansowany Bash

`echo "hello" >&${myproc[1]}`

Sources: linux_engineering_book.md, linux_engineering_book.pdf

1.10.149 `echo "log entry" >&3`

28.1 Bash — Zaawansowane Techniki

`echo "log entry" >&3`

Sources: linux_engineering_book.md, linux_engineering_book.pdf

1.10.150 `echo "mniejšie"`

10.2 Struktury Kontrolne

`echo "mniejšie"`

Sources: linux_engineering_book.md, linux_engineering_book.pdf

1.10.151 `echo "mq-deadline" > /sys/block/sda/queue/scheduler`

14.4 Tuning Systemowy

`echo "mq-deadline" > /sys/block/sda/queue/scheduler`

Sources: linux_engineering_book.md, linux_engineering_book.pdf

1.10.152 `echo "performance"`

tee

`echo "performance"`

Sources: linux_engineering_book.pdf

1.10.153 `echo "performance" | tee /sys/devices/system/cpu/cpu*/cpufreq/scaling_governor`

14.4 Tuning Systemowy

`echo "performance" | tee /sys/devices/system/cpu/cpu*/cpufreq/scaling_governor`

Sources: linux_engineering_book.md

1.10.154 `echo "równe"`

10.2 Struktury Kontrolne

`echo "równe"`

Sources: linux_engineering_book.md, linux_engineering_book.pdf

1.10.155 `echo "vm.swappiness = 10" >> /etc/sysctl.d/99-sys-perf.conf`

Zapewnienie trwałości konfiguracji po restarcie systemu poprzez wpis w katalogu sysctl.d

`echo "vm.swappiness = 10" >> /etc/sysctl.d/99-sys-perf.conf`

Sources: encyklopedia_inzynierii_linux.html, encyklopedia_inzynierii_linux.pdf

1.10.156 `echo "vm.swappiness=10" >> /etc/sysctl.conf`

Kontrola swappiness (0-200, domyślnie 60)

`echo "vm.swappiness=10" >> /etc/sysctl.conf`

Sources: linux_engineering_book.md, linux_engineering_book.pdf

1.10.157 `echo "większe"`

10.2 Struktury Kontrolne

`echo "większe"`

Sources: linux_engineering_book.md, linux_engineering_book.pdf

1.10.158 `echo $((2**10)) # potęgowanie`

10.5 Zaawansowany Bash

`echo $((2**10)) # potęgowanie`

Sources: linux_engineering_book.md, linux_engineering_book.pdf

1.10.159 `echo $((RANDOM %`

.name' Iteruj po tablicy JSON

`echo $((RANDOM %`

Sources: RHCSA_Command_Library.pdf

1.10.160 `echo $PID > /sys/fs/cgroup/mygroup/cgroup.procs`

4.6 Cgroups (Control Groups)

`echo $PID > /sys/fs/cgroup/mygroup/cgroup.procs`

Sources: linux_engineering_book.md, linux_engineering_book.pdf

1.10.161 `echo $VARIABLE`

Sekcja Komend

`echo $VARIABLE`

Sources: RHCSA_Command_Library.pdf

1.10.162 `echo $i`

10.2 Struktury Kontrolne

`echo $i`

Sources: linux_engineering_book.md, linux_engineering_book.pdf

1.10.163 `echo ${!map[@]} # wszystkie klucze`

10.1 Bash — Podstawy

`echo ${!map[@]} # wszystkie klucze`

Sources: linux_engineering_book.md, linux_engineering_book.pdf

1.10.164 `echo ${#ARRAY[@]}`

Sekcja Komend

`echo ${#ARRAY[@]}`

Sources: RHCSA_Command_Library.pdf

1.10.165 `echo ${#arr[@]} # długość tablicy`

10.1 Bash — Podstawy

`echo ${#arr[@]} # długość tablicy`

Sources: linux_engineering_book.md, linux_engineering_book.pdf

1.10.166 `echo ${ARRAY[0]}`

Sekcja Komend

`echo ${ARRAY[0]}`

Sources: RHCSA_Command_Library.pdf

1.10.167 `echo ${ARRAY[@]}`

Sekcja Komend

`echo ${ARRAY[@]}`

Sources: RHCSA_Command_Library.pdf

1.10.168 `echo ${arr[0]} # pierwszy element`

10.1 Bash — Podstawy

`echo ${arr[0]} # pierwszy element`

Sources: linux_engineering_book.md, linux_engineering_book.pdf

1.10.169 `echo ${arr[@]} # wszystkie elementy`

10.1 Bash — Podstawy

`echo ${arr[@]} # wszystkie elementy`

Sources: linux_engineering_book.md, linux_engineering_book.pdf

1.10.170 `echo ${map[key1]}`

10.1 Bash — Podstawy

`echo ${map[key1]}`

Sources: linux_engineering_book.md, linux_engineering_book.pdf

1.10.171 `echo '/swapfile none swap sw 0 0' >> /etc/fstab`

Trwałe dodanie (fstab)

`echo '/swapfile none swap sw 0 0' >> /etc/fstab`

Sources: linux_engineering_book.md, linux_engineering_book.pdf

1.10.172 `echo -1000 > /proc/[pid]/oom_score_adj`

Ochrona procesu przed OOM

`echo -1000 > /proc/[pid]/oom_score_adj`

Sources: linux_engineering_book.md, linux_engineering_book.pdf

1.10.173 `echo 0 > tracing_on`

15.2 Diagnostyka Jądra

`echo 0 > tracing_on`

Sources: linux_engineering_book.md, linux_engineering_book.pdf

1.10.174 `echo 1 > /proc/sys/kernel/sysrq`

Aktywacja obsługi wszystkich poleceń SysRq w jądrze

`echo 1 > /proc/sys/kernel/sysrq`

Sources: encyklopedia_inzynierii_linux.html, encyklopedia_inzynierii_linux.pdf

1.10.175 `echo 1 > /proc/sys/net/ipv4/ip_forward`

iptables (legacy)

`echo 1 > /proc/sys/net/ipv4/ip_forward`

Sources: linux_engineering_book.md, linux_engineering_book.pdf

1.10.176 `echo 1 > tracing_on`

15.2 Diagnostyka Jądra

`echo 1 > tracing_on`

Sources: linux_engineering_book.md, linux_engineering_book.pdf

1.10.177 `echo 1024 > /proc/sys/vm/nr_hugepages # zarezerwuj 1024 huge pages`

5.2 Strony Pamięci

`echo 1024 > /proc/sys/vm/nr_hugepages # zarezerwuj 1024 huge pages`

Sources: linux_engineering_book.md, linux_engineering_book.pdf

1.10.178 `echo b > /proc/sysrq-trigger #`

Sekwencja REISUB (Synchronizacja dysków, Odmontowanie, Natychmiastowy restart)

`echo b > /proc/sysrq-trigger #`

Sources: encyklopedia_inzynierii_linux.html, encyklopedia_inzynierii_linux.pdf

1.10.179 `echo file{1..3}.txt file1.txt file2.txt file3.txt jq '.' file.json`

Komentarz Komentarz w skrypcie function myfunc() { Definicja funkcji

`echo file{1..3}.txt file1.txt file2.txt file3.txt jq '.' file.json`

Sources: RHCSA_Command_Library.pdf

1.10.180 `echo function > current_tracer`

15.2 Diagnostyka Jądra

`echo function > current_tracer`

Sources: linux_engineering_book.md, linux_engineering_book.pdf

1.10.181 `echo madvise > /sys/kernel/mm/transparent_hugepage/enabled`

Transparent Huge Pages

`echo madvise > /sys/kernel/mm/transparent_hugepage/enabled`

Sources: linux_engineering_book.md, linux_engineering_book.pdf

1.10.182 `echo rm -rf "$VAR"`

print command before execution. **SAFE**

`echo rm -rf "$VAR"`

Sources: komendy

1.10.183 `echo s > /proc/sysrq-trigger #`

Sekwencja REISUB (Synchronizacja dysków, Odmontowanie, Natychmiastowy restart)

`echo s > /proc/sysrq-trigger #`

Sources: encyklopedia_inzynierii_linux.html, encyklopedia_inzynierii_linux.pdf

1.10.184 `echo u > /proc/sysrq-trigger #`

Sekwencja REISUB (Synchronizacja dysków, Odmontowanie, Natychmiastowy restart)

`echo u > /proc/sysrq-trigger #`

Sources: encyklopedia_inzynierii_linux.html, encyklopedia_inzynierii_linux.pdf

1.10.185 `echo {1..5}`

Komentarz Komentarz w skrypcie function myfunc() { Definicja funkcji

`echo {1..5}`

Sources: RHCSA_Command_Library.pdf

1.10.186 `echo {a,b,c}.log a.log b.log c.log jq '.key' file.json`

Komentarz Komentarz w skrypcie function myfunc() { Definicja funkcji

`echo {a,b,c}.log a.log b.log c.log jq '.key' file.json`

Sources: RHCSA_Command_Library.pdf

1.10.187 `emacs`

Edytor Emacs

`emacs file.txt`

Sources: LINUX COMMAND

1.10.188 env

display environment. **SAFE**

```
env \  
env
```

Sources: LINUX COMMAND, RHCSA_Command_Library.pdf, komendy

1.10.189 exit

Kończy skrypt

```
exit 1
```

Sources: LINUX COMMAND

1.10.190 expand

Zamienia tabulatory na spacje

```
expand file.txt
```

Sources: LINUX COMMAND

1.10.191 export

set env var. **SAFE**

```
export
```

Sources: LINUX COMMAND, RHCSA_Command_Library.pdf, komendy

1.10.192 export conn > user@example.com

Komentarz Komentarz w skrypcie function myfunc() { Definicja funkcji

```
export conn > user@example.com
```

Sources: RHCSA_Command_Library.pdf

1.10.193 fmt

Formatuje akapity

```
fmt -w 80 file.txt
```

Sources: LINUX COMMAND

1.10.194 fold

Zawija długie linie

```
fold -w 80 file.txt
```

Sources: LINUX COMMAND

1.10.195 getopt

Parsuje opcje

```
getopts
```

Sources: LINUX COMMAND

1.10.196 `getopts "a:b" opt`

parse short options. **SAFE**

```
getopts "a:b" opt
```

Sources: komendy

1.10.197 `grep`

Wyszukiwanie wzorca w tekście

```
grep "root" /etc/passwd  
grep
```

Sources: LINUX_COMMAND, RHCSA_Command_Library.pdf, komendy

1.10.198 `grep "pattern" <<< "$variable"`

10.5 Zaawansowany Bash

```
grep "pattern" <<< "$variable"
```

Sources: linux_engineering_book.md, linux_engineering_book.pdf

1.10.199 `grep '/etc'`

Sekcja Komend

```
grep '/etc'
```

Sources: RHCSA_Command_Library.pdf

1.10.200 `grep 'Accepted'`

Sekcja Komend

```
grep 'Accepted'
```

Sources: RHCSA_Command_Library.pdf

1.10.201 `grep 'Failed'`

Sekcja Komend

```
grep 'Failed'
```

Sources: RHCSA_Command_Library.pdf

1.10.202 `grep 'pattern' file`

Sekcja Komend

```
grep 'pattern' file
```

Sources: RHCSA_Command_Library.pdf

1.10.203 `grep -A 3 'pattern'`

Sekcja Komend

```
grep -A 3 'pattern'
```

Sources: RHCSA_Command_Library.pdf

1.10.204 `grep -B 3 'pattern'`

Sekcja Komend

```
grep -B 3 'pattern'
```

Sources: RHCSA_Command_Library.pdf

1.10.205 `grep -C 3 'pattern'`

Sekcja Komend

```
grep -C 3 'pattern'
```

Sources: RHCSA_Command_Library.pdf

1.10.206 `grep -E`

Rozszerzone wyrażenia regularne

```
grep -E "192\.168\.(1\
```

Sources: LINUX COMMAND

1.10.207 `grep -E "regex" file # extended regex`

`grep` — wyszukiwanie wzorców

```
grep -E "regex" file # extended regex
```

Sources: linux_engineering_book.md, linux_engineering_book.pdf

1.10.208 `grep -E '(ERROR`

FATAL

```
grep -E '(ERROR
```

Sources: linux_engineering_book.pdf

1.10.209 `grep -E '(ERROR|FATAL|CRITICAL)' /var/log/app.log`

29.2 Praktyczne Przykłady

```
grep -E '(ERROR|FATAL|CRITICAL)' /var/log/app.log
```

Sources: linux_engineering_book.md

1.10.210 `grep -E 'pat1`

pat2' Szukaj rozszerzonego wyrażenia sort -k2 file.txt Sortuj po 2. kolumnie

```
grep -E 'pat1
```

Sources: RHCSA_Command_Library.pdf

1.10.211 `grep -F 'literal'`

Sekcja Komend

```
grep -F 'literal'
```

Sources: RHCSA_Command_Library.pdf

1.10.212 `grep -P "perl_regex" file #`

`grep` — wyszukiwanie wzorców

`grep -P "perl_regex" file #`

Sources: linux_engineering_book.md, linux_engineering_book.pdf

1.10.213 `grep -P '\d+' file`

Sekcja Komend

`grep -P '\d+' file`

Sources: RHCSA_Command_Library.pdf

1.10.214 `grep -P '\d{3}-\d{2}'`

Perl-compatible regex. **SAFE**

`grep -P '\d{3}-\d{2}'`

Sources: komendy

1.10.215 `grep -P '^[a-zA-Z0-9._%+~]+@[a-zA-Z0-9.-]+\.[a-zA-Z]{2,}$' file`

29.2 Praktyczne Przykłady

`grep -P '^[a-zA-Z0-9._%+~]+@[a-zA-Z0-9.-]+\.[a-zA-Z]{2,}$' file`

Sources: linux_engineering_book.md, linux_engineering_book.pdf

1.10.216 `grep -P '^d{1,3}\.d{1,3}\.d{1,3}\.d{1,3}$' file`

29.2 Praktyczne Przykłady

`grep -P '^d{1,3}\.d{1,3}\.d{1,3}\.d{1,3}$' file`

Sources: linux_engineering_book.md, linux_engineering_book.pdf

1.10.217 `grep -c "pattern" file # liczba dopasowań`

`grep` — wyszukiwanie wzorców

`grep -c "pattern" file # liczba dopasowań`

Sources: linux_engineering_book.md, linux_engineering_book.pdf

1.10.218 `grep -c 'pattern'`

Sekcja Komend

`grep -c 'pattern'`

Sources: RHCSA_Command_Library.pdf

1.10.219 `grep -i`

Ignoruje wielkość liter

`grep -i "error" log.txt`

Sources: LINUX COMMAND

1.10.220 `grep -i "pattern" file # case-insensitive`

grep — wyszukiwanie wzorców

`grep -i "pattern" file # case-insensitive`

Sources: linux_engineering_book.md, linux_engineering_book.pdf

1.10.221 `grep -i 'pattern'`

Sekcja Komend

`grep -i 'pattern'`

Sources: RHCSA_Command_Library.pdf

1.10.222 `grep -i pattern file`

case-insensitive. **SAFE**

`grep -i pattern file`

Sources: komendy

1.10.223 `grep -l "pattern" *.txt # tylko nazwy plików`

grep — wyszukiwanie wzorców

`grep -l "pattern" *.txt # tylko nazwy plików`

Sources: linux_engineering_book.md, linux_engineering_book.pdf

1.10.224 `grep -l 'pattern'`

Sekcja Komend

`grep -l 'pattern'`

Sources: RHCSA_Command_Library.pdf

1.10.225 `grep -m 5 'pattern'`

Sekcja Komend

`grep -m 5 'pattern'`

Sources: RHCSA_Command_Library.pdf

1.10.226 `grep -n 'pattern'`

Sekcja Komend

`grep -n 'pattern'`

Sources: RHCSA_Command_Library.pdf

1.10.227 `grep -n pattern file`

line numbers. **SAFE**

`grep -n pattern file`

Sources: komendy

1.10.228 `grep -o`

Test regex with before .

```
grep -o
```

Sources: komendy

1.10.229 `grep -o 'pattern'`

Sekcja Komend

```
grep -o 'pattern'
```

Sources: RHCSA_Command_Library.pdf

1.10.230 `grep -oP 'https?:/[^\s"<>]+' file`

29.2 Praktyczne Przykłady

```
grep -oP 'https?:/[^\s"<>]+' file
```

Sources: linux_engineering_book.md, linux_engineering_book.pdf

1.10.231 `grep -oP 'prefix_\K\d+'`

output only match, drop prefix. **SAFE**

```
grep -oP 'prefix_\K\d+'
```

Sources: komendy

1.10.232 `grep -q 'pattern'`

Sekcja Komend

```
grep -q 'pattern'
```

Sources: RHCSA_Command_Library.pdf

1.10.233 `grep -q pattern file`

before append.

```
grep -q pattern file
```

Sources: komendy

1.10.234 `grep -q pattern file before sed append.`

Idempotency Checks

```
grep -q pattern file` before `sed` append.
```

Sources: komendy

1.10.235 `grep -r`

Rekurencyjne przeszukiwanie

```
grep -r "PermitRootLogin" /etc/ssh/
```

Sources: LINUX COMMAND

1.10.236 `grep -r "pattern" /path # rekursywne`

`grep` — wyszukiwanie wzorców

`grep -r "pattern" /path # rekursywne`

Sources: linux_engineering_book.md, linux_engineering_book.pdf

1.10.237 `grep -r 'pattern'`

Sekcja Komend

`grep -r 'pattern'`

Sources: RHCSA_Command_Library.pdf

1.10.238 `grep -r pattern dir/`

recursive. **SAFE**

`grep -r pattern dir/`

Sources: komendy

1.10.239 `grep -v`

Inwersja dopasowania

`grep -v "^#" config.txt`

Sources: LINUX COMMAND

1.10.240 `grep -v "pattern" file # odwrócone (linie`

`grep` — wyszukiwanie wzorców

`grep -v "pattern" file # odwrócone (linie`

Sources: linux_engineering_book.md, linux_engineering_book.pdf

1.10.241 `grep -v 'pattern'`

Sekcja Komend

`grep -v 'pattern'`

Sources: RHCSA_Command_Library.pdf

1.10.242 `grep -v pattern file`

invert match. **SAFE**

`grep -v pattern file`

Sources: komendy

1.10.243 `grep -w 'word' file`

Sekcja Komend

`grep -w 'word' file`

Sources: RHCSA_Command_Library.pdf

1.10.244 `grep cifs smb nmb`

Komentarz Komentarz w skrypcie function myfunc() { Definicja funkcji

`grep cifs smb nmb`

Sources: RHCSA_Command_Library.pdf

1.10.245 `grep http`

Sekcja Komend

`grep http`

Sources: RHCSA_Command_Library.pdf

1.10.246 `gunzip`

Dekompresuje gzip

`gunzip file.txt.gz`

Sources: LINUX COMMAND

1.10.247 `gunzip file.gz`

Sekcja Komend

`gunzip file.gz`

Sources: RHCSA_Command_Library.pdf

1.10.248 `gzip`

Kompresuje plik

`gzip file.txt`

Sources: LINUX COMMAND

1.10.249 `gzip -9 file`

Sekcja Komend

`gzip -9 file`

Sources: RHCSA_Command_Library.pdf

1.10.250 `gzip -d file.gz`

Archiwum cpio przez find

`gzip -d file.gz`

Sources: RHCSA_Command_Library.pdf

1.10.251 `gzip -k file`

Sekcja Komend

`gzip -k file`

Sources: RHCSA_Command_Library.pdf

1.10.252 `gzip -l file.gz`

Sekcja Komend

`gzip -l file.gz`

Sources: RHCSA_Command_Library.pdf

1.10.253 `gzip file`

Sekcja Komend

`gzip file`

Sources: RHCSA_Command_Library.pdf

1.10.254 `help`

Pomoc wbudowanych komend bash

`help cd`

Sources: LINUX COMMAND

1.10.255 `history`

with timestamps: . **SAFE**

`history \`

`history`

Sources: LINUX COMMAND, RHCSA_Command_Library.pdf, komendy

1.10.256 `history -c`

Sekcja Komend

`history -c`

Sources: RHCSA_Command_Library.pdf

1.10.257 `history -w`

Sekcja Komend

`history -w`

Sources: RHCSA_Command_Library.pdf

1.10.258 `iconv`

Konwersja kodowania znaków

`iconv -f`

Sources: LINUX COMMAND

1.10.259 `iconv -f`

encoding conversion. **SAFE**

`iconv -f`

Sources: komendy

1.10.260 info

Dokumentacja info

`info coreutils`

Sources: LINUX COMMAND

1.10.261 join

Łączy pliki po kluczu

`join file1 file2`

Sources: LINUX COMMAND

1.10.262 local

Lokalna zmienna w funkcji

`local var=value`

Sources: LINUX COMMAND

1.10.263 local name=\$1 # zmienna lokalna

10.3 Funkcje

`local name=$1 # zmienna lokalna`

Sources: linux_engineering_book.md, linux_engineering_book.pdf

1.10.264 local var=value

function-local scope. **SAFE**

`local var=value`

Sources: komendy

1.10.265 logger

Loguje do systemu

`logger -p local0.info "message"`
`logger "Test message"`

Sources: LINUX COMMAND

1.10.266 logger "Test message"

inject test log. **SAFE**

`logger "Test message"`

Sources: komendy

1.10.267 logger 'Test

Sekcja Komend

`logger 'Test`

Sources: RHCSA_Command_Library.pdf

1.10.268 `logger -p user.err`

Sekcja Komend

`logger -p user.err`

Sources: RHCSA_Command_Library.pdf

1.10.269 `logger -t myapp`

Sekcja Komend

`logger -t myapp`

Sources: RHCSA_Command_Library.pdf

1.10.270 `man`

Podręcznik systemowy

`man bash`

Sources: LINUX COMMAND

1.10.271 `mandb`

Indeksuje strony man

`mandb`

Sources: LINUX COMMAND

1.10.272 `nano`

Prosty edytor

`nano file.txt`

Sources: LINUX COMMAND

1.10.273 `paste`

Scala pliki w kolumny

`paste file1 file2`

Sources: LINUX COMMAND

1.10.274 `paste -d',' file1 file2`

merge side-by-side. **SAFE**

`paste -d',' file1 file2`

Sources: komendy

1.10.275 `printenv`

Drukuje zmienne środowiskowe

`printenv`

Sources: LINUX COMMAND, RHCSA_Command_Library.pdf

1.10.276 printf

Formatowane wyjście

```
printf "%s\n" "hello"
```

Sources: LINUX COMMAND

1.10.277 printf '%s ' {a..z}

Komentarz Komentarz w skrypcie function myfunc() { Definicja funkcji

```
printf '%s ' {a..z}
```

Sources: RHCSA_Command_Library.pdf

1.10.278 printf '%s\n' "\$var"

Use instead of for arbitrary data.

```
printf '%s\n' "$var"
```

Sources: komendy

1.10.279 read

Komentarz Komentarz w skrypcie function myfunc() { Definicja funkcji

```
read -p "Name: " name  
read
```

Sources: LINUX COMMAND, RHCSA_Command_Library.pdf

1.10.280 read -a

Komentarz Komentarz w skrypcie function myfunc() { Definicja funkcji

```
read -a
```

Sources: RHCSA_Command_Library.pdf

1.10.281 read -p 'Enter

Komentarz Komentarz w skrypcie function myfunc() { Definicja funkcji

```
read -p 'Enter
```

Sources: RHCSA_Command_Library.pdf

1.10.282 read -s

Komentarz Komentarz w skrypcie function myfunc() { Definicja funkcji

```
read -s
```

Sources: RHCSA_Command_Library.pdf

1.10.283 read -t 5

Komentarz Komentarz w skrypcie function myfunc() { Definicja funkcji

```
read -t 5
```

Sources: RHCSA_Command_Library.pdf

1.10.284 read line <&4

28.1 Bash — Zaawansowane Techniki

```
read line <&4
```

Sources: linux_engineering_book.md, linux_engineering_book.pdf

1.10.285 read response <&\${myproc[0]}

10.5 Zaawansowany Bash

```
read response <&${myproc[0]}
```

Sources: linux_engineering_book.md, linux_engineering_book.pdf

1.10.286 readonly

Sekcja Komend

```
readonly
```

Sources: LINUX COMMAND, RHCSA_Command_Library.pdf

1.10.287 readonly var=value

immutable. **SAFE**

```
readonly var=value
```

Sources: komendy

1.10.288 reset

Resetuje terminal

```
reset
```

Sources: LINUX COMMAND

1.10.289 rev

Odwraca linie

```
echo "abc" \
```

Sources: LINUX COMMAND

1.10.290 rev < file

reverse characters per line. **SAFE**

```
rev < file
```

Sources: komendy

1.10.291 sed

Use for field-based; for line-based.

```
sed 's/old/new/g' file  
sed
```

Sources: LINUX COMMAND, komendy

1.10.292 `sed '/^#/d' file`

delete comment lines. **SAFE**

`sed '/^#/d' file`

Sources: komendy

1.10.293 `sed '/^#/d; /^$/d' config.file`

29.2 Praktyczne Przykłady

`sed '/^#/d; /^$/d' config.file`

Sources: linux_engineering_book.md, linux_engineering_book.pdf

1.10.294 `sed '/^$/d' file # usuń puste linie`

10.4 Przetwarzanie Tekstu

`sed '/^$/d' file # usuń puste linie`

Sources: linux_engineering_book.md, linux_engineering_book.pdf

1.10.295 `sed '/pattern/d' file # usuń linie z wzorcem`

10.4 Przetwarzanie Tekstu

`sed '/pattern/d' file # usuń linie z wzorcem`

Sources: linux_engineering_book.md, linux_engineering_book.pdf

1.10.296 `sed 's/old/new/'`

Sekcja Komend

`sed 's/old/new/'`

Sources: RHCSA_Command_Library.pdf

1.10.297 `sed 's/old/new/' file # zamień pierwsze wystąpienie`

10.4 Przetwarzanie Tekstu

`sed 's/old/new/' file # zamień pierwsze wystąpienie`

Sources: linux_engineering_book.md, linux_engineering_book.pdf

1.10.298 `sed 's/old/new/g'`

Sekcja Komend

`sed 's/old/new/g'`

Sources: RHCSA_Command_Library.pdf

1.10.299 `sed 's/old/new/g' file # zamień wszystkie`

10.4 Przetwarzanie Tekstu

`sed 's/old/new/g' file # zamień wszystkie`

Sources: linux_engineering_book.md, linux_engineering_book.pdf

1.10.300 `sed -E 's/([0-9]{2})\1/([0-9]{4})/3-2-1/g' file`

29.2 Praktyczne Przykłady

`sed -E 's/([0-9]{2})\1/([0-9]{4})/3-2-1/g' file`

Sources: linux_engineering_book.md, linux_engineering_book.pdf

1.10.301 `sed -i`

Never on critical files without backup.

`sed -i`

Sources: komendy

1.10.302 `sed -i 's/old/new/g'`

Sekcja Komend

`sed -i 's/old/new/g'`

Sources: RHCSA_Command_Library.pdf

1.10.303 `sed -i 's/old/new/g' file`

in-place replace. **CAUTION**

`sed -i 's/old/new/g' file`

Sources: komendy

1.10.304 `sed -i 's/old/new/g' file # edycja in-place`

10.4 Przetwarzanie Tekstu

`sed -i 's/old/new/g' file # edycja in-place`

Sources: linux_engineering_book.md, linux_engineering_book.pdf

1.10.305 `sed -i.bak`

Sekcja Komend

`sed -i.bak`

Sources: RHCSA_Command_Library.pdf

1.10.306 `sed -i.bak 's/old/new/g' file`

in-place with backup. **CAUTION**

`sed -i.bak 's/old/new/g' file`

Sources: komendy

1.10.307 `sed -n '5,10p' file`

print lines 5-10. **SAFE**

`sed -n '5,10p' file`

Sources: RHCSA_Command_Library.pdf, komendy

1.10.308 `sed -n '5,10p' file # wyświetl linie 5-10`

10.4 Przetwarzanie Tekstu

`sed -n '5,10p' file # wyświetl linie 5-10`

Sources: linux_engineering_book.md, linux_engineering_book.pdf

1.10.309 `set`

all shell variables/functions. **SAFE**

`set -o vi`

`set -e`

`set`

Sources: LINUX_COMMAND, RHCSA_Command_Library.pdf, komendy

1.10.310 `set +x`

Sekcja Komend

`set +x`

Sources: RHCSA_Command_Library.pdf

1.10.311 `set -e`

Sekcja Komend

`set -e`

Sources: RHCSA_Command_Library.pdf

1.10.312 `set -e # zakończ przy błędzie`

Obsługa błędów

`set -e # zakończ przy błędzie`

Sources: linux_engineering_book.md, linux_engineering_book.pdf

1.10.313 `set -euo pipefail`

exit on error, unset vars, pipe failures. **SAFE**

`set -euo pipefail`

Sources: komendy

1.10.314 `set -o pipefail`

Sekcja Komend

`set -o pipefail`

Sources: RHCSA_Command_Library.pdf

1.10.315 `set -o pipefail # propaguj błędy w pipe`

Obsługa błędów

`set -o pipefail # propaguj błędy w pipe`

Sources: linux_engineering_book.md, linux_engineering_book.pdf

1.10.316 set -u

Sekcja Komend

```
set -u
```

Sources: RHCSA_Command_Library.pdf

1.10.317 set -u # błąd przy niezdefiniowanej zmiennej

Obsługa błędów

```
set -u # błąd przy niezdefiniowanej zmiennej
```

Sources: linux_engineering_book.md, linux_engineering_book.pdf

1.10.318 set -x

in scripts to trace actual commands. **SAFE**

```
set -x
```

Sources: RHCSA_Command_Library.pdf, komendy, linux_engineering_book.md, linux_engineering_book.pdf

1.10.319 set -x # debug - wyświetlaj polecenia

Obsługa błędów

```
set -x # debug - wyświetlaj polecenia
```

Sources: linux_engineering_book.md, linux_engineering_book.pdf

1.10.320 set -x in scripts to trace actual commands. **SAFE**

Verification Before Execution

```
set -x` in scripts to trace actual commands. **SAFE**
```

Sources: komendy

1.10.321 sh

POSIX shell

```
sh script.sh
```

Sources: LINUX COMMAND

1.10.322 sh script.sh

Komentarz Komentarz w skrypcie function myfunc() { Definicja funkcji

```
sh script.sh
```

Sources: RHCSA_Command_Library.pdf

1.10.323 shift

consume positional args. **SAFE**

```
shift 2
```

```
shift
```

Sources: LINUX COMMAND, komendy

1.10.324 sort

Sortuje linie

```
sort -t: -k3 -n /etc/passwd
```

Sources: LINUX COMMAND

1.10.325 sort -k2 -n file # sortuj po kolumnie 2 numerycznie

10.4 Przetwarzanie Tekstu

```
sort -k2 -n file # sortuj po kolumnie 2 numerycznie
```

Sources: linux_engineering_book.md, linux_engineering_book.pdf

1.10.326 sort -t: -k3 -n /etc/passwd

numeric sort by field. **SAFE**

```
sort -t: -k3 -n /etc/passwd
```

Sources: komendy

1.10.327 sort -u file # sortuj i usuń duplikaty

10.4 Przetwarzanie Tekstu

```
sort -u file # sortuj i usuń duplikaty
```

Sources: linux_engineering_book.md, linux_engineering_book.pdf

1.10.328 source

on untrusted files.

```
source
```

Sources: RHCSA_Command_Library.pdf, komendy

1.10.329 source "\$file"

execute file in current shell. **DANGEROUS** (if untrusted)

```
source "$file"
```

Sources: komendy

1.10.330 source / .

Wykonuje w obecnej powłoce

```
source script.sh
```

Sources: LINUX COMMAND

1.10.331 source /etc/profile

reload profile. **SAFE**

```
source /etc/profile
```

Sources: komendy

1.10.332 `source /opt/venv/bin/activate`

activate venv. **SAFE**

`source /opt/venv/bin/activate`

Sources: komendy

1.10.333 `source ~/.bashrc`

Sekcja Komend

`source ~/.bashrc`

Sources: RHCSA_Command_Library.pdf

1.10.334 `star`

Archiwizacja (xattr/ACL)

`star -c -f=backup.star /etc`

Sources: LINUX COMMAND

1.10.335 `stty`

Ustawienia terminala

`stty -a`

Sources: LINUX COMMAND

1.10.336 `tar`

Sekcja Komend

`tar`

Sources: RHCSA_Command_Library.pdf

1.10.337 `tar --delete -f`

Sekcja Komend

`tar --delete -f`

Sources: RHCSA_Command_Library.pdf

1.10.338 `tar --exclude-vcs`

Sekcja Komend

`tar --exclude-vcs`

Sources: RHCSA_Command_Library.pdf

1.10.339 `tar --listed-incremental=snapshot.snar -czf inc.tar.gz /data # incremental`

19.1 Narzędzia Backup

`tar --listed-incremental=snapshot.snar -czf inc.tar.gz /data # incremental`

Sources: linux_engineering_book.md, linux_engineering_book.pdf

1.10.340 tar -C

Rozpakowuje do katalogu

```
tar -xvf backup.tar -C /tmp
```

Sources: LINUX COMMAND

1.10.341 tar -cjvf

Tworzy archiwum bzip2

```
tar -cjvf backup.tar.bz2 /etc
```

Sources: LINUX COMMAND

1.10.342 tar -cvJf

Sekcja Komend

```
tar -cvJf
```

Sources: RHCSA_Command_Library.pdf

1.10.343 tar -cvf

Tworzy archiwum tar

```
tar -cvf backup.tar /etc
```

Sources: LINUX COMMAND

1.10.344 tar -cvf archive.tar

Sekcja Komend

```
tar -cvf archive.tar
```

Sources: RHCSA_Command_Library.pdf

1.10.345 tar -cvjf

Sekcja Komend

```
tar -cvjf
```

Sources: RHCSA_Command_Library.pdf

1.10.346 tar -cvzf

Sekcja Komend

```
tar -cvzf
```

Sources: RHCSA_Command_Library.pdf

1.10.347 tar -czf - /path

Archiwum przez SSH unzip -o archive.zip Nadpisz istniejące pliki

```
tar -czf - /path
```

Sources: RHCSA_Command_Library.pdf

1.10.348 `tar -czf arch.tar.gz`

Sekcja Komend

`tar -czf arch.tar.gz`

Sources: RHCSA_Command_Library.pdf

1.10.349 `tar -czvf`

Tworzy archiwum gzip

`tar -czvf backup.tar.gz /etc`

Sources: LINUX COMMAND

1.10.350 `tar -rvf archive.tar`

Sekcja Komend

`tar -rvf archive.tar`

Sources: RHCSA_Command_Library.pdf

1.10.351 `tar -tvf archive.tar`

Sekcja Komend

`tar -tvf archive.tar`

Sources: RHCSA_Command_Library.pdf

1.10.352 `tar -tvzf`

Sekcja Komend

`tar -tvzf`

Sources: RHCSA_Command_Library.pdf

1.10.353 `tar -uvf archive.tar`

Sekcja Komend

`tar -uvf archive.tar`

Sources: RHCSA_Command_Library.pdf

1.10.354 `tar -xvJf`

Sekcja Komend

`tar -xvJf`

Sources: RHCSA_Command_Library.pdf

1.10.355 `tar -xvf`

Rozpakowuje archiwum tar

`tar -xvf backup.tar`

Sources: LINUX COMMAND

1.10.356 `tar -xvf archive.tar`

Sekcja Komend

`tar -xvf archive.tar`

Sources: RHCSA_Command_Library.pdf

1.10.357 `tar -xvjf`

Sekcja Komend

`tar -xvjf`

Sources: RHCSA_Command_Library.pdf

1.10.358 `tar -xvzf`

Sekcja Komend

`tar -xvzf`

Sources: RHCSA_Command_Library.pdf

1.10.359 `tar -xzvf`

Rozpakowuje gzip tar

`tar -xzvf backup.tar.gz`

Sources: LINUX COMMAND

1.10.360 `tar czf /backup/$(hostname)-$(date +%Y%m%d-%H%M%S).tar.gz /important/data`

Dodatek A: Przydatne Jednowierszowce

`tar czf /backup/$(hostname)-$(date +%Y%m%d-%H%M%S).tar.gz /important/data`

Sources: linux_engineering_book.md, linux_engineering_book.pdf

1.10.361 `tar czf backup-$(date +%Y%m%d).tar.gz /data # z data`

19.1 Narzędzia Backup

`tar czf backup-$(date +%Y%m%d).tar.gz /data # z data`

Sources: linux_engineering_book.md, linux_engineering_book.pdf

1.10.362 `tar czf backup.tar.gz /data # tworzenie`

19.1 Narzędzia Backup

`tar czf backup.tar.gz /data # tworzenie`

Sources: linux_engineering_book.md, linux_engineering_book.pdf

1.10.363 `tar xf linux-6.8.tar.xz`

16.1 Pobieranie i Konfiguracja

`tar xf linux-6.8.tar.xz`

Sources: linux_engineering_book.md, linux_engineering_book.pdf

1.10.364 `tar xzf backup.tar.gz -C /restore # rozpakowywanie`

19.1 Narzędzia Backup

`tar xzf backup.tar.gz -C /restore # rozpakowywanie`

Sources: linux_engineering_book.md, linux_engineering_book.pdf

1.10.365 `tar xzvf backup.tar.gz -C /restore`

extract backup. **CAUTION**

`tar xzvf backup.tar.gz -C /restore`

Sources: komendy

1.10.366 `tee`

Zapisuje do pliku i pokazuje na ekranie

```
echo "test" \  
tee
```

Sources: LINUX COMMAND

1.10.367 `tee -a logfile`

append to file and stdout. **SAFE**

`tee -a logfile`

Sources: komendy

1.10.368 `test`

.gitlab-ci.yml (GitLab CI)

```
test
```

Sources: linux_engineering_book.pdf

1.10.369 `test -f file`

before .

```
test -f file
```

Sources: komendy

1.10.370 `test -f file before touch file.`

Idempotency Checks

```
test -f file` before `touch file`.
```

Sources: komendy

1.10.371 `test / [`

Testowanie warunków

```
test -f file
```

Sources: LINUX COMMAND

1.10.372 tput

Steruje terminalem

```
tput setaf 1
```

Sources: LINUX COMMAND

1.10.373 tr

Zamiana/zusuwanie znaków

```
cat file \
```

Sources: LINUX COMMAND

1.10.374 tr 'a-z' 'A-Z' < file

character translate. **SAFE**

```
tr 'a-z' 'A-Z' < file
```

Sources: komendy

1.10.375 tr -d '\r'

delete carriage returns. **SAFE**

```
tr -d '\r'
```

Sources: komendy

1.10.376 tr -s ' '

squeeze repeats. **SAFE**

```
tr -s ' '
```

Sources: komendy

1.10.377 trap

Obsługa sygnałów

```
trap "echo exit"
```

Sources: LINUX COMMAND

1.10.378 trap 'cleanup; exit'

Przechwytywanie sygnałów w bash

```
trap 'cleanup; exit'
```

Sources: linux_engineering_book.md, linux_engineering_book.pdf

1.10.379 trap 'echo "Caught

Przechwytywanie sygnałów w bash

```
trap 'echo "Caught
```

Sources: linux_engineering_book.md, linux_engineering_book.pdf

1.10.380 `trap 'rm -f "$tmpfile"'`

cleanup on exit. **SAFE**

`trap 'rm -f "$tmpfile"'`

Sources: komendy

1.10.381 `trap cleanup`

10.3 Funkcje

`trap cleanup`

Sources: linux_engineering_book.md, linux_engineering_book.pdf

1.10.382 `tty`

Nazwa terminala

`tty`

Sources: LINUX COMMAND

1.10.383 `unalias`

Usuwa alias

`unalias ll`

Sources: LINUX COMMAND

1.10.384 `unexpand`

Zamienia spacje na tabulatory

`unexpand -a file.txt`

Sources: LINUX COMMAND

1.10.385 `uniq`

Usuwa/raportuje duplikaty

`sort file \`

Sources: LINUX COMMAND

1.10.386 `uniq -c file # zlicz powtórzenia`

10.4 Przetwarzanie Tekstu

`uniq -c file # zlicz powtórzenia`

Sources: linux_engineering_book.md, linux_engineering_book.pdf

1.10.387 `unset`

remove env var. **SAFE**

`unset`

Sources: LINUX COMMAND, RHCSA_Command_Library.pdf, komendy

1.10.388 unxz

Dekompresuje xz

`unxz file.txt.xz`

Sources: LINUX COMMAND

1.10.389 unzip

Rozpakowuje zip

`unzip backup.zip`

Sources: LINUX COMMAND

1.10.390 vim +/pattern

Sekcja Komend

`vim +/pattern`

Sources: RHCSA_Command_Library.pdf

1.10.391 vim +10 file.txt

Sekcja Komend

`vim +10 file.txt`

Sources: RHCSA_Command_Library.pdf

1.10.392 vim -R file.txt

Sekcja Komend

`vim -R file.txt`

Sources: RHCSA_Command_Library.pdf

1.10.393 vim -d file1 file2

Sekcja Komend

`vim -d file1 file2`

Sources: RHCSA_Command_Library.pdf

1.10.394 vim -u

Sekcja Komend

`vim -u`

Sources: RHCSA_Command_Library.pdf

1.10.395 vim / vi

Zaawansowany edytor

`vim /etc/hosts`

Sources: LINUX COMMAND

1.10.396 vim file.txt

Sekcja Komend

vim file.txt

Sources: RHCSA_Command_Library.pdf

1.10.397 wc

Liczy linie, słowa, bajty

wc -l /etc/passwd

Sources: LINUX COMMAND

1.10.398 wc -l file # liczba linii

10.4 Przetwarzanie Tekstu

wc -l file # *liczba linii*

Sources: linux_engineering_book.md, linux_engineering_book.pdf

1.10.399 wc file.txt

Sekcja Komend

wc file.txt

Sources: RHCSA_Command_Library.pdf

1.10.400 whatis

Krótki opis komendy

whatis ls

Sources: LINUX COMMAND

1.10.401 xz

Kompresuje xz

xz file.txt

Sources: LINUX COMMAND

1.10.402 xzcat

Wyświetla xz

xzcat file.txt.xz

Sources: LINUX COMMAND

1.10.403 zcat

Wyświetla skompresowany plik

zcat file.txt.gz

Sources: LINUX COMMAND

1.10.404 zip

Tworzy zip

```
zip backup.zip file1 file2
```

Sources: LINUX COMMAND

1.11 SSH

1.11.1 /etc/ssh/ssh_config

Konfig klienta SSH

```
/etc/ssh/ssh_config
```

Sources: LINUX COMMAND

1.11.2 /etc/ssh/sshd_config

Always backup before edits.

```
/etc/ssh/sshd_config
```

Sources: LINUX COMMAND, RHCSA_Command_Library.pdf, komendy

1.11.3 nft add rule inet edge_firewall inbound tcp dport { 22, 443 } ct state new accept

Otwarcie portów dla usług SSH oraz HTTPS pod warunkiem inicjalizacji nowego połączenia (New)

```
nft add rule inet edge_firewall inbound tcp dport { 22, 443 } ct state new accept
```

Sources: encyklopedia_inzynierii_linux.html, encyklopedia_inzynierii_linux.pdf

1.11.4 scp

Kopiowanie przez SSH

```
scp file.txt user@host:/tmp
```

```
scp file user@server:/tmp
```

```
scp file user@host:/path
```

Sources: LINUX COMMAND

1.11.5 scp -P

Inny port SCP

```
scp -P 2222 file user@host:/tmp
```

Sources: LINUX COMMAND

1.11.6 scp -P 2222 file.txt user@host:/path/

secure copy. **SAFE**

```
scp -P 2222 file.txt user@host:/path/
```

Sources: komendy

1.11.7 scp -r

Kopiowanie katalogu

```
scp -r /local user@server:/remote
```

Sources: LINUX COMMAND

1.11.8 `scp -r /etc root@host:/tmp`

copying sensitive configs.

`scp -r /etc root@host:/tmp`

Sources: komendy

1.11.9 `scp -r /etc root@host:/tmp` copying sensitive configs.‘

Dangerous Mistakes

`scp -r /etc root@host:/tmp`` copying sensitive configs.

Sources: komendy

1.11.10 `sftp`

Interaktywny transfer SFTP

`sftp user@host`

`sftp user@server`

Sources: LINUX COMMAND

1.11.11 `sftp -P`

Inny port SFTP

`sftp -P 2222 user@host`

Sources: LINUX COMMAND

1.11.12 `sftp user@host`

interactive SFTP. **SAFE**

`sftp user@host`

Sources: komendy

1.11.13 `ssh`

Zdalna powłoka

`ssh user@server`

Sources: LINUX COMMAND

1.11.14 `ssh $host "uptime" &`

28.1 Bash — Zaawansowane Techniki

`ssh $host "uptime" &`

Sources: linux_engineering_book.md, linux_engineering_book.pdf

1.11.15 `ssh -A user@host`

Sekcja Komend

`ssh -A user@host`

Sources: RHCSA_Command_Library.pdf

1.11.16 `ssh -D`

Dynamiczny SOCKS proxy

```
ssh -D 1080 user@server
```

Sources: LINUX COMMAND

1.11.17 `ssh -D 1080`

Sekcja Komend

```
ssh -D 1080
```

Sources: RHCSA_Command_Library.pdf

1.11.18 `ssh -D 1080 user@host`

dynamic SOCKS proxy. **CAUTION**

```
ssh -D 1080 user@host
```

Sources: komendy

1.11.19 `ssh -J bastion user@internal`

jump host. **SAFE**

```
ssh -J bastion user@internal
```

Sources: komendy

1.11.20 `ssh -J jumphost`

Sekcja Komend

```
ssh -J jumphost
```

Sources: RHCSA_Command_Library.pdf

1.11.21 `ssh -L`

Lokalny port forwarding

```
ssh -L 8080:localhost:80 user@server
```

```
ssh -L
```

Sources: LINUX COMMAND, RHCSA_Command_Library.pdf

1.11.22 `ssh -L 8080:localhost:80 user@host`

local port forward. **CAUTION**

```
ssh -L 8080:localhost:80 user@host
```

Sources: komendy

1.11.23 `ssh -N -f -L`

Sekcja Komend

```
ssh -N -f -L
```

Sources: RHCSA_Command_Library.pdf

1.11.24 ssh -R

Zdalny port forwarding

```
ssh -R 9090:localhost:3000 user@server  
ssh -R
```

Sources: LINUX COMMAND, RHCSA_Command_Library.pdf

1.11.25 ssh -R 9090:localhost:3000 user@host

remote port forward. **CAUTION**

```
ssh -R 9090:localhost:3000 user@host
```

Sources: komendy

1.11.26 ssh -X

X11 forwarding

```
ssh -X user@server
```

Sources: LINUX COMMAND

1.11.27 ssh -X user@host

Sekcja Komend

```
ssh -X user@host
```

Sources: RHCSA_Command_Library.pdf

1.11.28 ssh -i

Klucz prywatny

```
ssh -i ~/.ssh/id_rsa user@server  
ssh -i
```

Sources: LINUX COMMAND, RHCSA_Command_Library.pdf

1.11.29 ssh -i ~/.ssh/id_ed25519 user@host

specific key. **SAFE**

```
ssh -i ~/.ssh/id_ed25519 user@host
```

Sources: komendy

1.11.30 ssh -o

bypass host key check. **DANGEROUS**

```
ssh -o
```

Sources: LINUX COMMAND, RHCSA_Command_Library.pdf, komendy

1.11.31 ssh -p

Inny port

```
ssh -p 2222 user@server
```

Sources: LINUX COMMAND

1.11.32 `ssh -p 2222`

Sekcja Komend

`ssh -p 2222`

Sources: RHCSA_Command_Library.pdf

1.11.33 `ssh -p 2222 user@host`

alternate port. **SAFE**

`ssh -p 2222 user@host`

Sources: komendy

1.11.34 `ssh -t user@host`

Sekcja Komend

`ssh -t user@host`

Sources: RHCSA_Command_Library.pdf

1.11.35 `ssh -v user@host`

Sekcja Komend

`ssh -v user@host`

Sources: RHCSA_Command_Library.pdf

1.11.36 `ssh -vvv user@host`

Sekcja Komend

`ssh -vvv user@host`

Sources: RHCSA_Command_Library.pdf

1.11.37 `ssh user@host 'cat >`

Sekcja Komend

`ssh user@host 'cat >`

Sources: RHCSA_Command_Library.pdf

1.11.38 `ssh user@hostname`

Sekcja Komend

`ssh user@hostname`

Sources: RHCSA_Command_Library.pdf

1.11.39 `ssh-add`

Dodaje klucz do agenta

`ssh-add ~/.ssh/id_rsa`

`ssh-add`

Sources: LINUX_COMMAND, RHCSA_Command_Library.pdf

1.11.40 `ssh-add -D`

Usuwa wszystkie klucze

`ssh-add -D`

Sources: LINUX COMMAND

1.11.41 `ssh-add -l`

list loaded keys. **SAFE**

`ssh-add -l`

Sources: LINUX COMMAND, RHCSA_Command_Library.pdf, komendy

1.11.42 `ssh-add ~/.ssh/id_ed25519`

add key to agent. **SAFE**

`ssh-add ~/.ssh/id_ed25519`

Sources: komendy

1.11.43 `ssh-agent`

Agent kluczy

`ssh-agent`

Sources: LINUX COMMAND

1.11.44 `ssh-agent -k`

Kill agent

`ssh-agent -k`

Sources: LINUX COMMAND

1.11.45 `ssh-agent -s`

start agent. **SAFE**

`ssh-agent -s`

Sources: komendy

1.11.46 `ssh-copy-id`

Kopiuje klucz publiczny

`ssh-copy-id user@server`

`ssh-copy-id`

Sources: LINUX COMMAND, RHCSA_Command_Library.pdf

1.11.47 `ssh-copy-id -i`

Sekcja Komend

`ssh-copy-id -i`

Sources: RHCSA_Command_Library.pdf

1.11.48 `ssh-copy-id -i ~/.ssh/id_ed25519.pub user@host`

install public key. **SAFE**

`ssh-copy-id -i ~/.ssh/id_ed25519.pub user@host`

Sources: komendy

1.11.49 `ssh-copy-id -i ~/.ssh/id_ed25519.pub user@server # kopiuj klucz`

9.2 SSH Hardening

`ssh-copy-id -i ~/.ssh/id_ed25519.pub user@server # kopiuj klucz`

Sources: linux_engineering_book.md, linux_engineering_book.pdf

1.11.50 `ssh-keygen`

Generuje klucz SSH

`ssh-keygen -t ed25519`

Sources: LINUX COMMAND

1.11.51 `ssh-keygen -R`

Usuwa z `known_hosts`

`ssh-keygen -R server`

`ssh-keygen -R`

Sources: LINUX COMMAND, RHCSA_Command_Library.pdf

1.11.52 `ssh-keygen -f`

Sekcja Komend

`ssh-keygen -f`

Sources: RHCSA_Command_Library.pdf

1.11.53 `ssh-keygen -l -f`

Sekcja Komend

`ssh-keygen -l -f`

Sources: RHCSA_Command_Library.pdf

1.11.54 `ssh-keygen -p -f`

Sekcja Komend

`ssh-keygen -p -f`

Sources: RHCSA_Command_Library.pdf

1.11.55 `ssh-keygen -t`

Sekcja Komend

`ssh-keygen -t`

Sources: RHCSA_Command_Library.pdf

1.11.56 `ssh-keygen -t ecdsa`

Sekcja Komend

`ssh-keygen -t ecdsa`

Sources: LINUX COMMAND, RHCSA_Command_Library.pdf

1.11.57 `ssh-keygen -t ed25519`

Ed25519 key

`ssh-keygen -t ed25519`

Sources: LINUX COMMAND

1.11.58 `ssh-keygen -t ed25519 -C "alice@example.com"`

generate modern key. **SAFE**

`ssh-keygen -t ed25519 -C "alice@example.com"`

Sources: komendy

1.11.59 `ssh-keygen -t ed25519 -C "user@host" # generuj klucz`

9.2 SSH Hardening

`ssh-keygen -t ed25519 -C "user@host" # generuj klucz`

Sources: linux_engineering_book.md, linux_engineering_book.pdf

1.11.60 `ssh-keygen -t rsa`

RSA key

`ssh-keygen -t rsa -b 4096`

Sources: LINUX COMMAND

1.11.61 `ssh-keygen -t rsa -b`

Sekcja Komend

`ssh-keygen -t rsa -b`

Sources: RHCSA_Command_Library.pdf

1.11.62 `ssh-keygen -t rsa -b 4096`

legacy RSA. **SAFE**

`ssh-keygen -t rsa -b 4096`

Sources: komendy

1.11.63 `ssh-keygen -y -f`

Sekcja Komend

`ssh-keygen -y -f`

Sources: RHCSA_Command_Library.pdf

1.11.64 ssh-keyscan

Zbiera host keys

[ssh-keyscan](#) [github.com](#)

Sources: LINUX COMMAND

1.11.65 sshpass

SSH z hasłem (niezalecane)

`sshpass -p pass ssh user@server`

Sources: LINUX COMMAND

1.12 Logs

1.12.1 abrt-cli

Analiza crashy

`abrt-cli`

Sources: LINUX COMMAND

1.12.2 audisp-remote

Remote logging

`audisp-remote`

Sources: LINUX COMMAND

1.12.3 audispd

Audit dispatcher

`systemctl`

Sources: LINUX COMMAND

1.12.4 auditctl

for system call auditing. **SAFE**

`auditctl -l`

`auditctl`

Sources: LINUX COMMAND, komendy

1.12.5 auditctl --backlog_wait_time

Backlog wait

`auditctl --backlog_wait_time 60000`

Sources: LINUX COMMAND

1.12.6 auditctl -A

Append rule

`auditctl -A exit,always -S openat`

Sources: LINUX COMMAND

1.12.7 auditctl -C

Compare fields

```
auditctl -a always,exit -C uid!=euid
```

Sources: LINUX COMMAND

1.12.8 auditctl -D

Delete all rules

```
auditctl -D
```

Sources: LINUX COMMAND

1.12.9 auditctl -F

Filter field

```
auditctl -a always,exit -F arch=b64 -S setuid
```

Sources: LINUX COMMAND

1.12.10 auditctl -I

Insert rule

```
auditctl -I 1 -w /etc/passwd -p wa
```

Sources: LINUX COMMAND

1.12.11 auditctl -R

Read rules from file

```
auditctl -R /etc/audit/rules.d/audit.rules
```

Sources: LINUX COMMAND

1.12.12 auditctl -S

Syscall

```
auditctl -a always,exit -S all -F uid=0
```

Sources: LINUX COMMAND

1.12.13 auditctl -a

Add rule

```
auditctl -a always,exit -F arch=b64 -S openat
```

Sources: LINUX COMMAND

1.12.14 auditctl -a always,exit -F arch=b64 -S execve -k exec_log

9.5 Hardening Systemu

```
auditctl -a always,exit -F arch=b64 -S execve -k exec_log
```

Sources: linux_engineering_book.md, linux_engineering_book.pdf

1.12.15 auditctl -b

Backlog limit

```
auditctl -b 8192
```

Sources: LINUX COMMAND

1.12.16 auditctl -d

Delete rule

```
auditctl -d always,exit -F arch=b64 -S openat
```

Sources: LINUX COMMAND

1.12.17 auditctl -e

Enable/disable

```
auditctl -e 1
```

Sources: LINUX COMMAND

1.12.18 auditctl -f

Failure mode

```
auditctl -f 1
```

Sources: LINUX COMMAND

1.12.19 auditctl -h

Help

```
auditctl -h
```

Sources: LINUX COMMAND

1.12.20 auditctl -k

Key for rule

```
auditctl -w /etc/shadow -p wa -k shadow_watch
```

Sources: LINUX COMMAND

1.12.21 auditctl -l

List rules

```
auditctl -l
```

Sources: LINUX COMMAND

1.12.22 auditctl -m

Send user message

```
auditctl -m "Message"
```

Sources: LINUX COMMAND

1.12.23 auditctl -p

Permissions (rwx)

```
auditctl -w /etc/passwd -p rwx
```

Sources: LINUX COMMAND

1.12.24 auditctl -q

Query

```
auditctl -q
```

Sources: LINUX COMMAND

1.12.25 auditctl -r

Rate limit

```
auditctl -r 0
```

Sources: LINUX COMMAND

1.12.26 auditctl -s

Status

```
auditctl -s
```

Sources: LINUX COMMAND

1.12.27 auditctl -t

Trim logs

```
auditctl -t
```

Sources: LINUX COMMAND

1.12.28 auditctl -v

Version

```
auditctl -v
```

Sources: LINUX COMMAND

1.12.29 auditctl -w

Watch file

```
auditctl -w /etc/passwd -p wa -k passwd_changes
```

Sources: LINUX COMMAND

1.12.30 auditctl -w /etc/passwd -p wa -k identity

watch file. **SAFE**

```
auditctl -w /etc/passwd -p wa -k identity
```

Sources: komendy

1.12.31 auditctl -w /etc/passwd -p wa -k passwd_changes

9.5 Hardening Systemu

`auditctl -w /etc/passwd -p wa -k passwd_changes`

Sources: linux_engineering_book.md, linux_engineering_book.pdf

1.12.32 auditctl -w /etc/shadow -p wa -k shadow_changes

9.5 Hardening Systemu

`auditctl -w /etc/shadow -p wa -k shadow_changes`

Sources: linux_engineering_book.md, linux_engineering_book.pdf

1.12.33 auditd

Audit daemon

`systemctl`

Sources: LINUX COMMAND

1.12.34 augenrules

Generate rules

`augenrules --load`

Sources: LINUX COMMAND

1.12.35 aureport

Audit report

`aureport --login`

Sources: LINUX COMMAND

1.12.36 aureport --login --summary

login report. **SAFE**

`aureport --login --summary`

Sources: komendy

1.12.37 aureport --summary

9.5 Hardening Systemu

`aureport --summary`

Sources: LINUX COMMAND, linux_engineering_book.md, linux_engineering_book.pdf

1.12.38 aureport -au

Authentication report

`aureport -au`

Sources: LINUX COMMAND

1.12.39 aureport -f

File report

```
aureport -f
```

Sources: LINUX COMMAND

1.12.40 aureport -l

Login report

```
aureport -l
```

Sources: LINUX COMMAND

1.12.41 aureport -x

Executable report

```
aureport -x
```

Sources: LINUX COMMAND

1.12.42 ausearch

Wyszukiwanie w audit.log

```
ausearch -m avc -ts recent
```

Sources: LINUX COMMAND

1.12.43 ausearch -c

By command

```
ausearch -c httpd
```

Sources: LINUX COMMAND

1.12.44 ausearch -f

By file

```
ausearch -f /etc/passwd
```

Sources: LINUX COMMAND

1.12.45 ausearch -ga

By GID

```
ausearch -ga 1000
```

Sources: LINUX COMMAND

1.12.46 ausearch -i

Interpret

```
ausearch -i
```

Sources: LINUX COMMAND

1.12.47 ausearch -k

By key

```
ausearch -k mykey
```

Sources: LINUX COMMAND

1.12.48 ausearch -k identity

search by key. **SAFE**

```
ausearch -k identity
```

Sources: komendy

1.12.49 ausearch -k passwd_changes

9.5 Hardening Systemu

```
ausearch -k passwd_changes
```

Sources: linux_engineering_book.md, linux_engineering_book.pdf

1.12.50 ausearch -m

24.1 SELinux — Tworzenie Polityk

```
ausearch -m
```

Sources: RHCSA_Command_Library.pdf, linux_engineering_book.md, linux_engineering_book.pdf

1.12.51 ausearch -m avc

AVC messages

```
ausearch -m avc -ts recent
```

Sources: LINUX COMMAND

1.12.52 ausearch -m avc -ts recent

recent denials. **SAFE**

```
ausearch -m avc -ts recent
```

Sources: komendy

1.12.53 ausearch -m selinux_err

SELinux errors

```
ausearch -m selinux_err
```

Sources: LINUX COMMAND

1.12.54 ausearch -m user_avc

User AVC

```
ausearch -m user_avc
```

Sources: LINUX COMMAND

1.12.55 ausearch -m user_login

Login messages

```
ausearch -m user_login
```

Sources: LINUX COMMAND

1.12.56 ausearch -m user_logout

Logout messages

```
ausearch -m user_logout
```

Sources: LINUX COMMAND

1.12.57 ausearch -pp

By PID

```
ausearch -pp 1234
```

Sources: LINUX COMMAND

1.12.58 ausearch -r

Raw

```
ausearch -r
```

Sources: LINUX COMMAND

1.12.59 ausearch -sc

By syscall

```
ausearch -sc openat
```

Sources: LINUX COMMAND

1.12.60 ausearch -te

Time end

```
ausearch -te now
```

Sources: LINUX COMMAND

1.12.61 ausearch -ts

Time start

```
ausearch -ts today
```

Sources: LINUX COMMAND

1.12.62 ausearch -ui

By UID

```
ausearch -ui 1000
```

Sources: LINUX COMMAND

1.12.63 ausearch -x

Exact match

```
ausearch -x /usr/bin/bash
```

Sources: LINUX COMMAND

1.12.64 dmesg

grep -i Komunikaty jdra o sobie shutdown -h +10 Wycz za 10 minut

```
dmesg -T
```

```
dmesg
```

Sources: LINUX COMMAND, RHCSA_Command_Library.pdf, komendy, linux_engineering_book.pdf

1.12.65 dmesg -H

Sekcja Komend

```
dmesg -H
```

Sources: RHCSA_Command_Library.pdf

1.12.66 dmesg -T

Czytelne timestamps

```
dmesg -T \
```

```
dmesg -T
```

Sources: LINUX COMMAND, RHCSA_Command_Library.pdf

1.12.67 dmesg -T # z timestampami

dmesg — ring buffer jądra

```
dmesg -T # z timestampami
```

Sources: linux_engineering_book.md, linux_engineering_book.pdf

1.12.68 dmesg -T --level=err,warn

kernel ring buffer, readable timestamps. **CAUTION** (verbose)

```
dmesg -T --level=err,warn
```

Sources: komendy

1.12.69 dmesg -c

Sekcja Komend

```
dmesg -c
```

Sources: RHCSA_Command_Library.pdf

1.12.70 dmesg -l err,warn # tylko błędy i ostrzeżenia

dmesg — ring buffer jądra

```
dmesg -l err,warn # tylko błędy i ostrzeżenia
```

Sources: linux_engineering_book.md, linux_engineering_book.pdf

1.12.71 dmesg -w

Na żywo

```
dmesg -w
```

Sources: LINUX COMMAND

1.12.72 dmesg -w # follow

dmesg — ring buffer jądra

```
dmesg -w # follow
```

Sources: linux_engineering_book.md, linux_engineering_book.pdf

1.12.73 dmesg -w ledzenie nowych komunikatów jdra cat /etc/audit/audit

Sekcja Komend

```
dmesg -w ledzenie nowych komunikatów jdra cat /etc/audit/audit
```

Sources: RHCSA_Command_Library.pdf

1.12.74 dmesg | grep -i "out of memory"

5.4 OOM Killer

```
dmesg | grep -i "out of memory"
```

Sources: linux_engineering_book.md

1.12.75 dmesg | tail

27.1 Prostý Modul

```
dmesg | tail
```

Sources: linux_engineering_book.md

1.12.76 dmesg → sprawdź

Dodatek B: Komendy Ratunkowe

```
dmesg → sprawdź
```

Sources: linux_engineering_book.md, linux_engineering_book.pdf

1.12.77 logrotate

Rotacja logów

```
logrotate -d /etc/logrotate.conf
```

Sources: LINUX COMMAND

1.12.78 logrotate -d /etc/logrotate.conf

debug mode (dry-run). **SAFE**

```
logrotate -d /etc/logrotate.conf
```

Sources: komendy

1.12.79 `logrotate -f /etc/logrotate.conf`

force rotation. **CAUTION**

`logrotate -f /etc/logrotate.conf`

Sources: komendy

1.12.80 `script`

record terminal session. **SAFE**

`script session.log`

`script`

Sources: LINUX COMMAND, komendy

1.12.81 `sealert`

Analiza alertów (SETroubleshoot)

`sealert -a /var/log/audit/audit.log`

Sources: LINUX COMMAND

1.12.82 `sealert -a /var/log/audit/audit.log`

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`sealert -a /var/log/audit/audit.log`

Sources: linux_engineering_book.md, linux_engineering_book.pdf

1.12.83 `sosreport`

Zbiera diagnostykę

`sosreport`

Sources: LINUX COMMAND

1.13 Security

1.13.1 `/etc/selinux/config`

Disabling SELinux in without reboot planning.

`/etc/selinux/config`

Sources: LINUX COMMAND, komendy

1.13.2 `/var/log/audit/audit.log`

Logi audytu

`tail -f /var/log/audit/audit.log`

Sources: LINUX COMMAND

1.13.3 `aide`

Intrusion detection

`aide --init`

Sources: LINUX COMMAND

1.13.4 aide --check

Sprawdza integralność

```
aide --check
```

Sources: LINUX COMMAND, komendy

1.13.5 aide --update

Aktualizuje bazę

```
aide --update
```

Sources: LINUX COMMAND

1.13.6 audit2allow

SELinux denials logged as ; use and for analysis.

```
audit2allow -a -M mymodule  
audit2allow
```

Sources: LINUX COMMAND, komendy

1.13.7 audit2allow -M mymodule

Running and blindly applying without review.

```
audit2allow -M mymodule
```

Sources: komendy

1.13.8 audit2allow -a

Generate allow from audit

```
audit2allow -a -M local
```

Sources: LINUX COMMAND

1.13.9 audit2allow -a # generuj politykę z odmów

9.3 SELinux

```
audit2allow -a # generuj politykę z odmów
```

Sources: linux_engineering_book.md, linux_engineering_book.pdf

1.13.10 audit2allow -i

From specific file

```
audit2allow -i /var/log/audit/audit.log
```

Sources: LINUX COMMAND

1.13.11 audit2why

SELinux denials logged as ; use and for analysis.

```
audit2why < /var/log/audit/audit.log  
audit2why
```

Sources: LINUX COMMAND, komendy

1.13.12 audit2why -w

Why denied

```
audit2why -w
```

Sources: LINUX COMMAND

1.13.13 authselect

Nowoczesna auth config

```
authselect select sssd
```

Sources: LINUX COMMAND

1.13.14 authselect current

Aktualny profil

```
authselect current
```

Sources: LINUX COMMAND

1.13.15 certmonger

Certificate management

```
certmonger
```

Sources: LINUX COMMAND

1.13.16 certtool

GnuTLS cert tool

```
certtool --generate-self-signed ...
```

Sources: LINUX COMMAND

1.13.17 chcon

Tymczasowa zmiana kontekstu

```
chcon -t httpd_sys_content_t file
```

Sources: LINUX COMMAND

1.13.18 chcon --reference

Reference file context

```
chcon --reference /etc/hosts file
```

Sources: LINUX COMMAND

1.13.19 chcon --reference=/e

Sekcja Komend

```
chcon --reference=/e
```

Sources: RHCSA_Command_Library.pdf

1.13.20 chcon -R

Rekurencyjnie

```
chcon -R -t httpd_sys_content_t /web
```

Sources: LINUX COMMAND

1.13.21 chcon -R -t

Sekcja Komend

```
chcon -R -t
```

Sources: RHCSA_Command_Library.pdf

1.13.22 chcon -r

SELinux role change

```
chcon -r object_r file
```

Sources: LINUX COMMAND

1.13.23 chcon -t

SELinux type change

```
chcon -t httpd_sys_content_t file
```

```
chcon -t
```

Sources: LINUX COMMAND, RHCSA_Command_Library.pdf

1.13.24 chcon -t httpd_sys_content_t file

temporary context change. **CAUTION**

```
chcon -t httpd_sys_content_t file
```

Sources: komendy

1.13.25 chcon -u

SELinux user change

```
chcon -u system_u file
```

Sources: LINUX COMMAND

1.13.26 checkmodule

Kompiluje modul

```
checkmodule -M -m -o mymod.mod mymod.te
```

Sources: LINUX COMMAND

1.13.27 checkmodule -M -m

Compile module

```
checkmodule -M -m -o mymod.mod mymod.te
```

Sources: LINUX COMMAND

1.13.28 `checkmodule -M -m -o mypolicy.mod mypolicy.te`

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`checkmodule -M -m -o mypolicy.mod mypolicy.te`

Sources: `linux_engineering_book.md`, `linux_engineering_book.pdf`

1.13.29 `checkpolicy`

Check policy

`checkpolicy -M -o policy.conf policy.te`

Sources: LINUX COMMAND

1.13.30 `chkrootkit`

Rootkit checker

`chkrootkit`

Sources: LINUX COMMAND

1.13.31 `fixfiles`

Naprawia konteksty masowo

`fixfiles onboot`

`fixfiles`

Sources: LINUX COMMAND

1.13.32 `fixfiles relabel`

Relabel

`fixfiles relabel`

Sources: LINUX COMMAND

1.13.33 `fixfiles restore`

Restore contexts

`fixfiles restore /var/www`

Sources: LINUX COMMAND

1.13.34 `genhomedircon`

Generuje konteksty home

`genhomedircon`

Sources: LINUX COMMAND

1.13.35 `getenforce`

current mode. **SAFE**

`getenforce`

Sources: LINUX COMMAND, `RHCSA_Command_Library.pdf`, komendy

1.13.36 `getenforce`

9.3 SELinux

`getenforce` #

Sources: `linux_engineering_book.md`, `linux_engineering_book.pdf`

1.13.37 `getsebool`

Wartość boolean

`getsebool httpd_enable_homedirs`

Sources: `LINUX COMMAND`

1.13.38 `getsebool httpd_enable_homedirs`

boolean value. **SAFE**

`getsebool httpd_enable_homedirs`

Sources: `komendy`

1.13.39 `gpg`

Komentarz Komentarz w skrypcie `function myfunc()` { Definicja funkcji

`gpg --gen-key`

`gpg`

Sources: `LINUX COMMAND`, `RHCSA_Command_Library.pdf`

1.13.40 `gpg --armor --export alice@example.com`

export public key. **SAFE**

`gpg --armor --export alice@example.com`

Sources: `komendy`

1.13.41 `gpg --decrypt`

Komentarz Komentarz w skrypcie `function myfunc()` { Definicja funkcji

`gpg --decrypt file.txt.gpg`

`gpg --decrypt`

Sources: `LINUX COMMAND`, `RHCSA_Command_Library.pdf`

1.13.42 `gpg --encrypt`

Szyfrowanie

`gpg --encrypt --recipient john file.txt`

Sources: `LINUX COMMAND`

1.13.43 `gpg --encrypt -r`

Komentarz Komentarz w skrypcie `function myfunc()` { Definicja funkcji

`gpg --encrypt -r`

Sources: `RHCSA_Command_Library.pdf`

1.13.44 `gpg --export`

Komentarz Komentarz w skrypcie function myfunc() { Definicja funkcji

```
gpg --export -a john > john.asc  
gpg --export
```

Sources: LINUX COMMAND, RHCSA_Command_Library.pdf

1.13.45 `gpg --full-generate-key`

generate GPG key. **SAFE**

```
gpg --full-generate-key
```

Sources: komendy

1.13.46 `gpg --gen-key`

Komentarz Komentarz w skrypcie function myfunc() { Definicja funkcji

```
gpg --gen-key
```

Sources: RHCSA_Command_Library.pdf

1.13.47 `gpg --import`

Import klucza

```
gpg --import key.asc
```

Sources: LINUX COMMAND

1.13.48 `gpg --import key.gpg`

Komentarz Komentarz w skrypcie function myfunc() { Definicja funkcji

```
gpg --import key.gpg
```

Sources: RHCSA_Command_Library.pdf

1.13.49 `gpg --list-keys`

Komentarz Komentarz w skrypcie function myfunc() { Definicja funkcji

```
gpg --list-keys
```

Sources: RHCSA_Command_Library.pdf

1.13.50 `gpg --sign`

Podpis

```
gpg --sign file.txt
```

Sources: LINUX COMMAND

1.13.51 `gpg --sign file`

Komentarz Komentarz w skrypcie function myfunc() { Definicja funkcji

```
gpg --sign file
```

Sources: RHCSA_Command_Library.pdf

1.13.52 **gpg --verify**

Komentarz Komentarz w skrypcie function myfunc() { Definicja funkcji

```
gpg --verify file.txt.sig  
gpg --verify
```

Sources: LINUX COMMAND, RHCSA_Command_Library.pdf

1.13.53 **gpg --verify file.sig file**

signature verification. **SAFE**

```
gpg --verify file.sig file
```

Sources: komendy

1.13.54 **ipa**

FreeIPA CLI

```
ipa user-add john
```

Sources: LINUX COMMAND

1.13.55 **kdestroy**

Niszczy tickety

```
kdestroy
```

Sources: LINUX COMMAND

1.13.56 **kinit**

Kerberos ticket

```
kinit john@EXAMPLE.COM
```

Sources: LINUX COMMAND

1.13.57 **klist**

Lista ticketów

```
klist
```

Sources: LINUX COMMAND

1.13.58 **load_policy**

Load policy into kernel

```
load_policy
```

Sources: LINUX COMMAND

1.13.59 **lynis**

Security audit

```
lynis audit system
```

Sources: LINUX COMMAND

1.13.60 openssl

Crypto toolkit

```
openssl genrsa -out key.pem 2048
```

Sources: LINUX COMMAND

1.13.61 openssl dgst

Digest

```
openssl dgst -sha256 file
```

Sources: LINUX COMMAND

1.13.62 openssl dgst -sha256

Komentarz Komentarz w skrypcie function myfunc() { Definicja funkcji

```
openssl dgst -sha256
```

Sources: RHCSA_Command_Library.pdf

1.13.63 openssl enc

Komentarz Komentarz w skrypcie function myfunc() { Definicja funkcji

```
openssl enc -aes-256-cbc -in file -out file.enc
```

```
openssl enc
```

Sources: LINUX COMMAND, RHCSA_Command_Library.pdf

1.13.64 openssl enc -d

Komentarz Komentarz w skrypcie function myfunc() { Definicja funkcji

```
openssl enc -d
```

Sources: RHCSA_Command_Library.pdf

1.13.65 openssl passwd

Hash hasła

```
openssl passwd -6
```

Sources: LINUX COMMAND

1.13.66 openssl passwd -6

Komentarz Komentarz w skrypcie function myfunc() { Definicja funkcji

```
openssl passwd -6
```

Sources: RHCSA_Command_Library.pdf

1.13.67 openssl pkcs12

PKCS#12 operations

```
openssl pkcs12 -export -out cert.p12 ...
```

Sources: LINUX COMMAND

1.13.68 openssl rand

Random bytes

```
openssl rand -base64 32
```

Sources: LINUX COMMAND

1.13.69 openssl rand -base64

Komentarz Komentarz w skrypcie function myfunc() { Definicja funkcji

```
openssl rand -base64
```

Sources: RHCSA_Command_Library.pdf

1.13.70 openssl rand -base64 32

Dodatek A: Przydatne Jednowierszowce

```
openssl rand -base64 32
```

Sources: linux_engineering_book.md, linux_engineering_book.pdf

1.13.71 openssl rand -hex 16 32 znaki hex losowe xmllint --xpath

Komentarz Komentarz w skrypcie function myfunc() { Definicja funkcji

```
openssl rand -hex 16 32 znaki hex losowe xmllint --xpath
```

Sources: RHCSA_Command_Library.pdf

1.13.72 openssl req

CSR generation

```
openssl req -new -key key.pem -out csr.pem
```

Sources: LINUX COMMAND

1.13.73 openssl req -new

Komentarz Komentarz w skrypcie function myfunc() { Definicja funkcji

```
openssl req -new
```

Sources: RHCSA_Command_Library.pdf

1.13.74 openssl s_client

Komentarz Komentarz w skrypcie function myfunc() { Definicja funkcji

```
openssl s_client -connect host:443
```

```
openssl s_client
```

Sources: LINUX COMMAND, RHCSA_Command_Library.pdf

1.13.75 openssl s_client -connect example.com:443 -servername example.com </dev/null

Sprawdź certyfikat SSL

```
openssl s_client -connect example.com:443 -servername example.com </dev/null
```

Sources: linux_engineering_book.pdf

1.13.76 `openssl s_client -connect example.com:443 -servername example.com </dev/null 2>/dev/null | openssl x509 -noout -dates`

Dodatek A: Przydatne Jednowierszowce

`openssl s_client -connect example.com:443 -servername example.com </dev/null 2>/dev/null | openssl x509 -noout -dates`

Sources: linux_engineering_book.md

1.13.77 `openssl s_client -connect host:443 -tls1_3`

test TLS. **SAFE**

`openssl s_client -connect host:443 -tls1_3`

Sources: komendy

1.13.78 `openssl s_server`

SSL server test

`openssl s_server -cert cert.pem -key key.pem`

Sources: LINUX COMMAND

1.13.79 `openssl speed`

Benchmark

`openssl speed rsa2048`

Sources: LINUX COMMAND

1.13.80 `openssl x509`

Cert operations

`openssl x509 -in cert.pem -text`

Sources: LINUX COMMAND

1.13.81 `openssl x509 -in`

Komentarz Komentarz w skrypcie function myfunc() { Definicja funkcji

`openssl x509 -in`

Sources: RHCSA_Command_Library.pdf

1.13.82 `openssl x509 -in cert.pem -text -noout`

inspect cert. **SAFE**

`openssl x509 -in cert.pem -text -noout`

Sources: komendy

1.13.83 `openssl x509 -req`

Komentarz Komentarz w skrypcie function myfunc() { Definicja funkcji

`openssl x509 -req`

Sources: RHCSA_Command_Library.pdf

1.13.84 openssl x509 -text

Komentarz Komentarz w skrypcie function myfunc() { Definicja funkcji

```
openssl x509 -text
```

Sources: RHCSA_Command_Library.pdf

1.13.85 oscap

OpenSCAP scanner

```
oscap xccdf eval --profile xccdf...
```

Sources: LINUX COMMAND

1.13.86 realm

Zarządzanie domeną

```
realm
```

Sources: LINUX COMMAND

1.13.87 realmd

Join do AD/FreeIPA

```
realm join ad.example.com
```

Sources: LINUX COMMAND

1.13.88 restorecon

Przywraca konteksty

```
restorecon -Rv /var/www
```

```
restorecon
```

Sources: LINUX COMMAND, RHCSA_Command_Library.pdf

1.13.89 restorecon -F

Force reset context

```
restorecon -F -R /var/www
```

Sources: LINUX COMMAND

1.13.90 restorecon -F /path/

Sekcja Komend

```
restorecon -F /path/
```

Sources: RHCSA_Command_Library.pdf

1.13.91 restorecon -R

Rekurencyjnie

```
restorecon -R /home
```

```
restorecon -R
```

Sources: LINUX COMMAND, RHCSA_Command_Library.pdf

1.13.92 restorecon -Rv

Sekcja Komend

`restorecon -Rv`

Sources: RHCSA_Command_Library.pdf

1.13.93 restorecon -Rv /web

restore contexts. **CAUTION**

`restorecon -Rv /web`

Sources: komendy

1.13.94 restorecon -Rv /web # przywróć konteksty

9.3 SELinux

`restorecon -Rv /web # przywróć konteksty`

Sources: linux_engineering_book.md, linux_engineering_book.pdf

1.13.95 restorecon -i

Ignore errors

`restorecon -i -R /var/www`

Sources: LINUX COMMAND

1.13.96 restorecon -v

Verbose

`restorecon -v /etc/hosts`

Sources: LINUX COMMAND

1.13.97 rkhunter

Rootkit hunter

`rkhunter --check`

Sources: LINUX COMMAND

1.13.98 seinfo

Informacje o polityce

`seinfo`

Sources: LINUX COMMAND

1.13.99 seinfo --class

Class info

`seinfo --class file`

Sources: LINUX COMMAND

1.13.100 seinfo --common

Common permissions

```
seinfo --common
```

Sources: LINUX COMMAND

1.13.101 seinfo -b

List booleans

```
seinfo -b
```

Sources: LINUX COMMAND

1.13.102 seinfo -c

List classes

```
seinfo -c
```

Sources: LINUX COMMAND

1.13.103 seinfo -r

List roles

```
seinfo -r
```

Sources: LINUX COMMAND

1.13.104 seinfo -t

List types

```
seinfo -t
```

Sources: LINUX COMMAND

1.13.105 seinfo -u

List users

```
seinfo -u
```

Sources: LINUX COMMAND

1.13.106 selinuxenabled

Check if enabled

```
selinuxenabled
```

Sources: LINUX COMMAND

1.13.107 semanage boolean

Lista boole'ów

```
semanage boolean -l
```

Sources: LINUX COMMAND

1.13.108 semanage boolean -C

Custom booleans

```
semanage boolean -C -l
```

Sources: LINUX COMMAND

1.13.109 semanage boolean -D

Delete custom boolean

```
semanage boolean -D
```

Sources: LINUX COMMAND

1.13.110 semanage boolean -l

List booleans

```
semanage boolean -l
```

Sources: LINUX COMMAND, RHCSA_Command_Library.pdf

1.13.111 semanage boolean -m

Modify boolean

```
semanage boolean -m -l httpd_enable_homedirs
```

Sources: LINUX COMMAND

1.13.112 semanage dontaudit

Toggle dontaudit

```
semanage dontaudit off
```

Sources: LINUX COMMAND

1.13.113 semanage export

Export config

```
semanage export -f backup.conf
```

Sources: LINUX COMMAND

1.13.114 semanage fcontext

Konteksty plików

```
semanage fcontext -a -t httpd_sys_content_t "/web(/.*)?"
```

Sources: LINUX COMMAND

1.13.115 semanage fcontext -C

Local customizations

```
semanage fcontext -C -l
```

Sources: LINUX COMMAND

1.13.116 semanage fcontext -a

Sekcja Komend

```
semanage fcontext -a -t httpd_sys_content_t "/web(/.*)?"  
semanage fcontext -a
```

Sources: LINUX COMMAND, RHCSA_Command_Library.pdf

1.13.117 semanage fcontext -a -t httpd_sys_content_t "/web(/.*)?"

add context rule. **CAUTION**

```
semanage fcontext -a -t httpd_sys_content_t "/web(/.*)?"
```

Sources: komendy, linux_engineering_book.md, linux_engineering_book.pdf

1.13.118 semanage fcontext -d

Delete fcontext

```
semanage fcontext -d "/web(/.*)?"  
semanage fcontext -d
```

Sources: LINUX COMMAND, RHCSA_Command_Library.pdf

1.13.119 semanage fcontext -l

file context rules. **SAFE**

```
semanage fcontext -l
```

Sources: LINUX COMMAND, RHCSA_Command_Library.pdf, komendy

1.13.120 semanage fcontext -m

Modify fcontext

```
semanage fcontext -m -t public_content_t "/web(/.*)?"  
semanage fcontext -m
```

Sources: LINUX COMMAND, RHCSA_Command_Library.pdf

1.13.121 semanage ibendport -a

Add ibendport

```
semanage ibendport -a -t ibendport_t mlx4_0 1
```

Sources: LINUX COMMAND

1.13.122 semanage ibpkey -a

Add ibpkey

```
semanage ibpkey -a -t ibpkey_t fe80:: 1
```

Sources: LINUX COMMAND

1.13.123 semanage import

Import config

```
semanage import -f backup.conf
```

Sources: LINUX COMMAND

1.13.124 semanage interface -a

Add interface

```
semanage interface -a -t netif_t eth1
```

Sources: LINUX COMMAND

1.13.125 semanage login

Mapowanie użytkowników

```
semanage login -l
```

Sources: LINUX COMMAND

1.13.126 semanage login -a

Add login mapping

```
semanage login -a -s user_u john
```

Sources: LINUX COMMAND

1.13.127 semanage login -d

Delete login mapping

```
semanage login -d john
```

Sources: LINUX COMMAND

1.13.128 semanage login -l

login mappings. **SAFE**

```
semanage login -l
```

Sources: LINUX COMMAND, komendy

1.13.129 semanage login -m

Modify login mapping

```
semanage login -m -s staff_u john
```

Sources: LINUX COMMAND

1.13.130 semanage module -a

Add module

```
semanage module -a mymodule
```

Sources: LINUX COMMAND

1.13.131 semanage module -d

Delete module

```
semanage module -d mymodule
```

Sources: LINUX COMMAND

1.13.132 semanage module -l

List modules

```
semanage module -l
```

Sources: LINUX COMMAND

1.13.133 semanage node -a

Add node

```
semanage node -a -t node_t -M 255.255.255.255 192.168.1.0
```

Sources: LINUX COMMAND

1.13.134 semanage permissive -a

Add permissive type

```
semanage permissive -a httpd_t
```

Sources: LINUX COMMAND

1.13.135 semanage permissive -d

Delete permissive type

```
semanage permissive -d httpd_t
```

Sources: LINUX COMMAND

1.13.136 semanage permissive -l

List permissive types

```
semanage permissive -l
```

Sources: LINUX COMMAND

1.13.137 semanage port

Mapowanie portów

```
semanage port -a -t http_port_t -p tcp 8080
```

Sources: LINUX COMMAND

1.13.138 semanage port -C

Custom ports

```
semanage port -C -l
```

Sources: LINUX COMMAND

1.13.139 semanage port -a

Add port

```
semanage port -a -t http_port_t -p tcp 8080
```

Sources: LINUX COMMAND

1.13.140 semanage port -a -t

grep Typy SELinux dla httpd

```
semanage port -a -t
```

Sources: RHCSA_Command_Library.pdf

1.13.141 semanage port -d

Delete port

```
semanage port -d -p tcp 8080
```

Sources: LINUX COMMAND

1.13.142 semanage port -d -t

Sekcja Komend

```
semanage port -d -t
```

Sources: RHCSA_Command_Library.pdf

1.13.143 semanage port -l

Porty dla usugi http seinfo -r Informacje o rolach SELinux

```
semanage port -l
```

Sources: LINUX COMMAND, RHCSA_Command_Library.pdf

1.13.144 semanage port -m

Modify port

```
semanage port -m -t http_port_t -p tcp 8080
```

Sources: LINUX COMMAND

1.13.145 semanage port -m -t

Sekcja Komend

```
semanage port -m -t
```

Sources: RHCSA_Command_Library.pdf

1.13.146 semanage translate

Translate context

```
semanage translate -T httpd_t
```

Sources: LINUX COMMAND

1.13.147 semanage user

Użytkownicy SELinux

```
semanage user -l
```

Sources: LINUX COMMAND

1.13.148 semanage user -a

Add SELinux user

```
semanage user -a -R "staff_r sysadm_r" -r s0-s0:c0.c1023 staff_u
```

Sources: LINUX COMMAND

1.13.149 semanage user -d

Delete SELinux user

```
semanage user -d staff_u
```

Sources: LINUX COMMAND

1.13.150 semanage user -l

List SELinux users

```
semanage user -l
```

Sources: LINUX COMMAND

1.13.151 semanage user -m

Modify SELinux user

```
semanage user -m -R "staff_r" staff_u
```

Sources: LINUX COMMAND

1.13.152 semodule

Zarządzanie modułami

```
semodule -i mymod.pp
```

Sources: LINUX COMMAND

1.13.153 semodule -B

Build and load

```
semodule -B
```

Sources: LINUX COMMAND

1.13.154 semodule -R

Reload policy

```
semodule -R
```

Sources: LINUX COMMAND

1.13.155 semodule -X

Priority

```
semodule -X 400 -i mymod.pp
```

Sources: LINUX COMMAND

1.13.156 `semodule -d`

Disable module

```
semodule -d module  
semodule -d mymod
```

Sources: LINUX COMMAND

1.13.157 `semodule -e`

Enable module

```
semodule -e module  
semodule -e mymod
```

Sources: LINUX COMMAND

1.13.158 `semodule -i mypolicy.pp`

24.1 SELinux — Tworzenie Polityk

```
semodule -i mypolicy.pp
```

Sources: linux_engineering_book.md, linux_engineering_book.pdf

1.13.159 `semodule -l`

List SELinux modules

```
semodule -l
```

Sources: LINUX COMMAND

1.13.160 `semodule -r`

Remove module

```
semodule -r mymodule  
semodule -r mymod
```

Sources: LINUX COMMAND

1.13.161 `semodule -u`

Upgrade module

```
semodule -u mymod.pp
```

Sources: LINUX COMMAND

1.13.162 `semodule_package`

Package module

```
semodule_package -o mymod.pp -m mymod.mod
```

Sources: LINUX COMMAND

1.13.163 `semodule_package -o`

Package module

```
semodule_package -o mymod.pp -m mymod.mod
```

Sources: LINUX COMMAND

1.13.164 semodule_package -o mypolicy.pp -m mypolicy.mod

24.1 SELinux — Tworzenie Polityk

semodule_package -o mypolicy.pp -m mypolicy.mod

Sources: linux_engineering_book.md, linux_engineering_book.pdf

1.13.165 sepolicy

SELinux policy analysis

sepolicy generate --init /usr/sbin/httpd

Sources: LINUX COMMAND

1.13.166 sepolicy booleans

Boolean analysis

sepolicy booleans -a

Sources: LINUX COMMAND

1.13.167 sepolicy communicate

Communication analysis

sepolicy communicate -s httpd_t -t mysqld_t

Sources: LINUX COMMAND

1.13.168 sepolicy gui

GUI analysis

sepolicy gui

Sources: LINUX COMMAND

1.13.169 sepolicy manpage

Generate man pages

sepolicy manpage -a

Sources: LINUX COMMAND

1.13.170 sepolicy network

Network analysis

sepolicy network -d httpd_t

Sources: LINUX COMMAND

1.13.171 sepolicy transition

Transition analysis

sepolicy transition -s init_t -t httpd_exec_t

Sources: LINUX COMMAND

1.13.172 sestatus

Wyszukiwanie reguł

```
sestatus -A -s httpd_t -t httpd_sys_content_t
```

Sources: LINUX COMMAND

1.13.173 sestatus --allow

Allow rules only

```
sestatus --allow
```

Sources: LINUX COMMAND

1.13.174 sestatus --auditallow

Auditallow rules

```
sestatus --auditallow
```

Sources: LINUX COMMAND

1.13.175 sestatus --dontaudit

Dontaudit rules

```
sestatus --dontaudit
```

Sources: LINUX COMMAND

1.13.176 sestatus --neverallow

Neverallow rules

```
sestatus --neverallow
```

Sources: LINUX COMMAND

1.13.177 sestatus -A

Allow rules

```
sestatus -A -s httpd_t -t httpd_sys_content_t
```

Sources: LINUX COMMAND

1.13.178 sestatus -T

Type transition

```
sestatus -T -s initrc_t -t httpd_exec_t
```

Sources: LINUX COMMAND

1.13.179 sestatus -c

Class

```
sestatus -A -c file
```

Sources: LINUX COMMAND

1.13.180 `sesearch -p`

Permission

```
sesearch -A -p read
```

Sources: LINUX COMMAND

1.13.181 `sesearch -s`

Source type

```
sesearch -A -s httpd_t
```

Sources: LINUX COMMAND

1.13.182 `sesearch -t`

Target type

```
sesearch -A -t httpd_sys_content_t
```

Sources: LINUX COMMAND

1.13.183 `sestatus`

detailed status. **SAFE**

```
sestatus
```

Sources: LINUX COMMAND, RHCSA_Command_Library.pdf, komendy

1.13.184 `sestatus -v`

Verbose status

```
sestatus -v
```

Sources: LINUX COMMAND, RHCSA_Command_Library.pdf

1.13.185 `setenforce`

Zmienia tryb (0/1)

```
setenforce 1
```

Sources: LINUX COMMAND

1.13.186 `setenforce 0`

permissive mode. **CAUTION** (temporarily disables enforcement)

```
setenforce 0
```

Sources: RHCSA_Command_Library.pdf, komendy

1.13.187 `setenforce 0` and forgetting to re-enable.

Dangerous Mistakes

```
setenforce 0` and forgetting to re-enable.
```

Sources: komendy

1.13.188 **setenforce 1**

enforcing mode. **SAFE**

```
setenforce 1
```

Sources: RHCSA_Command_Library.pdf, komendy

1.13.189 **setenforce 1 # włącz enforcing (tymczasowo)**

9.3 SELinux

```
setenforce 1 # włącz enforcing (tymczasowo)
```

Sources: linux_engineering_book.md, linux_engineering_book.pdf

1.13.190 **setsebool**

Set boolean (temp)

```
setsebool httpd_enable_homedirs on
```

```
setsebool httpd_enable_homedirs 1
```

Sources: LINUX COMMAND

1.13.191 **setsebool -P**

Persist boolean

```
setsebool -P httpd_enable_homedirs on
```

```
setsebool -P httpd_enable_homedirs 1
```

Sources: LINUX COMMAND

1.13.192 **setsebool -P httpd_can_network_connect on # boolean**

9.3 SELinux

```
setsebool -P httpd_can_network_connect on # boolean
```

Sources: linux_engineering_book.md, linux_engineering_book.pdf

1.13.193 **setsebool -P httpd_enable_homedirs on**

persistent boolean. **CAUTION**

```
setsebool -P httpd_enable_homedirs on
```

Sources: komendy

1.13.194 **sssctl**

Zarządzanie SSSD

```
sssctl logs-remove
```

Sources: LINUX COMMAND

1.13.195 **sssd**

Komentarz Komentarz w skrypcie function myfunc() { Definicja funkcji

```
systemctl
```

```
sssd
```

Sources: LINUX COMMAND, RHCSA_Command_Library.pdf

1.14 Packages

1.14.1 apt

remove (keep config). **CAUTION**

apt

Sources: komendy, linux_engineering_book.md, linux_engineering_book.pdf

1.14.2 apt autoremove

remove unused deps. **CAUTION**

apt autoremove

Sources: komendy

1.14.3 apt autoremove # usuń niepotrzebne zależności

12.1 APT (Debian/Ubuntu)

apt autoremove # *usuń niepotrzebne zależności*

Sources: linux_engineering_book.md, linux_engineering_book.pdf

1.14.4 apt clean / dnf clean all

Dodatek B: Komendy Ratunkowe

apt clean / dnf clean all

Sources: linux_engineering_book.md, linux_engineering_book.pdf

1.14.5 apt full-upgrade

upgrade with removals. **CAUTION**

apt full-upgrade

Sources: komendy

1.14.6 apt full-upgrade # aktualizuj z usuwaniem

12.1 APT (Debian/Ubuntu)

apt full-upgrade # *aktualizuj z usuwaniem*

Sources: linux_engineering_book.md, linux_engineering_book.pdf

1.14.7 apt install nfs-kernel-server

18.3 NFS

apt install nfs-kernel-server

Sources: linux_engineering_book.md, linux_engineering_book.pdf

1.14.8 apt install open-iscsi

Initiator (klient)

apt install open-iscsi

Sources: linux_engineering_book.md, linux_engineering_book.pdf

1.14.9 apt install pacemaker corosync pcs

22.1 Pacemaker/Corosync

```
apt install pacemaker corosync pcs
```

Sources: linux_engineering_book.md, linux_engineering_book.pdf

1.14.10 apt install package # instaluj

12.1 APT (Debian/Ubuntu)

```
apt install package # instaluj
```

Sources: linux_engineering_book.md, linux_engineering_book.pdf

1.14.11 apt install pkg

install. **CAUTION**

```
apt install pkg
```

Sources: komendy

1.14.12 apt install pkg=1.2.3

Pin all package versions: .

```
apt install pkg=1.2.3
```

Sources: komendy

1.14.13 apt install qemu-kvm libvirt-daemon-system virtinst

13.4 KVM/QEMU (virtualizacja)

```
apt install qemu-kvm libvirt-daemon-system virtinst
```

Sources: linux_engineering_book.md, linux_engineering_book.pdf

1.14.14 apt install targetcli-fb

Target (serwer)

```
apt install targetcli-fb
```

Sources: linux_engineering_book.md, linux_engineering_book.pdf

1.14.15 apt purge package # usuń z configiem

12.1 APT (Debian/Ubuntu)

```
apt purge package # usuń z configiem
```

Sources: linux_engineering_book.md, linux_engineering_book.pdf

1.14.16 apt purge pkg

remove with config. **CAUTION**

```
apt purge pkg
```

Sources: komendy

1.14.17 apt search keyword # szukaj

12.1 APT (Debian/Ubuntu)

apt search keyword # *szukaj*

Sources: linux_engineering_book.md, linux_engineering_book.pdf

1.14.18 apt search term

search. **SAFE**

apt search term

Sources: komendy

1.14.19 apt update

refresh index. **SAFE**

apt update

Sources: komendy

1.14.20 apt update # aktualizuj listę pakietów

12.1 APT (Debian/Ubuntu)

apt update # *aktualizuj listę pakietów*

Sources: linux_engineering_book.md, linux_engineering_book.pdf

1.14.21 apt upgrade

upgrade installed. **CAUTION**

apt upgrade

Sources: komendy

1.14.22 apt upgrade # aktualizuj pakiety

12.1 APT (Debian/Ubuntu)

apt upgrade # *aktualizuj pakiety*

Sources: linux_engineering_book.md, linux_engineering_book.pdf

1.14.23 apt-get

Use in scripts (stable CLI).

apt-get

Sources: komendy

1.14.24 createrepo

Tworzy repozytorium YUM

createrepo /var/www/html/repo

Sources: LINUX COMMAND

1.14.25 dnf

remove. **CAUTION**

dnf

Sources: LINUX COMMAND, RHCSA_Command_Library.pdf, komendy, linux_engineering_book.md, linux_engineering_book.pdf

1.14.26 dnf autoremove

Sekcja Komend

dnf autoremove

Sources: RHCSA_Command_Library.pdf

1.14.27 dnf check-update # sprawdź aktualizacje

12.2 DNF/YUM (RHEL/Fedora)

dnf check-update # *sprawdź aktualizacje*

Sources: linux_engineering_book.md, linux_engineering_book.pdf

1.14.28 dnf clean all

Sekcja Komend

dnf clean all

Sources: RHCSA_Command_Library.pdf

1.14.29 dnf clean metadata

Sekcja Komend

dnf clean metadata

Sources: RHCSA_Command_Library.pdf

1.14.30 dnf clean packages

Sekcja Komend

dnf clean packages

Sources: RHCSA_Command_Library.pdf

1.14.31 dnf config-manager

Zarządzanie repo

dnf config-manager --add-repo url

dnf config-manager

Sources: LINUX COMMAND, RHCSA_Command_Library.pdf

1.14.32 dnf downgrade

Sekcja Komend

dnf downgrade

Sources: RHCSA_Command_Library.pdf

1.14.33 dnf erase package

Sekcja Komend

`dnf erase package`

Sources: RHCSA_Command_Library.pdf

1.14.34 dnf groupinfo 'Group

Sekcja Komend

`dnf groupinfo 'Group`

Sources: RHCSA_Command_Library.pdf

1.14.35 dnf groupinstall

Instalacja grupy pakietów

`dnf groupinstall "Development`

`dnf groupinstall`

Sources: LINUX_COMMAND, RHCSA_Command_Library.pdf

1.14.36 dnf grouplist

Sekcja Komend

`dnf grouplist`

Sources: RHCSA_Command_Library.pdf

1.14.37 dnf groupremove

Sekcja Komend

`dnf groupremove`

Sources: RHCSA_Command_Library.pdf

1.14.38 dnf history

Historia transakcji

`dnf history`

Sources: LINUX_COMMAND, RHCSA_Command_Library.pdf

1.14.39 dnf history # historia transakcji

12.2 DNF/YUM (RHEL/Fedora)

`dnf history # historia transakcji`

Sources: linux_engineering_book.md, linux_engineering_book.pdf

1.14.40 dnf history info 5

Sekcja Komend

`dnf history info 5`

Sources: RHCSA_Command_Library.pdf

1.14.41 dnf history redo 5

Sekcja Komend

dnf history redo 5

Sources: RHCSA_Command_Library.pdf

1.14.42 dnf history rollback

Sekcja Komend

dnf history rollback

Sources: RHCSA_Command_Library.pdf

1.14.43 dnf history undo

Cofa transakcję

dnf history undo 5

Sources: LINUX COMMAND

1.14.44 dnf history undo 5

Sekcja Komend

dnf history undo 5

Sources: RHCSA_Command_Library.pdf

1.14.45 dnf history undo 5 # cofnij transakcję

12.2 DNF/YUM (RHEL/Fedora)

dnf history undo 5 # *cofnij transakcję*

Sources: linux_engineering_book.md, linux_engineering_book.pdf

1.14.46 dnf info

Informacje o pakiecie

dnf info httpd

Sources: LINUX COMMAND

1.14.47 dnf info package

Sekcja Komend

dnf info package

Sources: RHCSA_Command_Library.pdf

1.14.48 dnf info package # informacje

12.2 DNF/YUM (RHEL/Fedora)

dnf info package # *informacje*

Sources: linux_engineering_book.md, linux_engineering_book.pdf

1.14.49 dnf info pkg

details. **SAFE**

dnf info pkg

Sources: komendy

1.14.50 dnf install

Instaluje pakiet

dnf install httpd

Sources: LINUX COMMAND

1.14.51 dnf install -y

Sekcja Komend

dnf install -y

Sources: RHCSA_Command_Library.pdf

1.14.52 dnf install kernel

Sekcja Komend

dnf install kernel

Sources: RHCSA_Command_Library.pdf

1.14.53 dnf install package

Sekcja Komend

dnf install package

Sources: RHCSA_Command_Library.pdf

1.14.54 dnf install package # instaluj

12.2 DNF/YUM (RHEL/Fedora)

dnf install package # *instaluj*

Sources: linux_engineering_book.md, linux_engineering_book.pdf

1.14.55 dnf install pkg

install. **CAUTION**

dnf install pkg

Sources: komendy

1.14.56 dnf makecache

Sekcja Komend

dnf makecache

Sources: RHCSA_Command_Library.pdf

1.14.57 dnf module

Moduły aplikacji (RHEL 8+)

`dnf module`

Sources: LINUX COMMAND

1.14.58 dnf provides

Sekcja Komend

`dnf provides`

Sources: RHCSA_Command_Library.pdf

1.14.59 dnf provides /path/file

owning package. **SAFE**

`dnf provides /path/file`

Sources: komendy

1.14.60 dnf provides /path/to/file # który pakiet zawiera plik

12.2 DNF/YUM (RHEL/Fedora)

`dnf provides /path/to/file # który pakiet zawiera plik`

Sources: linux_engineering_book.md, linux_engineering_book.pdf

1.14.61 dnf reinstall

Sekcja Komend

`dnf reinstall`

Sources: RHCSA_Command_Library.pdf

1.14.62 dnf repoinfo repo-id

Wyodrbnij zawarto pakietu RPM

`dnf repoinfo repo-id`

Sources: RHCSA_Command_Library.pdf

1.14.63 dnf repolist

Lista repozytoriów

`dnf repolist all`

`dnf repolist`

Sources: LINUX COMMAND, RHCSA_Command_Library.pdf

1.14.64 dnf repolist all

Sekcja Komend

`dnf repolist all`

Sources: RHCSA_Command_Library.pdf

1.14.65 `dnf repoquery -l pkg`

files in package. **SAFE**

`dnf repoquery -l pkg`

Sources: komendy

1.14.66 `dnf search`

Wyszukuje pakiet

`dnf search web server`

Sources: LINUX COMMAND

1.14.67 `dnf search keyword`

Sekcja Komend

`dnf search keyword`

Sources: RHCSA_Command_Library.pdf

1.14.68 `dnf search keyword # szukaj`

12.2 DNF/YUM (RHEL/Fedora)

`dnf search keyword # szukaj`

Sources: linux_engineering_book.md, linux_engineering_book.pdf

1.14.69 `dnf search term`

search. **SAFE**

`dnf search term`

Sources: komendy

1.14.70 `dnf update`

refresh and upgrade. **CAUTION**

`dnf update`

Sources: LINUX COMMAND, RHCSA_Command_Library.pdf, komendy

1.14.71 `dnf update package`

Sekcja Komend

`dnf update package`

Sources: RHCSA_Command_Library.pdf

1.14.72 `dnf upgrade`

Ulepsz system

`dnf upgrade`

Sources: LINUX COMMAND, RHCSA_Command_Library.pdf

1.14.73 dnf upgrade # aktualizuj

12.2 DNF/YUM (RHEL/Fedora)

`dnf upgrade # aktualizuj`

Sources: linux_engineering_book.md, linux_engineering_book.pdf

1.14.74 dnf upgrade-minimal

Sekcja Komend

`dnf upgrade-minimal`

Sources: RHCSA_Command_Library.pdf

1.14.75 dnf whatprovides

Sekcja Komend

`dnf whatprovides`

Sources: RHCSA_Command_Library.pdf

1.14.76 yum

Legacy wrapper (alias dnf)

`yum install httpd`

Sources: LINUX COMMAND

1.14.77 dnf-automatic

Automatyczne aktualizacje

`systemctl enable dnf-automatic.timer`

Sources: LINUX COMMAND

1.14.78 dpkg -L package # pliki pakietu

12.1 APT (Debian/Ubuntu)

`dpkg -L package # pliki pakietu`

Sources: linux_engineering_book.md, linux_engineering_book.pdf

1.14.79 dpkg -L pkg

files in package. **SAFE**

`dpkg -L pkg`

Sources: komendy

1.14.80 dpkg -S /path/file

owning package. **SAFE**

`dpkg -S /path/file`

Sources: komendy

1.14.81 `dpkg -S /path/to/file # który pakiet zawiera plik`

12.1 APT (Debian/Ubuntu)

`dpkg -S /path/to/file # który pakiet zawiera plik`

Sources: linux_engineering_book.md, linux_engineering_book.pdf

1.14.82 `dpkg -i`

on incompatible architecture.

`dpkg -i`

Sources: komendy

1.14.83 `dpkg -i file.deb`

install local deb. **CAUTION**

`dpkg -i file.deb`

Sources: komendy

1.14.84 `dpkg -i package.deb # instaluj .deb`

12.1 APT (Debian/Ubuntu)

`dpkg -i package.deb # instaluj .deb`

Sources: linux_engineering_book.md, linux_engineering_book.pdf

1.14.85 `dpkg -i on incompatible architecture.`

Dangerous Mistakes

`dpkg -i` on incompatible architecture.`

Sources: komendy

1.14.86 `dpkg -l`

all installed packages. **SAFE**

`dpkg -l`

Sources: komendy

1.14.87 `dpkg -l #`

12.1 APT (Debian/Ubuntu)

`dpkg -l #`

Sources: linux_engineering_book.md, linux_engineering_book.pdf

1.14.88 `dpkg -l > baseline.txt`

Baseline: for comparison.

`dpkg -l > baseline.txt`

Sources: komendy

1.14.89 dpkg -r package # usuń

12.1 APT (Debian/Ubuntu)

`dpkg -r package # usuń`

Sources: linux_engineering_book.md, linux_engineering_book.pdf

1.14.90 flatpak

installed flatpaks. **SAFE**

`flatpak`

Sources: LINUX COMMAND, komendy, linux_engineering_book.md, linux_engineering_book.pdf

1.14.91 flatpak install

Instaluje Flatpak (RHEL 10)

`flatpak install flathub org.gimp.GIMP`

Sources: LINUX COMMAND

1.14.92 flatpak install flathub org.mozilla.firefox

12.4 Flatpak i Snap

`flatpak install flathub org.mozilla.firefox`

Sources: linux_engineering_book.md, linux_engineering_book.pdf

1.14.93 flatpak remote-add

Dodaje repozytorium

`flatpak remote-add flathub https://...`

Sources: LINUX COMMAND

1.14.94 flatpak run

Uruchamia Flatpak

`flatpak run org.gimp.GIMP`

Sources: LINUX COMMAND

1.14.95 flatpak run org.mozilla.firefox

12.4 Flatpak i Snap

`flatpak run org.mozilla.firefox`

Sources: linux_engineering_book.md, linux_engineering_book.pdf

1.14.96 flatpak update

12.4 Flatpak i Snap

`flatpak update`

Sources: linux_engineering_book.md, linux_engineering_book.pdf

1.14.97 reposync

Synchronizuje repozytorium

```
reposync -p /var/repo
```

Sources: LINUX COMMAND

1.14.98 rpm --import

Importuje klucz GPG

```
rpm --import /etc/pki/rpm-gpg/RPM-GPG-KEY-redhat-release
```

Sources: LINUX COMMAND

1.14.99 rpm -Uvh

on running kernel without reboot plan.

```
rpm -Uvh
```

Sources: komendy

1.14.100 rpm -Uvh on running kernel without reboot plan.

Dangerous Mistakes

```
rpm -Uvh` on running kernel without reboot plan.
```

Sources: komendy

1.14.101 rpm -V pkg

verify integrity. **SAFE**

```
rpm -V pkg
```

Sources: komendy

1.14.102 rpm -Va

verify all installed packages. **SAFE**

```
rpm -Va
```

Sources: komendy

1.14.103 rpm -e

Usuwa pakiet (bez zależności)

```
rpm -e httpd
```

Sources: LINUX COMMAND

1.14.104 rpm -ivh

Instaluje lokalny RPM

```
rpm -ivh package.rpm
```

Sources: LINUX COMMAND

1.14.105 rpm -ivh file.rpm

install local rpm. **CAUTION**

`rpm -ivh file.rpm`

Sources: komendy

1.14.106 rpm -ivh package.rpm # instaluj

12.2 DNF/YUM (RHEL/Fedora)

`rpm -ivh package.rpm # instaluj`

Sources: linux_engineering_book.md, linux_engineering_book.pdf

1.14.107 rpm -qa

all packages. **SAFE**

`rpm -qa \`

`rpm -qa`

Sources: LINUX COMMAND, komendy

1.14.108 rpm -qa #

12.2 DNF/YUM (RHEL/Fedora)

`rpm -qa #`

Sources: linux_engineering_book.md, linux_engineering_book.pdf

1.14.109 rpm -qa kernel

Sekcja Komend

`rpm -qa kernel`

Sources: RHCSA_Command_Library.pdf

1.14.110 rpm -qf

Który pakiet zawiera plik

`rpm -qf /bin/bash`

Sources: LINUX COMMAND

1.14.111 rpm -qf /path/file

file owner. **SAFE**

`rpm -qf /path/file`

Sources: komendy

1.14.112 rpm -qf /path/to/file # który pakiet

12.2 DNF/YUM (RHEL/Fedora)

`rpm -qf /path/to/file # który pakiet`

Sources: linux_engineering_book.md, linux_engineering_book.pdf

1.14.113 rpm -qi

Info o pakiecie

```
rpm -qi httpd
```

Sources: LINUX COMMAND

1.14.114 rpm -ql

Pliki w pakiecie

```
rpm -ql httpd
```

Sources: LINUX COMMAND

1.14.115 rpm -ql package # pliki pakietu

12.2 DNF/YUM (RHEL/Fedora)

```
rpm -ql package # pliki pakietu
```

Sources: linux_engineering_book.md, linux_engineering_book.pdf

1.14.116 rpm -ql pkg

package files. **SAFE**

```
rpm -ql pkg
```

Sources: komendy

1.14.117 rpm2cpio

Ekstrakcja plików z RPM

```
rpm2cpio package.rpm \
```

Sources: LINUX COMMAND

1.14.118 subscription-manager

Lista subskrypcji

```
subscription-manager
```

Sources: LINUX COMMAND

1.14.119 subscription-manager attach

Przyłącza subskrypcję

```
subscription-manager attach --auto
```

Sources: LINUX COMMAND

1.14.120 subscription-manager enable

Włącza repo

```
subscription-manager repos --enable rhel-9-for-x86_64-baseos-rpms
```

Sources: LINUX COMMAND

1.14.121 subscription-manager register

Rejestracja RHN

```
subscription-manager register --username user
```

Sources: LINUX COMMAND

1.14.122 subscription-manager repos

Lista repozytoriów

```
subscription-manager repos --list
```

Sources: LINUX COMMAND

1.15 Automation

1.15.1 ansible

Automation tool

ansible

Sources: LINUX COMMAND

1.15.2 ansible --check

flags: , , .

```
ansible --check
```

Sources: komendy

1.15.3 ansible webserver -a "uptime"

20.1 Ansible

```
ansible webserver -a "uptime"
```

Sources: linux_engineering_book.md, linux_engineering_book.pdf

1.15.4 ansible webserver -m ping

20.1 Ansible

```
ansible webserver -m ping
```

Sources: linux_engineering_book.md, linux_engineering_book.pdf

1.15.5 ansible-doc

Dokumentacja modułów

```
ansible-doc yum
```

Sources: LINUX COMMAND

1.15.6 ansible-galaxy

Role i kolekcje

```
ansible-galaxy install role
```

Sources: LINUX COMMAND

1.15.7 ansible-inventory

Zarządzanie inwentarzem

`ansible-inventory --list`

Sources: LINUX COMMAND

1.15.8 ansible-vault

Szyfrowanie

`ansible-vault encrypt secrets.yml`

Sources: LINUX COMMAND

1.15.9 ansible-vault encrypt secrets.yml

20.1 Ansible

`ansible-vault encrypt secrets.yml`

Sources: linux_engineering_book.md, linux_engineering_book.pdf

1.15.10 at

Jednorazowe zadanie

`echo "reboot" \`

Sources: LINUX COMMAND

1.15.11 atq

list jobs. **SAFE**

`atq`

Sources: LINUX COMMAND, komendy

1.15.12 atrm

Usuwa zadanie at

`atrm 1`

Sources: LINUX COMMAND

1.15.13 atrm 1

remove job. **CAUTION**

`atrm 1`

Sources: komendy

1.15.14 autoconf

Generuje configure

`autoconf`

Sources: LINUX COMMAND

1.15.15 automake

Generuje Makefile.in

```
automake --add-missing
```

Sources: LINUX COMMAND

1.15.16 batch

Wykonuje gdy load spadnie

```
echo "cmd" \
```

Sources: LINUX COMMAND

1.15.17 buildah

Build tool dla kontenerów

```
buildah from scratch
```

Sources: LINUX COMMAND

1.15.18 buildah bud

Build Dockerfile

```
buildah bud -t myimg .
```

Sources: LINUX COMMAND

1.15.19 chef-client

Chef client

```
chef-client
```

Sources: LINUX COMMAND

1.15.20 cmake

Build system

```
cmake
```

Sources: LINUX COMMAND

1.15.21 crictl

CRI CLI (kubernetes)

```
crictl ps
```

Sources: LINUX COMMAND

1.15.22 crontab -e

edit user crontab. **CAUTION**

```
crontab -e
```

Sources: LINUX COMMAND, RHCSA_Command_Library.pdf, komendy

1.15.23 `crontab -e # edytuj`

11.2 Cron i Harmonogramowanie

`crontab -e # edytuj`

Sources: linux_engineering_book.md, linux_engineering_book.pdf

1.15.24 `crontab -e -u user`

Sekcja Komend

`crontab -e -u user`

Sources: RHCSA_Command_Library.pdf

1.15.25 `crontab -l`

list user crontab. **SAFE**

`crontab -l`

Sources: LINUX_COMMAND, RHCSA_Command_Library.pdf, komendy

1.15.26 `crontab -l # wyświetl`

11.2 Cron i Harmonogramowanie

`crontab -l # wyświetl`

Sources: linux_engineering_book.md, linux_engineering_book.pdf

1.15.27 `crontab -l -u user`

Sekcja Komend

`crontab -l -u user`

Sources: RHCSA_Command_Library.pdf

1.15.28 `crontab -r`

remove user crontab. **DANGEROUS**

`crontab -r`

Sources: LINUX_COMMAND, RHCSA_Command_Library.pdf, komendy

1.15.29 `crontab -r # usuń`

11.2 Cron i Harmonogramowanie

`crontab -r # usuń`

Sources: linux_engineering_book.md, linux_engineering_book.pdf

1.15.30 `crontab -r -u user`

Sekcja Komend

`crontab -r -u user`

Sources: RHCSA_Command_Library.pdf

1.15.31 crontab -r instead of crontab -e

Dangerous Mistakes

`crontab -r` instead of `crontab -e`.

Sources: komendy

1.15.32 expect

Automatyzacja interaktywna

`expect script.exp`

Sources: LINUX COMMAND

1.15.33 git

working tree status. **SAFE**

`git clone url`

`git`

Sources: LINUX COMMAND, komendy

1.15.34 git add

Dodaje do staging

`git add file.txt`

Sources: LINUX COMMAND

1.15.35 git add -p

interactive patch staging. **SAFE**

`git add -p`

Sources: komendy

1.15.36 git bisect

21.1 Git — Zaawansowane

`git bisect`

Sources: LINUX COMMAND, linux_engineering_book.md, linux_engineering_book.pdf

1.15.37 git bisect bad # aktualny jest zły

21.1 Git — Zaawansowane

`git bisect bad # aktualny jest zły`

Sources: linux_engineering_book.md, linux_engineering_book.pdf

1.15.38 git bisect good v1.0 # v1.0 był dobry

21.1 Git — Zaawansowane

`git bisect good v1.0 # v1.0 był dobry`

Sources: linux_engineering_book.md, linux_engineering_book.pdf

1.15.39 git blame file.txt

line annotations. **SAFE**

```
git blame file.txt
```

Sources: komendy

1.15.40 git branch

Zarządzanie branchami

```
git branch -a
```

Sources: LINUX COMMAND

1.15.41 git branch -a

list branches. **SAFE**

```
git branch -a
```

Sources: komendy

1.15.42 git branch feature/new-api

21.1 Git — Zaawansowane

```
git branch feature/new-api
```

Sources: linux_engineering_book.md, linux_engineering_book.pdf

1.15.43 git checkout

Przełącza branch

```
git checkout -b feature
```

Sources: LINUX COMMAND

1.15.44 git checkout -b feature

create and switch. **SAFE**

```
git checkout -b feature
```

Sources: komendy

1.15.45 git checkout -b feature/new-api # utwórz i przełącz

21.1 Git — Zaawansowane

```
git checkout -b feature/new-api # utwórz i przełącz
```

Sources: linux_engineering_book.md, linux_engineering_book.pdf

1.15.46 git cherry-pick

Przenosi commit

```
git cherry-pick abc123
```

Sources: LINUX COMMAND

1.15.47 `git cherry-pick abc123`

apply specific commit. **CAUTION**

```
git cherry-pick abc123
```

Sources: komendy

1.15.48 `git cherry-pick abc123 # wybierz konkretny commit`

21.1 Git — Zaawansowane

```
git cherry-pick abc123 # wybierz konkretny commit
```

Sources: linux_engineering_book.md, linux_engineering_book.pdf

1.15.49 `git clone`

Klonuje repo

```
git clone https://github.com/...
```

Sources: LINUX COMMAND

1.15.50 `git clone https://repo.git`

clone. **SAFE**

```
git clone https://repo.git
```

Sources: komendy

1.15.51 `git commit`

Commituje zmiany

```
git commit -m "fix"
```

Sources: LINUX COMMAND

1.15.52 `git commit --amend`

amend last commit. **CAUTION** (rewrites history)

```
git commit --amend
```

Sources: komendy

1.15.53 `git commit -m "msg"`

commit. **SAFE**

```
git commit -m "msg"
```

Sources: komendy

1.15.54 `git config`

Konfiguracja

```
git config --global user.name "John"
```

Sources: LINUX COMMAND

1.15.55 git diff

compare with previous. **SAFE**

`git diff`

Sources: LINUX COMMAND, komendy

1.15.56 git fetch

Pobiera bez merge

`git fetch origin`

Sources: LINUX COMMAND

1.15.57 git fetch origin

download refs. **SAFE**

`git fetch origin`

Sources: komendy

1.15.58 git init

initialize repo. **SAFE**

`git init`

Sources: LINUX COMMAND, komendy

1.15.59 git log

Historia commitów

`git log --oneline`

Sources: LINUX COMMAND

1.15.60 git log --oneline --graph --all

compact history. **SAFE**

`git log --oneline --graph --all`

Sources: komendy

1.15.61 git merge

Łączy branch

`git merge feature`

Sources: LINUX COMMAND

1.15.62 git merge feature

merge branch. **CAUTION**

`git merge feature`

Sources: komendy

1.15.63 `git merge feature/new-api # merge`

21.1 Git — Zaawansowane

`git merge feature/new-api # merge`

Sources: linux_engineering_book.md, linux_engineering_book.pdf

1.15.64 `git pull`

Pobiera zmiany

`git pull origin main`

Sources: LINUX COMMAND

1.15.65 `git pull origin main`

fetch and merge. **CAUTION**

`git pull origin main`

Sources: komendy

1.15.66 `git push`

Wysła na zdalne

`git push origin main`

Sources: LINUX COMMAND

1.15.67 `git push --force`

force push. **DANGEROUS** (overwrites remote history)

`git push --force`

Sources: komendy

1.15.68 `git push --force-with-lease`

safer force push. **CAUTION**

`git push --force-with-lease`

Sources: komendy

1.15.69 `git push --force on main branch`

Dangerous Mistakes

`git push --force` on main branch.`

Sources: komendy

1.15.70 `git push origin main`

push commits. **CAUTION**

`git push origin main`

Sources: komendy

1.15.71 `git push origin v1.0`

push tag. **SAFE**

`git push origin v1.0`

Sources: komendy

1.15.72 `git rebase`

Rebase branch

`git rebase main`

Sources: LINUX COMMAND

1.15.73 `git rebase main`

rebase branch. **CAUTION** (rewrites history)

`git rebase main`

Sources: komendy

1.15.74 `git rebase main # rebase`

21.1 Git — Zaawansowane

`git rebase main # rebase`

Sources: linux_engineering_book.md, linux_engineering_book.pdf

1.15.75 `git remote`

Zdalne repozytoria

`git remote -v`

Sources: LINUX COMMAND

1.15.76 `git remote -v`

remotes. **SAFE**

`git remote -v`

Sources: komendy

1.15.77 `git reset`

Resetuje stan

`git reset --hard`

Sources: LINUX COMMAND

1.15.78 `git reset --hard`

undo commit discard changes. **DANGEROUS**

`git reset --hard`

Sources: komendy

1.15.79 `git reset --hard` losing uncommitted work:

Dangerous Mistakes

```
git reset --hard` losing uncommitted work.
```

Sources: komendy

1.15.80 `git reset --soft`

undo commit keep changes. **CAUTION**

```
git reset --soft
```

Sources: komendy

1.15.81 `git revert`

Revert commit

```
git revert
```

Sources: LINUX COMMAND

1.15.82 `git stash`

21.1 Git — Zaawansowane

```
git stash push
```

```
git stash
```

Sources: LINUX COMMAND, linux_engineering_book.md, linux_engineering_book.pdf

1.15.83 `git stash # schowaj zmiany`

21.1 Git — Zaawansowane

```
git stash # schowaj zmiany
```

Sources: linux_engineering_book.md, linux_engineering_book.pdf

1.15.84 `git stash pop`

Przywraca ze schowka

```
git stash pop
```

Sources: LINUX COMMAND

1.15.85 `git stash pop # przywróć`

21.1 Git — Zaawansowane

```
git stash pop # przywróć
```

Sources: linux_engineering_book.md, linux_engineering_book.pdf

1.15.86 `git submodule add https://github.com/lib/lib.git`

21.1 Git — Zaawansowane

```
git submodule add https://github.com/lib/lib.git
```

Sources: linux_engineering_book.md, linux_engineering_book.pdf

1.15.87 `git submodule update --init --recursive`

21.1 Git — Zaawansowane

`git submodule update --init --recursive`

Sources: linux_engineering_book.md, linux_engineering_book.pdf

1.15.88 `git tag`

Tagi

`git tag -a v1.0 -m "Release"`

Sources: LINUX COMMAND

1.15.89 `git tag -a v1.0 -m "Release"`

annotated tag. **SAFE**

`git tag -a v1.0 -m "Release"`

Sources: komendy

1.15.90 `helm`

Package manager K8s

`helm install myapp ./chart`

Sources: LINUX COMMAND

1.15.91 `kubectl`

create namespace. **SAFE**

`kubectl get nodes`

`kubectl`

Sources: LINUX COMMAND, komendy, linux_engineering_book.md, linux_engineering_book.pdf

1.15.92 `kubectl apply -f`

Use instead of for idempotency.

`kubectl apply -f`

Sources: komendy

1.15.93 `kubectl apply -f deployment.yaml`

13.3 Kubernetes (podstawy)

`kubectl apply -f deployment.yaml`

Sources: linux_engineering_book.md, linux_engineering_book.pdf

1.15.94 `kubectl autoscale deployment nginx --min=2 --max=10 --cpu-percent=80`

13.3 Kubernetes (podstawy)

`kubectl autoscale deployment nginx --min=2 --max=10 --cpu-percent=80`

Sources: linux_engineering_book.md, linux_engineering_book.pdf

1.15.95 `kubectl cluster-info`

cluster endpoint info. **SAFE**

```
kubectl cluster-info
```

Sources: komendy

1.15.96 `kubectl config set-context --current --namespace=prod`

set default ns. **SAFE**

```
kubectl config set-context --current --namespace=prod
```

Sources: komendy

1.15.97 `kubectl delete -f .`

deleting entire namespaces.

```
kubectl delete -f .
```

Sources: komendy

1.15.98 `kubectl delete -f .` deleting entire namespaces:

Dangerous Mistakes

```
kubectl delete -f .` deleting entire namespaces.
```

Sources: komendy

1.15.99 `kubectl delete pod pod-name`

delete pod. **CAUTION**

```
kubectl delete pod pod-name
```

Sources: komendy

1.15.100 `kubectl describe node node1`

node details. **SAFE**

```
kubectl describe node node1
```

Sources: komendy

1.15.101 `kubectl describe pod pod-name`

pod events and status. **SAFE**

```
kubectl describe pod pod-name
```

Sources: komendy, linux_engineering_book.md, linux_engineering_book.pdf

1.15.102 `kubectl drain`

Use before node maintenance.

```
kubectl drain
```

Sources: komendy

1.15.103 kubectl exec

into production pods without audit trail.

```
kubectl exec
```

Sources: komendy

1.15.104 kubectl exec -it pod-name -- /bin/sh

shell into pod. **CAUTION**

```
kubectl exec -it pod-name -- /bin/sh
```

Sources: komendy

1.15.105 kubectl exec -it pod-name -- bash

13.3 Kubernetes (podstawy)

```
kubectl exec -it pod-name -- bash
```

Sources: linux_engineering_book.md, linux_engineering_book.pdf

1.15.106 kubectl expose deployment nginx --port=80 --type=LoadBalancer

13.3 Kubernetes (podstawy)

```
kubectl expose deployment nginx --port=80 --type=LoadBalancer
```

Sources: linux_engineering_book.md, linux_engineering_book.pdf

1.15.107 kubectl get configmap

list configmaps. **SAFE**

```
kubectl get configmap
```

Sources: komendy

1.15.108 kubectl get deployments

deployment list. **SAFE**

```
kubectl get deployments
```

Sources: komendy

1.15.109 kubectl get deployments # deploymenty

kubectl — zarządzanie klastrem

```
kubectl get deployments # deploymenty
```

Sources: linux_engineering_book.md, linux_engineering_book.pdf

1.15.110 kubectl get endpoints

service endpoints. **SAFE**

```
kubectl get endpoints
```

Sources: komendy

1.15.111 `kubectl get ingress`

list ingresses. **SAFE**

`kubectl get ingress`

Sources: komendy

1.15.112 `kubectl get namespaces`

list namespaces. **SAFE**

`kubectl get namespaces`

Sources: komendy

1.15.113 `kubectl get nodes` # węzły klastra

`kubectl` — zarządzanie klastrem

`kubectl get nodes` # *węzły klastra*

Sources: linux_engineering_book.md, linux_engineering_book.pdf

1.15.114 `kubectl get nodes -o wide`

node list with details. **SAFE**

`kubectl get nodes -o wide`

Sources: komendy

1.15.115 `kubectl get pods -A` # wszystkie pody

`kubectl` — zarządzanie klastrem

`kubectl get pods -A` # *wszystkie pody*

Sources: linux_engineering_book.md, linux_engineering_book.pdf

1.15.116 `kubectl get pods -o wide`

pod list. **SAFE**

`kubectl get pods -o wide`

Sources: komendy

1.15.117 `kubectl get secret`

list secrets. **SAFE**

`kubectl get secret`

Sources: komendy

1.15.118 `kubectl get services` # usługi

`kubectl` — zarządzanie klastrem

`kubectl get services` # *usługi*

Sources: linux_engineering_book.md, linux_engineering_book.pdf

1.15.119 `kubectl get svc`

list services. **SAFE**

`kubectl get svc`

Sources: komendy

1.15.120 `kubectl logs --previous pod-name`

previous container logs. **SAFE**

`kubectl logs --previous pod-name`

Sources: komendy

1.15.121 `kubectl logs -f pod-name`

follow logs. **SAFE**

`kubectl logs -f pod-name`

Sources: komendy

1.15.122 `kubectl logs pod-name`

pod logs. **SAFE**

`kubectl logs pod-name`

Sources: komendy

1.15.123 `kubectl logs pod-name -f`

13.3 Kubernetes (podstawy)

`kubectl logs pod-name -f`

Sources: linux_engineering_book.md, linux_engineering_book.pdf

1.15.124 `kubectl port-forward svc/app 8080:80`

local port forward. **SAFE**

`kubectl port-forward svc/app 8080:80`

Sources: komendy

1.15.125 `kubectl rollout`

watch rollout. **SAFE**

`kubectl rollout`

Sources: komendy

1.15.126 `kubectl rollout history deployment/app`

revision history. **SAFE**

`kubectl rollout history deployment/app`

Sources: komendy

1.15.127 `kubectl rollout undo deployment/app`

rollback. **CAUTION**

`kubectl rollout undo deployment/app`

Sources: komendy

1.15.128 `kubectl scale deployment app --replicas=3`

scale. **CAUTION**

`kubectl scale deployment app --replicas=3`

Sources: komendy

1.15.129 `kubectl scale deployment nginx --replicas=5`

13.3 Kubernetes (podstawy)

`kubectl scale deployment nginx --replicas=5`

Sources: linux_engineering_book.md, linux_engineering_book.pdf

1.15.130 `kubectl set image deployment/app app=nginx:1.21`

update image. **CAUTION**

`kubectl set image deployment/app app=nginx:1.21`

Sources: komendy

1.15.131 `kubectl top pods # zużycie zasobów`

13.3 Kubernetes (podstawy)

`kubectl top pods # zużycie zasobów`

Sources: linux_engineering_book.md, linux_engineering_book.pdf

1.15.132 `kubectl version`

client/server versions. **SAFE**

`kubectl version`

Sources: komendy

1.15.133 `make`

27.1 Prosty Modul

`make -j4`

`make`

Sources: LINUX COMMAND, linux_engineering_book.md, linux_engineering_book.pdf

1.15.134 `make -C /lib/modules/$(shell uname -r)/build`

27.1 Prosty Modul

`make -C /lib/modules/$(shell uname -r)/build`

Sources: linux_engineering_book.md, linux_engineering_book.pdf

1.15.135 `make -j$(nproc) # jądro (vmlinux/bzImage)`

16.2 Kompilacja i Instalacja

`make -j$(nproc) # jądro (vmlinux/bzImage)`

Sources: linux_engineering_book.md, linux_engineering_book.pdf

1.15.136 `make install`

Instaluje z Makefile

`make install`

Sources: LINUX COMMAND

1.15.137 `make install # /boot/vmlinuz-6.8.0 + initramfs`

16.2 Kompilacja i Instalacja

`make install # /boot/vmlinuz-6.8.0 + initramfs`

Sources: linux_engineering_book.md, linux_engineering_book.pdf

1.15.138 `make localmodconfig # config z aktualnie załadowanych modułów`

16.1 Pobieranie i Konfiguracja

`make localmodconfig # config z aktualnie załadowanych modułów`

Sources: linux_engineering_book.md, linux_engineering_book.pdf

1.15.139 `make menuconfig # interaktywne menu (ncurses)`

16.1 Pobieranie i Konfiguracja

`make menuconfig # interaktywne menu (ncurses)`

Sources: linux_engineering_book.md, linux_engineering_book.pdf

1.15.140 `make modules -j$(nproc) # moduły`

16.2 Kompilacja i Instalacja

`make modules -j$(nproc) # moduły`

Sources: linux_engineering_book.md, linux_engineering_book.pdf

1.15.141 `make modules_install # /lib/modules/6.8.0/`

16.2 Kompilacja i Instalacja

`make modules_install # /lib/modules/6.8.0/`

Sources: linux_engineering_book.md, linux_engineering_book.pdf

1.15.142 `make oldconfig # aktualizacja starego .config`

16.1 Pobieranie i Konfiguracja

`make oldconfig # aktualizacja starego .config`

Sources: linux_engineering_book.md, linux_engineering_book.pdf

1.15.143 make olddefconfig

Kopiowanie aktualnej konfiguracji

```
make olddefconfig
```

Sources: linux_engineering_book.md, linux_engineering_book.pdf

1.15.144 make xconfig

16.1 Pobieranie i Konfiguracja

```
make xconfig #
```

Sources: linux_engineering_book.md, linux_engineering_book.pdf

1.15.145 meson

Build system

```
meson setup builddir
```

Sources: LINUX COMMAND

1.15.146 ninja

Build tool

```
ninja -C builddir
```

Sources: LINUX COMMAND

1.15.147 oc

OpenShift CLI

```
oc get pods
```

Sources: LINUX COMMAND

1.15.148 patch

Aplikuje patch

```
patch -p1 < fix.patch  
patch
```

Sources: LINUX COMMAND, komendy

1.15.149 patch -p1 < patch-file.patch

16.3 Patching Jądra

```
patch -p1 < patch-file.patch
```

Sources: linux_engineering_book.md, linux_engineering_book.pdf

1.15.150 podman

Uruchomienie kontenera z pełnymi uprawnieniami roota hosta i dostępem do surowego systemu

```
podman run -it rhel bash  
podman
```

Sources: LINUX COMMAND, RHCSA_Command_Library.pdf, encyklopedia_inzynierii_linux.pdf

1.15.151 podman --version

Sekcja Komend

```
podman --version
```

Sources: RHCSA_Command_Library.pdf

1.15.152 podman build

Build image

```
podman build -t myapp .
```

Sources: LINUX COMMAND

1.15.153 podman container

Sekcja Komend

```
podman container
```

Sources: RHCSA_Command_Library.pdf

1.15.154 podman exec

Wykonuje w kontenerze

```
podman exec -it ctr_id bash
```

Sources: LINUX COMMAND

1.15.155 podman exec -it

Sekcja Komend

```
podman exec -it
```

Sources: RHCSA_Command_Library.pdf

1.15.156 podman generate

Generuje YAML/Kube

```
podman generate kube ctr_id
```

Sources: LINUX COMMAND

1.15.157 podman generate systemd --new --name web >

Podman — drop-in replacement dla Docker (bez daemona)

```
podman generate systemd --new --name web >
```

Sources: linux_engineering_book.pdf

1.15.158 podman generate systemd --new --name web > ~/.config/systemd/user/web.service

13.2 Podman (rootless containers)

```
podman generate systemd --new --name web > ~/.config/systemd/user/web.service
```

Sources: linux_engineering_book.md

1.15.159 podman image history

Sekcja Komend

podman image history

Sources: RHCSA_Command_Library.pdf

1.15.160 podman image inspect

Sekcja Komend

podman image inspect

Sources: RHCSA_Command_Library.pdf

1.15.161 podman image prune

Sekcja Komend

podman image prune

Sources: RHCSA_Command_Library.pdf

1.15.162 podman images

Lista obrazów

podman images

Sources: LINUX COMMAND, RHCSA_Command_Library.pdf

1.15.163 podman images -a

Sekcja Komend

podman images -a

Sources: RHCSA_Command_Library.pdf

1.15.164 podman info

Sekcja Komend

podman info

Sources: RHCSA_Command_Library.pdf

1.15.165 podman inspect prod_web_app

grep -i ipaddress

podman inspect prod_web_app

Sources: encyklopedia_inzynierii_linux.html, encyklopedia_inzynierii_linux.pdf

1.15.166 podman kill

Sekcja Komend

podman kill

Sources: RHCSA_Command_Library.pdf

1.15.167 podman kill -s

Sekcja Komend

podman kill -s

Sources: RHCSA_Command_Library.pdf

1.15.168 podman load -i

Sekcja Komend

podman load -i

Sources: RHCSA_Command_Library.pdf

1.15.169 podman login

Logowanie do registry

podman login registry.redhat.io

Sources: LINUX COMMAND

1.15.170 podman network

Zarządzanie sieciami

podman network

Sources: LINUX COMMAND

1.15.171 podman play

Uruchamia z YAML

podman play kube pod.yaml

Sources: LINUX COMMAND

1.15.172 podman pod

13.2 Podman (rootless containers)

podman pod

Sources: LINUX COMMAND, linux_engineering_book.md, linux_engineering_book.pdf

1.15.173 podman ps

Lista kontenerów

podman ps -a

podman ps

Sources: LINUX COMMAND, RHCSA_Command_Library.pdf

1.15.174 podman ps --all

OCI w trybie rootless lub rootful. * Przykład produkcyjny: #

podman ps --all

Sources: encyklopedia_inzynierii_linux.html, encyklopedia_inzynierii_linux.pdf

1.15.175 podman ps --format

Sekcja Komend

```
podman ps --format
```

Sources: RHCSA_Command_Library.pdf

1.15.176 podman ps -a

Sekcja Komend

```
podman ps -a
```

Sources: RHCSA_Command_Library.pdf

1.15.177 podman ps -q

Sekcja Komend

```
podman ps -q
```

Sources: RHCSA_Command_Library.pdf

1.15.178 podman ps -qa

Sekcja Komend

```
podman ps -qa
```

Sources: RHCSA_Command_Library.pdf

1.15.179 podman pull

Pobiera obraz

```
podman pull registry.redhat.io/rhel9
```

```
podman pull
```

Sources: LINUX COMMAND, RHCSA_Command_Library.pdf

1.15.180 podman pull registry

Sekcja Komend

```
podman pull registry
```

Sources: RHCSA_Command_Library.pdf

1.15.181 podman push

Sekcja Komend

```
podman push myimg registry.io/myimg
```

```
podman push
```

Sources: LINUX COMMAND, RHCSA_Command_Library.pdf

1.15.182 podman rm

Usuwa kontener

```
podman rm ctr_id
```

Sources: LINUX COMMAND

1.15.183 podman rm -a

Sekcja Komend

```
podman rm -a
```

Sources: RHCSA_Command_Library.pdf

1.15.184 podman rm -f

Sekcja Komend

```
podman rm -f
```

Sources: RHCSA_Command_Library.pdf

1.15.185 podman rm container

Sekcja Komend

```
podman rm container
```

Sources: RHCSA_Command_Library.pdf

1.15.186 podman rmi

Usuwa obraz

```
podman rmi img_id
```

Sources: LINUX COMMAND

1.15.187 podman rmi -a

Sekcja Komend

```
podman rmi -a
```

Sources: RHCSA_Command_Library.pdf

1.15.188 podman rmi -f image

Sekcja Komend

```
podman rmi -f image
```

Sources: RHCSA_Command_Library.pdf

1.15.189 podman rmi image

Sekcja Komend

```
podman rmi image
```

Sources: RHCSA_Command_Library.pdf

1.15.190 podman run

Sekcja Komend

```
podman run
```

Sources: RHCSA_Command_Library.pdf

1.15.191 podman run --cap-add

Sekcja Komend

```
podman run --cap-add
```

Sources: RHCSA_Command_Library.pdf

1.15.192 podman run --cpus

Sekcja Komend

```
podman run --cpus
```

Sources: RHCSA_Command_Library.pdf

1.15.193 podman run --memory

Sekcja Komend

```
podman run --memory
```

Sources: RHCSA_Command_Library.pdf

1.15.194 podman run --name

Sekcja Komend

```
podman run --name
```

Sources: RHCSA_Command_Library.pdf

1.15.195 podman run --network

Sekcja Komend

```
podman run --network
```

Sources: RHCSA_Command_Library.pdf

1.15.196 podman run --privileged --net=host --pid=host -v /:/host ubuntu:latest chroot /host

Uruchomienie kontenera z pełnymi uprawnieniami roota hosta i dostępem do surowego systemu plików

```
podman run --privileged --net=host --pid=host -v /:/host ubuntu:latest chroot /host
```

Sources: encyklopedia_inzynierii_linux.html, encyklopedia_inzynierii_linux.pdf

1.15.197 podman run --restart

Sekcja Komend

```
podman run --restart
```

Sources: RHCSA_Command_Library.pdf

1.15.198 podman run --rm

Sekcja Komend

```
podman run --rm
```

Sources: RHCSA_Command_Library.pdf

1.15.199 `podman run -d --name secure_sandbox --memory="1g" --memory-swap="1g" --cpus="1.5" --`

Inicjalizacja kontenera z twardym limitem pamięci RAM, procesora oraz dedykowanym kontekstem

`podman run -d --name secure_sandbox --memory="1g" --memory-swap="1g" --cpus="1.5" --`

Sources: [encyklopedia_inzynierii_linux.pdf](#)

1.15.200 `podman run -d --name secure_sandbox --memory="1g" --memory-swap="1g" --cpus="1.5" --pids-limit=100
--security-opt label=type:container_kvm_t docker.io/library/nginx:alpine`

Inicjalizacja kontenera z twardym limitem pamięci RAM, procesora oraz dedykowanym kontekstem zabezpieczeń

`podman run -d --name secure_sandbox --memory="1g" --memory-swap="1g" --cpus="1.5" --pids-limit=100 --security-opt`

Sources: [encyklopedia_inzynierii_linux.html](#)

1.15.201 `podman run -d --name web -p 8080:80 nginx`

Podman — drop-in replacement dla Docker (bez daemona)

`podman run -d --name web -p 8080:80 nginx`

Sources: [linux_engineering_book.md](#), [linux_engineering_book.pdf](#)

1.15.202 `podman run -d --pod mypod nginx`

13.2 Podman (rootless containers)

`podman run -d --pod mypod nginx`

Sources: [linux_engineering_book.md](#), [linux_engineering_book.pdf](#)

1.15.203 `podman run -d --pod mypod redis`

13.2 Podman (rootless containers)

`podman run -d --pod mypod redis`

Sources: [linux_engineering_book.md](#), [linux_engineering_book.pdf](#)

1.15.204 `podman run -d image`

Sekcja Komend

`podman run -d image`

Sources: [RHCSA_Command_Library.pdf](#)

1.15.205 `podman run -e`

Sekcja Komend

`podman run -e`

Sources: [RHCSA_Command_Library.pdf](#)

1.15.206 `podman run -it image`

Sekcja Komend

`podman run -it image`

Sources: [RHCSA_Command_Library.pdf](#)

1.15.207 podman run -p

Sekcja Komend

podman run -p

Sources: RHCSA_Command_Library.pdf

1.15.208 podman run -u user

Sekcja Komend

podman run -u user

Sources: RHCSA_Command_Library.pdf

1.15.209 podman run -v

Sekcja Komend

podman run -v

Sources: RHCSA_Command_Library.pdf

1.15.210 podman run image

Sekcja Komend

podman run image

Sources: RHCSA_Command_Library.pdf

1.15.211 podman save image -o

Sekcja Komend

podman save image -o

Sources: RHCSA_Command_Library.pdf

1.15.212 podman search

Sekcja Komend

podman search

Sources: RHCSA_Command_Library.pdf

1.15.213 podman search nginx

Sekcja Komend

podman search nginx

Sources: RHCSA_Command_Library.pdf

1.15.214 podman tag src:tag

Sekcja Komend

podman tag src:tag

Sources: RHCSA_Command_Library.pdf

1.15.215 podman volume

Zarządzanie wolumenami

podman volume

Sources: LINUX COMMAND

1.15.216 puppet

Configuration management

puppet apply manifest.pp

Sources: LINUX COMMAND

1.15.217 salt

SaltStack

salt '*' state.apply

Sources: LINUX COMMAND

1.15.218 skopeo

Inspekcja obrazów zdalnych

skopeo inspect docker://registry.io/img

Sources: LINUX COMMAND

1.15.219 vagrant

VM management

vagrant up

Sources: LINUX COMMAND

1.16 RHCSA Exam Tasks

1.16.1 chmod

Zmienia uprawnienia

chmod 755 script.sh

chmod

Sources: LINUX COMMAND, RHCSA_Command_Library.pdf

1.16.2 chmod +t /tmp

sticky bit. **SAFE**

chmod +t /tmp

Sources: komendy

1.16.3 chmod +t dir

Sekcja Komend

chmod +t dir

Sources: RHCSA_Command_Library.pdf

1.16.4 `chmod +t directory #`

Uprawnienia plików

```
chmod +t directory #
```

Sources: linux_engineering_book.md, linux_engineering_book.pdf

1.16.5 `chmod +x`

Nadaje prawo wykonania

```
chmod +x script.sh
```

Sources: LINUX COMMAND

1.16.6 `chmod +x script.sh`

Komentarz Komentarz w skrypcie function myfunc() { Definicja funkcji

```
chmod +x script.sh
```

Sources: RHCSA_Command_Library.pdf

1.16.7 `chmod -R`

Rekurencyjnie

```
chmod -R 755 /var/www
```

```
chmod -R
```

Sources: LINUX COMMAND, komendy

1.16.8 `chmod -R 750 directory # rekursywnie`

Uprawnienia plików

```
chmod -R 750 directory # rekursywnie
```

Sources: linux_engineering_book.md, linux_engineering_book.pdf

1.16.9 `chmod -R 755 dir/`

Sekcja Komend

```
chmod -R 755 dir/
```

Sources: RHCSA_Command_Library.pdf

1.16.10 `chmod 1755 dir`

Sekcja Komend

```
chmod 1755 dir
```

Sources: RHCSA_Command_Library.pdf

1.16.11 `chmod 2755 dir`

Sekcja Komend

```
chmod 2755 dir
```

Sources: RHCSA_Command_Library.pdf

1.16.12 `chmod 4755 file`

Sekcja Komend

`chmod 4755 file`

Sources: RHCSA_Command_Library.pdf

1.16.13 `chmod 600 /swapfile`

Tworzenie swap

`chmod 600 /swapfile`

Sources: linux_engineering_book.md, linux_engineering_book.pdf

1.16.14 `chmod 600 file`

Sekcja Komend

`chmod 600 file`

Sources: RHCSA_Command_Library.pdf

1.16.15 `chmod 644 file`

Sekcja Komend

`chmod 644 file`

Sources: RHCSA_Command_Library.pdf

1.16.16 `chmod 700 /home/*`

enforce private home dirs. **CAUTION**

`chmod 700 /home/*`

Sources: komendy

1.16.17 `chmod 750 dir`

`rwxr-x—`. **CAUTION**

`chmod 750 dir`

Sources: komendy

1.16.18 `chmod 755 file`

Sekcja Komend

`chmod 755 file`

Sources: RHCSA_Command_Library.pdf

1.16.19 `chmod 755 file # rwxr-xr-x`

Uprawnienia plików

`chmod 755 file # rwxr-xr-x`

Sources: linux_engineering_book.md, linux_engineering_book.pdf

1.16.20 `chmod 777`

anywhere in production.

`chmod 777`

Sources: komendy

1.16.21 `chmod 777 /etc`

or .

`chmod 777 /etc`

Sources: komendy

1.16.22 `chmod 777 /etc` or `chown -R user:user /`

Dangerous Mistakes

`chmod 777 /etc`` or ``chown -R user:user /``.

Sources: komendy

1.16.23 `chmod 777 file`

Sekcja Komend

`chmod 777 file`

Sources: RHCSA_Command_Library.pdf

1.16.24 `chmod a+r file`

Sekcja Komend

`chmod a+r file`

Sources: RHCSA_Command_Library.pdf

1.16.25 `chmod g+s`

Setgid bit

`chmod g+s /shared`

Sources: LINUX COMMAND

1.16.26 `chmod g+s /shared`

setgid on directory. **CAUTION**

`chmod g+s /shared`

Sources: komendy

1.16.27 `chmod g+s dir`

Sekcja Komend

`chmod g+s dir`

Sources: RHCSA_Command_Library.pdf

1.16.28 `chmod g+s directory #`

Uprawnienia plików

`chmod g+s directory #`

Sources: linux_engineering_book.md, linux_engineering_book.pdf

1.16.29 `chmod g-w`

Odbiera write dla grupy

`chmod g-w file.txt`

Sources: LINUX COMMAND

1.16.30 `chmod g-w file`

Sekcja Komend

`chmod g-w file`

Sources: RHCSA_Command_Library.pdf

1.16.31 `chmod g-w file # usuń write dla grupy`

Uprawnienia plików

`chmod g-w file # usuń write dla grupy`

Sources: linux_engineering_book.md, linux_engineering_book.pdf

1.16.32 `chmod o+t`

Sticky bit

`chmod o+t /tmp`

Sources: LINUX COMMAND

1.16.33 `chmod o-r file`

Sekcja Komend

`chmod o-r file`

Sources: RHCSA_Command_Library.pdf

1.16.34 `chmod o= file # usuń wszystko dla others`

Uprawnienia plików

`chmod o= file # usuń wszystko dla others`

Sources: linux_engineering_book.md, linux_engineering_book.pdf

1.16.35 `chmod o=r`

Ustawia read dla innych

`chmod o=r file.txt`

Sources: LINUX COMMAND

1.16.36 `chmod u+s`

Setuid bit

```
chmod u+s /usr/bin/passwd
```

Sources: LINUX COMMAND

1.16.37 `chmod u+s /bin/ping`

setuid. **DANGEROUS**

```
chmod u+s /bin/ping
```

Sources: komendy

1.16.38 `chmod u+s file`

Kopiuj ACL między katalogami

```
chmod u+s file
```

Sources: RHCSA_Command_Library.pdf

1.16.39 `chmod u+s file #`

Uprawnienia plików

```
chmod u+s file #
```

Sources: linux_engineering_book.md, linux_engineering_book.pdf

1.16.40 `chmod u+x`

Dodaje execute dla właściciela

```
chmod u+x script.sh
```

Sources: LINUX COMMAND

1.16.41 `chmod u+x file`

add owner execute. **SAFE**

```
chmod u+x file
```

Sources: RHCSA_Command_Library.pdf, komendy

1.16.42 `chmod u+x file # dodaj execute dla właściciela`

Uprawnienia plików

```
chmod u+x file # dodaj execute dla właściciela
```

Sources: linux_engineering_book.md, linux_engineering_book.pdf

1.16.43 `chmod u=rwx,g=rx,o=r`

Sekcja Komend

```
chmod u=rwx,g=rx,o=r
```

Sources: RHCSA_Command_Library.pdf

1.16.44 chown

Zmienia właściciela

```
chown john file.txt
```

Sources: LINUX COMMAND

1.16.45 chown -R

Rekurencyjnie

```
chown -R apache:apache /var/www
```

Sources: LINUX COMMAND

1.16.46 chown -R user:group

Sekcja Komend

```
chown -R user:group
```

Sources: RHCSA_Command_Library.pdf

1.16.47 chown -R user:user /

or .

```
chown -R user:user /
```

Sources: komendy

1.16.48 chown :group file

Sekcja Komend

```
chown :group file
```

Sources: RHCSA_Command_Library.pdf

1.16.49 chown user file

Sekcja Komend

```
chown user file
```

Sources: RHCSA_Command_Library.pdf

1.16.50 chown user:group

Właściciel i grupa

```
chown john:developers file.txt
```

```
chown user:group
```

Sources: LINUX COMMAND, RHCSA_Command_Library.pdf

1.16.51 chown user:group file

change owner and group. **CAUTION**

```
chown user:group file
```

Sources: komendy

1.16.52 crontab -e

edit user crontab. **CAUTION**

```
crontab -e
```

Sources: LINUX COMMAND, RHCSA_Command_Library.pdf, komendy

1.16.53 crontab -e # edytuj

11.2 Cron i Harmonogramowanie

```
crontab -e # edytuj
```

Sources: linux_engineering_book.md, linux_engineering_book.pdf

1.16.54 crontab -e -u user

Sekcja Komend

```
crontab -e -u user
```

Sources: RHCSA_Command_Library.pdf

1.16.55 crontab -l

list user crontab. **SAFE**

```
crontab -l
```

Sources: LINUX COMMAND, RHCSA_Command_Library.pdf, komendy

1.16.56 crontab -l # wyświetl

11.2 Cron i Harmonogramowanie

```
crontab -l # wyświetl
```

Sources: linux_engineering_book.md, linux_engineering_book.pdf

1.16.57 crontab -l -u user

Sekcja Komend

```
crontab -l -u user
```

Sources: RHCSA_Command_Library.pdf

1.16.58 crontab -r

remove user crontab. **DANGEROUS**

```
crontab -r
```

Sources: LINUX COMMAND, RHCSA_Command_Library.pdf, komendy

1.16.59 crontab -r # usuń

11.2 Cron i Harmonogramowanie

```
crontab -r # usuń
```

Sources: linux_engineering_book.md, linux_engineering_book.pdf

1.16.60 crontab -r -u user

Sekcja Komend

```
crontab -r -u user
```

Sources: RHCSA_Command_Library.pdf

1.16.61 crontab -r instead of crontab -e.

Dangerous Mistakes

```
crontab -r` instead of `crontab -e`.
```

Sources: komendy

1.16.62 dnf

remove. **CAUTION**

dnf

Sources: LINUX COMMAND, RHCSA_Command_Library.pdf, komendy, linux_engineering_book.md, linux_engineering_book.pdf

1.16.63 dnf autoremove

Sekcja Komend

```
dnf autoremove
```

Sources: RHCSA_Command_Library.pdf

1.16.64 dnf check-update # sprawdź aktualizacje

12.2 DNF/YUM (RHEL/Fedora)

```
dnf check-update # sprawdź aktualizacje
```

Sources: linux_engineering_book.md, linux_engineering_book.pdf

1.16.65 dnf clean all

Sekcja Komend

```
dnf clean all
```

Sources: RHCSA_Command_Library.pdf

1.16.66 dnf clean metadata

Sekcja Komend

```
dnf clean metadata
```

Sources: RHCSA_Command_Library.pdf

1.16.67 dnf clean packages

Sekcja Komend

```
dnf clean packages
```

Sources: RHCSA_Command_Library.pdf

1.16.68 dnf config-manager

Zarządzanie repo

```
dnf config-manager --add-repo url
dnf config-manager
```

Sources: LINUX COMMAND, RHCSA_Command_Library.pdf

1.16.69 dnf downgrade

Sekcja Komend

```
dnf downgrade
```

Sources: RHCSA_Command_Library.pdf

1.16.70 dnf erase package

Sekcja Komend

```
dnf erase package
```

Sources: RHCSA_Command_Library.pdf

1.16.71 dnf groupinfo 'Group

Sekcja Komend

```
dnf groupinfo 'Group
```

Sources: RHCSA_Command_Library.pdf

1.16.72 dnf groupinstall

Instalacja grupy pakietów

```
dnf groupinstall "Development
dnf groupinstall
```

Sources: LINUX COMMAND, RHCSA_Command_Library.pdf

1.16.73 dnf grouplist

Sekcja Komend

```
dnf grouplist
```

Sources: RHCSA_Command_Library.pdf

1.16.74 dnf groupremove

Sekcja Komend

```
dnf groupremove
```

Sources: RHCSA_Command_Library.pdf

1.16.75 dnf history

Historia transakcji

```
dnf history
```

Sources: LINUX COMMAND, RHCSA_Command_Library.pdf

1.16.76 `dnf history # historia transakcji`

12.2 DNF/YUM (RHEL/Fedora)

`dnf history # historia transakcji`

Sources: linux_engineering_book.md, linux_engineering_book.pdf

1.16.77 `dnf history info 5`

Sekcja Komend

`dnf history info 5`

Sources: RHCSA_Command_Library.pdf

1.16.78 `dnf history redo 5`

Sekcja Komend

`dnf history redo 5`

Sources: RHCSA_Command_Library.pdf

1.16.79 `dnf history rollback`

Sekcja Komend

`dnf history rollback`

Sources: RHCSA_Command_Library.pdf

1.16.80 `dnf history undo`

Cofa transakcję

`dnf history undo 5`

Sources: LINUX COMMAND

1.16.81 `dnf history undo 5`

Sekcja Komend

`dnf history undo 5`

Sources: RHCSA_Command_Library.pdf

1.16.82 `dnf history undo 5 # cofnij transakcję`

12.2 DNF/YUM (RHEL/Fedora)

`dnf history undo 5 # cofnij transakcję`

Sources: linux_engineering_book.md, linux_engineering_book.pdf

1.16.83 `dnf info`

Informacje o pakiecie

`dnf info httpd`

Sources: LINUX COMMAND

1.16.84 dnf info package

Sekcja Komend

dnf info package

Sources: RHCSA_Command_Library.pdf

1.16.85 dnf info package # informacje

12.2 DNF/YUM (RHEL/Fedora)

dnf info package # *informacje*

Sources: linux_engineering_book.md, linux_engineering_book.pdf

1.16.86 dnf info pkg

details. **SAFE**

dnf info pkg

Sources: komendy

1.16.87 dnf install

Instaluje pakiet

dnf install httpd

Sources: LINUX COMMAND

1.16.88 dnf install -y

Sekcja Komend

dnf install *-y*

Sources: RHCSA_Command_Library.pdf

1.16.89 dnf install kernel

Sekcja Komend

dnf install kernel

Sources: RHCSA_Command_Library.pdf

1.16.90 dnf install package

Sekcja Komend

dnf install package

Sources: RHCSA_Command_Library.pdf

1.16.91 dnf install package # instaluj

12.2 DNF/YUM (RHEL/Fedora)

dnf install package # *instaluj*

Sources: linux_engineering_book.md, linux_engineering_book.pdf

1.16.92 dnf install pkg

install. **CAUTION**

dnf install pkg

Sources: komendy

1.16.93 dnf makecache

Sekcja Komend

dnf makecache

Sources: RHCSA_Command_Library.pdf

1.16.94 dnf module

Moduły aplikacji (RHEL 8+)

dnf module

Sources: LINUX COMMAND

1.16.95 dnf provides

Sekcja Komend

dnf provides

Sources: RHCSA_Command_Library.pdf

1.16.96 dnf provides /path/file

owning package. **SAFE**

dnf provides /path/file

Sources: komendy

1.16.97 dnf provides /path/to/file # który pakiet zawiera plik

12.2 DNF/YUM (RHEL/Fedora)

dnf provides /path/to/file *# który pakiet zawiera plik*

Sources: linux_engineering_book.md, linux_engineering_book.pdf

1.16.98 dnf reinstall

Sekcja Komend

dnf reinstall

Sources: RHCSA_Command_Library.pdf

1.16.99 dnf repoinfo repo-id

Wyodrbnij zawarto pakietu RPM

dnf repoinfo repo-id

Sources: RHCSA_Command_Library.pdf

1.16.100 dnf repolist

Lista repozytoriów

```
dnf repolist all
dnf repolist
```

Sources: LINUX COMMAND, RHCSA_Command_Library.pdf

1.16.101 dnf repolist all

Sekcja Komend

```
dnf repolist all
```

Sources: RHCSA_Command_Library.pdf

1.16.102 dnf repoquery -l pkg

files in package. **SAFE**

```
dnf repoquery -l pkg
```

Sources: komendy

1.16.103 dnf search

Wyszukuje pakiet

```
dnf search web server
```

Sources: LINUX COMMAND

1.16.104 dnf search keyword

Sekcja Komend

```
dnf search keyword
```

Sources: RHCSA_Command_Library.pdf

1.16.105 dnf search keyword # szukaj

12.2 DNF/YUM (RHEL/Fedora)

```
dnf search keyword # szukaj
```

Sources: linux_engineering_book.md, linux_engineering_book.pdf

1.16.106 dnf search term

search. **SAFE**

```
dnf search term
```

Sources: komendy

1.16.107 dnf update

refresh and upgrade. **CAUTION**

```
dnf update
```

Sources: LINUX COMMAND, RHCSA_Command_Library.pdf, komendy

1.16.108 dnf update package

Sekcja Komend

dnf update package

Sources: RHCSA_Command_Library.pdf

1.16.109 dnf upgrade

Ulepsza system

dnf upgrade

Sources: LINUX COMMAND, RHCSA_Command_Library.pdf

1.16.110 dnf upgrade # aktualizuj

12.2 DNF/YUM (RHEL/Fedora)

dnf upgrade # *aktualizuj*

Sources: linux_engineering_book.md, linux_engineering_book.pdf

1.16.111 dnf upgrade-minimal

Sekcja Komend

dnf upgrade-minimal

Sources: RHCSA_Command_Library.pdf

1.16.112 dnf whatprovides

Sekcja Komend

dnf whatprovides

Sources: RHCSA_Command_Library.pdf

1.16.113 yum

Legacy wrapper (alias dnf)

yum install httpd

Sources: LINUX COMMAND

1.16.114 fdisk

Partycjonowanie MBR/GPT

fdisk /dev/sda

Sources: LINUX COMMAND

1.16.115 fdisk -l

Lista partycji

fdisk -l

Sources: LINUX COMMAND, RHCSA_Command_Library.pdf

1.16.116 `fdisk -l /dev/sda`

list partitions. **SAFE**

`fdisk -l /dev/sda`

Sources: RHCSA_Command_Library.pdf, komendy

1.16.117 `fdisk -u /dev/sda` `fdisk` z jednostkami sektorów `hdparm -I /dev/sda`

Sekcja Komend

`fdisk -u /dev/sda` `fdisk` z jednostkami sektorów `hdparm -I /dev/sda`

Sources: RHCSA_Command_Library.pdf

1.16.118 `fdisk /dev/sda`

interactive MBR/GPT. **DANGEROUS**

`fdisk /dev/sda`

Sources: RHCSA_Command_Library.pdf, komendy

1.16.119 `fdisk /dev/sdb #`

11.4 Zarządzanie Dyskami

`fdisk /dev/sdb #`

Sources: linux_engineering_book.md, linux_engineering_book.pdf

1.16.120 `findmnt`

mounted filesystems as tree. **SAFE**

`findmnt`

Sources: LINUX COMMAND, RHCSA_Command_Library.pdf, komendy

1.16.121 `findmnt --fstab`

Sekcja Komend

`findmnt --fstab`

Sources: RHCSA_Command_Library.pdf

1.16.122 `findmnt --kernel`

Kernel mountinfo

`findmnt --kernel`

Sources: LINUX COMMAND

1.16.123 `findmnt --poll`

Monitor mounts

`findmnt --poll`

Sources: LINUX COMMAND

1.16.124 findmnt --target

Find mount for path

```
findmnt --target /var
```

Sources: LINUX COMMAND

1.16.125 findmnt -t xfs

Sekcja Komend

```
findmnt -t xfs
```

Sources: RHCSA_Command_Library.pdf

1.16.126 findmnt /mnt

verify mount options. **SAFE**

```
findmnt /mnt
```

Sources: RHCSA_Command_Library.pdf, komendy

1.16.127 firewall-cmd

Komentarz Komentarz w skrypcie function myfunc() { Definicja funkcji

```
firewall-cmd --state
```

```
firewall-cmd
```

Sources: LINUX COMMAND, RHCSA_Command_Library.pdf

1.16.128 firewall-cmd --add-interface

Dodaje interfejs do strefy

```
firewall-cmd --add-interface=eth1 --zone=internal
```

Sources: LINUX COMMAND

1.16.129 firewall-cmd --add-p

Sekcja Komend

```
firewall-cmd --add-p
```

Sources: RHCSA_Command_Library.pdf

1.16.130 firewall-cmd --add-port

Dodaje port

```
firewall-cmd --add-port=8080/tcp
```

Sources: LINUX COMMAND

1.16.131 firewall-cmd --add-service

Dodaje usługę (runtime)

```
firewall-cmd --add-service=http
```

```
firewall-cmd --add-service=https
```

Sources: LINUX COMMAND

1.16.132 `firewall-cmd --add-service=ssh`

temporary add. **CAUTION**

```
firewall-cmd --add-service=ssh
```

Sources: komendy

1.16.133 `firewall-cmd --add-service=ssh --permanent`

permanent add. **CAUTION**

```
firewall-cmd --add-service=ssh --permanent
```

Sources: komendy

1.16.134 `firewall-cmd --add-source`

Dodaje źródło do strefy

```
firewall-cmd --add-source=192.168.1.0/24 --zone=trusted
```

Sources: LINUX COMMAND

1.16.135 `firewall-cmd --change-interface`

Zmienia strefę interfejsu

```
firewall-cmd --change-interface=eth1 --zone=dmz
```

Sources: LINUX COMMAND

1.16.136 `firewall-cmd --get-active-zones`

firewalld (RHEL/Fedora)

```
firewall-cmd --get-active-zones
```

Sources: linux_engineering_book.md, linux_engineering_book.pdf

1.16.137 `firewall-cmd --get-default-zone`

default zone. **SAFE**

```
firewall-cmd --get-default-zone
```

Sources: LINUX COMMAND, komendy

1.16.138 `firewall-cmd --get-services`

Lista znanych usług

```
firewall-cmd --get-services
```

Sources: LINUX COMMAND

1.16.139 `firewall-cmd --get-zones`

Dostępne strefy

```
firewall-cmd --get-zones
```

Sources: LINUX COMMAND

1.16.140 firewall-cmd --list-all

active zone rules. **SAFE**

```
firewall-cmd --list-all
```

Sources: LINUX COMMAND, komendy, linux_engineering_book.md, linux_engineering_book.pdf

1.16.141 firewall-cmd --list-services

Lista aktywnych usług

```
firewall-cmd --list-services
```

Sources: LINUX COMMAND

1.16.142 firewall-cmd --panic-off

Wyłącza panic

```
firewall-cmd --panic-off
```

Sources: LINUX COMMAND

1.16.143 firewall-cmd --panic-on

Panic mode (blokuje wszystko)

```
firewall-cmd --panic-on
```

Sources: LINUX COMMAND

1.16.144 firewall-cmd --permanent

Zmiana trwała

```
firewall-cmd --add-service=http --permanent
```

Sources: LINUX COMMAND

1.16.145 firewall-cmd --query-panic

Stan panic

```
firewall-cmd --query-panic
```

Sources: LINUX COMMAND

1.16.146 firewall-cmd --query-service

Sprawdza usługę

```
firewall-cmd --query-service=ssh
```

Sources: LINUX COMMAND

1.16.147 firewall-cmd --reload

reload. **CAUTION**

```
firewall-cmd --reload
```

Sources: LINUX COMMAND, komendy, linux_engineering_book.md, linux_engineering_book.pdf

1.16.148 firewall-cmd --reload losing temporary rescue rules:

Dangerous Mistakes

```
firewall-cmd --reload` losing temporary rescue rules.
```

Sources: komendy

1.16.149 firewall-cmd --remov

Sekcja Komend

```
firewall-cmd --remov
```

Sources: RHCSA_Command_Library.pdf

1.16.150 firewall-cmd --remove-service

Usuwa usługę

```
firewall-cmd --remove-service=http
```

Sources: LINUX COMMAND

1.16.151 firewall-cmd --remove-source

Usuwa źródło

```
firewall-cmd --remove-source=192.168.1.0/24
```

Sources: LINUX COMMAND

1.16.152 firewall-cmd --runtime-to-permanent

Zapisuje runtime na stałe

```
firewall-cmd --runtime-to-permanent
```

Sources: LINUX COMMAND

1.16.153 firewall-cmd --set-d

Sekcja Komend

```
firewall-cmd --set-d
```

Sources: RHCSA_Command_Library.pdf

1.16.154 firewall-cmd --set-default-zone

Ustawia domyślną

```
firewall-cmd --set-default-zone=drop
```

Sources: LINUX COMMAND

1.16.155 firewall-cmd --state

firewalld (RHEL/Fedora)

```
firewall-cmd --state
```

Sources: RHCSA_Command_Library.pdf, linux_engineering_book.md, linux_engineering_book.pdf

1.16.156 `firewall-cmd --zone=public`

Wybór strefy

`firewall-cmd --zone=public --list-all`

Sources: LINUX COMMAND

1.16.157 `firewall-cmd --zone=public --add-port=8080/tcp --permanent`

firewalld (RHEL/Fedora)

`firewall-cmd --zone=public --add-port=8080/tcp --permanent`

Sources: linux_engineering_book.md, linux_engineering_book.pdf

1.16.158 `firewall-cmd --zone=public --add-service=http --permanent`

firewalld (RHEL/Fedora)

`firewall-cmd --zone=public --add-service=http --permanent`

Sources: linux_engineering_book.md, linux_engineering_book.pdf

1.16.159 `getfacl`

Wyświetla ACL

`getfacl file.txt`

Sources: LINUX COMMAND

1.16.160 `getfacl file`

display ACL. **SAFE**

`getfacl file`

Sources: komendy

1.16.161 `getfacl file # pokaż`

ACL (Access Control Lists)

`getfacl file # pokaż`

Sources: linux_engineering_book.md, linux_engineering_book.pdf

1.16.162 `journalctl`

all logs since boot. **SAFE**

`journalctl -xe`

`journalctl`

Sources: LINUX COMMAND, RHCSA_Command_Library.pdf, komendy

1.16.163 `journalctl --disk-usage`

journal size. **SAFE**

`journalctl --disk-usage`

Sources: komendy

1.16.164 journalctl --disk-usage # zużycie dysku

Journald (logi)

journalctl --disk-usage # zużycie dysku

Sources: linux_engineering_book.md, linux_engineering_book.pdf

1.16.165 journalctl --list-boots

available boots. **SAFE**

journalctl --list-boots

Sources: komendy

1.16.166 journalctl --since

Od konkretnego czasu

journalctl --since "1 hour ago"

journalctl --since

Sources: LINUX COMMAND, RHCSA_Command_Library.pdf

1.16.167 journalctl --since "1 hour ago"

time window. **SAFE**

journalctl --since "1 hour ago"

Sources: komendy

1.16.168 journalctl --since "1 hour ago" # od godziny

Journald (logi)

journalctl --since "1 hour ago" # od godziny

Sources: linux_engineering_book.md, linux_engineering_book.pdf

1.16.169 journalctl --since today --until "2 hours ago"

range. **SAFE**

journalctl --since today --until "2 hours ago"

Sources: komendy

1.16.170 journalctl --vacuum-size=100M

Dodatek B: Komendy Ratunkowe

journalctl --vacuum-size=100M

Sources: linux_engineering_book.md, linux_engineering_book.pdf

1.16.171 journalctl --vacuum-size=1G

limit size. **CAUTION**

journalctl --vacuum-size=1G

Sources: komendy

1.16.172 `journalctl --vacuum-size=500M # ogranicz rozmiar`

Journald (logi)

`journalctl --vacuum-size=500M # ogranicz rozmiar`

Sources: linux_engineering_book.md, linux_engineering_book.pdf

1.16.173 `journalctl --vacuum-time=1s`

deleting all logs.

`journalctl --vacuum-time=1s`

Sources: komendy

1.16.174 `journalctl --vacuum-time=1s deleting all logs.`

Dangerous Mistakes

`journalctl --vacuum-time=1s` deleting all logs.`

Sources: komendy

1.16.175 `journalctl --vacuum-time=30d`

retain 30 days. **CAUTION**

`journalctl --vacuum-time=30d`

Sources: komendy

1.16.176 `journalctl -b`

current boot only. **SAFE**

`journalctl -b`

Sources: LINUX COMMAND, RHCSA_Command_Library.pdf, komendy

1.16.177 `journalctl -b -1`

previous boot. **SAFE**

`journalctl -b -1`

Sources: komendy

1.16.178 `journalctl -b -1 # poprzedni boot`

Journald (logi)

`journalctl -b -1 # poprzedni boot`

Sources: linux_engineering_book.md, linux_engineering_book.pdf

1.16.179 `journalctl -f`

Na żywo (follow)

`journalctl -f`

Sources: LINUX COMMAND

1.16.180 `journalctl -f # follow (jak tail -f)`

Journald (logi)

`journalctl -f # follow (jak tail -f)`

Sources: linux_engineering_book.md, linux_engineering_book.pdf

1.16.181 `journalctl -f` ledzenie logów na ywo aureport -l

Sekcja Komend

`journalctl -f` ledzenie logów na ywo aureport -l

Sources: RHCSA_Command_Library.pdf

1.16.182 `journalctl -k`

kernel logs only. **SAFE**

`journalctl -k`

Sources: komendy, linux_engineering_book.pdf

1.16.183 `journalctl -k | grep -i oom`

5.4 OOM Killer

`journalctl -k | grep -i oom`

Sources: linux_engineering_book.md

1.16.184 `journalctl -o`

Sekcja Komend

`journalctl -o`

Sources: RHCSA_Command_Library.pdf

1.16.185 `journalctl -p`

Po prioritycie (err, warning)

`journalctl -p err`

Sources: LINUX COMMAND

1.16.186 `journalctl -p err`

priority filter. **SAFE**

`journalctl -p err`

Sources: RHCSA_Command_Library.pdf, komendy

1.16.187 `journalctl -p err # tylko błędy`

Journald (logi)

`journalctl -p err # tylko błędy`

Sources: linux_engineering_book.md, linux_engineering_book.pdf

1.16.188 journalctl -u

Logi konkretnej usługi

```
journalctl -u sshd
```

Sources: LINUX COMMAND

1.16.189 journalctl -u nginx

unit-specific logs. **SAFE**

```
journalctl -u nginx
```

Sources: komendy

1.16.190 journalctl -u nginx # logi usługi

Journald (logi)

```
journalctl -u nginx # logi usługi
```

Sources: linux_engineering_book.md, linux_engineering_book.pdf

1.16.191 journalctl -u nginx -f

follow unit logs. **SAFE**

```
journalctl -u nginx -f
```

Sources: komendy

1.16.192 journalctl -u sshd

Sekcja Komend

```
journalctl -u sshd
```

Sources: RHCSA_Command_Library.pdf

1.16.193 journalctl -xb

current boot logs. **SAFE**

```
journalctl -xb
```

Sources: komendy

1.16.194 journalctl -xe

recent errors with context. **SAFE**

```
journalctl -xe
```

Sources: RHCSA_Command_Library.pdf, komendy

1.16.195 lvcreate

Tworzy Logical Volume

```
lvcreate -L 10G -n lv_www vg_data
```

Sources: LINUX COMMAND

1.16.196 `lvcreate -L 100G -n lv_data vg0`

create LV. **SAFE**

`lvcreate -L 100G -n lv_data vg0`

Sources: komendy

1.16.197 `lvcreate -L 100G -n lv_data vg_data`

6.4 LVM (Logical Volume Manager)

`lvcreate -L 100G -n lv_data vg_data`

Sources: linux_engineering_book.md, linux_engineering_book.pdf

1.16.198 `lvcreate -L 10G -s -n lv_snap /dev/vg0/lv_data`

snapshot. **SAFE**

`lvcreate -L 10G -s -n lv_snap /dev/vg0/lv_data`

Sources: komendy

1.16.199 `lvcreate -L 10G -s -n snap_data /dev/vg_data/lv_data`

6.4 LVM (Logical Volume Manager)

`lvcreate -L 10G -s -n snap_data /dev/vg_data/lv_data`

Sources: linux_engineering_book.md, linux_engineering_book.pdf

1.16.200 `lvcreate -l`

Używa %VG lub %FREE

`lvcreate -l 100%FREE -n lv_tmp vg_data`

Sources: LINUX COMMAND

1.16.201 `lvextend`

Rozszerza LV

`lvextend -L +5G /dev/vg_data/lv_www`

Sources: LINUX COMMAND

1.16.202 `lvextend --resizefs -L +500G /dev/mapper/vg_prod_storage-lv_db`

Dynamiczne rozszerzenie wolumenu logicznego wraz z jednoczesnym przeskalowaniem systemu plików XFS online

`lvextend --resizefs -L +500G /dev/mapper/vg_prod_storage-lv_db`

Sources: encyklopedia_inzynierii_linux.html, encyklopedia_inzynierii_linux.pdf

1.16.203 `lvextend -L +50G /dev/vg0/lv_data`

extend LV. **CAUTION**

`lvextend -L +50G /dev/vg0/lv_data`

Sources: komendy

1.16.204 `lvextend -L +50G /dev/vg_data/lv_data`

6.4 LVM (Logical Volume Manager)

`lvextend -L +50G /dev/vg_data/lv_data`

Sources: linux_engineering_book.md, linux_engineering_book.pdf

1.16.205 `lvextend -r`

Use to extend filesystem in one step.

`lvextend -r`

Sources: komendy

1.16.206 `mkfs.xfs`

Formatuje XFS

`mkfs.xfs /dev/sda1`

Sources: LINUX COMMAND

1.16.207 `mkfs.xfs -L`

Sekcja Komend

`mkfs.xfs -L`

Sources: RHCSA_Command_Library.pdf

1.16.208 `mkfs.xfs -L "data" /dev/sdb1`

Zaawansowane

`mkfs.xfs -L "data" /dev/sdb1`

Sources: linux_engineering_book.md, linux_engineering_book.pdf

1.16.209 `mkfs.xfs -L data /dev/sda1`

create XFS. **DANGEROUS**

`mkfs.xfs -L data /dev/sda1`

Sources: komendy

1.16.210 `mkfs.xfs -f`

Sekcja Komend

`mkfs.xfs -f`

Sources: RHCSA_Command_Library.pdf

1.16.211 `mkfs.xfs /dev/sdb1`

Sekcja Komend

`mkfs.xfs /dev/sdb1`

Sources: RHCSA_Command_Library.pdf

1.16.212 mount

Montuje filesystem

```
mount /dev/sda1 /mnt  
mount
```

Sources: LINUX COMMAND, RHCSA_Command_Library.pdf

1.16.213 mount --bind

Bind mount

```
mount --bind /var/www /home/user/www
```

Sources: LINUX COMMAND

1.16.214 mount --bind /dev /mnt/dev

19.2 Disaster Recovery

```
mount --bind /dev /mnt/dev
```

Sources: linux_engineering_book.md, linux_engineering_book.pdf

1.16.215 mount --bind /dev /mnt/sysimage/dev

bind mount devices. **SAFE**

```
mount --bind /dev /mnt/sysimage/dev
```

Sources: komendy

1.16.216 mount --bind /proc /mnt/proc

19.2 Disaster Recovery

```
mount --bind /proc /mnt/proc
```

Sources: linux_engineering_book.md, linux_engineering_book.pdf

1.16.217 mount --bind /proc /mnt/sysimage/proc

bind mount proc. **SAFE**

```
mount --bind /proc /mnt/sysimage/proc
```

Sources: komendy

1.16.218 mount --bind /source /destination

6.3 Montowanie i fstab

```
mount --bind /source /destination
```

Sources: linux_engineering_book.md, linux_engineering_book.pdf

1.16.219 mount --bind /sys /mnt/sys

19.2 Disaster Recovery

```
mount --bind /sys /mnt/sys
```

Sources: linux_engineering_book.md, linux_engineering_book.pdf

1.16.220 `mount --bind /sys /mnt/sysimage/sys`

bind mount sys. **SAFE**

`mount --bind /sys /mnt/sysimage/sys`

Sources: komendy

1.16.221 `mount --make-private`

Private mount

`mount --make-private /mnt`

Sources: LINUX COMMAND

1.16.222 `mount --make-shared`

Shared mount

`mount --make-shared /mnt`

Sources: LINUX COMMAND

1.16.223 `mount --make-slave`

Slave mount

`mount --make-slave /mnt`

Sources: LINUX COMMAND

1.16.224 `mount --make-unbindable`

Unbindable

`mount --make-unbindable /mnt`

Sources: LINUX COMMAND

1.16.225 `mount --move`

Move mount

`mount --move /old /new`

Sources: LINUX COMMAND

1.16.226 `mount --rbind`

Recursive bind

`mount --rbind /dev /mnt/chroot/dev`

Sources: LINUX COMMAND

1.16.227 `mount -a`

Montuje wszystko z fstab

`mount -a`

Sources: LINUX COMMAND, RHCSA_Command_Library.pdf

1.16.228 `mount -o`

Sekcja Komend

`mount -o`

Sources: RHCSA_Command_Library.pdf

1.16.229 `mount -o loop image.iso /mnt/iso`

6.3 Montowanie i fstab

`mount -o loop image.iso /mnt/iso`

Sources: linux_engineering_book.md, linux_engineering_book.pdf

1.16.230 `mount -o remount,ro`

Sekcja Komend

`mount -o remount,ro`

Sources: RHCSA_Command_Library.pdf

1.16.231 `mount -o remount,rw`

Przymontowuje do zapisu

`mount -o remount,rw /`

`mount -o remount,rw`

Sources: LINUX_COMMAND, RHCSA_Command_Library.pdf

1.16.232 `mount -o remount,rw /`

Po uruchomieniu powłoki awaryjnej, przemontowanie głównego systemu plików w tryb zapisu/odczytu

`mount -o remount,rw /`

Sources: encyklopedia_inzynierii_linux.html, encyklopedia_inzynierii_linux.pdf, komendy, linux_engineering_book.m
linux_engineering_book.pdf

1.16.233 `mount -o remount,rw,nosuid,nodev,noexec /tmp`

restrict /tmp. **CAUTION**

`mount -o remount,rw,nosuid,nodev,noexec /tmp`

Sources: komendy

1.16.234 `mount -o ro`

Sekcja Komend

`mount -o ro`

Sources: RHCSA_Command_Library.pdf

1.16.235 `mount -o ro,soft`

Sekcja Komend

`mount -o ro,soft`

Sources: RHCSA_Command_Library.pdf

1.16.236 `mount -o rw,noexec`

Sekcja Komend

`mount -o rw,noexec`

Sources: RHCSA_Command_Library.pdf

1.16.237 `mount -t cifs`

Komentarz Komentarz w skrypcie function myfunc() { Definicja funkcji

`mount -t cifs`

Sources: RHCSA_Command_Library.pdf

1.16.238 `mount -t ext4`

Sekcja Komend

`mount -t ext4`

Sources: RHCSA_Command_Library.pdf

1.16.239 `mount -t nfs`

Sekcja Komend

`mount -t nfs`

Sources: RHCSA_Command_Library.pdf

1.16.240 `mount -t nfs server:/share /mnt/nfs`

6.3 Montowanie i fstab

`mount -t nfs server:/share /mnt/nfs`

Sources: linux_engineering_book.md, linux_engineering_book.pdf

1.16.241 `mount -t nfs server:/srv/nfs /mnt/nfs`

18.3 NFS

`mount -t nfs server:/srv/nfs /mnt/nfs`

Sources: linux_engineering_book.md, linux_engineering_book.pdf

1.16.242 `mount -t nfs4`

Sekcja Komend

`mount -t nfs4`

Sources: RHCSA_Command_Library.pdf

1.16.243 `mount -t xfs`

Sekcja Komend

`mount -t xfs`

Sources: RHCSA_Command_Library.pdf

1.16.244 `mount /dev/sda1 /mnt`

mount filesystem. **CAUTION**

`mount /dev/sda1 /mnt`

Sources: komendy

1.16.245 `mount /dev/sda1 /mnt/boot`

19.2 Disaster Recovery

`mount /dev/sda1 /mnt/boot`

Sources: linux_engineering_book.md, linux_engineering_book.pdf

1.16.246 `mount /dev/sda2 /mnt`

19.2 Disaster Recovery

`mount /dev/sda2 /mnt`

Sources: linux_engineering_book.md, linux_engineering_book.pdf

1.16.247 `mount /dev/sdb1 /mnt`

Sekcja Komend

`mount /dev/sdb1 /mnt`

Sources: RHCSA_Command_Library.pdf

1.16.248 `mount /dev/sdb1 /mnt/data`

6.3 Montowanie i fstab

`mount /dev/sdb1 /mnt/data`

Sources: linux_engineering_book.md, linux_engineering_book.pdf

1.16.249 `nmcli`

NetworkManager CLI

`nmcli dev`

Sources: LINUX COMMAND

1.16.250 `nmcli con`

Lista połączeń

`nmcli con`

Sources: LINUX COMMAND

1.16.251 `nmcli con add type bond`

Tworzy bond

`nmcli con add type bond con-name bond0 ifname bond0`

Sources: LINUX COMMAND

1.16.252 nmcli con add type bridge

Tworzy bridge NM

```
nmcli con add type bridge con-name br0 ifname br0
```

Sources: LINUX COMMAND

1.16.253 nmcli con add type team

Tworzy team w NM

```
nmcli con add type team con-name team0 ifname team0
```

Sources: LINUX COMMAND

1.16.254 nmcli con add type vlan

Tworzy VLAN NM

```
nmcli con add type vlan con-name vlan10 ifname eth0.10 dev eth0 id 10
```

Sources: LINUX COMMAND

1.16.255 nmcli con down

Deaktywuje

```
nmcli con down "Wired connection 1"
```

Sources: LINUX COMMAND

1.16.256 nmcli con mod

Modyfikuje połączenie

```
nmcli con mod "eth0" ipv4.addresses 192.168.1.10/24
```

Sources: LINUX COMMAND

1.16.257 nmcli con up

Aktywuje połączenie

```
nmcli con up "Wired connection 1"
```

Sources: LINUX COMMAND

1.16.258 nmcli con up id "Wired connection 1"

activate connection. **CAUTION**

```
nmcli con up id "Wired connection 1"
```

Sources: komendy

1.16.259 nmcli connection

Komentarz Komentarz w skrypcie function myfunc() { Definicja funkcji

```
nmcli connection
```

Sources: RHCSA_Command_Library.pdf, linux_engineering_book.md, linux_engineering_book.pdf

1.16.260 nmcli connection add type ethernet con-name "LAN" ifname eth0

NetworkManager

nmcli connection add type ethernet con-name "LAN" ifname eth0

Sources: linux_engineering_book.md, linux_engineering_book.pdf

1.16.261 nmcli connection modify "LAN" ipv4.addresses 192.168.1.10/24

NetworkManager

nmcli connection modify "LAN" ipv4.addresses 192.168.1.10/24

Sources: linux_engineering_book.md, linux_engineering_book.pdf

1.16.262 nmcli connection modify "LAN" ipv4.dns "8.8.8.8 8.8.4.4"

NetworkManager

nmcli connection modify "LAN" ipv4.dns "8.8.8.8 8.8.4.4"

Sources: linux_engineering_book.md, linux_engineering_book.pdf

1.16.263 nmcli connection modify "LAN" ipv4.gateway 192.168.1.1

NetworkManager

nmcli connection modify "LAN" ipv4.gateway 192.168.1.1

Sources: linux_engineering_book.md, linux_engineering_book.pdf

1.16.264 nmcli connection modify "LAN" ipv4.method manual

NetworkManager

nmcli connection modify "LAN" ipv4.method manual

Sources: linux_engineering_book.md, linux_engineering_book.pdf

1.16.265 nmcli connection up "LAN"

NetworkManager

nmcli connection up "LAN"

Sources: linux_engineering_book.md, linux_engineering_book.pdf

1.16.266 nmcli dev

Szczegóły urządzenia

nmcli dev

Sources: LINUX COMMAND

1.16.267 nmcli device

NetworkManager status. **SAFE**

nmcli device

Sources: komendy, linux_engineering_book.md, linux_engineering_book.pdf

1.16.268 nmcli device connect

Łączy

```
nmcli device connect eth0
```

Sources: LINUX COMMAND

1.16.269 nmcli device disconnect

Rozłącza

```
nmcli device disconnect eth0
```

Sources: LINUX COMMAND

1.16.270 nmcli device reapply

Reaplikuje konfigurację

```
nmcli device reapply eth0
```

Sources: LINUX COMMAND

1.16.271 nmcli device wifi

WiFi scan

```
nmcli device wifi
```

Sources: LINUX COMMAND

1.16.272 nmcli device wifi connect

WiFi connect

```
nmcli device wifi connect "SSID" password "pass"
```

Sources: LINUX COMMAND

1.16.273 nmcli networking

Włącza/wyłącza networking

```
nmcli networking off
```

Sources: LINUX COMMAND

1.16.274 nmcli radio

Zarządzanie radiem

```
nmcli radio wifi off
```

Sources: LINUX COMMAND

1.16.275 parted

Zaawansowane partycjonowanie

```
parted /dev/sda print
```

Sources: LINUX COMMAND

1.16.276 parted -l

Sekcja Komend

parted -l

Sources: RHCSA_Command_Library.pdf

1.16.277 parted /dev/sda

interactive partitioning. **DANGEROUS**

parted /dev/sda

Sources: RHCSA_Command_Library.pdf, komendy

1.16.278 parted /dev/sda print

print table. **SAFE**

parted /dev/sda print

Sources: komendy

1.16.279 parted /dev/sda rm 1

Sekcja Komend

parted /dev/sda rm 1

Sources: RHCSA_Command_Library.pdf

1.16.280 parted /dev/sdb #

11.4 Zarządzanie Dyskami

parted /dev/sdb #

Sources: linux_engineering_book.md, linux_engineering_book.pdf

1.16.281 passwd

Zmienia hasło

passwd john

Sources: LINUX COMMAND

1.16.282 passwd --status user

Sekcja Komend

passwd --status user

Sources: RHCSA_Command_Library.pdf

1.16.283 passwd -d

Usuwa hasło (puste)

passwd -d john

Sources: LINUX COMMAND

1.16.284 `passwd -d user`

Sekcja Komend

`passwd -d user`

Sources: RHCSA_Command_Library.pdf

1.16.285 `passwd -e user`

Sekcja Komend

`passwd -e user`

Sources: RHCSA_Command_Library.pdf

1.16.286 `passwd -i 10 user`

Sekcja Komend

`passwd -i 10 user`

Sources: RHCSA_Command_Library.pdf

1.16.287 `passwd -l`

Blokuje hasło

`passwd -l john`

Sources: LINUX COMMAND

1.16.288 `passwd -l alice`

lock password. **CAUTION**

`passwd -l alice`

Sources: komendy

1.16.289 `passwd -l user`

Sekcja Komend

`passwd -l user`

Sources: RHCSA_Command_Library.pdf

1.16.290 `passwd -n 7 user`

Sekcja Komend

`passwd -n 7 user`

Sources: RHCSA_Command_Library.pdf

1.16.291 `passwd -u`

Odblokowuje hasło

`passwd -u john`

Sources: LINUX COMMAND

1.16.292 `passwd -u user`

Sekcja Komend

`passwd -u user`

Sources: RHCSA_Command_Library.pdf

1.16.293 `passwd -w 14 user`

Sekcja Komend

`passwd -w 14 user`

Sources: RHCSA_Command_Library.pdf

1.16.294 `passwd -x 90 user`

Sekcja Komend

`passwd -x 90 user`

Sources: RHCSA_Command_Library.pdf

1.16.295 `passwd root`

change root password. **DANGEROUS**

`passwd root`

Sources: encyklopedia_inzynierii_linux.html, encyklopedia_inzynierii_linux.pdf, komendy, linux_engineering_book.md, linux_engineering_book.pdf

1.16.296 `passwd user`

Sekcja Komend

`passwd user`

Sources: RHCSA_Command_Library.pdf

1.16.297 `passwd user1 # zmień hasło`

/etc/modprobe.d/bonding.conf

`passwd user1 # zmień hasło`

Sources: linux_engineering_book.md, linux_engineering_book.pdf

1.16.298 `pvcreate`

on active filesystem partition.

`pvcreate /dev/sdb1`

`pvcreate`

Sources: LINUX_COMMAND, komendy

1.16.299 `pvcreate /dev/sda1`

initialize PV. **DANGEROUS**

`pvcreate /dev/sda1`

Sources: komendy

1.16.300 pvcreate /dev/sdb

Sekcja Komend

```
pvcreate /dev/sdb
```

Sources: RHCSA_Command_Library.pdf

1.16.301 pvcreate /dev/sdb /dev/sdc

6.4 LVM (Logical Volume Manager)

```
pvcreate /dev/sdb /dev/sdc
```

Sources: linux_engineering_book.md, linux_engineering_book.pdf

1.16.302 pvcreate /dev/sdc

Inicjalizacja nowego fizycznego urządzenia dyskowego dla menedżera LVM

```
pvcreate /dev/sdc
```

Sources: encyklopedia_inzynierii_linux.html, encyklopedia_inzynierii_linux.pdf

1.16.303 restorecon

Przywraca konteksty

```
restorecon -Rv /var/www
```

```
restorecon
```

Sources: LINUX COMMAND, RHCSA_Command_Library.pdf

1.16.304 restorecon -F

Force reset context

```
restorecon -F -R /var/www
```

Sources: LINUX COMMAND

1.16.305 restorecon -F /path/

Sekcja Komend

```
restorecon -F /path/
```

Sources: RHCSA_Command_Library.pdf

1.16.306 restorecon -R

Rekurencyjnie

```
restorecon -R /home
```

```
restorecon -R
```

Sources: LINUX COMMAND, RHCSA_Command_Library.pdf

1.16.307 restorecon -Rv

Sekcja Komend

```
restorecon -Rv
```

Sources: RHCSA_Command_Library.pdf

1.16.308 `restorecon -Rv /web`

restore contexts. **CAUTION**

`restorecon -Rv /web`

Sources: komendy

1.16.309 `restorecon -Rv /web # przywróć konteksty`

9.3 SELinux

`restorecon -Rv /web # przywróć konteksty`

Sources: linux_engineering_book.md, linux_engineering_book.pdf

1.16.310 `restorecon -i`

Ignore errors

`restorecon -i -R /var/www`

Sources: LINUX COMMAND

1.16.311 `restorecon -v`

Verbose

`restorecon -v /etc/hosts`

Sources: LINUX COMMAND

1.16.312 `rpm --import`

Importuje klucz GPG

`rpm --import /etc/pki/rpm-gpg/RPM-GPG-KEY-redhat-release`

Sources: LINUX COMMAND

1.16.313 `rpm -Uvh`

on running kernel without reboot plan.

`rpm -Uvh`

Sources: komendy

1.16.314 `rpm -Uvh on running kernel without reboot plan.`

Dangerous Mistakes

`rpm -Uvh` on running kernel without reboot plan.`

Sources: komendy

1.16.315 `rpm -V pkg`

verify integrity. **SAFE**

`rpm -V pkg`

Sources: komendy

1.16.316 rpm -Va

verify all installed packages. **SAFE**

rpm -Va

Sources: komendy

1.16.317 rpm -e

Usuwa pakiet (bez zależności)

rpm -e httpd

Sources: LINUX COMMAND

1.16.318 rpm -ivh

Instaluje lokalny RPM

rpm -ivh package.rpm

Sources: LINUX COMMAND

1.16.319 rpm -ivh file.rpm

install local rpm. **CAUTION**

rpm -ivh file.rpm

Sources: komendy

1.16.320 rpm -ivh package.rpm # instaluj

12.2 DNF/YUM (RHEL/Fedora)

rpm -ivh package.rpm # *instaluj*

Sources: linux_engineering_book.md, linux_engineering_book.pdf

1.16.321 rpm -qa

all packages. **SAFE**

rpm -qa \

rpm -qa

Sources: LINUX COMMAND, komendy

1.16.322 rpm -qa

12.2 DNF/YUM (RHEL/Fedora)

rpm -qa #

Sources: linux_engineering_book.md, linux_engineering_book.pdf

1.16.323 rpm -qa kernel

Sekcja Komend

rpm -qa kernel

Sources: RHCSA_Command_Library.pdf

1.16.324 rpm -qf

Który pakiet zawiera plik

```
rpm -qf /bin/bash
```

Sources: LINUX COMMAND

1.16.325 rpm -qf /path/file

file owner. **SAFE**

```
rpm -qf /path/file
```

Sources: komendy

1.16.326 rpm -qf /path/to/file # który pakiet

12.2 DNF/YUM (RHEL/Fedora)

```
rpm -qf /path/to/file # który pakiet
```

Sources: linux_engineering_book.md, linux_engineering_book.pdf

1.16.327 rpm -qi

Info o pakiecie

```
rpm -qi httpd
```

Sources: LINUX COMMAND

1.16.328 rpm -ql

Pliki w pakiecie

```
rpm -ql httpd
```

Sources: LINUX COMMAND

1.16.329 rpm -ql package # pliki pakietu

12.2 DNF/YUM (RHEL/Fedora)

```
rpm -ql package # pliki pakietu
```

Sources: linux_engineering_book.md, linux_engineering_book.pdf

1.16.330 rpm -ql pkg

package files. **SAFE**

```
rpm -ql pkg
```

Sources: komendy

1.16.331 rsync

Synchronizacja (różnicowa)

```
rsync -avz /src/ /dest/
```

```
rsync -avz --delete /src/ user@server:/dest/
```

Sources: LINUX COMMAND

1.16.332 rsync --backup

Backup changed files

```
rsync --backup /src/ /dest/
```

Sources: LINUX COMMAND

1.16.333 rsync --bwlimit

Bandwidth limit

```
rsync --bwlimit=1000 /src/ /dest/
```

Sources: LINUX COMMAND

1.16.334 rsync --delete

Usuwa pliki nieistniejące w źródle

```
rsync -avz --delete /src/ /dest/  
rsync --delete /src/ /dest/
```

Sources: LINUX COMMAND

1.16.335 rsync --dry-run

flags: , , .

```
rsync --dry-run -avz /src/ /dest/  
rsync --dry-run
```

Sources: LINUX COMMAND, komendy

1.16.336 rsync --exclude

Exclude pattern

```
rsync --exclude='*.tmp' /src/ /dest/
```

Sources: LINUX COMMAND

1.16.337 rsync --inplace

Atomic updates with or .

```
rsync --inplace /src/ /dest/  
rsync --inplace
```

Sources: LINUX COMMAND, komendy

1.16.338 rsync --link-dest

Hardlink backups

```
rsync --link-dest=/prev /src/ /dest/
```

Sources: LINUX COMMAND

1.16.339 rsync --partial

Resume partial

```
rsync --partial /src/ /dest/
```

Sources: LINUX COMMAND

1.16.340 `rsync --progress`

Show progress

```
rsync --progress /src/ /dest/
```

Sources: LINUX COMMAND

1.16.341 `rsync -a /backup/ /`

restoring without verification.

```
rsync -a /backup/ /
```

Sources: komendy

1.16.342 `rsync -a /backup/ / restoring without verification.`

Dangerous Mistakes

```
rsync -a /backup/ /` restoring without verification.
```

Sources: komendy

1.16.343 `rsync -av --dry-run`

before real sync.

```
rsync -av --dry-run
```

Sources: komendy

1.16.344 `rsync -av --dry-run before real sync.`

Production-Safe Patterns

```
rsync -av --dry-run` before real sync.
```

Sources: komendy

1.16.345 `rsync -av /backup/ /`

restore from backup. **DANGEROUS** (overwrites live system)

```
rsync -av /backup/ /
```

Sources: komendy

1.16.346 `rsync -avz`

Archive, verbose, compress

```
rsync -avz /src/ /dest/
```

Sources: LINUX COMMAND

1.16.347 `rsync -avz --delete /source/ /dest/ # mirror (usuń nadmiarowe)`

rsync — synchronizacja plików

```
rsync -avz --delete /source/ /dest/ # mirror (usuń nadmiarowe)
```

Sources: linux_engineering_book.md, linux_engineering_book.pdf

1.16.348 `rsync -avz --delete src/ dest/`

sync with deletion. **DANGEROUS**

`rsync -avz --delete src/ dest/`

Sources: komendy

1.16.349 `rsync -avz --exclude='*.log' /source/ /dest/`

rsync — synchronizacja plików

`rsync -avz --exclude='*.log' /source/ /dest/`

Sources: linux_engineering_book.md, linux_engineering_book.pdf

1.16.350 `rsync -avz -e "ssh -p 2222" /local/ user@host:/remote/`

sync over SSH. **CAUTION**

`rsync -avz -e "ssh -p 2222" /local/ user@host:/remote/`

Sources: komendy

1.16.351 `rsync -avz -e ssh /source/ user@host:/dest/ # zdalna kopia`

rsync — synchronizacja plików

`rsync -avz -e ssh /source/ user@host:/dest/ # zdalna kopia`

Sources: linux_engineering_book.md, linux_engineering_book.pdf

1.16.352 `rsync -avz /source/ /destination/ # lokalna kopia`

rsync — synchronizacja plików

`rsync -avz /source/ /destination/ # lokalna kopia`

Sources: linux_engineering_book.md, linux_engineering_book.pdf

1.16.353 `rsync -e ssh`

Rsync przez SSH

`rsync -avz -e "ssh -p 2222" /src/ host:/dest/`

Sources: LINUX COMMAND

1.16.354 `semanage boolean`

Lista boole'ów

`semanage boolean -l`

Sources: LINUX COMMAND

1.16.355 `semanage boolean -C`

Custom booleans

`semanage boolean -C -l`

Sources: LINUX COMMAND

1.16.356 `semanage boolean -D`

Delete custom boolean

```
semanage boolean -D
```

Sources: LINUX COMMAND

1.16.357 `semanage boolean -l`

List booleans

```
semanage boolean -l
```

Sources: LINUX COMMAND, RHCSA_Command_Library.pdf

1.16.358 `semanage boolean -m`

Modify boolean

```
semanage boolean -m -l httpd_enable_homedirs
```

Sources: LINUX COMMAND

1.16.359 `semanage dontaudit`

Toggle dontaudit

```
semanage dontaudit off
```

Sources: LINUX COMMAND

1.16.360 `semanage export`

Export config

```
semanage export -f backup.conf
```

Sources: LINUX COMMAND

1.16.361 `semanage fcontext`

Konteksty plików

```
semanage fcontext -a -t httpd_sys_content_t "/web(/.*)?"
```

Sources: LINUX COMMAND

1.16.362 `semanage fcontext -C`

Local customizations

```
semanage fcontext -C -l
```

Sources: LINUX COMMAND

1.16.363 `semanage fcontext -a`

Sekcja Komend

```
semanage fcontext -a -t httpd_sys_content_t "/web(/.*)?"
```

```
semanage fcontext -a
```

Sources: LINUX COMMAND, RHCSA_Command_Library.pdf

1.16.364 `semanage fcontext -a -t httpd_sys_content_t "/web(/.*)?"`

add context rule. **CAUTION**

`semanage fcontext -a -t httpd_sys_content_t "/web(/.*)?"`

Sources: komendy, linux_engineering_book.md, linux_engineering_book.pdf

1.16.365 `semanage fcontext -d`

Delete fcontext

`semanage fcontext -d "/web(/.*)?"`

`semanage fcontext -d`

Sources: LINUX COMMAND, RHCSA_Command_Library.pdf

1.16.366 `semanage fcontext -l`

file context rules. **SAFE**

`semanage fcontext -l`

Sources: LINUX COMMAND, RHCSA_Command_Library.pdf, komendy

1.16.367 `semanage fcontext -m`

Modify fcontext

`semanage fcontext -m -t public_content_t "/web(/.*)?"`

`semanage fcontext -m`

Sources: LINUX COMMAND, RHCSA_Command_Library.pdf

1.16.368 `semanage ibendport -a`

Add ibendport

`semanage ibendport -a -t ibendport_t mlx4_0 1`

Sources: LINUX COMMAND

1.16.369 `semanage ibpkey -a`

Add ibpkey

`semanage ibpkey -a -t ibpkey_t fe80:: 1`

Sources: LINUX COMMAND

1.16.370 `semanage import`

Import config

`semanage import -f backup.conf`

Sources: LINUX COMMAND

1.16.371 `semanage interface -a`

Add interface

`semanage interface -a -t netif_t eth1`

Sources: LINUX COMMAND

1.16.372 semanage login

Mapowanie użytkowników

```
semanage login -l
```

Sources: LINUX COMMAND

1.16.373 semanage login -a

Add login mapping

```
semanage login -a -s user_u john
```

Sources: LINUX COMMAND

1.16.374 semanage login -d

Delete login mapping

```
semanage login -d john
```

Sources: LINUX COMMAND

1.16.375 semanage login -l

login mappings. **SAFE**

```
semanage login -l
```

Sources: LINUX COMMAND, komendy

1.16.376 semanage login -m

Modify login mapping

```
semanage login -m -s staff_u john
```

Sources: LINUX COMMAND

1.16.377 semanage module -a

Add module

```
semanage module -a mymodule
```

Sources: LINUX COMMAND

1.16.378 semanage module -d

Delete module

```
semanage module -d mymodule
```

Sources: LINUX COMMAND

1.16.379 semanage module -l

List modules

```
semanage module -l
```

Sources: LINUX COMMAND

1.16.380 semanage node -a

Add node

```
semanage node -a -t node_t -M 255.255.255.255 192.168.1.0
```

Sources: LINUX COMMAND

1.16.381 semanage permissive -a

Add permissive type

```
semanage permissive -a httpd_t
```

Sources: LINUX COMMAND

1.16.382 semanage permissive -d

Delete permissive type

```
semanage permissive -d httpd_t
```

Sources: LINUX COMMAND

1.16.383 semanage permissive -l

List permissive types

```
semanage permissive -l
```

Sources: LINUX COMMAND

1.16.384 semanage port

Mapowanie portów

```
semanage port -a -t http_port_t -p tcp 8080
```

Sources: LINUX COMMAND

1.16.385 semanage port -C

Custom ports

```
semanage port -C -l
```

Sources: LINUX COMMAND

1.16.386 semanage port -a

Add port

```
semanage port -a -t http_port_t -p tcp 8080
```

Sources: LINUX COMMAND

1.16.387 semanage port -a -t

grep Typy SELinux dla httpd

```
semanage port -a -t
```

Sources: RHCSA_Command_Library.pdf

1.16.388 semanage port -d

Delete port

```
semanage port -d -p tcp 8080
```

Sources: LINUX COMMAND

1.16.389 semanage port -d -t

Sekcja Komend

```
semanage port -d -t
```

Sources: RHCSA_Command_Library.pdf

1.16.390 semanage port -l

Porty dla usugi http seinfo -r Informacje o rolach SELinux

```
semanage port -l
```

Sources: LINUX COMMAND, RHCSA_Command_Library.pdf

1.16.391 semanage port -m

Modify port

```
semanage port -m -t http_port_t -p tcp 8080
```

Sources: LINUX COMMAND

1.16.392 semanage port -m -t

Sekcja Komend

```
semanage port -m -t
```

Sources: RHCSA_Command_Library.pdf

1.16.393 semanage translate

Translate context

```
semanage translate -T httpd_t
```

Sources: LINUX COMMAND

1.16.394 semanage user

Użytkownicy SELinux

```
semanage user -l
```

Sources: LINUX COMMAND

1.16.395 semanage user -a

Add SELinux user

```
semanage user -a -R "staff_r sysadm_r" -r s0-s0:c0.c1023 staff_u
```

Sources: LINUX COMMAND

1.16.396 semanage user -d

Delete SELinux user

```
semanage user -d staff_u
```

Sources: LINUX COMMAND

1.16.397 semanage user -l

List SELinux users

```
semanage user -l
```

Sources: LINUX COMMAND

1.16.398 semanage user -m

Modify SELinux user

```
semanage user -m -R "staff_r" staff_u
```

Sources: LINUX COMMAND

1.16.399 setfacl

Ustawia ACL

```
setfacl -m u:john:rw file.txt
```

Sources: LINUX COMMAND

1.16.400 setfacl --set-file=

Sekcja Komend

```
setfacl --set-file=
```

Sources: RHCSA_Command_Library.pdf

1.16.401 setfacl -R

Rekurencyjne ACL

```
setfacl -R -m u:john:rx /shared
```

Sources: LINUX COMMAND

1.16.402 setfacl -R -m d:u:alice:rwX dir

default ACL recursive. **CAUTION**

```
setfacl -R -m d:u:alice:rwX dir
```

Sources: komendy

1.16.403 setfacl -b

Usuwa wszystkie ACL

```
setfacl -b file.txt
```

Sources: LINUX COMMAND

1.16.404 `setfacl -b file`

remove all ACLs. **CAUTION**

```
setfacl -b file
```

Sources: komendy

1.16.405 `setfacl -b file # usuń wszystkie`

ACL (Access Control Lists)

```
setfacl -b file # usuń wszystkie
```

Sources: linux_engineering_book.md, linux_engineering_book.pdf

1.16.406 `setfacl -d`

Domyślne ACL (katalogi)

```
setfacl -d -m u:john:rx /shared
```

Sources: LINUX COMMAND

1.16.407 `setfacl -d -m g:sec_auditors:rX /var/log/audit/`

Ustawienie domyślnej maski ACL (Inheritance), aby każdy nowy plik automatycznie dziedziczył dostęp

```
setfacl -d -m g:sec_auditors:rX /var/log/audit/
```

Sources: encyklopedia_inzynierii_linux.html, encyklopedia_inzynierii_linux.pdf

1.16.408 `setfacl -m`

Modyfikuje ACL

```
setfacl -m g:developers:rwx dir
```

Sources: LINUX COMMAND

1.16.409 `setfacl -m g:devs:rx file #`

ACL (Access Control Lists)

```
setfacl -m g:devs:rx file #
```

Sources: linux_engineering_book.md, linux_engineering_book.pdf

1.16.410 `setfacl -m g:sec_auditors:rwX /var/log/audit/`

Nadanie uprawnień odczytu i zapisu dla konkretnej grupy audytorów do katalogu logów

```
setfacl -m g:sec_auditors:rwX /var/log/audit/
```

Sources: encyklopedia_inzynierii_linux.html, encyklopedia_inzynierii_linux.pdf

1.16.411 `setfacl -m u:alice:rw file`

grant user read-write. **CAUTION**

```
setfacl -m u:alice:rw file
```

Sources: komendy

1.16.412 `setfacl -m u:user1:rwx file # dodaj`

ACL (Access Control Lists)

`setfacl -m u:user1:rwx file # dodaj`

Sources: linux_engineering_book.md, linux_engineering_book.pdf

1.16.413 `setfacl -x`

Usuwa wpis ACL

`setfacl -x u:john file.txt`

Sources: LINUX COMMAND

1.16.414 `setsebool`

Set boolean (temp)

`setsebool httpd_enable_homedirs on`

`setsebool httpd_enable_homedirs 1`

Sources: LINUX COMMAND

1.16.415 `setsebool -P`

Persist boolean

`setsebool -P httpd_enable_homedirs on`

`setsebool -P httpd_enable_homedirs 1`

Sources: LINUX COMMAND

1.16.416 `setsebool -P httpd_can_network_connect on # boolean`

9.3 SELinux

`setsebool -P httpd_can_network_connect on # boolean`

Sources: linux_engineering_book.md, linux_engineering_book.pdf

1.16.417 `setsebool -P httpd_enable_homedirs on`

persistent boolean. **CAUTION**

`setsebool -P httpd_enable_homedirs on`

Sources: komendy

1.16.418 `ssh`

Zdalna powłoka

`ssh user@server`

Sources: LINUX COMMAND

1.16.419 `ssh $host "uptime" &`

28.1 Bash — Zaawansowane Techniki

`ssh $host "uptime" &`

Sources: linux_engineering_book.md, linux_engineering_book.pdf

1.16.420 `ssh -A user@host`

Sekcja Komend

`ssh -A user@host`

Sources: RHCSA_Command_Library.pdf

1.16.421 `ssh -D`

Dynamiczny SOCKS proxy

`ssh -D 1080 user@server`

Sources: LINUX COMMAND

1.16.422 `ssh -D 1080`

Sekcja Komend

`ssh -D 1080`

Sources: RHCSA_Command_Library.pdf

1.16.423 `ssh -D 1080 user@host`

dynamic SOCKS proxy. **CAUTION**

`ssh -D 1080 user@host`

Sources: komendy

1.16.424 `ssh -J bastion user@internal`

jump host. **SAFE**

`ssh -J bastion user@internal`

Sources: komendy

1.16.425 `ssh -J jumphost`

Sekcja Komend

`ssh -J jumphost`

Sources: RHCSA_Command_Library.pdf

1.16.426 `ssh -L`

Lokalny port forwarding

`ssh -L 8080:localhost:80 user@server`

`ssh -L`

Sources: LINUX COMMAND, RHCSA_Command_Library.pdf

1.16.427 `ssh -L 8080:localhost:80 user@host`

local port forward. **CAUTION**

`ssh -L 8080:localhost:80 user@host`

Sources: komendy

1.16.428 `ssh -N -f -L`

Sekcja Komend

`ssh -N -f -L`

Sources: RHCSA_Command_Library.pdf

1.16.429 `ssh -R`

Zdalny port forwarding

`ssh -R 9090:localhost:3000 user@server`

`ssh -R`

Sources: LINUX COMMAND, RHCSA_Command_Library.pdf

1.16.430 `ssh -R 9090:localhost:3000 user@host`

remote port forward. **CAUTION**

`ssh -R 9090:localhost:3000 user@host`

Sources: komendy

1.16.431 `ssh -X`

X11 forwarding

`ssh -X user@server`

Sources: LINUX COMMAND

1.16.432 `ssh -X user@host`

Sekcja Komend

`ssh -X user@host`

Sources: RHCSA_Command_Library.pdf

1.16.433 `ssh -i`

Klucz prywatny

`ssh -i ~/.ssh/id_rsa user@server`

`ssh -i`

Sources: LINUX COMMAND, RHCSA_Command_Library.pdf

1.16.434 `ssh -i ~/.ssh/id_ed25519 user@host`

specific key. **SAFE**

`ssh -i ~/.ssh/id_ed25519 user@host`

Sources: komendy

1.16.435 `ssh -o`

bypass host key check. **DANGEROUS**

`ssh -o`

Sources: LINUX COMMAND, RHCSA_Command_Library.pdf, komendy

1.16.436 `ssh -p`

Inny port

`ssh -p 2222 user@server`

Sources: LINUX COMMAND

1.16.437 `ssh -p 2222`

Sekcja Komend

`ssh -p 2222`

Sources: RHCSA_Command_Library.pdf

1.16.438 `ssh -p 2222 user@host`

alternate port. **SAFE**

`ssh -p 2222 user@host`

Sources: komendy

1.16.439 `ssh -t user@host`

Sekcja Komend

`ssh -t user@host`

Sources: RHCSA_Command_Library.pdf

1.16.440 `ssh -v user@host`

Sekcja Komend

`ssh -v user@host`

Sources: RHCSA_Command_Library.pdf

1.16.441 `ssh -vvv user@host`

Sekcja Komend

`ssh -vvv user@host`

Sources: RHCSA_Command_Library.pdf

1.16.442 `ssh user@host 'cat >`

Sekcja Komend

`ssh user@host 'cat >`

Sources: RHCSA_Command_Library.pdf

1.16.443 `ssh user@hostname`

Sekcja Komend

`ssh user@hostname`

Sources: RHCSA_Command_Library.pdf

1.16.444 `systemctl`

Uruchomienie kontenera z pełnymi uprawnieniami roota hosta i dostępem do surowego systemu

`systemctl`

Sources: LINUX COMMAND, RHCSA_Command_Library.pdf, encyklopedia_inzynierii_linux.html, encyklopedia_inzynierii_linux.pdf, komendy, linux_engineering_book.md, linux_engineering_book.pdf

1.16.445 `systemctl --failed`

failed units. **SAFE**

`systemctl --failed`

Sources: komendy

1.16.446 `systemctl --user enable --now web.service`

Podman — drop-in replacement dla Docker (bez daemona)

`systemctl --user enable --now web.service`

Sources: linux_engineering_book.md, linux_engineering_book.pdf

1.16.447 `systemctl cat`

Wyświetla definicję usługi

`systemctl cat httpd`

`systemctl cat`

Sources: LINUX COMMAND, RHCSA_Command_Library.pdf

1.16.448 `systemctl cat nginx`

show unit file. **SAFE**

`systemctl cat nginx`

Sources: komendy

1.16.449 `systemctl daemon-reload`

Przeładowanie konfiguracji menedżera systemd i uruchomienie nowej usługi bezpiecznej

`systemctl daemon-reload`

Sources: LINUX COMMAND, encyklopedia_inzynierii_linux.html, encyklopedia_inzynierii_linux.pdf, komendy

1.16.450 `systemctl daemon-reload # przeładuj konfigurację`

Zarządzanie usługami

`systemctl daemon-reload # przeładuj konfigurację`

Sources: linux_engineering_book.md, linux_engineering_book.pdf

1.16.451 `systemctl disable`

Wyłącza autostart

`systemctl disable httpd`

`systemctl disable`

Sources: LINUX COMMAND, RHCSA_Command_Library.pdf

1.16.452 `systemctl disable nginx`

disable autostart. **CAUTION**

```
systemctl disable nginx
```

Sources: komendy

1.16.453 `systemctl edit`

Edytuje override usługi

```
systemctl edit --full httpd
```

```
systemctl edit
```

Sources: LINUX COMMAND, RHCSA_Command_Library.pdf, komendy

1.16.454 `systemctl edit --full nginx`

override unit. **CAUTION**

```
systemctl edit --full nginx
```

Sources: komendy

1.16.455 `'systemctl edit'`

Operational Notes

```
systemctl edit`
```

Sources: komendy

1.16.456 `systemctl emergency`

emergency shell. **DANGEROUS**

```
systemctl emergency
```

Sources: komendy

1.16.457 `systemctl enable`

Włącza autostart

```
systemctl enable httpd
```

```
systemctl enable
```

Sources: LINUX COMMAND, RHCSA_Command_Library.pdf

1.16.458 `systemctl enable --now secure_processor.service`

Przeładowanie konfiguracji menedżera systemd i uruchomienie nowej usługi bezpiecznej

```
systemctl enable --now secure_processor.service
```

Sources: encyklopedia_inzynierii_linux.html, encyklopedia_inzynierii_linux.pdf

1.16.459 `systemctl enable --now systemd-networkd`

17.4 Networkd i Resolved

```
systemctl enable --now systemd-networkd
```

Sources: linux_engineering_book.md, linux_engineering_book.pdf

1.16.460 `systemctl enable backup.timer`

enable timer. **CAUTION**

`systemctl enable backup.timer`

Sources: komendy

1.16.461 `systemctl enable kdump`

15.2 Diagnostyka Jądra

`systemctl enable kdump`

Sources: linux_engineering_book.md, linux_engineering_book.pdf

1.16.462 `systemctl enable nginx`

autostart on boot. **CAUTION**

`systemctl enable nginx`

Sources: komendy

1.16.463 `systemctl enable wg-quick@wg0`

23.3 WireGuard VPN

`systemctl enable wg-quick@wg0`

Sources: linux_engineering_book.md, linux_engineering_book.pdf

1.16.464 `systemctl enable/disable nginx # autostart`

Zarządzanie usługami

`systemctl enable/disable nginx # autostart`

Sources: linux_engineering_book.md, linux_engineering_book.pdf

1.16.465 `systemctl get-default`

current target. **SAFE**

`systemctl get-default`

Sources: komendy

1.16.466 `systemctl halt`

Sekcja Komend

`systemctl halt`

Sources: RHCSA_Command_Library.pdf

1.16.467 `systemctl hibernate`

Sekcja Komend

`systemctl hibernate`

Sources: RHCSA_Command_Library.pdf

1.16.468 `systemctl is-active`

Czy usługa jest aktywna?

```
systemctl is-active sshd
systemctl is-active
```

Sources: LINUX COMMAND, RHCSA_Command_Library.pdf, komendy

1.16.469 `systemctl is-active nginx`

boolean active check. **SAFE**

```
systemctl is-active nginx
```

Sources: komendy

1.16.470 `systemctl is-active nginx # czy działa`

Zarządzanie usługami

```
systemctl is-active nginx # czy działa
```

Sources: linux_engineering_book.md, linux_engineering_book.pdf

1.16.471 `systemctl is-active before restart`

Idempotency Checks

```
systemctl is-active before restart
```

Sources: komendy

1.16.472 `systemctl is-enabled`

Czy włączona do autostartu?

```
systemctl is-enabled httpd
systemctl is-enabled
```

Sources: LINUX COMMAND, RHCSA_Command_Library.pdf

1.16.473 `systemctl is-enabled nginx`

boolean enabled check. **SAFE**

```
systemctl is-enabled nginx
```

Sources: komendy

1.16.474 `systemctl is-enabled nginx # czy autostart`

Zarządzanie usługami

```
systemctl is-enabled nginx # czy autostart
```

Sources: linux_engineering_book.md, linux_engineering_book.pdf

1.16.475 `systemctl is-failed`

Czy usługa jest w stanie failed?

```
systemctl is-failed httpd
systemctl is-failed
```

Sources: LINUX COMMAND, RHCSA_Command_Library.pdf

1.16.476 `systemctl isolate`

Przechodzi do targetu

```
systemctl isolate multi-user.target
systemctl isolate
```

Sources: LINUX COMMAND, RHCSA_Command_Library.pdf

1.16.477 `systemctl list-timers`

active timers. **SAFE**

```
systemctl list-timers --all
systemctl list-timers
```

Sources: LINUX COMMAND, komendy

1.16.478 `systemctl list-timers --all`

scheduled timers. **SAFE**

```
systemctl list-timers --all
```

Sources: komendy

1.16.479 `systemctl list-unit-files`

Lista plików jednostek

```
systemctl list-unit-files
```

Sources: LINUX COMMAND

1.16.480 `systemctl list-unit-files #`

Zarządzanie usługami

```
systemctl list-unit-files #
```

Sources: linux_engineering_book.md, linux_engineering_book.pdf

1.16.481 `systemctl list-unit-files --state=enabled`

enabled units. **SAFE**

```
systemctl list-unit-files --state=enabled
```

Sources: komendy

1.16.482 `systemctl list-units`

Lista aktywnych jednostek

```
systemctl list-units --type=service
systemctl list-units
```

Sources: LINUX COMMAND, RHCSA_Command_Library.pdf

1.16.483 `systemctl list-units --failed`

failed boot units. **SAFE**

```
systemctl list-units --failed
```

Sources: komendy

1.16.484 `systemctl list-units --type=service #`

Zarządzanie usługami

`systemctl list-units --type=service #`

Sources: linux_engineering_book.md, linux_engineering_book.pdf

1.16.485 `systemctl list-units --type=service --state=running`

active services. **SAFE**

`systemctl list-units --type=service --state=running`

Sources: komendy

1.16.486 `systemctl mask`

Maskuje usługę (uniemożliwia start)

`systemctl mask cups`

`systemctl mask`

Sources: LINUX_COMMAND, RHCSA_Command_Library.pdf

1.16.487 `systemctl mask nginx`

prevent start entirely. **CAUTION**

`systemctl mask nginx`

Sources: komendy

1.16.488 `systemctl poweroff`

Sekcja Komend

`systemctl poweroff`

Sources: RHCSA_Command_Library.pdf

1.16.489 `systemctl reboot`

Sekcja Komend

`systemctl reboot`

Sources: RHCSA_Command_Library.pdf

1.16.490 `systemctl rescue`

rescue target. **DANGEROUS**

`systemctl rescue`

Sources: komendy

1.16.491 `systemctl set-default multi-user.target`

CLI boot. **CAUTION**

`systemctl set-default multi-user.target`

Sources: komendy

1.16.492 `systemctl set-property myservice.service`

Systemd slice (automatyczne cgroups)

```
systemctl set-property myservice.service
```

Sources: linux_engineering_book.md, linux_engineering_book.pdf

1.16.493 `systemctl start/stop/restart/reload nginx`

Zarządzanie usługami

```
systemctl start/stop/restart/reload nginx
```

Sources: linux_engineering_book.md, linux_engineering_book.pdf

1.16.494 `systemctl suspend`

Sekcja Komend

```
systemctl suspend
```

Sources: RHCSA_Command_Library.pdf

1.16.495 `systemctl unmask`

Sekcja Komend

```
systemctl unmask cups
```

```
systemctl unmask
```

Sources: LINUX COMMAND, RHCSA_Command_Library.pdf

1.16.496 `systemctl unmask nginx`

restore. **CAUTION**

```
systemctl unmask nginx
```

Sources: komendy

1.16.497 `tar`

Sekcja Komend

```
tar
```

Sources: RHCSA_Command_Library.pdf

1.16.498 `tar --delete -f`

Sekcja Komend

```
tar --delete -f
```

Sources: RHCSA_Command_Library.pdf

1.16.499 `tar --exclude-vcs`

Sekcja Komend

```
tar --exclude-vcs
```

Sources: RHCSA_Command_Library.pdf

1.16.500 `tar --listed-incremental=snapshot.snar -czf inc.tar.gz /data # incremental`

19.1 Narzędzia Backup

`tar --listed-incremental=snapshot.snar -czf inc.tar.gz /data # incremental`

Sources: linux_engineering_book.md, linux_engineering_book.pdf

1.16.501 `tar -C`

Rozpakowuje do katalogu

`tar -xvf backup.tar -C /tmp`

Sources: LINUX COMMAND

1.16.502 `tar -cjvf`

Tworzy archiwum bzip2

`tar -cjvf backup.tar.bz2 /etc`

Sources: LINUX COMMAND

1.16.503 `tar -cvJf`

Sekcja Komend

`tar -cvJf`

Sources: RHCSA_Command_Library.pdf

1.16.504 `tar -cvf`

Tworzy archiwum tar

`tar -cvf backup.tar /etc`

Sources: LINUX COMMAND

1.16.505 `tar -cvf archive.tar`

Sekcja Komend

`tar -cvf archive.tar`

Sources: RHCSA_Command_Library.pdf

1.16.506 `tar -cvjf`

Sekcja Komend

`tar -cvjf`

Sources: RHCSA_Command_Library.pdf

1.16.507 `tar -cvzf`

Sekcja Komend

`tar -cvzf`

Sources: RHCSA_Command_Library.pdf

1.16.508 `tar -czf - /path`

Archiwum przez SSH unzip -o archive.zip Nadpisz istniejące pliki

`tar -czf - /path`

Sources: RHCSA_Command_Library.pdf

1.16.509 `tar -czf arch.tar.gz`

Sekcja Komend

`tar -czf arch.tar.gz`

Sources: RHCSA_Command_Library.pdf

1.16.510 `tar -czvf`

Tworzy archiwum gzip

`tar -czvf backup.tar.gz /etc`

Sources: LINUX COMMAND

1.16.511 `tar -rvf archive.tar`

Sekcja Komend

`tar -rvf archive.tar`

Sources: RHCSA_Command_Library.pdf

1.16.512 `tar -tvf archive.tar`

Sekcja Komend

`tar -tvf archive.tar`

Sources: RHCSA_Command_Library.pdf

1.16.513 `tar -tvzf`

Sekcja Komend

`tar -tvzf`

Sources: RHCSA_Command_Library.pdf

1.16.514 `tar -uvf archive.tar`

Sekcja Komend

`tar -uvf archive.tar`

Sources: RHCSA_Command_Library.pdf

1.16.515 `tar -xvJf`

Sekcja Komend

`tar -xvJf`

Sources: RHCSA_Command_Library.pdf

1.16.516 `tar -xvf`

Rozpakowuje archiwum tar

`tar -xvf backup.tar`

Sources: LINUX COMMAND

1.16.517 `tar -xvf archive.tar`

Sekcja Komend

`tar -xvf archive.tar`

Sources: RHCSA_Command_Library.pdf

1.16.518 `tar -xvjf`

Sekcja Komend

`tar -xvjf`

Sources: RHCSA_Command_Library.pdf

1.16.519 `tar -xvzf`

Sekcja Komend

`tar -xvzf`

Sources: RHCSA_Command_Library.pdf

1.16.520 `tar -xzvf`

Rozpakowuje gzip tar

`tar -xzvf backup.tar.gz`

Sources: LINUX COMMAND

1.16.521 `tar czf /backup/$(hostname)-$(date +%Y%m%d-%H%M%S).tar.gz /important/data`

Dodatek A: Przydatne Jednowierszowce

`tar czf /backup/$(hostname)-$(date +%Y%m%d-%H%M%S).tar.gz /important/data`

Sources: linux_engineering_book.md, linux_engineering_book.pdf

1.16.522 `tar czf backup-$(date +%Y%m%d).tar.gz /data # z data`

19.1 Narzędzia Backup

`tar czf backup-$(date +%Y%m%d).tar.gz /data # z data`

Sources: linux_engineering_book.md, linux_engineering_book.pdf

1.16.523 `tar czf backup.tar.gz /data # tworzenie`

19.1 Narzędzia Backup

`tar czf backup.tar.gz /data # tworzenie`

Sources: linux_engineering_book.md, linux_engineering_book.pdf

1.16.524 `tar xf linux-6.8.tar.xz`

16.1 Pobieranie i Konfiguracja

`tar xf linux-6.8.tar.xz`

Sources: linux_engineering_book.md, linux_engineering_book.pdf

1.16.525 `tar xzf backup.tar.gz -C /restore # rozpakowywanie`

19.1 Narzędzia Backup

`tar xzf backup.tar.gz -C /restore # rozpakowywanie`

Sources: linux_engineering_book.md, linux_engineering_book.pdf

1.16.526 `tar xzvf backup.tar.gz -C /restore`

extract backup. **CAUTION**

`tar xzvf backup.tar.gz -C /restore`

Sources: komendy

1.16.527 `useradd`

Tworzy użytkownika

`useradd -m -s /bin/bash john`

Sources: LINUX COMMAND

1.16.528 `useradd -G`

Dodatkowe grupy

`useradd -G wheel john`

Sources: LINUX COMMAND

1.16.529 `useradd -G g1,g2`

Sekcja Komend

`useradd -G g1,g2`

Sources: RHCSA_Command_Library.pdf

1.16.530 `useradd -M user`

Sekcja Komend

`useradd -M user`

Sources: RHCSA_Command_Library.pdf

1.16.531 `useradd -N user`

Sekcja Komend

`useradd -N user`

Sources: RHCSA_Command_Library.pdf

1.16.532 `useradd -c 'Imi`

Sekcja Komend

`useradd -c 'Imi`

Sources: RHCSA_Command_Library.pdf

1.16.533 `useradd -d`

Sekcja Komend

`useradd -d`

Sources: RHCSA_Command_Library.pdf

1.16.534 `useradd -e`

Sekcja Komend

`useradd -e`

Sources: RHCSA_Command_Library.pdf

1.16.535 `useradd -f 30 user`

Sekcja Komend

`useradd -f 30 user`

Sources: RHCSA_Command_Library.pdf

1.16.536 `useradd -g group`

Sekcja Komend

`useradd -g group`

Sources: RHCSA_Command_Library.pdf

1.16.537 `useradd -m -s`

Sekcja Komend

`useradd -m -s`

Sources: RHCSA_Command_Library.pdf

1.16.538 `useradd -m -s /bin/bash -G sudo,docker user1 # dodaj użytkownika`

`/etc/modprobe.d/bonding.conf`

`useradd -m -s /bin/bash -G sudo,docker user1 # dodaj użytkownika`

Sources: linux_engineering_book.md, linux_engineering_book.pdf

1.16.539 `useradd -m -s /bin/bash -U alice`

create user with home and group. **SAFE**

`useradd -m -s /bin/bash -U alice`

Sources: komendy

1.16.540 `useradd -m username`

Sekcja Komend

`useradd -m username`

Sources: RHCSA_Command_Library.pdf

1.16.541 `useradd -r -s /usr/sbin/nologin svc_name`

Separate service accounts: .

`useradd -r -s /usr/sbin/nologin svc_name`

Sources: komendy

1.16.542 `useradd -r sysuser`

Sekcja Komend

`useradd -r sysuser`

Sources: RHCSA_Command_Library.pdf

1.16.543 `useradd -s`

Określa powłokę

`useradd -s /sbin/nologin service`

Sources: LINUX COMMAND

1.16.544 `useradd -u 1500 user`

Sekcja Komend

`useradd -u 1500 user`

Sources: RHCSA_Command_Library.pdf

1.16.545 `useradd username`

Sekcja Komend

`useradd username`

Sources: RHCSA_Command_Library.pdf

1.16.546 `usermod`

Modyfikuje użytkownika

`usermod -aG wheel john`

Sources: LINUX COMMAND

1.16.547 `usermod -G g1,g2`

Sekcja Komend

`usermod -G g1,g2`

Sources: RHCSA_Command_Library.pdf

1.16.548 usermod -L

Blokuje konto (zablokuj hasło)

```
usermod -L john
```

Sources: LINUX COMMAND

1.16.549 usermod -L user

Sekcja Komend

```
usermod -L user
```

Sources: RHCSA_Command_Library.pdf

1.16.550 usermod -U

Odblokowuje konto

```
usermod -U john
```

Sources: LINUX COMMAND

1.16.551 usermod -U user

Sekcja Komend

```
usermod -U user
```

Sources: RHCSA_Command_Library.pdf

1.16.552 usermod -aG

Dodaje do grupy (append)

```
usermod -aG docker john
```

Sources: LINUX COMMAND

1.16.553 usermod -aG group

Sekcja Komend

```
usermod -aG group
```

Sources: RHCSA_Command_Library.pdf

1.16.554 usermod -aG sudo alice

add to group. **CAUTION**

```
usermod -aG sudo alice
```

Sources: komendy

1.16.555 usermod -aG wheel user1 # dodaj do grupy

/etc/modprobe.d/bonding.conf

```
usermod -aG wheel user1 # dodaj do grupy
```

Sources: linux_engineering_book.md, linux_engineering_book.pdf

1.16.556 `usermod -c 'Nowy`

Sekcja Komend

`usermod -c 'Nowy`

Sources: RHCSA_Command_Library.pdf

1.16.557 `usermod -d /new/home`

Sekcja Komend

`usermod -d /new/home`

Sources: RHCSA_Command_Library.pdf

1.16.558 `usermod -e`

Sekcja Komend

`usermod -e`

Sources: RHCSA_Command_Library.pdf

1.16.559 `usermod -e '' user`

Sekcja Komend

`usermod -e '' user`

Sources: RHCSA_Command_Library.pdf

1.16.560 `usermod -g group`

Sekcja Komend

`usermod -g group`

Sources: RHCSA_Command_Library.pdf

1.16.561 `usermod -l newname`

Sekcja Komend

`usermod -l newname`

Sources: RHCSA_Command_Library.pdf

1.16.562 `usermod -s /bin/zsh`

Sekcja Komend

`usermod -s /bin/zsh`

Sources: RHCSA_Command_Library.pdf

1.16.563 `usermod -u 1600 user`

Sekcja Komend

`usermod -u 1600 user`

Sources: RHCSA_Command_Library.pdf

1.16.564 `vgcreate`

Tworzy Volume Group

```
vgcreate vg_data /dev/sdb1 /dev/sdc1
```

Sources: LINUX COMMAND

1.16.565 `vgcreate -s 16M`

Sekcja Komend

```
vgcreate -s 16M
```

Sources: RHCSA_Command_Library.pdf

1.16.566 `vgcreate vg0 /dev/sda1 /dev/sdb1`

create VG. **CAUTION**

```
vgcreate vg0 /dev/sda1 /dev/sdb1
```

Sources: komendy

1.16.567 `vgcreate vg_data /dev/sdb /dev/sdc`

6.4 LVM (Logical Volume Manager)

```
vgcreate vg_data /dev/sdb /dev/sdc
```

Sources: linux_engineering_book.md, linux_engineering_book.pdf

1.16.568 `vgcreate vgname`

Sekcja Komend

```
vgcreate vgname
```

1.17 Sources: RHCSA_Command_Library.pdf

1.18 Gemini Expansion Additions

These entries were appended without shrinking or rewriting the original master library.

1.18.1 `ctr`

- Command name: `ctr`
- Short description: CLI for containerd.
- Category: Automation
- Difficulty level: Advanced / Hard
- Ecosystem tags: Kubernetes; containerd; V. Cloud-Native & Low-Level Container Ecosystem
- Enterprise relevance: Raw interaction with containerd when Kubelet is dead or malfunctioning.
- Related commands: None detected in source merge
- Rarity score: Advanced / Hard

```
ctr
```

Sources: GEMINI_EXPANSION

1.18.2 `k9s`

- Command name: `k9s`
- Short description: Terminal UI to interact with Kubernetes.
- Category: Automation
- Difficulty level: Advanced / Medium
- Ecosystem tags: K8s; DevOps; V. Cloud-Native & Low-Level Container Ecosystem

- Enterprise relevance: Drastically speeds up navigating multi-cluster environments compared to raw kubectl.
- Related commands: None detected in source merge
- Rarity score: Advanced / Medium

k9s

Sources: GEMINI_EXPANSION

1.18.3 **nerdctl**

- Command name: **nerdctl**
- Short description: Docker-compatible CLI for containerd.
- Category: Automation
- Difficulty level: Advanced / Medium
- Ecosystem tags: Containerd; V. Cloud-Native & Low-Level Container Ecosystem
- Enterprise relevance: Used in modern rootless environments where Docker is deprecated but Docker-like syntax is required.
- Related commands: None detected in source merge
- Rarity score: Advanced / Medium

nerdctl

Sources: GEMINI_EXPANSION

1.18.4 **nsenter**

- Command name: **nsenter**
- Short description: Run program with namespaces of other processes.
- Category: Automation
- Difficulty level: Rare / Expert
- Ecosystem tags: Containers; Deep Debugging; V. Cloud-Native & Low-Level Container Ecosystem
- Enterprise relevance: The ultimate “break-glass” tool. Allows you to inject a shell into a container that has no shell binary installed.
- Related commands: None detected in source merge
- Rarity score: Rare / Expert

nsenter

Sources: GEMINI_EXPANSION

1.18.5 **ply**

- Command name: **ply**
- Short description: Light-weight dynamic tracer for Linux.
- Category: Automation
- Difficulty level: Ultra-rare / Hard
- Ecosystem tags: Embedded Linux; I. eBPF, Kernel Observability & Performance Telemetry
- Enterprise relevance: eBPF-based tracer designed specifically for embedded devices with limited memory.
- Related commands: None detected in source merge
- Rarity score: Ultra-rare / Hard

ply

Sources: GEMINI_EXPANSION

1.18.6 **systemtap**

- Command name: **systemtap**
- Short description: Scripting language and tool for dynamic kernel instrumentation.
- Category: Logs
- Difficulty level: Rare / Expert
- Ecosystem tags: Red Hat; SUSE; I. eBPF, Kernel Observability & Performance Telemetry

- Enterprise relevance: A legacy predecessor to eBPF, still heavily used in RHEL enterprise environments for live patching analysis.
- Related commands: None detected in source merge
- Rarity score: Rare / Expert

systemtap

Sources: GEMINI_EXPANSION

1.18.7 trace-cmd

- Command name: **trace-cmd**
- Short description: Command-line frontend for Ftrace.
- Category: Logs
- Difficulty level: Obscure / Expert
- Ecosystem tags: Kernel; I. eBPF, Kernel Observability & Performance Telemetry
- Enterprise relevance: Allows deep-dive latency tracking and capturing of context switches at the microsecond level.
- Related commands: None detected in source merge
- Rarity score: Obscure / Expert

trace-cmd

Sources: GEMINI_EXPANSION

1.18.8 cilium

- Command name: **cilium**
- Short description: CLI to inspect Cilium eBPF network state.
- Category: Networking
- Difficulty level: Ultra-rare / Expert
- Ecosystem tags: K8s; CNI; V. Cloud-Native & Low-Level Container Ecosystem
- Enterprise relevance: Required for debugging modern eBPF-based Kubernetes network policies and routing.
- Related commands: None detected in source merge
- Rarity score: Ultra-rare / Expert

cilium

Sources: GEMINI_EXPANSION

1.18.9 rdma

- Command name: **rdma**
- Short description: RDMA tool for InfiniBand and RoCE.
- Category: Networking
- Difficulty level: Ultra-rare / Expert
- Ecosystem tags: HPC; Networking; III. Hardware, Firmware & Bare-Metal Orchestration
- Enterprise relevance: Essential for zero-copy networking in ultra-low latency trading or supercomputing environments.
- Related commands: None detected in source merge
- Rarity score: Ultra-rare / Expert

rdma

Sources: GEMINI_EXPANSION

1.18.10 numactl

- Command name: **numactl**
- Short description: Controls NUMA policy for processes.
- Category: Processes
- Difficulty level: Rare / Expert
- Ecosystem tags: HPC; Database Admin; III. Hardware, Firmware & Bare-Metal Orchestration

- Enterprise relevance: Pins database memory/CPU's to specific physical sockets to drastically reduce memory latency.
- Related commands: None detected in source merge
- Rarity score: Rare / Expert

numactl

Sources: GEMINI_EXPANSION

1.18.11 perf

- Command name: **perf**
- Short description: Linux profiling with performance counters.
- Category: Processes
- Difficulty level: Advanced / Hard
- Ecosystem tags: Kernel; HPC; I. eBPF, Kernel Observability & Performance Telemetry
- Enterprise relevance: The absolute gold standard for CPU profiling, flame graph generation, and cache-miss analysis.
- Related commands: None detected in source merge
- Rarity score: Advanced / Hard

perf

Sources: GEMINI_EXPANSION

1.18.12 binwalk

- Command name: **binwalk**
- Short description: Firmware analysis tool.
- Category: Security
- Difficulty level: Rare / Hard
- Ecosystem tags: Forensics; Embedded; IV. DFIR (Digital Forensics & Incident Response)
- Enterprise relevance: Scans firmware images for embedded files and executable code (essential for IoT hacking).
- Related commands: None detected in source merge
- Rarity score: Rare / Hard

binwalk

Sources: GEMINI_EXPANSION

1.18.13 bpftool

- Command name: **bpftool**
- Short description: Inspection and manipulation of eBPF programs and maps.
- Category: Security
- Difficulty level: Ultra-rare / Expert
- Ecosystem tags: eBPF; Kernel; I. eBPF, Kernel Observability & Performance Telemetry
- Enterprise relevance: Essential for debugging eBPF pipelines (like Cilium) at the raw kernel level.
- Related commands: None detected in source merge
- Rarity score: Ultra-rare / Expert

bpftool

Sources: GEMINI_EXPANSION

1.18.14 bpftrace

- Command name: **bpftrace**
- Short description: High-level tracing language for Linux eBPF.
- Category: Security
- Difficulty level: Ultra-rare / Expert
- Ecosystem tags: eBPF; Kernel; SRE; I. eBPF, Kernel Observability & Performance Telemetry

- Enterprise relevance: Replaces complex kernel modules; allows real-time tracing of specific kernel functions without recompilation.
- Related commands: None detected in source merge
- Rarity score: Ultra-rare / Expert

bpftrace

Sources: GEMINI_EXPANSION

1.18.15 foremost

- Command name: **foremost**
- Short description: Console program to recover files based on headers/footers.
- Category: Security
- Difficulty level: Obscure / Hard
- Ecosystem tags: DFIR; Recovery; IV. DFIR (Digital Forensics & Incident Response)
- Enterprise relevance: Carves deleted files directly out of raw disk space without relying on the filesystem index.
- Related commands: None detected in source merge
- Rarity score: Obscure / Hard

foremost

Sources: GEMINI_EXPANSION

1.18.16 volatility

- Command name: **volatility**
- Short description: Advanced memory forensics framework.
- Category: Security
- Difficulty level: Rare / Expert
- Ecosystem tags: DFIR; Security; IV. DFIR (Digital Forensics & Incident Response)
- Enterprise relevance: Extracts passwords, bash histories, and rootkits directly from raw RAM dumps of compromised machines.
- Related commands: None detected in source merge
- Rarity score: Rare / Expert

volatility

Sources: GEMINI_EXPANSION

1.18.17 btrfs-balance

- Command name: **btrfs-balance**
- Short description: Balances block groups across devices.
- Category: Storage
- Difficulty level: Advanced / Hard
- Ecosystem tags: Btrfs; SUSE; II. Advanced Storage Orchestration (ZFS & Btrfs)
- Enterprise relevance: Required maintenance for multi-device Btrfs arrays to prevent false “disk full” scenarios.
- Related commands: None detected in source merge
- Rarity score: Advanced / Hard

btrfs-balance

Sources: GEMINI_EXPANSION

1.18.18 btrfs-scrub

- Command name: **btrfs-scrub**
- Short description: Reads all data and metadata on a Btrfs filesystem to verify checksums.
- Category: Storage
- Difficulty level: Advanced / Medium
- Ecosystem tags: Btrfs; SUSE; II. Advanced Storage Orchestration (ZFS & Btrfs)
- Enterprise relevance: Critical for detecting and automatically repairing silent data corruption (bit rot).

- Related commands: None detected in source merge
- Rarity score: Advanced / Medium

btrfs-scrub

Sources: GEMINI_EXPANSION

1.18.19 ddrescue

- Command name: **ddrescue**
- Short description: Data recovery tool for failing block devices.
- Category: Storage
- Difficulty level: Advanced / Medium
- Ecosystem tags: Recovery; IV. DFIR (Digital Forensics & Incident Response)
- Enterprise relevance: Succeeds where standard dd fails; intelligently skips bad sectors to salvage maximum data from dying disks.
- Related commands: None detected in source merge
- Rarity score: Advanced / Medium

ddrescue

Sources: GEMINI_EXPANSION

1.18.20 dmidecode

- Command name: **dmidecode**
- Short description: Dumps computer's DMI (SMBIOS) table.
- Category: Storage
- Difficulty level: Common / Easy
- Ecosystem tags: Sysadmin; Hardware; III. Hardware, Firmware & Bare-Metal Orchestration
- Enterprise relevance: The quickest way to extract physical RAM serial numbers, BIOS versions, and CPU stepping data.
- Related commands: None detected in source merge
- Rarity score: Common / Easy

dmidecode

Sources: GEMINI_EXPANSION

1.18.21 ipmitool

- Command name: **ipmitool**
- Short description: Utility for IPMI control.
- Category: Storage
- Difficulty level: Rare / Hard
- Ecosystem tags: Bare-Metal; Data Center; III. Hardware, Firmware & Bare-Metal Orchestration
- Enterprise relevance: Allows out-of-band management: rebooting frozen servers, reading hardware sensors remotely.
- Related commands: None detected in source merge
- Rarity score: Rare / Hard

ipmitool

Sources: GEMINI_EXPANSION

1.18.22 nvme-cli

- Command name: **nvme-cli**
- Short description: NVM-Express user space tooling.
- Category: Storage
- Difficulty level: Advanced / Medium
- Ecosystem tags: Storage; Bare-Metal; III. Hardware, Firmware & Bare-Metal Orchestration
- Enterprise relevance: Manages NVMe namespaces, formats drives securely, and reads raw flash controller health data.

- Related commands: None detected in source merge
- Rarity score: Advanced / Medium

nvme-cli

Sources: GEMINI_EXPANSION

1.18.23 zdb

- Command name: **zdb**
- Short description: ZFS debugger.
- Category: Storage
- Difficulty level: Ultra-rare / Expert
- Ecosystem tags: ZFS; Forensics; II. Advanced Storage Orchestration (ZFS & Btrfs)
- Enterprise relevance: Inspects raw ZFS structures; used only when a pool is completely corrupted and standard tools fail.
- Related commands: None detected in source merge
- Rarity score: Ultra-rare / Expert

zdb

Sources: GEMINI_EXPANSION

1.18.24 zfs

- Command name: **zfs**
- Short description: Configures ZFS datasets.
- Category: Storage
- Difficulty level: Advanced / Medium
- Ecosystem tags: ZFS; Enterprise Storage; II. Advanced Storage Orchestration (ZFS & Btrfs)
- Enterprise relevance: Manages snapshots, clones, compression, and deduplication at the filesystem level.
- Related commands: None detected in source merge
- Rarity score: Advanced / Medium

zfs

Sources: GEMINI_EXPANSION

1.18.25 zpool

- Command name: **zpool**
- Short description: Configures ZFS storage pools.
- Category: Storage
- Difficulty level: Advanced / Medium
- Ecosystem tags: ZFS; Enterprise Storage; II. Advanced Storage Orchestration (ZFS & Btrfs)
- Enterprise relevance: The core command for managing vdevs, RAID-Z configurations, and storage pool health.
- Related commands: None detected in source merge
- Rarity score: Advanced / Medium

zpool

Sources: GEMINI_EXPANSION